



SAT

MATHEMATICS

The Official

BEEC

Guide To The
Mathematics Sections

Includes:

16 Practice Tests

16 Practice Work Sheets



Contents

0	Know Thy Enemy	1
0.1	The Four Horsemen of the New SAT	1
0.2	All About the Heart	3
0.3	99 Problems but the SAT Math Ain't One	3
0.4	Travels in the Land of Advanced Math	4
0.5	Additional, but Not Quite Miscellaneous	4
0.6	Covering the Commandments	5
0.7	Last Minute Logistics and FAQ's	5
1	Let's Get Linear: All About the Algebra	7
1.1	Beginning with a Single Equation	7
1.2	Double the Trouble	11
1.3	All or Nothing ¹	13
1.4	Of Absolute Importance	15
1.5	Adding Inequalities	17
2	Let's Get Linear: Linearity Under a Different Lens	23
2.1	A Geometric Lens	23
2.2	Looking at Inequalities	25
2.3	A Real World Lens	30
3	Careers in Modeling: Being a Little Scatter Minded	35
3.1	Starting from Scratch	35
3.2	Talking Tables	40
3.3	Hysterical Histograms	41
3.4	Keeping Confident: Random Thoughts on Randomness	44
4	Ratios, Rates, Percents, Oh My!	49
4.1	Talking about Change	49
4.2	Peaceful Proportions and Radical Ratios	50
4.3	Interesting Percentages	53
4.4	Units, they Really Do Matter	57

¹Technically, I should say Infinite or Nothing, but this title sounds better

5	Advanced Math: Fun with Functions	61
5.1	The Most Important Mathematical Creature	61
5.2	Functions, Visualized	64
5.3	Looking at Quadratics	69
5.4	When Functions Collide	74
6	Advanced Math: Playing with Polynomials	79
6.1	An Aside on Exponents	79
6.2	Numbers in Disguise: Addition and Subtraction	81
6.3	Numbers in Disguise: Multiplication and Division	82
6.4	Getting a Little Rational	85
6.5	Getting a Little Radical	87
7	Plane and Simple: Angles to Triangles	91
7.1	Begin the Blitzkrieg	91
7.2	Life Begins on a Line	91
7.3	All About Angles	95
7.4	All About Triangles	97
7.5	A Touch of Trigonometry	103
8	Plane and Simple: Circles and Beyond	109
8.1	Arcs and Areas	109
8.2	Going on a Tangent	111
8.3	Adding Another Dimension	113
9	Once in a Blue Moon	117
9.1	More Advanced Circle Properties	117
9.2	More Advanced Root Theorems	120
9.3	Not so Complex Numbers	122
10	Appendix: The SAT Math Commandments	127
10.1	Heart of Algebra	127
10.2	Problem Solving and Data Analysis	127
10.3	Passport to Advanced Math	128
10.4	Additional Topics	129

Chapter 0

Know Thy Enemy

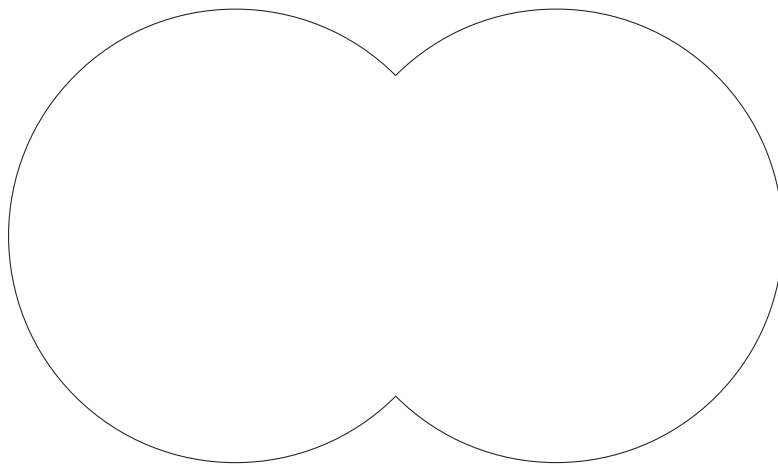
*The more things change, the more they
stay the same.*

-Jean-Baptiste Alphonse Karr

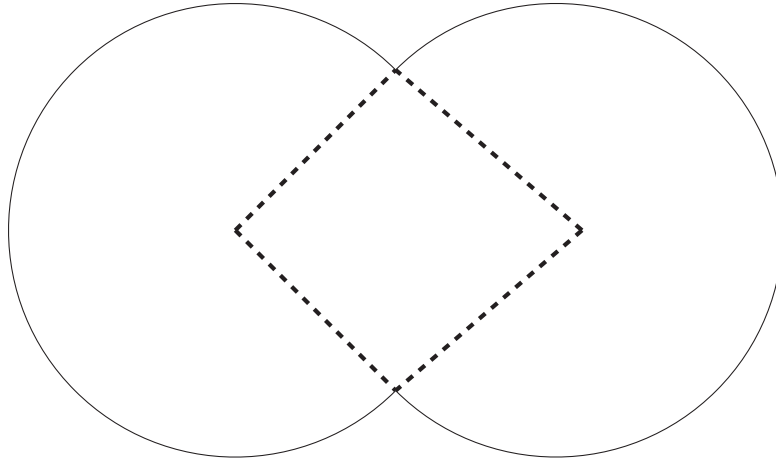
0.1 The Four Horsemen of the New SAT

Historically, the SAT Math has been a test of one's logical abilities in which many problems were solved by some "cute trick." For example,

If two 270° sectors of radius 3 are connected as shown below, then what is the area of the resulting figure?



To solve this problem, one was expected to draw in two auxillary lines:



Voila! The area is the sum of two sectors and a square. No advanced math, just a little bit of creativity. Simple and fair.

But as of 2016, the entire test has been revamped to test *content*. But what content can you test? Every student has a different pace in mathematics: some students will have Calculus under their belts, while others will be entering Algebra II. So to make the exam fair for everyone, you will only be tested on content **relevant to the real world**.

The College Board grouped the content under four categories:

- **Heart of Algebra** (32%)
- **Problem Solving and Data Analysis**(29%)
- **Passport to Advanced Math** (29%)
- **Additional Topics in Mathematics**(10%)

Passport? Heart? The new SAT sounds like more of a romantic getaway than a mathematics test. Luckily, the new SAT has written in depth about each of these new topics. 50+ pages worth. Here are the links:

- <https://collegereadiness.collegeboard.org/pdf/official-sat-study-guide-ch-19-heart-algebra.pdf>
- <https://collegereadiness.collegeboard.org/pdf/official-sat-study-guide-ch-20-problem-solving-data-analysis.pdf>
- <https://collegereadiness.collegeboard.org/pdf/official-sat-study-guide-ch-21-passport-advanced-math.pdf>
- <https://collegereadiness.collegeboard.org/pdf/official-sat-study-guide-ch-22-additional-topics-math.pdf>

But instead of having to sift through all those pages (though you definitely should), we have distilled the vital information:

0.2 All About the Heart

The **Heart of Algebra**, is the study of *linear* equations. No roots, radicals, or squares. Just simple things like

$$3x + 7 = 12.$$

Now if the world was only composed of puppy dogs and Pikachus, we would stop right there. Everyone would get an 800 on the SAT Math. But of course, like Donald Trump, we have to make things more complicated. You also have to know how to

- Solve multiple linear equations with multiple unknowns.
- Find the solution sets of linear inequalities
- Interpret linear equations and inequalities geometrically.

These will be covered in the first two chapters of the book, **Let's Get Linear: All About the Algebra** and **Let's Get Linear: Linearity Under a Different Lens**

0.3 99 Problems but the SAT Math Ain't One

Like the old SAT, the **Problem Solving and Data Analysis Section** focuses on

- Ratios
- Proportions
- Percentages
- Units

But unlike the old SAT,¹ there is also a *heavy* emphasis on

- Understanding Data Plots, Graphs, and Tables
- Modeling and Interpreting Data
- Understanding Basic Statistics²

These will be covered in two chapters, **Careers in Modeling: Being a Little Scatter Minded** and **Ratios, Rates, Proportions, Oh My!**

¹In fact, this bares some resemblance to ACT Science

²Yes, even standard deviation (though only the meaning, not the formula).

0.4 Travels in the Land of Advanced Math

The **Passport to Advanced Math** section takes the Heart of Algebra and kicks up the difficulty several notches. From linear equations, we move on to radical, rational, and polynomial functions. In particular, you need to know how

- Transform Functions.
- Solve Quadratics and Put them in Vertex Form.
- Work with Polynomials and Solve Rational and Radical Equations.

These more common advanced math topics will be covered in **Advanced Math: Fun with Functions** and **Advanced Math: Playing with Polynomials**.

You are also expected to know some rare topics that have shown up on either the Khan Academy tests or the College Board website:

- Sum of Roots and Product of Roots Theorem
- Rational Roots Theorem
- Remainder Theorem

These rare topics will be cursorily studied in the last chapter, **Once in a Blue Moon**.

0.5 Additional, but Not Quite Miscellaneous

Even though it is ok to be afraid of the unknown, it is not ok to be afraid of the **Additional Topics in Mathematics** section. This is because, to be frank, the College Board only mentions

- Coordinate Geometry
- Planar Geometry
- Complex Numbers

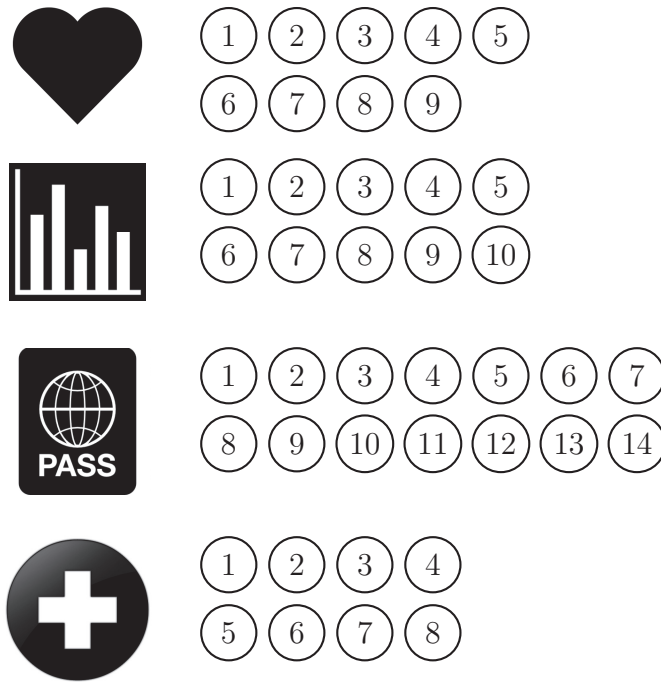
Moreover, the College Board claims that

The Additional Topics in Mathematics only comprises 6 out of the 58 questions.

Now, the author of this text is unwilling to believe there are only 6 geometry problems (in fact, a big portion of Heart of Algebra is studying lines!) But he definitely believes there is *at most* 1 question on complex numbers. Therefore, two full chapter will be dedicated to Geometry, **Plane and Simple: Angle to Triangles** and **Plane and Simple: Circles and Beyond**. Complex numbers will be another topic briefly covered in **Once in a Blue Moon**.

0.6 Covering the Commandments

If it wasn't enough to provide 50 pages of documentation, the fine people at the SAT have been generous enough to also give us a list of bullet points of each topic, which we've attached to the back of the book. It would be *remiss* of us at iQ3R to not use these bullet points to prove to you that we have hit all points above and beyond. So at the end of each chapter, we are going to include a list of what topics were covered and a progression chart:



By the time the course is finished, you would have mastered every iota of the SAT and beyond.

0.7 Last Minute Logistics and FAQ's

Now that we understand the content, let's cover the logistics. First, the test consists of **58 questions** distributed across **two sections**:

- Calculator Section
- Non-Calculator Section

45 questions will be devoted to **multiple choice** and the remaining **13 questions** are **grid-in**. It is very important to note that

You will not be given a penalty for getting a question wrong.

So do not leave any questions blank!

Finally, here are some common questions that we usually get:

- **What Calculator is Allowed on the Test?**

You can use any calculator, from scientific to graphing, as long as it doesn't have any sort of internet connection(in other words, no iPhones!). I recommend **TI-84** or **TI-89**.³

- **What Math Courses Should I have Taken?**

At the very least you should have the beginning of Algebra II, but you can get away with just having finished Algebra I, knowing a few basic trig and geometry facts, and being comfortable with the quadratic equation. We will teach you the rest.

- **Do I need Precalculus?**

Absolutely not.

- **Do I need to know Trigonometry?**

Only on the *surface level* i.e **SOHCAHTOA** and a few basic identities. No double angles, no law of cosines, no De Moivre's Theorem. Just understand the basic functions and basic identities.

- **Will this Book be Emphasizing Test Taking Techniques and Strategies ?** Absolutely!

Math theory is fun and every math instructor at iQ3R is madly in love with mathematics. But truth be told, our sole emphasis is acing the test. In fact,

We at iQ3R believe that every student can break 700.

But this will require a mixture of two things: knowledge of the basic material and the many *gaming tricks* to break everything else. In particular, pay close attention to the symbols:



The advice that we provide here are strategies and insight on common mistakes⁴ acquired over two decades of SAT Prep.

Now, without further ado, let's begin our conquest of SAT Mathematics.

³If you are also taking the ACT, it is better to get accustomed to the TI-84 as it is accepted on both exams, unlike the TI-89.

⁴Some of these notes may seem silly, but it's just like the warnings at the beginning of Pokemon Go: there is always one student who makes the mistake :P.

Chapter 1

Let's Get Linear: All About the Algebra

The journey of a thousand miles begins with a single x.

- Confucius

1.1 Beginning with a Single Equation

The most basic SAT question is solving for x in a linear equation:

Solve for x in

$$3x + 14 = 2$$

A) -4

B) -2

C) 4

D) 6

Solution: You've probably been doing this since elementary school, but let's write out the solution step by step just in case you suffer from Dory-ism.

First, isolate the x term on one side by subtracting 14:

$$3x = 2 - 14$$

Dividing both sides by 3,

$$x = \frac{-12}{3}$$

we get -4 (**A**).

But most of the time, you will be asked to solve for an expression *involving* x rather than x directly.

Given that

$$\frac{4x + 6}{2} = 5,$$

solve for $2x$.

- A) -1
- B) 0
- C) 1
- D) 2

Solution: First, multiple both sides by 2 to yield

$$4x + 6 = 10$$

Subtract 6

$$4x = 4$$

and finally divide by 2 to yield $2x = 2$ **(D)**.

With these types of question, you need to be really careful.



Make sure you submit the answer they are looking for and NOT the intermediate answer.

Since the question asks for $2x$, you need to make sure you submit this answer and NOT the value of x . It would be a shame to make such an error after doing all the work to solve for x !

Now that we're good with one variable, let's kick the difficulty up a notch and add a y :

Given that

$$3x + 9y = 6$$

find x in terms of y

- A) $6 - 3y$
- B) $2 - 3y$
- C) $6 - y$
- D) $2 - y$

Solution: Dividing both sides by 3 yields

$$x + 3y = 2$$

Now subtract $3y$ to get

$$x = 2 - 3y$$

So **(B)** is the answer.

And of course, we can make the problem more interesting by adding as many variables as we like:

If

$$8w + 2x - 4y + 4z = 12$$

find $x + 2z$ in terms of w and y

A) $3 - 4w + 2y$

B) $6 - 4w + y$

C) $6 - 4w + 2y$

D) Cannot be determined

Solution: Again, isolate the terms you are interested in (namely x and z) and put them to one side:

$$2x + 4z = 12 - 8w + 4y$$

Dividing by 2,

$$\frac{2x + 4z}{2} = \frac{12 - 8w + 4y}{2}$$

the equation reduces to

$$x + 2z = 6 - 4w + 2y.$$

Therefore, the answer is **(C)**.

Be careful with questions that have the choice *Cannot be determined*:



DO NOT Choose “Cannot be determined” just because you do not know how to solve the problem!

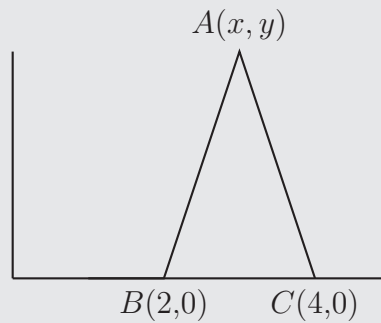
The SAT will only put an answer as *Cannot be determined* if there is a mathematical *reason* why it cannot be determined. For example, in the next section, we will have system of equations where there are *multiple* answers (and hence the original solution cannot be determined).

Here are some of the author's favorite examples of College Board problems whose solution are *Cannot be determined*.¹

Suppose 3 vertices of a parallelogram are located at $(0, 0)$, $(2, 0)$, and $(1, 1)$. Then what is the coordinate of the fourth vertex?

- A) $(1, 3)$
- B) $(1, -1)$
- C) $(2, 1)$
- D) Cannot be determined

Consider the triangle below:



If $AB = AC$, then what is the area of $\triangle ABC$?

- A) 2
- B) 4
- C) 10
- D) Cannot be determined

Suppose $\cos(x) = 0$. Then what is x ?

- A) 0°
- B) 45°
- C) 90°
- D) Cannot be determined

¹Technically, these are SAT II problems, but the author feels they are definitely SAT I valid. And as we instructors have seen, SAT problems are like the Dali Lama: they tend to be reincarnated.

1.2 Double the Trouble

From a single equation, we can take it one step further and add a second one. A *solution* is then an x, y pair that satisfies each equation in our collection.

Find the solution to the following pair of equations:

$$\begin{array}{rcl} 3 & = & 2x - y \\ 5 & = & 2x + y \end{array}$$

A) $x = 1, y = 1$

B) $x = 1, y = 2$

C) $x = 2, y = 1$

D) $x = 2, y = 2$

Solution: Take the two equations and add them together:

$$\begin{array}{rcl} 3 & = & 2x - y \\ + 5 & = & 2x + y \\ \hline 8 & = & 4x \end{array}$$

Solving for x , we get $x = 2$. Substituting back in yields $y = 1$ **(C)**.

Most of time, the trick to solving a system of equations is to

Add a multiple of one equation to the other.

In fact, here is a question from the old SAT that is completely fair game for the new SAT:

Consider the system of equations

$$\begin{array}{rcl} 3 & = & -a + b + c + d \\ 4 & = & a - b + c + d \\ 5 & = & a + b - c + d \\ 6 & = & a + b + c - d \end{array}$$

What is $a + b + c + d$?

A) 6

B) 9

C) 18

D) Cannot be determined

Solution: Again, we add all the equations together:

$$\begin{array}{rcl}
 3 & = & -a + b + c + d \\
 4 & = & a - b + c + d \\
 5 & = & a + b - c + d \\
 + \quad 6 & = & a + b + c - d \\
 \hline
 18 & = & 2a + 2b + 2c + 2d
 \end{array}$$

Dividing the last line by 2, we get

$$9 = a + b + c + d.$$

So **(B)** is the answer.

Even though adding equations is the way to go, once in a blue moon, you will need to use the method of *substitution*. Here is a problem from the **May 2016** experimental section which thwarts adding or subtracting equations:

What is the solution to the system:

$$\begin{array}{rcl}
 17 & = & 7x + 5y \\
 19 & = & 5x + 7y
 \end{array}$$

- A) $x = 1, y = 1$
- B) $x = 1, y = 2$
- C) $x = 2, y = 1$
- D) $x = 2, y = 2$

Solution: If your name was Sheldon Cooper, you would easily be able to find a multiple of the second equation to subtract from a multiple of the first. But we are not genius physicists, so instead we isolate x in the first equation

$$17 = 7x + 5y \quad \rightarrow \quad x = \frac{17 - 5y}{7}$$

and plug it into the second

$$19 = 5 \underbrace{\left(\frac{17 - 5y}{7} \right)}_x + 7y$$

Isolating y yields 2 and isolating x yields 1 **(B)**.

The above example where substitution is the key step is a rarity, and again, I've only seen it on one experimental section. So my advice is to *understand* substitution. But for the most part:



When attacking a system of equations try to solve it by adding or subtracting equations, not substitution.

Now, suppose that you forgot how to solve the system using either method. Alternatively, you can use one of the time proven SAT techniques:



Plug and chug the answer choices into the original equation.

Let's revisit the second example:

What is the solution to the system:

$$\begin{aligned} 17 &= 7x + 5y \\ 19 &= 5x + 7y \end{aligned}$$

- A) $x = 1, y = 1$
- B) $x = 1, y = 2$
- C) $x = 2, y = 1$
- D) $x = 2, y = 2$

Solution: Test each solution one at a time. When we test (A), we get

$$\begin{aligned} 17 &= 7(1) + 5(1) = 12 \\ 19 &= 5(1) + 7(1) = 12 \end{aligned}$$

which is absolutely incorrect. But when we try (B), we get

$$\begin{aligned} 17 &= 7(1) + 5(2) = 17 \\ 19 &= 5(1) + 7(2) = 19 \end{aligned}$$

which is indeed true (B).

1.3 All or Nothing²

In the preceding section, we looked at system of equations that had exactly one solution. However, no one ever said this had to be the case! For any system of equations we can have³

²Technically, I should say Infinite or Nothing, but this title sounds better

³How about exactly 2 solutions or 5 solutions? Well, there is a *geometric* reason why this is never the case and we will investigate this in the next section.

- 0 solutions
- 1 solution
- Infinitely many solutions

How does a system of equations yield 0 solutions? This phenomena arises when the equations are *inconsistent*, meaning that together, they form an impossible statement.

As an analogy, suppose we can deduce the following two statements:

Klaus hates onions.

Klaus does not hate onions.

But it cannot be the case that Klaus both hates and does not hate onions, this is purely *contradictory*! And the contradictory statement we are going to create when working with inconsistent system of equations is showing that two distinct numbers are equal.

How many solutions does the system

$$\begin{aligned} 5 &= 2x + 2y \\ 2 &= x + y \end{aligned}$$

have?

- A) 0
- B) 1
- C) 2
- D) Infinitely Many

Solution: Suppose there exists such an x and y that makes the above true. Then subtracting double the second equation from the first yields

$$1 = 0$$

But 1 is not equal to 0! So there are no solutions **(A)**.

On the opposite side of the spectrum, we can have infinitely many solutions. This occurs when we have redundant equations. For example,

$$\begin{aligned} 8 &= 2x + 2y \\ 12 &= 3x + 3y \\ 16 &= 4x + 4y \\ 20 &= 5x + 5y \\ &\vdots \end{aligned}$$

are really the same as

$$4 = x + y$$

And as you can see, there are infinitely many pairs like

$$\begin{array}{lcl} x = 1 & , & y = 3 \\ x = 2 & , & y = 2 \\ x = 2.01 & , & y = 1.99 \\ & \vdots & \end{array}$$

that satisfies this equation.

Suppose x and y are chosen so they satisfy the system

$$\begin{array}{rcl} 3 & = & x + y \\ 6 & = & 2x + 2y \end{array}$$

What was the original x and y pair?

- A) $x = 1, y = 2$
- B) $x = 3, y = 3$
- C) $x = 5, y = 1$
- D) Cannot be determined

Solution: By plug and chug, you may suspect that (A) is the answer. But the second equation is unnecessary since it is double the first, hence any x and y that satisfies

$$3 = x + y$$

is a solution. Therefore, the original pair chosen to satisfy this equation cannot be determined (**D**).

1.4 Of Absolute Importance

You've definitely seen the absolute value symbol $|x|$ in Pre-Algebra, but you may have forgotten some facts as you rarely use it in school. So let's do a brief review.

Formally, on the College Board website, they define $|x|$ as the piecewise function

$$|x| = \begin{cases} -x & x \leq 0 \\ x & x > 0 \end{cases}$$

But this rigorous definition is unnecessary. For the SAT, what is really important is the knowledge that absolute value outputs the *magnitude* of a number. For example,

$$\begin{array}{rcl} |-5| & = & 5 \\ |0| & = & 0 \\ |12| & = & 12 \end{array}$$

Solving an absolute value equation is pretty straightforward:

For what values of x is

$$|x - 3| = 5$$

- A) $x = -2$
- B) $x = 8$
- C) $x = -2$ or $x = 8$
- D) $x = -8$ or $x = 2$

Solution: Note that the absolute value of a number is only equal to 5 when the original number is 5 or -5 . Hence, we have two possibilities:

$$\begin{aligned}x - 3 &= 5 \\x - 3 &= -5\end{aligned}$$

Solving both equations yield two possibilities, $x = 8$ or $x = -2$ (C).

Be careful! In middle school, one is often taught the rote technique of just equating the absolute value to $\pm$ the right hand side. But this is absolutely false! Consider:

Find all x such that

$$|x - 2| = -1$$

- A) $x = 1$
- B) $x = 3$
- C) $x = 1$ or $x = 3$
- D) No solutions

Solution: Remember,



The absolute value of a number can never be negative!

Hence, there are no solutions (D).

1.5 Adding Inequalities

Now that we've mastered solving

- One Linear *Equation*
- A System of Linear *Equations*
- Absolute Value *Equations*

it is time to move on to inequalities:

- One Linear *Inequality*
- A System of Linear *Inequalities*
- Absolute Value *Inequalities*

The move from equalities to inequalities may seem scary, but the inequality case is just an application of the equality case. The only real difference is that with inequalities, your solution tends to be a *set*.

Let's start with one linear inequality:

Which set contains all real numbers x such that

$$7x + 2 \leq 3x + 6?$$

- A) $(-\infty, 1]$
- B) $(-\infty, 2]$
- C) $(1, \infty)$
- D) $(2, \infty)$

Solution: Just like for an equation, isolate x on one side to get

$$4x \leq 4$$

Dividing both sides by 4, you get the set $x < 1$, or all x in the set $(-\infty, 1]$ **(A)**.

Now be very careful! Suppose instead we moved the x to the right hand side of the inequality to get

$$-4 \leq -4x$$

and divide by -4 to get

$$1 \leq x$$

This is absolutely incorrect.



When you divide or multiply an inequality by a negative number, you must flip the $>$ sign!

This is because negative numbers that are bigger in “size ⁵” are actually smaller. Certainly,

$$3 \leq 5$$

and by definition of “smaller” in terms of negative numbers,

$$-5 \leq -3$$

Now suppose we have two inequalities. Just like in the two equation case, we require that x satisfies BOTH inequalities.

Find the set containing all x such that

$$\begin{aligned} 3x + 1 &\leq 7 \\ -2x + 3 &\leq 1 \end{aligned}$$

- A) $[-2, -1]$
- B) $[-2, 1]$
- C) $[1, 2]$
- D) $[1, 5]$

Solution: Isolating x in the first equation yields

$$x \leq 2$$

and in the second,

$$x \geq 1$$

So x needs to satisfy BOTH

$$x \leq 2 \quad \text{AND} \quad x \geq 1$$

This means x has to be at most 2 and at least 1, or in other words, x is any real number in $[1, 2]$ (C).

Be careful: don't just look at the 2 and 1 of $x \leq 2$ and $x \geq 1$ and say say $[1, 2]$. You really need to take a moment to stop and think about the *meaning*. Otherwise, you will get a completely wrong answer. Here's an example:

⁵Precisely I should say magnitude.

Find the set containing all x such that

$$\begin{aligned} 3x + 3 &\leq 12 \\ 2x + 4 &\leq 14 \end{aligned}$$

- A) $(-\infty, 3]$
- B) $(-\infty, 5]$
- C) $[-5, -3]$
- D) $[3, 5]$

Solution: Again, by isolating x on one side, we get the pair

$$\begin{aligned} x &\leq 3 \\ x &\leq 5 \end{aligned}$$

Certainly if $x \leq 3$, then $x \leq 5$. Hence the answer is just $x \leq 3$ or $(-\infty, 3]$ **(A)**.

Lastly, let's cover absolute value inequalities.

The trick to solve any absolute value inequality is to

- Solve the corresponding absolute value **equation** for 2 solutions.⁶
- The solution of the absolute value **inequality** is either between the 2 solutions of the equation case or outside.

Expanding on the last bullet point, if we have an equation of the form

$$|\dots| \leq \dots$$

then the set of all solutions is *in between* the two solutions of the absolute value equation.



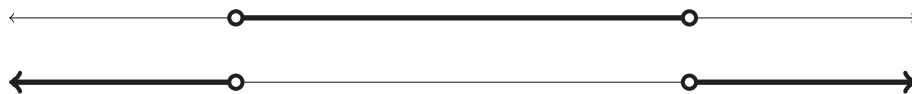
And if we have an equation of the form

$$|\dots| \geq \dots$$

then the set of all solutions is *outside* the two solutions of the absolute value equation.



If $\geq$ and $\leq$ is replaced with $>$ and $<$, then we do not include the solutions of the absolute value equation in our solution of the absolute value inequality:



⁶Let's only cover the 2 solution case since the other 2 are trivial.

Find the set of all solutions to

$$|2x + 1| \leq 5$$

- A) $(-2, 3)$
- B) $[-3, 2]$
- C) $(-3, 2)$
- D) $[-2, 3]$

Solution: Solve the equation

$$|2x + 1| = 5$$

to get $x = 2$ or $x = -3$. Since we have a $\leq$ the solution set of the inequality is in between these two solutions, or $[-3, 2]$ **(B)**.

Again, if you are not sure, just plug in points!

Find the set of all solutions to

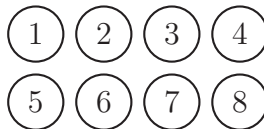
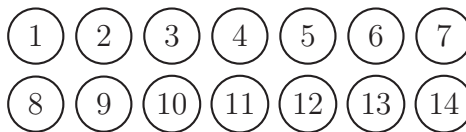
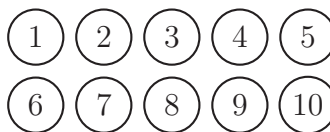
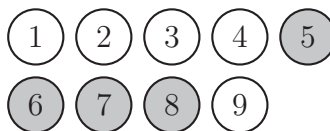
$$|3x + 6| > 9$$

- A) $(-1, 5)$
- B) $(-5, 1)$
- C) $(-\infty, -5) \cup (1, \infty)$
- D) $(-\infty, -1) \cup (5, \infty)$

Solution: Looking at choice (A), we can try plugging in 0. Since $3 \cdot 0 + 6 = 6$ is not greater than 9, (A) cannot be the answer since it contains 0. But (B) also contains 0, leaving just (C) and (D). Notice that (C) contains 2 whereas (D) does not, so trying 2, we get $3 \cdot 2 + 6 = 12$ which is indeed greater than 9. Hence, **(C)** is the answer.

Commandments Covered

- ♡5. Create, solve, and interpret systems of linear inequalities in two variables.
- ♡6. Create, solve, and interpret systems of two linear equations in two variables.
- ♡7. Algebraically solve linear equations (or inequalities) in one variable.
- ♡8. Algebraically solve systems of two linear equations in two variables.



Chapter 2

Let's Get Linear: Linearity Under a Different Lens

*If you listen to the math that's all around you,
you will know things you never knew, that you never knew.*

- Sohcahtoa

2.1 A Geometric Lens

One of the biggest changes to the SAT is its *philosophy*: not only is the algebra important, but also the *interpretation* of the algebra. To quote the College Board SAT Math page:

Questions on the Math Test are designed to mirror the problem solving and modeling you'll do in:

- *College math, science, and social science courses*
- *The jobs that you hold*
- *Your personal life.*

So now that we've mastered the *algebra* of linear equations, let's interpret the algebra in a different context, beginning with a *geometric* one. Pay close attention as a huge portion of the next chapter on modeling relies on this interpretation.

Starting with a linear equation, say

$$3x + y = 7$$

we can solve for y in terms of x :

$$y = 7 - 3x$$

When we let y vary as a function of x and graph all the corresponding (x, y) solutions, this forms a *line*.

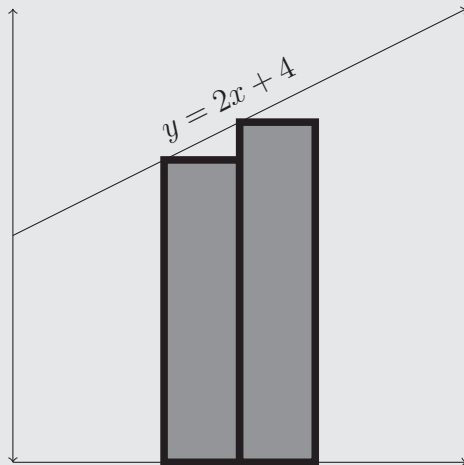
No surprise, right? You've probably done this since middle school. But you still have to know the formula:

$$y = mx + b$$

Just in case you forgot:

- Slope $m = \frac{\Delta y}{\Delta x}$
- y -intercept b is the value obtained from setting $x = 0$.

In the diagram below, two rectangles are inscribed under a line:



If the base of both rectangles are 1, then how much more area does the larger rectangle have?

- A) 2
- B) 4
- C) 6
- D) Cannot be determined

Solution: Let the left rectangle have height h . Since the base of the left rectangle is 1 and slope is 2, we can use the slope formula to calculate the change in height:

$$2 = \frac{\Delta y}{1}$$

Since $\Delta y = 2$, the height of the second rectangle is $h + 2$. Hence the difference of areas is

$$(h + 2) \cdot 1 - h \cdot 1 = 2$$

So **(A)** is the answer.

Now suppose that you have a system of two equations and two unknowns:

$$\begin{aligned} x - y &= -1 \\ x + y &= 2 \end{aligned}$$

For an (x, y) pair to satisfy just one of these equations, it needs to lie on the corresponding line. So to satisfy *both* of these equations, they need to lie on *both* lines. Or in another words, the solution(s)

of a system of equations is the *intersection* of the lines.

On the geometry of a system of equations, here are a few concepts that the SAT *loves* to test: a system of equations will have:

- Infinitely many solutions if they describe the same line.
- No solutions if they describe two parallel lines.
- Exactly one solution if they describe two non-parallel lines.

$$\begin{aligned} 2x + 6y &= 7 \\ x + 3y &= 1 \end{aligned}$$

Which of the following describes the system of equations above?

- A) Two lines that intersect at only one point
- B) Two parallel but distinct lines
- C) Two identical and overlapping lines
- D) Two perpendicular lines

Multiplying the second equation by 2 yields

$$\begin{aligned} 2x + 6y &= 7 \\ 2x + 6y &= 2 \end{aligned}$$

But this describes a system with no solution, hence we have two parallel but distinct lines **(B)**.

2.2 Looking at Inequalities

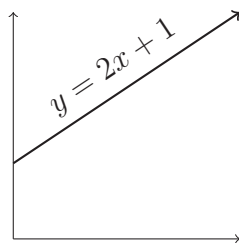
In the preceding chapter, we first studied the algebra of *equations* and then the algebra of *inequalities*. Again, we are going to progress from the geometric interpretation of *equations* to the geometric interpretation of *inequalities*. And just like last time, we are going to look at

- One Linear *Inequality*
- A System of Linear *Inequalities*
- Absolute Value *Inequalities*

Starting with one linear inequality, say

$$y \leq 2x + 1$$

we first draw the corresponding line $y = 2x + 1$:

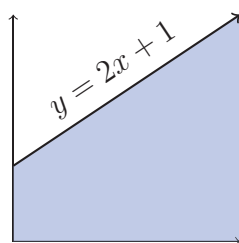


The solution set is then either the region above the line or below the line. And what you'll need to remember is

$y \leq \dots$ means **shade down**.

$y \geq \dots$ means **shade up**.

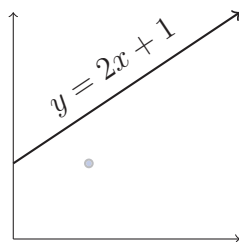
So in our case, we shade down:



In case you don't believe me, plug in points: consider

$$y \leq 2x + 1.$$

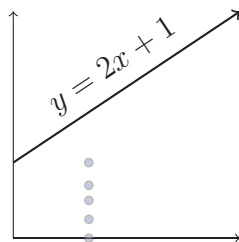
$(1, 1)$ is a solution since $1 \leq 3$



But we can also see that

$$\begin{aligned} (1, .5) \\ (1, .25) \\ (1, 0) \\ \vdots \end{aligned}$$

are solutions as well:

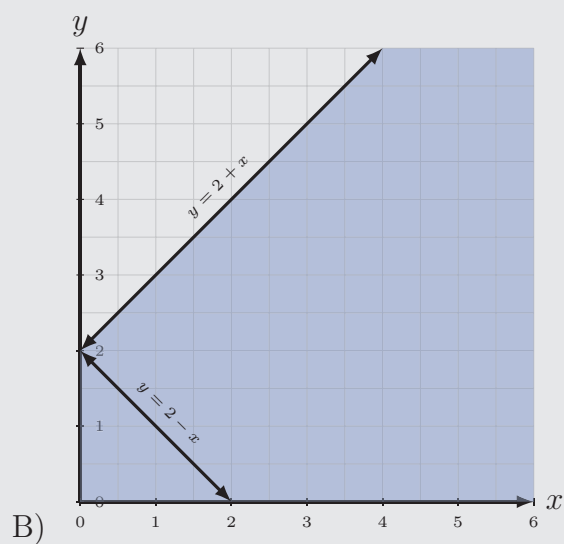
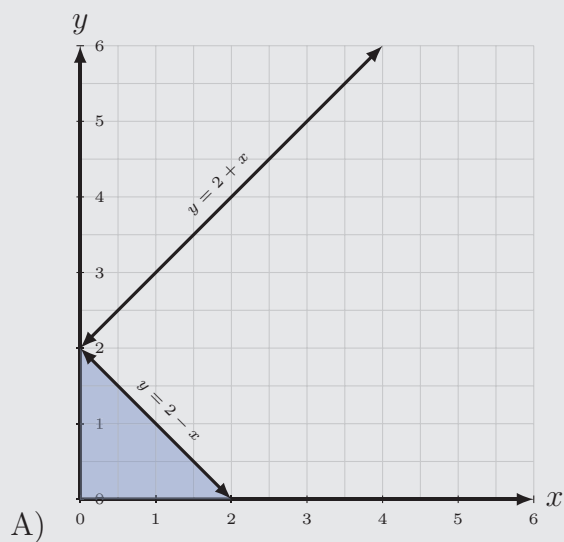


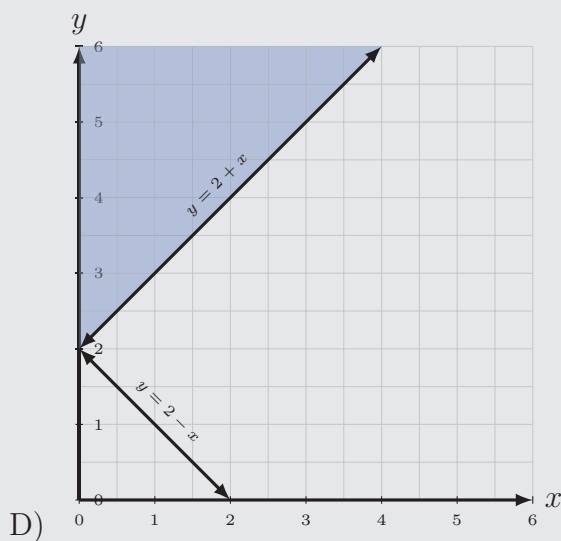
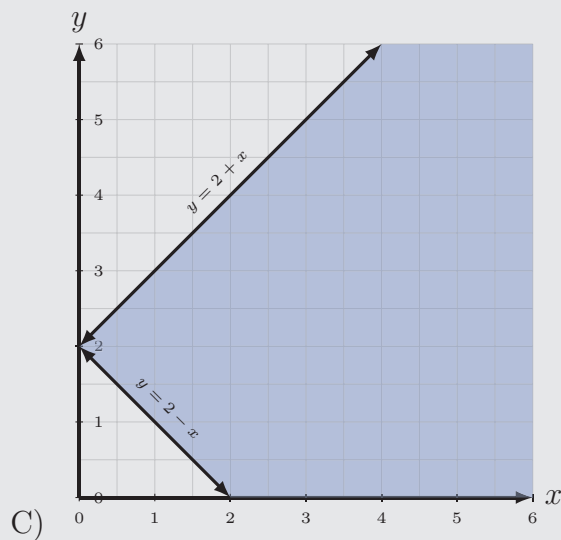
Since the y coordinate can be as negative as we want, the solutions need to lie *below* the line.

For a system of inequalities, the extension is easy: just take the *intersection* of the shaded regions:

Which of the graphs below show the set of all solutions of

$$\begin{aligned} y - x &\leq 2 \\ x &\geq 2 - y \end{aligned}$$





Solution: Isolating y on one side yields

$$\begin{aligned} y &\leq 2 + x \\ y &\geq 2 - x \end{aligned}$$

The top inequality means shade everything below the line $y = 2 + x$ and the bottom inequality means graph above $y = 2 - x$. Together, the intersection of their solutions form the region in between two lines (C).

Finally, let's cover absolute value inequalities. The expression

$$|x - a|$$

can be interpreted geometrically as

The distance from x to a .

Therefore, for a fixed a and r ,

$$|x - a| \leq r$$

describes the set of all x that are within a distance r from a .

Which of the following equations represent the set of all points that are within a distance 2 away from 3?

A) $|x - 3| \leq 2$

B) $|x - 2| \leq 3$

C) $|x - 3| \geq 2$

D) $|x - 2| \geq 3$

Solution: Using the formula, we get the answer **(A)**. But suppose we did not know the formula. Then we can use our SAT Math Mantra:



Try plugging and chugging values into the answer choices!

Since 4 is within a distance of 2 from 3, we can try plugging it in first:

A) $|x - 3| \leq 2 \rightarrow 1 \leq 2$

B) $|x - 2| \leq 3 \rightarrow 2 \leq 3$

C) $|x - 3| \geq 2 \rightarrow 1 \geq 2$

D) $|x - 2| \geq 3 \rightarrow 2 \geq 3$

But only (A) and (B) are true, leaving only one of these as the possible answer. Now, let's plug in a value that *doesn't* work, say 0. Then,

A) $|x - 3| \leq 2 \rightarrow 3 \leq 2$

B) $|x - 2| \leq 3 \rightarrow 2 \leq 3$

Since the inequality is false in only **(A)**, this is our answer.

2.3 A Real World Lens

To reiterate, one of the major changes to the SAT is its emphasize on real world applications. In this section, we are going to cover how to

- Extract a *linear equation or inequality model* from *real world data*.
- Extract *real world data* from a *linear equation or inequality model*.

Let's begin:

The simplest type of question is one in which you identify a scenario with a single linear function. The trick here is to associate quantity change with *slope* and the initial value with the *y*-intercept:

Klaus's bank account currently has \$12. If the money in his account increases by \$5 a day, then which function can be used to model how much money is in his account after d days?

- A) $P(d) = 5d + 12$
- B) $P(d) = 12d + 5$
- C) $P(d) = 5d - 12$
- D) $P(d) = 12d - 5$

Solution: Using $y = mx + b$, we know $b = \$12$ since the bank account is \$12 on day 0. Moreover, each day the money increases by \$5, hence $m = \$5$. Therefore the answer is (A).

Careful with these modeling questions though:



In a linear modeling question, do not assume that a given value is the y -intercept!

Carlo drives a taxi and charges \$10 for the first mile and \$5 for every mile after the first. Which of the following functions can be used to model the total cost of a m mile trip?

- A) $C(m) = 10m + 5$
- B) $C(m) = 5m + 10$
- C) $C(m) = 5m + 5$
- D) $C(m) = 5m - 5$

Solution: The answer is not (B)! You must take into account that when $m = 1$, the cost is \$10. Hence, the answer is (C).

Regardless of how easy a modeling question looks, I do have the following recommendation:



After constructing a model, verify that it is correct by plugging in the given data.

The SAT also tends to combine these translation problems with solving a system of equations. Here's a classic that you can find in any SAT text:

A cost of a pizza is composed of a base cost plus an additional charge for each topping. If a 3-topping pizza costs \$11 and a 5-topping pizza cost is \$15. How much is an 8-topping pizza?

- A) 5
- B) 20
- C) 21
- D) 26

Solution: Do not add the cost of a 3 topping pizza with the cost of a 5 topping pizza! That's nuttier than squirrel poo. Instead, we translate each line as a linear equation:

$$\begin{aligned} 3t + b &= 11 \\ 5t + b &= 15 \end{aligned}$$

Subtracting the equations, we get

$$-2t = -4$$

and hence $t = 2$. Plugging back into the first equation, we can solve for $b = 5$. The cost of a 8 topping pizza is therefore $8 \cdot 2 + 5 = 21$ (C).

Here is another classic SAT problem that has carried over to the new SAT. This one involves translating a real world scenario into an absolute value inequality:

In order to get into Stanford, one must have a SAT Math score between 720 and 800. Which of the following inequalities represents all permissible SAT Math scores to get into Stanford?

- A) $|x - 720| \leq 80$
- B) $|x - 760| \leq 80$
- C) $|x - 760| \leq 20$
- D) $|x - 760| \leq 40$

Solution: First, we find the midpoint of 720 and 800 as

$$\frac{720 + 800}{2} = 760$$

The set of all acceptable values is then all x that are 40 points from 760, or

$$|x - 760| \leq 40.$$

Therefore, the answer is **(D)**.

Now that we've gone through several examples in which we look at a real world phenomena to get a model, let's go the other direction: let's start with a model and use that to understand a real world scenario. Again, the typical trick is that if we have an equation of the form

$$y = mx + b,$$

we interpret m as the *change* and b as the starting value. Here's the last problem from the College Board's *Heart of Algebra* pdf:

A voter registration drive was held in Town Y. The number of voters, V , registered T days after the drive began can be estimated by the equation

$$V = 3,450 + 65T$$

What is the best interpretation of the number 65 in this equation?

- A) The number of registered voters at the beginning of the registration drive
- B) The number of registered voters at the end of the registration drive
- C) The total number of voters registered during the drive
- D) The number of voters registered each day during the drive

Solution: Since 65 is the slope, increasing T by 1 causes V to increase by 65. Hence, the answer is **(D)**.

And of course, our model can involve more than one equation:

Suppose the price, in dollars, of sending an object weighing x pounds through Fedex and UPS can be modeled, respectively, by the equations

$$\begin{aligned}y &= 2x + 10 \\y &= 3x + 5\end{aligned}$$

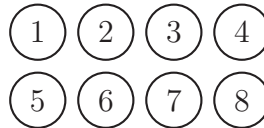
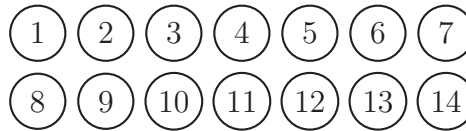
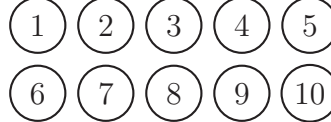
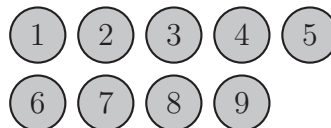
What is the best interpretation of the solution to this system of equations?

- A) The amount of dollars Fedex makes more than UPS
- B) The amount of dollars UPS makes more than Fedex
- C) The weight such that the price of sending the object through either company is free.
- D) The weight such that the price of sending the object through either company is the same.

Solution: A solution means that the x coordinate and y coordinate are the same in either equation. Hence, the weight and price are the same **(D)**.

Commandments Covered

- ♡1. Create, solve, or interpret a linear expression or equation in one variable that represents a context.
- ♡2. Create, solve, or interpret linear inequalities in one variable that represent a context.
- ♡3. Build a linear function that models a linear relationship between two quantities.
- ♡4. Describe a linear relationship that models a context using either an equation in two variables or function notation.
- ♡9. Understand connections between algebraic and graphical representations.



Chapter 3

Careers in Modeling: Being a Little Scatter Minded

Those who don't understand histograms are doomed to repeat the SAT.

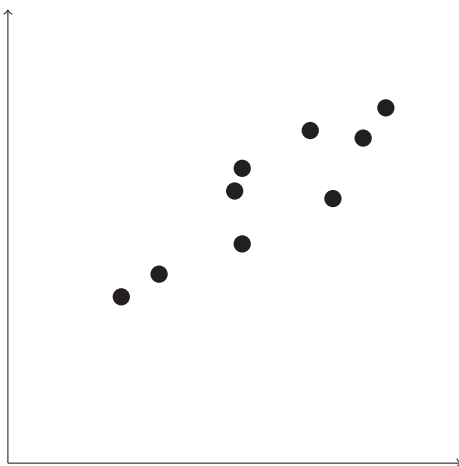
- Edmund Burke

3.1 Starting from Scratch

In the preceding section, we've seen linear equations and inequalities applied to modeling real world scenarios. We either started with a full mathematical model or we took a real world scenario and directly translated it into mathematics. But in the real world, we often don't have full equations. Instead, we are given a bunch of measured points like

(1.5, 2.2)	(2, 2.5)
(3.1, 3.9)	(3, 3.6)
(4.7, 4.3)	(5, 4.7)
(4.3, 3.5)	(3.1, 2.9)

which we can then graph in a scatterplot:



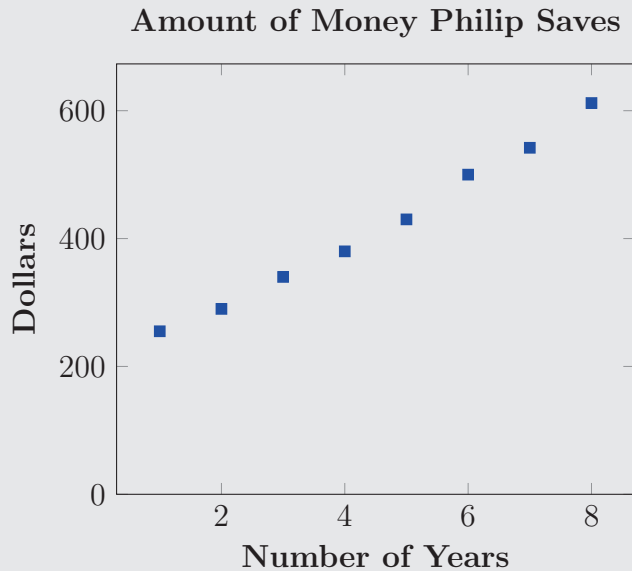
Now there are *many* questions that the SAT can ask you about scatterplots:

- Find the line of best fit

- Interpret the line of best fit
- Use the line of best fit to predict some unknown value

In fact, according to the College Board, the scatterplot question is the most common type of multiple part question.

The amount of money Philip saves each year after he started working for iQ3R is shown below:



Which of the following linear models best fit the above data?

- A) $y = 200 + 5x$
- B) $y = 200 + 50x$
- C) $y = 100 + 100x$
- D) $y = 100 + 10x$

Assuming a linear model, which of the following is the most reasonable approximation for the amount of money Philip's saves 20 years from when he first started working for iQ3R?

- A) \$900
- B) \$1200
- C) \$2000
- D) \$3000

Solution: Eyeballing the points, we see that (2, 300) and (4, 400) lie on the graph. Using the slope formula, we get

$$m = \frac{\Delta y}{\Delta x} = \frac{400 - 300}{4 - 2} = 50.$$

Then, plugging in the first point into $y = mx + b$, we get

$$300 = 50 \cdot 2 + b$$

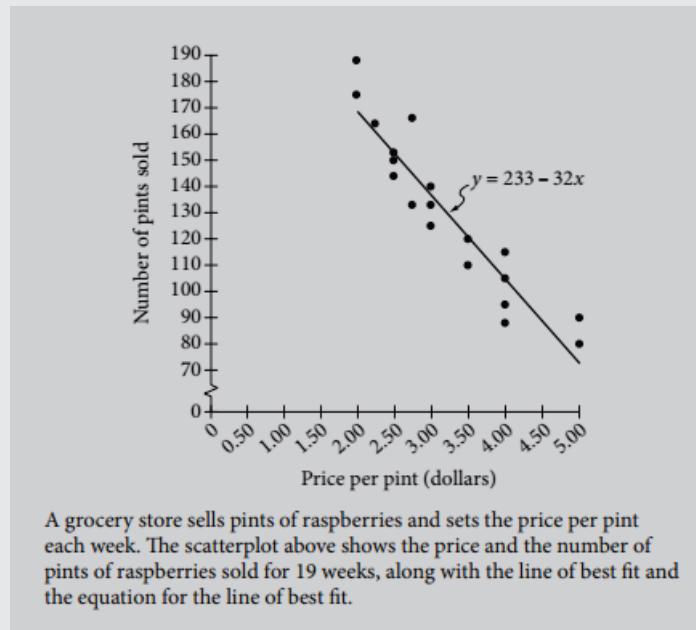
This gives us $b = 200$ and hence the line of best fit is $y = 200 + 50x$ **(B)**. We can then use this model to solve the second question by plugging in $x = 20$:

$$50 \cdot 20 + 200 = 1200$$

Hence, Philip saves \$1200 in 20 years.

Just for fun, let's even do the multipart question from the College Board's Data Analysis pdf:

The following series of questions use the graph below:



According to the line of best fit, how many pints of raspberries would the grocery store expect to sell in a week when the price of raspberries is \$4.50 per pint?

- A) 86
- B) 88
- C) 89
- D) 92

⁰In the pdf, this “question” mostly talked about the types of questions that could be asked. I just took the liberty of creating the actual questions :).

For how many of the 19 weeks shown was the number of pints of raspberries sold greater than the amount predicted by the line of best fit?

- A) 8
- B) 9
- C) 10
- D) 12

What is the best interpretation of the meaning of the slope of the line of best fit?

- A) The price per pint is \$32.
- B) For every 32 pints sold, the price of per pint increases by \$1 dollar.
- C) For every \$1 increase in pint cost, the number of pints sold increase by 32.
- D) For every \$1 increase in pint cost, the number of pints sold decrease by 32.

What is the best interpretation of the meaning of the y-intercept of the line of best fit?

- A) The cost of a single pint of raspberries.
- B) The starting price of a pint of raspberries.
- C) The minimal number of pints sold.
- D) The number of pints sold if the raspberries were free.

Solution: For the first question, we just need to input 4.5 for x . This gives us

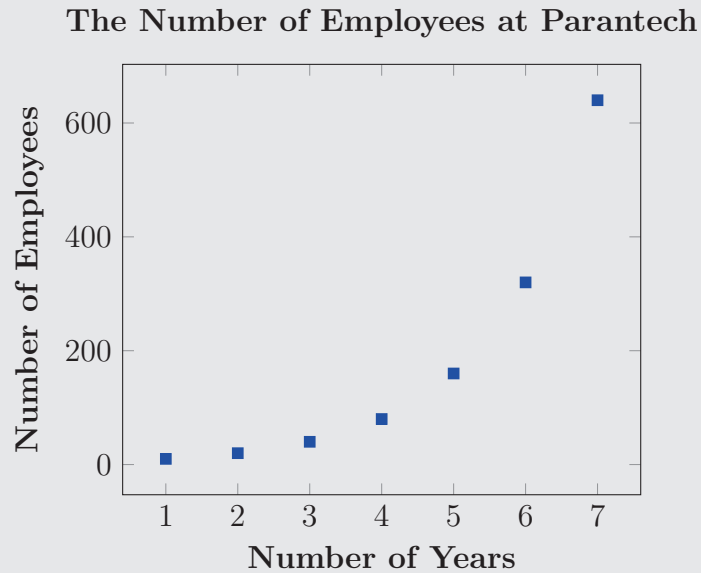
$$233 - 32 \cdot 4.5 = 89$$

So (C) is the first answer. For the second question, we need to find the number of dots *strictly above* the line of best fit. There are 8 of them (A). As for the third question, the slope is -32. Since the y -axis represents the number of pints sold and the x -axis represents price, the numbers of pints decrease by 32 when the cost increases by \$1 (D). Finally, for the last question, the y -intercept occurs when x is 0. Since x represents the price, the y -intercept represents the number of pints sold when each pint is free (D).

Of course no one said that every modeling question is going to be linear. It can also be exponential or quadratic. Luckily, the rules are the same:

- Find the model of best fit
- Interpret the model of best fit
- Use the model to predict some unknown value

Parantech is an electric car company in Silicon Valley. The number of employees at Parantech each year since the company's inception is shown below:



Which of the models below best fit the data?

- A) $y = 5^x$
- B) $y = 2 \cdot 5^x$
- C) $y = 2^x$
- D) $y = 5 \cdot 2^x$

Solution: Looking at year 6 to year 7, the number of employees increase from approximately 300 to 600. Hence, each year, the number of employees is doubling, so our equation should be of the form

$$y = P \cdot 2^x$$

Plugging in (6, 200),

$$200 = P \cdot 2^6$$

Solving for P , we get $P \approx 5$. Hence **(D)** is the answer.

3.2 Talking Tables

Instead of storing data on a scatterplot, we can count the frequency of each data point's occurrence and store the frequencies in a table. And from this table, you should know how to compute a few basic quantities:

- **Mean:** The *average* of the data.
- **Median:** The *middlemost* value when ordered. In the case of an even number of data points, it is the average of the middle two.
- **Mode:** The most *frequent* value.
- **Range:** The *difference* between the highest and lowest data value.

The SAT Math scores of a class of students were stored in the frequency table below:

Score	Frequency
660	1
670	2
700	3
790	2
800	1

How much greater is the mean compared to the median?

- A) 0
- B) 20
- C) 40
- D) 60

Solution: The median is the fifth data point, so it is 700. The mean is calculated by multiplying both columns together, adding the values, and then dividing by the total number of students:

$$\frac{660 + 670 \cdot 2 + 700 \cdot 3 + 790 \cdot 2 + 800}{9} = 720$$

The answer is therefore $720 - 700 = 20$ (**B**).

On the new SAT, you also need to know how to compute a not so basic operation: **conditional probability**.

Don't worry, you don't need to know the formula. To solve a conditional probability problem, simply focus on *part* of the table. Here is the example from the College Board's *Data Analysis* pdf:

A store is deciding whether to install a new security system to prevent shoplifting. The security manager of the store estimates that 10,000 customers enter the store each week, 24 of whom will attempt to shoplift. The manager estimates the results of the new security system in detecting shoplifters would be as shown in the table below.

	Alarm sounds	Alarm doesn't sound	Total
Customer attempts to shoplift	21	3	24
Customer does not attempt to shoplift	35	9941	9976
Total	56	9944	10000

According to the managers estimates, if the alarm sounds for a customer, what is the probability that the customer did not attempt to shoplift?

- A) 0.03%
- B) .35%
- C) .56%
- D) 62.5%

Solution: Since we are *given* the alarm sounds for a customer, we only need to look at 56 customers, **not** 10000. We then calculate the probability as

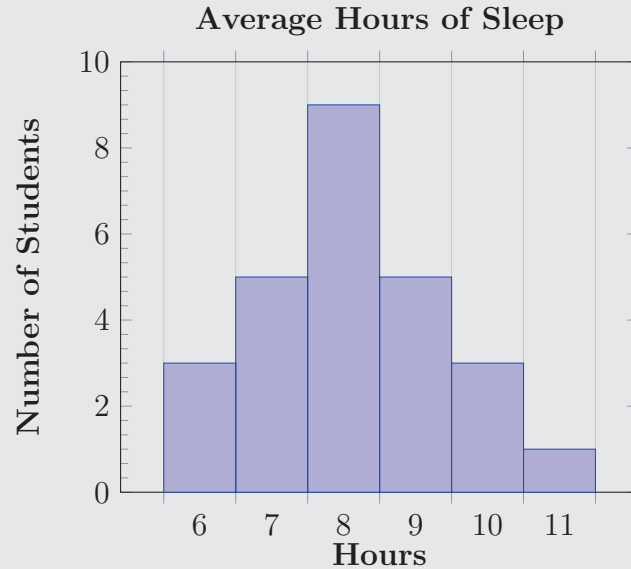
$$\frac{\text{Number of customers who did not shoplift } \textit{given} \text{ the alarm sounds}}{\text{Number of customers where the alarm sounds}} = \frac{35}{56} = .625$$

The correct answer is therefore **(D)**.

3.3 Hysterical Histograms

We can also look at a table visually in terms of a histogram. And just like before, you still need to know how to compute mean, median, and mode.

In a survey on how many hours of sleep one gets in a night, the following data was recorded below:



What is the probability that a randomly chosen student sleeps more than the mean?

- A) $\frac{1}{2}$
- B) $\frac{11}{26}$
- C) $\frac{9}{26}$
- D) $\frac{1}{26}$

Solution: First, we calculate the number of students in the survey:

$$\text{Number of Students} = 3 + 5 + 9 + 5 + 3 + 1 = 26.$$

The mean is then

$$\frac{6 \cdot 3 + 7 \cdot 5 + 8 \cdot 9 + 9 \cdot 5 + 10 \cdot 3 + 11 \cdot 1}{26} \approx 8.1$$

Since there are 9 people who sleep more than 9 hours a day, the probability is $\frac{9}{26}$ (C).

Careful: the answer is not necessarily $\frac{1}{2}$!



It need not be the case that half the data is greater than the mean!

We can easily have an outlier that pushes the mean upwards.¹ For example, if one was to look at the daily income of four Palo Alto residents and Bill Gates:

\$100
\$300
\$400
\$500
\$1000000

the mean is massive

$$\frac{1000000 + 1300}{5} = 200260$$

and only one man is above the mean.

There is, however, one intimidating histogram topic that has been added to the new SAT. And that is *standard deviation*.

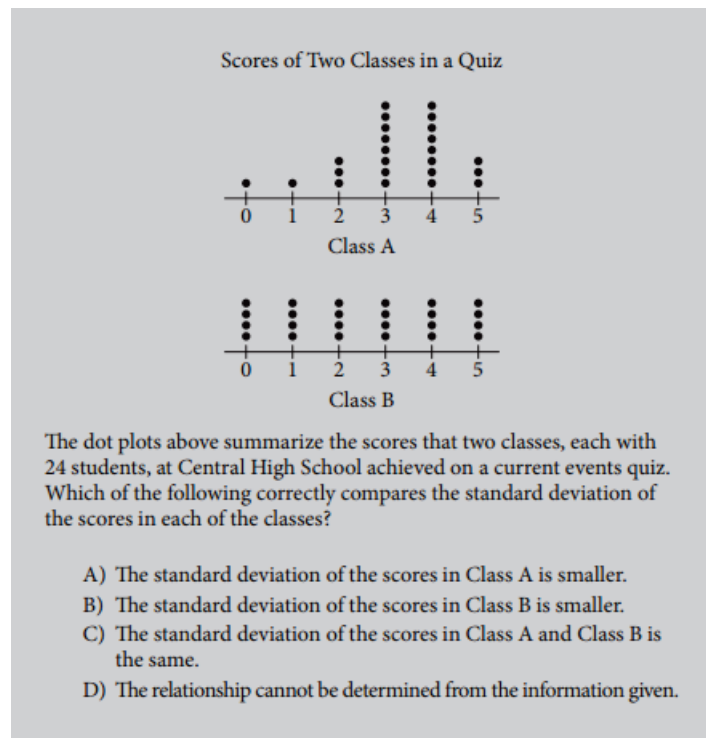
The standard deviation formula is like some of Donald Trump's ideas. Absolutely horrendous:

$$\sigma = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2}$$

Luckily, you will never be tested on the actual calculation of the standard deviation, only its *meaning*. Just know that the standard deviation is a measurement of *spread* and that

The greater the spread of data, the greater the standard deviation.

Seriously, that's all you need to know. If you don't believe me, here is the collegeboard's standard deviation example problem:



¹Just like how there is always one kid on a test that messes up the curve.

Solution: The second histogram is more spread out whereas the first histogram is clustered more around 3 and 4. So the answer is **(A)**. Done.

So just to reiterate,



The greater the spread,
the greater the standard deviation.

3.4 Keeping Confident: Random Thoughts on Randomness

Another new concept on the SAT is that of *random sampling*. A random sampling is a random sample of a large population; the theory is that if you study this sample, you can hypothesize facts about the entire population. For example, one can give a survey to thirty random students at Gunn to guess the average number of hours a Gunn student gets.

Now don't worry, I know you've never studied random sampling: there is a high probability² that you never took Stats AP or even seen words like *margin of error* or *confidence interval*. But all you have to know is three simple facts.

Here's the first:

When one conducts a random sampling, it needs to be *random*.

Otherwise, you *cannot* generalize your findings to the entire population.

From a recent survey given to all Gunn students, it was that discovered that the average GPA is 3.8. Which of the following is an appropriate conclusion.

- A) It is plausible that the average student GPA is 3.8
- B) It is plausible that the average student GPA is between 3.6 and 4.0
- C) Gunn students are smarter than most students in America.
- D) None of the above.

Solution: If the students were randomly selected from a large sample of all students, then *(A)* could be the answer. However, they were all selected from Gunn, and as we all know, Gunn students are smarter than the average bear. Hence, the answer is **(D)**.

Secondly,

Just because you see a pattern from a sample, you can not make a definite conclusion.

²In fact, I am 95% confident.

Or in other words,



Correlation does not imply causation!

Here is a College Board question that directly tests this idea:

A community center offers a Spanish course. This year, all students in the course were offered additional audio lessons they could take at home. The students who took these additional audio lessons did better in the course than students who didn't take the additional audio lessons. Which of the following is an appropriate conclusion?

- A) Taking additional audio lessons will cause an improvement for any student who takes any foreign language course.
- B) Taking additional audio lessons will cause an improvement for any student who takes a Spanish course.
- C) Taking additional audio lessons was the cause of the improvement for the students at the community center who took the Spanish course.
- D) No conclusion about cause and effect can be made regarding students at the community center who took the additional audio lessons at home and their performance in the Spanish course.

Solution: Choice (A), (B), and (C) all state a causation. But you cannot assume this! Just because the days you get a 100 are the same as the days that it rains in Calcutta, you cannot assume they are related. Hence the answer is (D).

The last thing you need to know about random sampling is:

The expression *the average is A with a 95% confidence and margin of error of E* means that it is *plausible* that the average is between $A - E$ and $A + E$.

Let's end the chapter with one more College Board example:

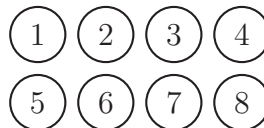
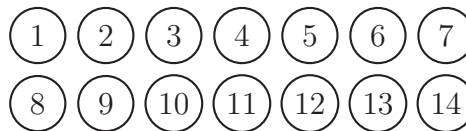
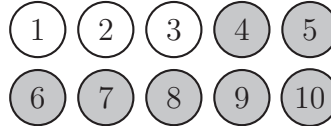
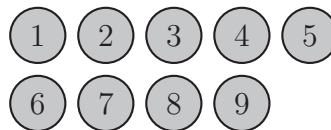
A quality control researcher at an electronics company is testing the life of the companys batteries in a certain camera. The researcher selects 100 batteries at random from the daily output of the batteries and finds that the average life of the batteries has a 95% confidence interval of 324 to 360 camera pictures. Which of the following conclusions is the most reasonable based on the confidence interval?

- A) 95% of all the batteries produced by the company that day have a life between 324 and 360 pictures.
- B) 95% of all the batteries ever produced by the company have a life between 324 and 360 pictures.
- C) It is plausible that the true average life of batteries produced by the company that day is between 324 and 360 pictures.
- D) It is plausible that the true average life of all the batteries ever produced by the company is between 324 and 360 pictures.

Solution: (A) and (B) are just silly, we can't know for sure that every battery life will lie within this fixed range. Since we are talking about daily output, the answer is **(C)**.

Commandments Covered

- D4. Given a scatterplot, use linear, quadratic, or exponential models to describe how the variables are related. Select the equation of a line or curve of best fit; interpret the line in the context of the situation; or use the line or curve of best fit to make a prediction.
- D5. Use the relationship between two variables to investigate key features of the graph.
- D6. Compare linear growth with exponential growth.
- D7. Use two-way tables to summarize categorical data and relative frequencies, and calculate conditional probability, association of variables, or independence of events.
- D8. Make inferences about population parameters based on sample data. The sample statistics may mention confidence intervals and measurement error that the student should understand and make use of, but need not calculate.
- D9. Use statistics to investigate measures of center of data and analyze shape, center, and spread. The measures of center include mean, median, and mode, and the measures of spread range (may be calculated) and standard deviation (non-calculated).
- D10. Evaluate reports to make inferences, justify conclusions, and determine appropriateness of data collection methods. The reports may consist of tables, graphs, or text summaries.



Chapter 4

Ratios, Rates, Percents, Oh My!

Not understanding proportions is just asking for a disaster of epic proportions.

- Idiom

4.1 Talking about Change

Now that we've finished talking about the *Data Analysis* portion of the SAT's *Problem Solving and Data Analysis* category, let's now focus on the *Problem Solving* side.

One of the most basic questions on the SAT *Problem Solving* category is that which involves *rate*. Most of the time, you will calculate rate as

$$\text{Rate} = \frac{\text{Distance}}{\text{Time}}$$

Or generally,

$$\text{Rate} = \frac{\text{Quantity}}{\text{Time}}$$

And if you are given a rate, of course you can calculate:

$$\text{Quantity} = \text{Rate} \times \text{Time}$$

Suppose Tiffany can travel 20 miles in 15 minutes. How many hours does it take her to travel 200 miles?

- A) 2
- B) 2.5
- C) 100
- D) 120

Solution: Before you begin solving this problem, you should take note:



Make sure your answer is in the correct units!

Since the problem is asking for the number of hours, we should convert the 15 minutes into .25 hours. Applying our rate formula, we see the rate is

$$\frac{20 \text{ miles}}{.25 \text{ hours}} = \frac{80 \text{ miles}}{1 \text{ hour}}$$

To solve for the time it takes to drive 200 miles, we set up the equation

$$\frac{80 \text{ miles}}{1 \text{ hour}} \cdot t = 200$$

Dividing both sides by 80, we get $t = 2.5$ **(B)**.

One rate property that you should know is that rates combine *additively*. For example, if one person can complete a job at a rate of R_1 and another person can complete a job at a rate of R_2 , then *together* they work at a rate of $R_1 + R_2$.

Printer A can print 20 pages in 15 minutes and Printer B can print 30 pages in 20 minutes. Working together, how many hours does it take to print 595 pages?

- A) 2
- B) 2.5
- C) 3
- D) 3.5

Solution: Converting the minutes to hours and adding the rates together, the combined rate is

$$\frac{20 \text{ pages}}{.25 \text{ hours}} + \frac{30 \text{ pages}}{\frac{1}{3} \text{ hours}} = \frac{170 \text{ pages}}{1 \text{ hour}}$$

Solving the equation

$$\frac{170 \text{ pages}}{1 \text{ hour}} \cdot t = 595,$$

we get $t = 3.5$ **(D)**.

4.2 Peaceful Proportions and Radical Ratios

Most things in the world are in *proportion*. For example, suppose producing 1 pound of oatmeal cookies requires 2 dozen eggs:

$$\frac{1 \text{ pound}}{2 \text{ dozen}}$$

Then logically, 2 pounds of oatmeal would require 4 dozen eggs. But notice that the proportion is maintained:

$$\frac{1 \text{ pound}}{2 \text{ dozen}} = \frac{2 \text{ pound}}{4 \text{ dozen}}$$

So if you wanted to see how many eggs are required in, say, 5.2 pounds of oatmeal, we just need to solve the proportion

$$\frac{1 \text{ pound}}{2 \text{ dozen}} = \frac{5.2 \text{ pound}}{x \text{ dozen}}$$

Too easy? Let's do the example on the College Board's *Problem Solving and Data Analysis* pdf:

On an architect's drawing of the floor plan for a house, 1 inch represents 3 feet. If a room is represented on the floor plan by a rectangle that has sides of lengths 3.5 inches and 5 inches, what is the actual floor area of the room in square feet?

- A) 17.5
- B) 51.0
- C) 52.5
- D) 157.5

Solution: Since 1 inch represents 3 feet, we can solve for the number of feet that 3.5 inches represents by setting up the proportion

$$\frac{1 \text{ inch}}{3 \text{ feet}} = \frac{3.5 \text{ inch}}{x \text{ feet}}$$

Solving for x yields 10.5 feet. Likewise, we can solve for the number of feet that 5 inches represent by using the proportion

$$\frac{1 \text{ inch}}{3 \text{ feet}} = \frac{5 \text{ inch}}{x \text{ feet}}.$$

This yields 15 feet. Hence the total area is $10.5 \times 15 = 157.5$ square feet **(D)**.

One particular type of proportion that is especially emphasized¹ on the new SAT is that of **Population Density**:

$$\text{Population Density} = \frac{\text{Total Population}}{\text{Total Area}}$$

In the port city of Magnolia, the population density is 320 people per square mile. If the city is 50 square miles in area, then how many people live in Magnolia?

- A) 16000
- B) 17000
- C) 18000
- D) 20000

¹In fact, it is a commandment

Solution: We set up the proportion

$$\frac{320 \text{ people}}{1 \text{ square mile}} = \frac{x \text{ people}}{50 \text{ square mile}}$$

Solving for x yields $x = 320 \cdot 50 = 16000$ people **(A)**.

You also need to know the jargon *directly proportional* and *inversely proportional*.²

We say that two quantities x and y are *directly proportional* if their quotient is always equal to a constant (just like a normal proportion):

$$\frac{x}{y} = k$$

Two quantities x and y are *inversely proportional* if x is proportional to the *inverse* of y :

$$\frac{x}{\frac{1}{y}} = k$$

Or equivalently, their product is equal to a constant:

$$xy = k$$

Suppose a person's IQ is inversely proportional to the number of hours spent playing Hearthstone. If a person with an IQ of 120 plays 4 hours of Hearthstone, then what is the IQ of someone who plays 12 hours of Hearthstone?

- A) 10
- B) 30
- C) 40
- D) 60

Solution: We first solve for the proportionality constant k by setting up the equation

$$(\text{IQ}) \cdot (\text{Hours of Hearthstone}) = 120 \cdot 4 = k$$

So $k = 480$. Using this k , we then solve

$$(\text{IQ}) \cdot 12 = 480$$

to find that the IQ is 40 **(C)**.

²Oddly enough, on the College Board *Problem Solving and Data Analysis* pdf, only direct proportionality is taught. Since inverse proportionality has often appeared on the old SAT and I feel that it is also fair game, I have included it here.

Like proportions, a *ratio* is another way to relate two quantities. We use the notation

$$a : b$$

to mean that for every a objects, you have b objects. So if I say the ratio of boys to girls in a class is $3 : 4$, then for every 3 boys, there are 4 girls.

If the ratio of boys to girls in a class of 70 students is $3 : 4$, how many more girls are there than boys?

- A) 5
- B) 10
- C) 30
- D) 40

Solution: For every 3 boys, there are 4 girls. Hence, they are $3x$ boys and $4x$ girls, meaning there are a total of $7x$ students. Since,

$$7x = 70$$

we can deduce $x = 10$ making $4 \cdot 10 - 3 \cdot 10 = 10$ more girls than boys. **(B)**.

Here is a classic twist to the same problem:

Suppose the rate of boys to girls in a class is $3 : 5$. Which cannot be the number of students in a class?

- A) 40
- B) 60
- C) 80
- D) 200

Solution: Since the rate is $3 : 5$, there are $3x$ boy and $5x$ girls. Hence there are $3x + 5x = 8x$ students in class. The only number that is not a multiple of 8 is 60**(B)**.

4.3 Interesting Percentages

Just like the original SAT, the new SAT will test you on percentages.

Let's start off with the basics:

To compute p **percent of** x , we use

$$\frac{p}{100} \cdot x$$

To increase x by p percent, we use

$$x \left(1 + \frac{p}{100} \right)$$

or equivalently³

$$x + \frac{p}{100}$$

To decrease x by p percent, we use

$$x \left(1 - \frac{p}{100} \right)$$

or equivalently

$$x - \frac{p}{100}$$

Suppose the cost of a car increased by 10% and then decreased by 10%. If the final price of the car was \$9900, what was the original price in dollars?

A 10000

B 9900

C 9000

D 1000

Solution: Be careful:



Increasing some number by a percent and then decreasing by the same percent need not yield the original number!

Solution: When we increase the initial price x by 10%, we get

$$1.1x.$$

Decreasing $1.1x$ by 10% yields

$$.9 \cdot (1.1)x = .99x$$

Solving

$$.99x = 9900$$

we get 10000 (**A**).

You also need to be careful when working with questions that mix percents and variables: often times you will have to multiply your answer by 100.

³They are obviously the same, but the equivalent form has better intuition: you are increasing the original amount via addition by some percent

s is what percent of n ?

- A) $\frac{n}{s}$
- B) $\frac{s}{n}$
- C) $\frac{100s}{n}$
- D) $\frac{100n}{s}$

Solution: The answer is not $\frac{s}{n}$. This only give the correct “decimal.” But the question is asking for the actual percentage, so we need to multiply by 100. Hence, the answer is $\frac{100s}{n}$ (C).

So the moral of the story is



When working with questions involving percent, make sure your answer is not off by a factor of 100.

Alternatively, if you did not know how to solve the preceding problem directly, you can also plug in numbers. And my advice is:



When it comes to percentages, the best numbers to plug in are 100 and multiples of 10.

Let's try this mantra in the preceding problem.

Solution: Let $n = 100$, $s = 20$. Then S should be 20 percent of 100. Plugging in, we get:

- A) $\frac{n}{s} \rightarrow 5$
- B) $\frac{s}{n} \rightarrow .2$
- C) $\frac{100s}{n} \rightarrow 20$
- D) $\frac{100n}{s} \rightarrow 500$

The answer is therefore (C).

Now that we covered the basics of percentages, let's cover the not-so basics.

One of the new fangled SAT topics is **compound interest**:

$$P \left(1 + \frac{r}{n} \right)^{nt}$$

where P is the initial amount, r is the rate, t is the number of years, and n is the number of time one compounds in a year.

Don't be intimidated by this formula. It's just multiple applications of percent increase: consider the simple case of $n = 1$. We start with the initial amount P and increase our wealth by r percent at the end of each year:

$$\begin{aligned} P \\ P(1+r) \\ P(1+r)(1+r) \\ P(1+r)(1+r)(1+r) \end{aligned}$$

Or generally, in t years,

$$P \underbrace{(1+r)(1+r)\dots(1+r)}_{t \text{ years}} = P(1+r)^t$$

Now, if we are compounding n times a year, we adjust the rate by dividing by n

$$r \rightarrow \frac{r}{n}$$

and for each of the t years, we compound n times for a total of nt times:

$$P \left(1 + \frac{r}{n} \right)^{nt}$$

Let's try out a compound interest problem from the College Board's *Problem Solving and Data Analysis* pdf:

A bank has opened a new branch and, as part of a promotion, the bank branch is offering \$1,000 certificates of deposit at an interest rate of 4% per year, compounded semiannually. The bank is selling certificates with terms of 1, 2, 3, or 4 years. Which of the following functions gives the total amount, A , in dollars, a customer will receive when a certificate with a term of k years is finally paid?

- A) $A = 1000(1.04k)$
- B) $A = 1000(1 + 0.08k)$
- C) $A = 1000(1.04)^k$
- D) $A = 1000(1.02)^{2k}$

Solution: Plugging in $n = 2$, $r = .04$, $P = 1000$ and $t = k$ into the compound interest formula yields (D).

You will also be tested on the formula for **simple interest**:

$$A = P(1 + rt)$$

Chances are, you've probably never studied this in school. But don't worry: it is really *simple*.

Suppose you leave P dollars at an interest rate of r . Simple compound interest just adds r percent of your *original deposit* each year.

$$\begin{aligned} P \\ P + Pr \\ P + Pr + Pr \\ P + Pr + Pr + Pr \end{aligned}$$

Or generally, in t years,

$$P + \underbrace{(Pr + Pr + Pr \dots + Pr)}_{t \text{ years}} = P(1 + rt)$$

Again, let's try another College Board Problem:

A bank has opened a new branch and, as part of a promotion, the bank branch is offering \$1,000 certificates of deposit at simple interest of 4% per year. The bank is selling certificates with terms of 1, 2, 3, or 4 years. Which of the following functions gives the total amount, A , in dollars, a customer will receive when a certificate with a term of k years is finally paid?

- A) $A = 1000(1.04k)$
- B) $A = 1000(1 + 0.04k)$
- C) $A = 1000(1.04)^k$
- D) $A = 1000(1 + 0.04^k)$

Solution: Simply apply the simple interest formula for $P = 1000$ and $r = .04$ (**B**).

Be careful of one simple misunderstanding:



Simple interest is not the same as compound interest in the case $n = 1$.

If it were, then (C) would be the answer, which it clearly is not.

4.4 Units, they Really Do Matter

As one final note, let's talk about *units*. Throughout this chapter, we've emphasized making sure one is in the correct units. But in addition to this, as a surprising new SAT topic, you also need to know your metric system:

Prefix	Factor
kilo	1000
deci	.1
centi	.01
milli	.001

Let's do a College Board Problem:

Scientists estimate that the Pacific Plate, one of Earth's tectonic plates, has moved about 1,060 kilometers in the past 10.3 million years. What was the average speed of the Pacific Plate during that time period, in centimeters per year?

- A) 1.03
- B) 10.3
- C) 103
- D) 1030

Solution: 1060 kilometers is the same as 1,060,000 meters. But 1,060,000 meters is the same as 106,000,000 centimeters. The average speed is then

$$\frac{106,000,000 \text{ centimeters}}{10,300,000 \text{ years}} \approx \frac{10.3 \text{ centimeters}}{\text{year}}$$

So the answer is **(B)**.

Commandments Covered

- D1. Use ratios, rates, proportional relationships, and scale drawings to solve single- and multistep problems.
- D2. Solve single- and multistep problems involving percentages.
- D3. Solve single- and multistep problems involving measurement quantities, units, and unit conversion. Calculate density; or use the concept of density to solve a multistep problem.



12345

6789



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Chapter 5

Advanced Math: Fun with Functions

People go to the gym to work out their quads. People go to SAT class to work out their quadratics.

- Anonymous SAT Math Teacher

5.1 The Most Important Mathematical Creature

One of the most important objects in Algebra and, indeed, the SAT is that of a *function*. A function is a *mapping* and the typical notation we use to represent this mapping is $f(x)$:

$$\begin{array}{rcl} x & \leftarrow & \text{input} \\ f(x) & \leftarrow & \text{output given input } x \end{array}$$

Given a function, the set of all possible inputs¹ is known as the domain.

On the SAT, a function is undefined when

- There is a fraction with denominator 0.
- There is a negative number under the square root²

What is the domain of

$$\frac{1}{\sqrt{x}}?$$

- A) $(0, \infty)$
- B) $[0, \infty)$
- C) $(1, \infty)$
- D) All real numbers.

¹In upper level math courses, this is not quite the definition of domain as we have a notion of a “restricted domain,” but this is the SAT, not rocket science :P

²Yes, there are complex numbers on the new SAT, but I guarantee, if a question asks about domain, it will only be considering the reals.

Solution: The function is undefined at 0:

$$\frac{1}{\sqrt{0}}$$

Moreover, $\sqrt{x}$ is undefined when $x < 0$. Hence the domain is $(0, \infty)$ **(A)**.

The *range* is the set of all possible outputs. In general, it is much more difficult to identify the range than it is to identify the domain. But you should be able to *verify* if a particular value is in the range.

Which of the following is not in the range of

$$x^2 + 1$$

- A) 0
- B) 1
- C) 37
- D) 101

Solution: x^2 is minimally 0 so $x^2 + 1$ is minimally 1. Hence **(A)** is the answer.

The *intercepts* are the values where the function intersects an axis. Precisely,

- The ***x*-intercepts**, or *zeros* is where a function intersects the *x*-axis. This occurs at the *x* values where $f(x) = 0$.
- The ***y*-intercept** is where a function intersects the *y*-axis. This occurs at $f(0)$.

Which function has *x*-intercepts at 2 and 3?

- A) $f(x) = (x - 3)(x + 2)$
- B) $f(x) = (x + 3)(x + 2)$
- C) $f(x) = (2x - 6)(3x + 3)$
- D) $f(x) = (2x - 6)(3x - 6)$

Solution: Plugging in $x = 2$ and $x = 3$, we only get $f(x) = 0$ in both cases in choice **(D)**.

Finally, you should know how to *create* functions. And one way to do so is via *composition*³:

$$f \circ g$$

This just means plug in g for all the x terms of f :

³Even though I loathe the term, students tend to remember composition as another name: *fog*

If

$$\begin{aligned} f(x) &= 3x + 5 \\ g(x) &= 2x + 1 \end{aligned}$$

Compute $f \circ g$.

- A) $6x + 8$
- B) $6x + 11$
- C) $8x + 6$
- D) $8x + 11$

Solution: Plugging g into the x terms of f yields

$$f \circ g = 3(2x + 1) + 5 = 6x + 3 + 5 = 6x + 8$$

Thus, the answer is (A).

Make sure that you plug g into f and not the other way around. This is because:



$f \circ g$ is NOT the same thing as $g \circ f$!

If

$$\begin{aligned} f(x) &= 2x + 5 \\ g(x) &= 4x - 1 \end{aligned}$$

Compute $(f \circ g) - (g \circ f)$.

- A) -16
- B) 16
- C) -19
- D) 19

Solution: First, we compute $f \circ g$ as

$$f \circ g = 2(4x - 1) + 5 = 8x + 3$$

We then compute

$$g \circ f = 4(2x + 5) - 1 = 8x + 19$$

The difference is then

$$8x + 3 - (8x + 19) = -16$$

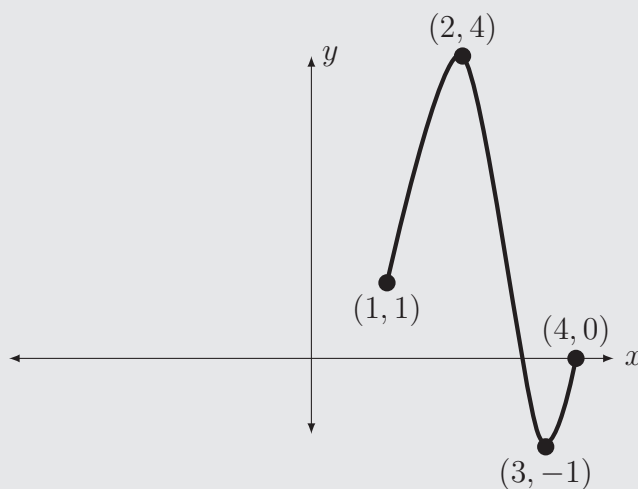
So (A) is the answer.

5.2 Functions, Visualized

One of the things we like to do with functions is to graph them on the Cartesian plane:

$$(x, y)$$

This should be no surprise, you've probably been doing this since Middle School.⁴ And if you are given the full visual graph of the function, you should be able to identify the range, domain, and intercepts:



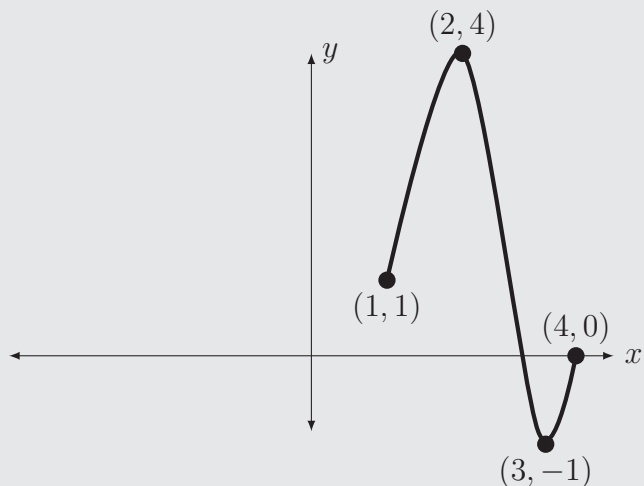
Consider the graph of the function above. What is the range of this function?

- A) $[1, 4]$
- B) $[0, 1]$
- C) $[-1, 4]$
- D) $[3, 4]$

Solution: Looking at the y -coordinates of the highest and lowest points, we see that the range is $[-1, 4]$.

Too easy? Let's try another common variation of the same graph.

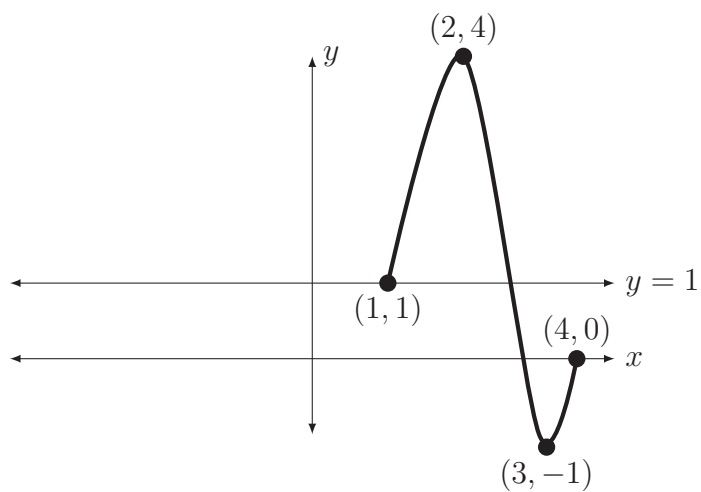
⁴In fact, we've done this multiple times in Chapter 2.



Consider the graph of the function above. For how many values of x does $f(x) = 1$?

- A) 1
- B) 2
- C) 3
- D) 4

Solution: Draw the line $y = 1$:



We see that the line $y = 1$ intersects the graph twice (**B**).

You also need to know how certain function manipulations transform a graph. First let's review *translations*. If h, k are positive numbers, we can shift the graph vertically or horizontally using the following operations:

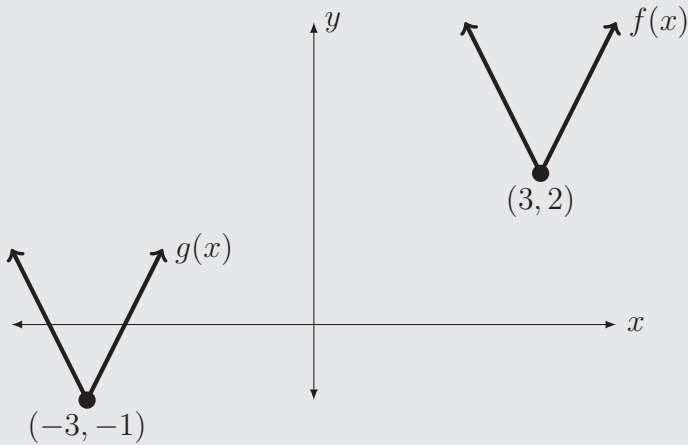
Operation	How the Graph is affected Visually
$f(x) + k$	Vertical Shift up by k
$f(x) - k$	Vertical Shift down by k
$f(x + h)$	Horizontal Shift left by h
$f(x - h)$	Horizontal Shift right by h

Be careful,



$f(x - h)$ is a right shift for positive h !

I know this sounds unintuitive, but think of the $-h$ as a *delay* by h . We have to wait h units in order to return to our normal input.



Consider the graphs of f and g above. If $g(x) = f(x + h) + k$, then what is hk ?

- A) -18
- B) -9
- C) 9
- D) 18

Solution: To get g from f , we have to move f to the left by 6 and down by 3. Therefore $h = 6$ and $k = -3$. Therefore, the answer is -18 (**A**).

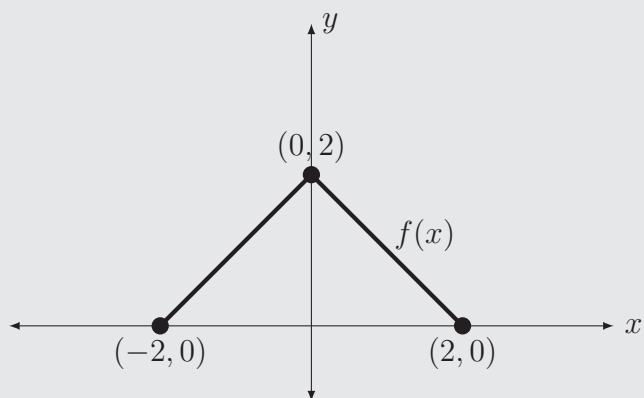
Next, let’s review *dilations*. This time, let a be positive:

Operation	How the Graph is affected Visually
$af(x)$	Vertical Scale by a factor of a
$f(ax)$	Horizontal Scale by a factor of $\frac{1}{a}$

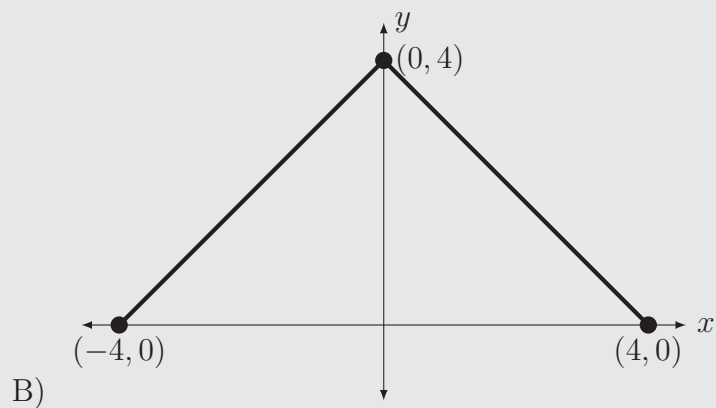
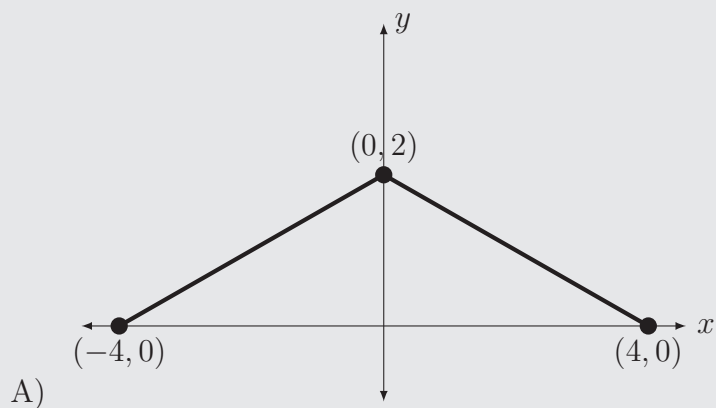
So, for example,

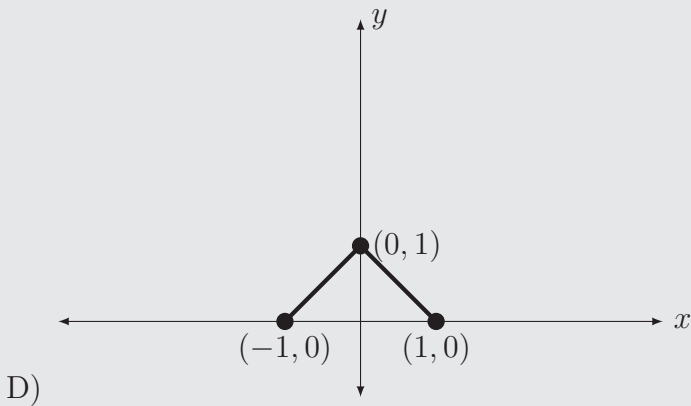
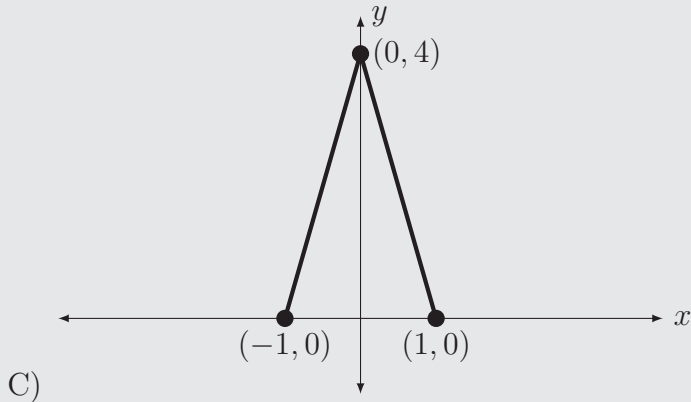
$$f(2x)$$

scales the function horizontally by a factor of $\frac{1}{2}$. Or in other words, the function is being squeezed by a factor of 2.



Which of the following is a graph of $2f(2x)$?



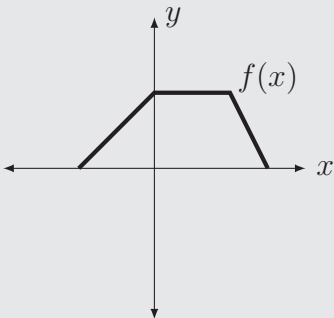


Solution: By our transformation properties, we are squeezing the graph horizontally by a factor of 2 while stretching vertically by a factor of 2 (**C**).

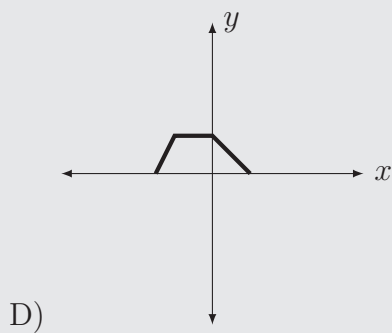
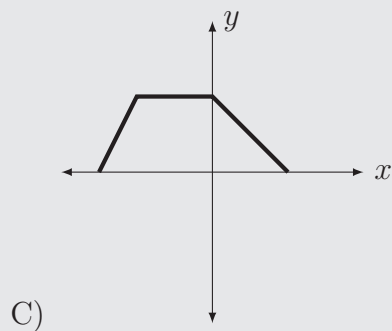
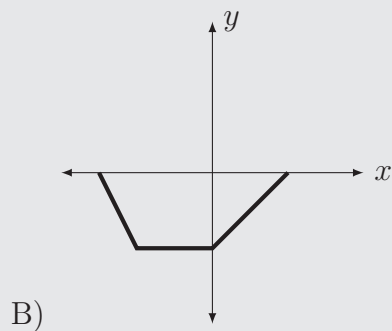
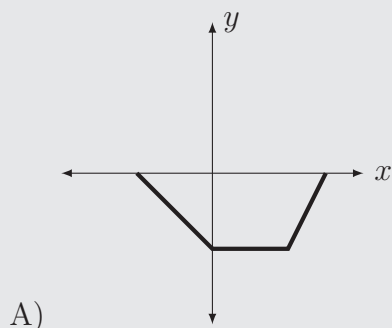
Lastly, let’s talk about *reflections*:

Operation	How the Graph is affected Visually
$f(-x)$	Left-Right flip
$-f(x)$	Up-Down flip

Consider $f(x)$ below:



Which of the following is a graph of $-f(-x)$



Solution: The graph is being flipped up-down and left-right. The answer is then **(B)**.

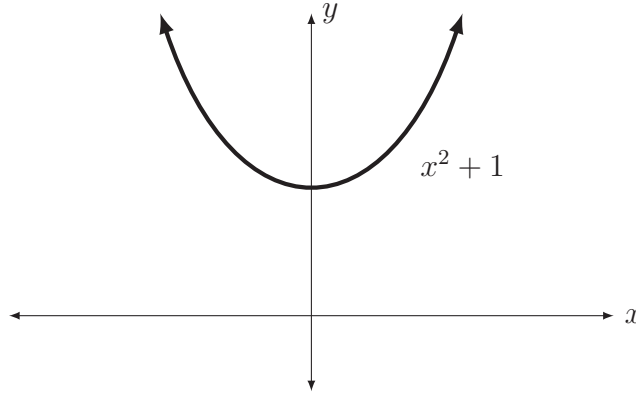
5.3 Looking at Quadratics

Now that we've gone over visualizing functions, let's look at one of the prettiest types of functions: *quadratics*.

A quadratic is a function of the form

$$ax^2 + bx + c$$

Visually, a quadratic has a parabolic shape:



There are tons of amazing facts about quadratics, but you will mainly be tested on two key concepts:

- Rewriting a Quadratic in terms of the **Vertex**
- **Quadratic Equation**

Let's go through each bullet point.

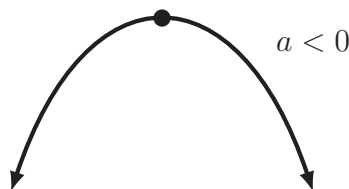
The **vertex** of a quadratic is the point at which the parabola begins to *turn*. And for any quadratic

$$ax^2 + bx + c$$

if $a > 0$, the **minimum** of the function is achieved at the vertex:



Whereas if $a < 0$, the **maximum** is achieved at the vertex:



To write a quadratic in terms of its vertex, we use

$$a(x - h)^2 + k$$

where (h, k) are the coordinates of the vertex.

To compute the coordinates of the vertex, you *could* memorize that the x -coordinate of the vertex occurs at

$$x = \frac{-b}{2a}$$

However, I feel that SAT questions involving the vertex can be solved using the fact that

The x coordinate of the vertex occurs in between the two roots.

Let's do an example from the fourth College Board SAT test:

$$f(x) = (x + 6)(x - 4)$$

Which of the following is an equivalent form of the function f above in which the minimum value of f appears as a constant or coefficient?

- A) $f(x) = x^2 - 24$
- B) $f(x) = x^2 + 2x - 24$
- C) $f(x) = (x - 1)^2 - 21$
- D) $f(x) = (x + 1)^2 - 25$

Solution: Since the roots occur at -6 and 4 , the x -coordinate of the vertex occurs at the midpoint, -1 . At this point we must be careful:



The x coordinate of the vertex is where the minimum⁵ is achieved whereas the y coordinate is the minimum value!

So to get the minimum value, we have to plug -1 into $f(x)$ to get -25 . The equation is then $f(x) = (x + 1)^2 - 25$ **(B)**.

Too easy? Let's do one that involves a real world scenario. This is from the College Board's *Passport to Advanced Math* pdf:

A store manager estimates that if a video game is sold at a price of p dollars, the store will have weekly revenue, in dollars, of $r(p) = 4p^2 + 200p$ from the sale of the video game. Which of the following equivalent forms of shows, as constants or coefficients, the maximum possible weekly revenue and the price that results in the maximum revenue?

- A) $r(p) = 200p - 4p^2$
- B) $r(p) = -2(2p^2 - 100p)$
- C) $r(p) = -4(p^2 - 50p)$
- D) $r(p) = -4(p - 25)^2 + 2500$

Solution: Solving

$$-4p^2 + 200p = 0$$

we see that $p = 0$ and $p = 50$, therefore the x -coordinate of the vertex is at 25. Plugging 25 back in, we see the maximum occurs at 2500. Therefore, the answer is $r(p) = -4(p - 25)^2 + 2500$ **(D)**.

Now that we've finished talking about the vertex, let's move on to the **quadratic equation**. Just in case you forgot, here is the formula:

Quadratic Equation: For a quadratic $ax^2 + bx + c$, the roots occur when

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

What are the roots of $6x^2 - 11x + 4$?

- A) $\frac{1}{2}, \frac{4}{3}$
- B) $2, \frac{4}{3}$
- C) $\frac{1}{2}, 2$
- D) $\frac{1}{2}, 3$

Solution: Plugging $a = 6$, $b = -9$, and $c = 4$ into the quadratic formula, we get

$$x = \frac{11 \pm \sqrt{(-11)^2 - 4 \cdot 6 \cdot 4}}{2 \cdot 6} = \frac{11 \pm 5}{12}$$

Therefore, the roots are $\frac{1}{2}, \frac{4}{3}$ **(A)**.

On the old SAT, you did not need to know the quadratic equation. But now,

There are some SAT questions you cannot solve without the quadratic equation.

Let's do a frequently missed quadratic equation problem from the fourth College Board SAT test:

$$x^2 - \frac{k}{2}x = 2p$$

In the quadratic equation above, k and p are constants. What are the solutions for x ?

A) $x = \frac{k}{4} \pm \frac{\sqrt{k^2 + 2p}}{4}$

B) $x = \frac{k}{4} \pm \frac{\sqrt{k^2 + 32p}}{4}$

C) $x = \frac{k}{2} \pm \frac{\sqrt{k^2 + 2p}}{2}$

D) $x = \frac{k}{2} \pm \frac{\sqrt{k^2 + 32p}}{4}$

Solution: Moving all the terms to one side, we get

$$x^2 - \frac{k}{2}x - 2p = 0$$

Applying quadratic equation with $a = 1$, $b = -\frac{k}{2}$, $c = -2p$, we get

$$\frac{\frac{k}{2} \pm \sqrt{\left(-\frac{k}{2}\right)^2 - 4 \cdot 1 \cdot -2p}}{2 \cdot 1}$$

Simplifying,

$$\frac{\frac{k}{2} \pm \sqrt{\frac{k^2}{4} + 8p}}{2} = \frac{\frac{k}{2} \pm \sqrt{\frac{k^2 + 32p}{4}}}{2} = \frac{\frac{k}{2} \pm \frac{1}{2}\sqrt{k^2 + 32p}}{2} = \frac{k}{4} \pm \frac{1}{4}\sqrt{k^2 + 32p}$$

Therefore, **(B)** is the answer.

Moreover,

There are several different ways that quadratic equations can show up on the exam.

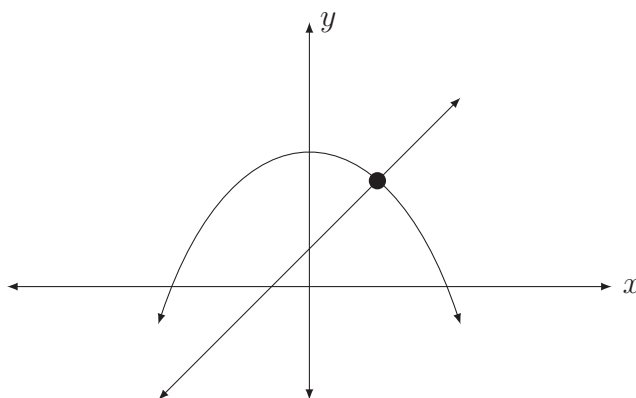
For example, you will see quadratic equations arise when

- Solving a System involving a Quadratic and a Line.
- Solving a Rational Equation
- Solving a Radical Equation.

We'll cover the first bullet point in the next section, and save the other two for the next chapter.

5.4 When Functions Collide

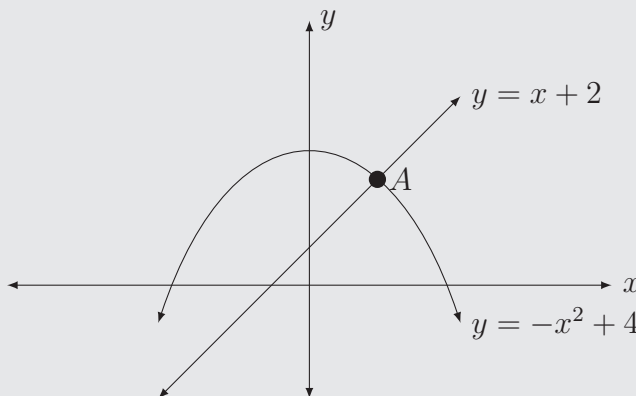
Often times, when we visualize two functions as graphs, we would like to solve for their intersection point:



And the key idea is that the (x, y) point needs to lie on *both* graphs. So in reality, you are actually solving a *system of equations*. But unlike the first two chapters, our system may contain *non-linear* equations.

First let's try a system of one linear equation and one quadratic:

Consider the graphs below:



What are the coordinates of A ?

- A) $(1, 2)$
- B) $(-2, -1)$
- C) $(-2, 1)$
- D) $(2, 1)$

Solution: Since A is the intersection of the graphs, we just need to solve the system

$$\begin{aligned}y &= -x^2 + 4 \\y &= x + 2\end{aligned}$$

Equating the y , we get

$$x + 2 = -x^2 + 4$$

or

$$x^2 + x - 2 = 0$$

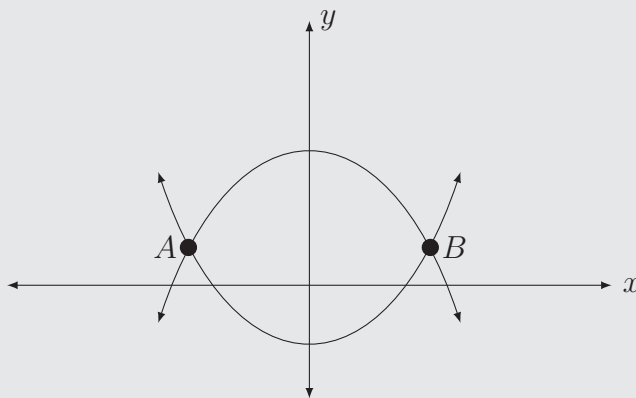
Either through factoring or quadratic equation, we get $x = -2, 1$. Since A is in the first quadrant, $x = 1$ and hence $y = 2$ (**A**).

Finally, let's try a system containing two quadratics:

The functions

$$\begin{aligned}f(x) &= x^2 - 2 \\g(x) &= -x^2 + 4\end{aligned}$$

are graphed below:



What is the distance from A to B ?

- A) 3
- B) 6
- C) $\sqrt{3}$
- D) $2\sqrt{3}$

Solution: To solve this problem, we just need to find the difference between the two x -coordinates of intersection. Equating the two functions,

$$x^2 - 2 = -x^2 + 4$$

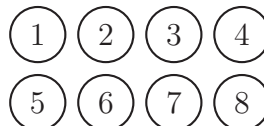
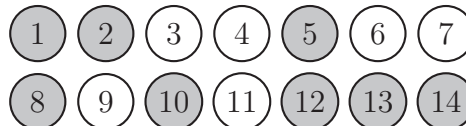
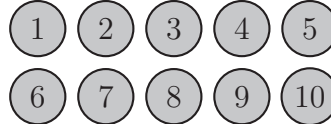
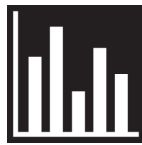
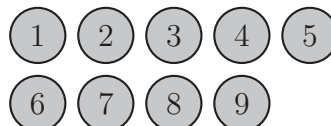
we can isolate x to get

$$2x^2 = 6$$

or $x = \pm\sqrt{3}$. Hence, the answer is $2\sqrt{3}$ (D).

Commandments Covered

- P1. Create a quadratic or exponential function or equation that models a context.
- P2. Determine the most suitable form of an expression or equation to reveal a particular trait.
- P5. Solve a quadratic equation having rational coefficients.
- P8. Solve a system of one linear equation and one quadratic equation.
- P10. Interpret parts of nonlinear expressions in terms of their context. Identify or describe the real-life meaning of a constant term, a variable, or a feature of the given equation.
- P12. Understand a nonlinear relationship between two variables by making connections between their algebraic and graphical representations. Determine the equation of a curve given a verbal description of a graph; determine key features of the graph of a linear function from its equation; or determine the impact on a graph of a change in the defining equation.
- P13. Use function notation, and interpret statements using function notation. Use function notation to solve conceptual problems related to transformations and compositions of functions.
- P14. Use structure to isolate or identify a quantity of interest in an expression. Rearrange an equation or formula to isolate a single variable.



Chapter 6

Advanced Math: Playing with Polynomials

Some things are rational. Some things are radical. The SAT Math is a little bit of both.

- Anonymous SAT Math Teacher

6.1 An Aside on Exponents

Last chapter, we began our discussion on functions and looked at one important class: quadratics. Today, we are going to go from quadratics to *general polynomials*. And once we understand polynomials, we can extend our understanding to rational functions and radicals.

But before we can begin any of that, let's do a quick review of basic exponent properties.

As you've known since elementary school, we can define for a positive whole number a

$$x^a = \underbrace{x \cdot x \cdot x \dots x}_{a \text{ times}}$$

We can also extend this definition to non-positive integer powers of a . If the exponent is zero, we define

$$x^0 = 1$$

If the exponent is rational, we define

$$x^{\frac{p}{q}} = \sqrt[q]{x^p}$$

And if the exponent is negative, we define

$$x^{-a} = \frac{1}{x^a}.$$

Evaluate:

$$3^0 + 4^{\frac{3}{2}} + 4^{\frac{-1}{2}}$$

- A) 8.5
- B) 9.5
- C) 10.5
- D) None of the above

Solution: From the definition of exponentiation, we get

$$3^0 + 4^{\frac{3}{2}} + 4^{\frac{-1}{2}} = 1 + \sqrt{4^3} + \frac{1}{\sqrt{4}} = 1 + 8 + .5 = 9.5$$

The answer is therefore **(B)**.

The exponent properties you will need to remember are

- When multiplying exponents of the same base, we add the powers.

$$x^a x^b = x^{a+b}$$

- When dividing exponents of the same base, we subtract the powers.

$$\frac{x^a}{x^b} = x^{a-b}$$

- When raising a power to a power, we multiply the powers

$$(x^a)^b = x^{ab}$$

- When multiplying exponents of the same power, we multiply the bases.

$$b^a \cdot c^a = (bc)^a$$

Even though these properties may seem a bit unintuitive, they are really quite obvious if you just write them in terms of products. For example

$$x^a x^b$$

just means multiply a copies of x with b copies of x :

$$x^a \cdot x^b = \underbrace{x \cdot x \cdot x \dots x}_{a \text{ times}} \cdot \underbrace{x \cdot x \cdot x \dots x}_{b \text{ times}}$$

But of course this just gives us $a + b$ copies of x :

$$\underbrace{x \cdot x \cdot x \dots x}_{a + b \text{ times}} = x^{a+b}$$

Reduce the following expression:

$$\frac{3^a 2^a}{(6^b)^2}$$

- A) 3^{a-2b}
- B) 3^{a-b}
- C) 6^{a-b}
- D) 6^{a-2b}

Solution: Since we are multiplying two exponents with the same power, the numerator becomes

$$\frac{6^a}{(6^b)^2}$$

But raising a power to a power yields

$$\frac{6^a}{6^{2b}}$$

Finally, dividing exponents gives us 6^{a-2b} **(D)**.

Now that we've mastered exponent properties, let's go back to polynomials:

6.2 Numbers in Disguise: Addition and Subtraction

There are a multitude of polynomial theorems that the SAT can test you on. In particular,

- Sum and Product of Roots Theorem
- Factor Theorem
- Rational Roots Theorem

We will talk about these in the last chapter **Once in a Blue Moon**. But for the majority of polynomial problems, it is more important to understand the algebraic *basics*. So let's talk about what a polynomial is and how we can manipulate them algebraically.

When we define a variable x , we are defining some unknown number. And since it is just a number, it should obey all the normal rules. For example, when we add x to itself, we should have double x :

$$x + x = 2x$$

And when we consider an expressions that can be formed from taking x and adding, scaling, subtracting, or multiplying, we create a polynomial:

$$a_n x^n + a_{n-1} x^{n-1} + \dots + a_2 x^2 + a_1 x + a_0$$

Now, adding and subtracting polynomials is easy, just treat them as you would numbers¹. Distribute any signs or multiplications and then group by common terms.

Reduce the following expression:

$$(3x^2 + 2x - 5) - (x^2 - x - 3)$$

- A) $2x^2 + x - 2$
- B) $2x^2 + x - 8$
- C) $2x^2 + 3x - 8$
- D) $2x^2 + 3x - 2$

Solution: Distributing, the subtraction, we get

$$3x^2 + 2x - 5 - x^2 + x + 3$$

Grouping common terms,

$$3x^2 - x^2 + 2x + x + -5 + 3$$

the expression reduces to

$$2x^2 + 3x - 2$$

So the answer is **(D)**.

Be very careful when performing polynomial subtraction:



Don't forget to distribute the negatives!

Everyone makes this mistake, so if you see a subtraction, be sure to double check your work! You'll thank me later.

6.3 Numbers in Disguise: Multiplication and Division

To multiply polynomials, you will need to know, for starters, the basic case of multiplying two binomials:

$$(a + b)(c + d)$$

Just FOIL: First, outer, inner, last:

$$(a + b)(c + d) = \underbrace{ac}_{\text{first}} + \underbrace{ad}_{\text{outer}} + \underbrace{bc}_{\text{inner}} + \underbrace{bd}_{\text{last}}$$

¹Because this is precisely what they are!

Expand the following:

$$(x^3 + 2x)(x^2 + 1)$$

- A) $x^5 + 3x^2 + 2$
- B) $x^5 + 3x^2 + 2x$
- C) $x^5 + 3x^3 + 2x$
- D) $x^5 + 3x^3 + 2$

Solution: Applying FOIL, we get

$$(x^3 + 2x)(x^2 + 1) = \underbrace{x^5}_{\text{first}} + \underbrace{x^3}_{\text{outer}} + \underbrace{2x^3}_{\text{inner}} + \underbrace{2x}_{\text{last}}$$

which reduces to $x^5 + 3x^3 + 2x$ (C).

But there is more to life than just FOIL: you are also expected to multiply other types of polynomials, say

$$(a + b)(c + e + f)$$

The extension is easy: just multiply each term on the left by each term on the right and add them together:

$$\begin{aligned} (a + b)(c + d + e) &= ac + ad + ae \\ &+ bc + bd + be \end{aligned}$$

Let's try the example from the College Board's *Passport to Advanced Math* pdf:

$$(x^2 + bx - 2)(x + 3) = x^3 + 6x^2 + 7x - 6$$

In the equation above, b is a constant. If the equation is true for all values of x , what is the value of b ?

- A) 2
- B) 3
- C) 7
- D) 9

Solution: We evaluate

$$(x^2 + bx - 2)(x + 3)$$

First, we multiply each term on the left by x

$$(x^2 + bx - 2)(x + 3) = x^3 + bx^2 - 2x + \dots$$

and then by 3:

$$(x^2 + bx - 2)(x + 3) = x^3 + bx^2 - 2x + 3x^2 + 3bx - 6$$

Grouping the right hand side,

$$x^3 + bx^2 + 3x^2 + (-2x + 3bx) - 6$$

we get

$$x^3 + (b + 3)x^2 + (-2 + 3b)x - 6$$

Equating coefficients to

$$x^3 + 6x^2 + 7x - 6$$

we see $b = 3$ (**B**).

Lastly, let's cover polynomial division.

Suppose you wanted to divide a polynomial, say $2x^2 - x - 1$, by $x - 1$. The trick is to set up the long division as you would normally:

$$x - 1 \overline{) 2x^2 - x - 1}$$

Now compute the biggest multiple of x that goes into $2x^2$. This is $2x$.

$$x - 1 \overline{) 2x^2 - x - 1} \quad \begin{array}{r} 2x \\ \hline \end{array}$$

Multiplying our divisor by $2x$ and subtracting, we get

$$x - 1 \overline{) 2x^2 - x - 1} \quad \begin{array}{r} 2x \\ \hline - 2x^2 + 2x \\ \hline x - 1 \end{array}$$

Repeating the above steps once more,

$$x - 1 \overline{) 2x^2 - x - 1} \quad \begin{array}{r} 2x + 1 \\ \hline - 2x^2 + 2x \\ \hline x - 1 \\ - x + 1 \\ \hline 0 \end{array}$$

which yields $2x + 1$ as the quotient.

Compute the following:

$$(x^3 + x^2 - 2x - 8) \div (x - 2)$$

- A) $x^2 + 4x - 3$
- B) $x^2 + 4x + 3$
- C) $x^2 + 3x - 4$
- D) $x^2 + 3x + 4$

Solution: When we perform the long division,

$$\begin{array}{r}
 x^2 + 3x + 4 \\
 x-2 \overline{) x^3 + x^2 - 2x - 8} \\
 \underline{-x^3 + 2x^2} \\
 3x^2 - 2x \\
 \underline{-3x^2 + 6x} \\
 4x - 8 \\
 \underline{-4x + 8} \\
 0
 \end{array}$$

we see that the quotient is $x^2 + 3x + 4$ (D).

Again, be careful during the subtraction step in long division:



Don't forget to distribute the negatives!

To reiterate, *everyone* makes this mistake.

6.4 Getting a Little Rational

The new SAT also places an emphasis on rational functions. A rational function is a *quotient* of polynomials, like

$$\frac{x+1}{x-1}, \quad \frac{x^2+1}{x^3-1}, \quad \frac{x^5+2x+1}{x^2-1}$$

Like polynomials, you will be asked to add, subtract, and multiply them. And just like polynomials, the trick is to treat them like normal numbers. In particular:

- When adding or subtracting rational functions, use a common denominator.
- When multiplying rational functions, multiply the numerator and denominator separately

Which of the following expressions is the same as

$$\frac{1}{x+1} - \frac{1}{x-1}$$

A) 0

B) $\frac{-2}{(x+1)(x-1)}$

C) $\frac{2}{(x+1)(x-1)}$

D) $\frac{2x}{(x+1)(x-1)}$

Solution:

$$\frac{x-1}{(x+1)(x-1)} - \frac{x+1}{(x-1)(x+1)} = \frac{(x-1) - (x+1)}{(x+1)(x-1)} = \frac{-2}{(x+1)(x-1)}$$

Therefore the answer is **(B)**.

You will also be asked to simplify rational expressions. But that's easy, just cancel out factors:

Which of the following functions has the same value as

$$\frac{x^2 - x - 6}{x^2 + x - 12}$$

for every x where both functions are defined?

A) $\frac{x+2}{x+4}$

B) $\frac{x+2}{x+3}$

C) $\frac{x+3}{x+4}$

D) $\frac{x+4}{x+2}$

Solution: Factoring the numerator and denominator, we get

$$\frac{(x+2)(x-3)}{(x+4)(x-3)}$$

Cancelling out the $x - 3$ term, we get

$$\frac{x+2}{x+4}$$

Hence, the answer is **(A)**.

Lastly, you will be asked to solve rational equations. Again, you use the same manipulations as you would with numbers, but you have to be careful:



Remember to remove any extraneous solutions!

In particular, if a solution yields a denominator that is zero, then that solution is extraneous.

What are the solutions to

$$\frac{x^2 - 4}{x - 2} = 0$$

- A) $-2, 2$
- B) -2
- C) 2
- D) None of the above

Solution: Multiplying both sides by $x - 2$ yields

$$x^2 - 4 = 0$$

so x is either 2 or -2. But plugging 2 back into the original equation makes the denominator undefined. Hence **(B)** is the answer.

6.5 Getting a Little Radical

You will also be asked to work with expressions involving radicals:

$$\sqrt{x}$$

Again, like with polynomials and rationals, just treat radicals like normal numbers. For example, we can combine like terms:

$$\sqrt{x} + 2\sqrt{x} = 3\sqrt{x}$$

We do have one additional property though. When we multiply two radicals, the multiplication takes place inside the radical sign:

$$\sqrt{a}\sqrt{b} = \sqrt{ab}$$

This should be no surprise: it is just an exponent property:

$$a^{\frac{1}{2}}b^{\frac{1}{2}} = (ab)^{\frac{1}{2}}$$

Which expression is the same as

$$\sqrt{2}\sqrt{3} + 3\sqrt{6}$$

- A) $4\sqrt{6}$
- B) $5\sqrt{6}$
- C) $6\sqrt{6}$
- D) None of the above

Solution: By exponent properties, $\sqrt{2}\sqrt{3} = \sqrt{6}$. Therefore

$$\sqrt{2}\sqrt{3} + 3\sqrt{6} = \sqrt{6} + 3\sqrt{6} = 4\sqrt{6}$$

The answer is **(A)**.

Just like how there are rational equations, there are also *radical* equations. And like their rational counterparts, you still have to be careful about *extraneous* solutions.

Let's end the chapter with a rational equation from the College Board *Passport to Advanced Math* pdf:

$$x - 12 = \sqrt{x + 44}$$

What is the solution set for the above equation?

- A) $\{5\}$
- B) $\{20\}$
- C) $\{-5, 20\}$
- D) $\{5, 20\}$

Solution: Squaring both sides, we get

$$(x - 12)^2 = x + 12$$

Expanding and isolating to one side, we get

$$x^2 - 25x + 100 = 0$$

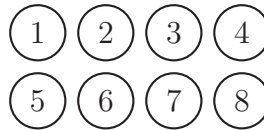
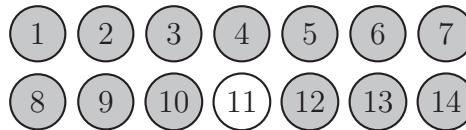
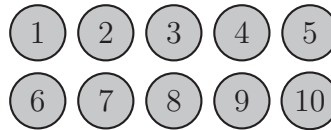
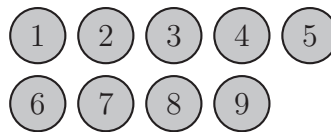
which has roots 5 and 20. But when we plug in 5 into the original equation, we get

$$-7 = \sqrt{49}$$

so 5 is extraneous. Therefore, 20 is the only solution **(B)**.

Commandments Covered

- P3. Create equivalent expressions involving rational exponents and radicals, including simplifying or rewriting in other forms.
- P4. Create an equivalent form of an algebraic expression by using structure and fluency with operations.
- P6. Add, subtract, and multiply polynomial expressions and simplify the result.
- P7. Solve an equation in one variable that contains radicals or contains the variable in the denominator of a fraction. The student may be required to identify when a resulting solution is extraneous.
- P9. Rewrite simple rational expressions. Students will add, subtract, multiply, or divide two rational expressions or divide two polynomial expressions and simplify the result.



Chapter 7

Plane and Simple: Angles to Triangles

It doesn't make sense to say that there is no point in studying Geometry. There is at least one line.

- Anonymous SAT Math Teacher

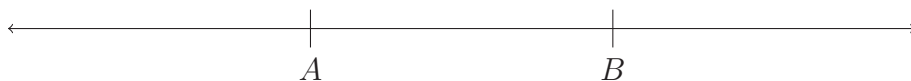
7.1 Begin the Blitzkrieg

Today, we begin our blitzkrieg of planar geometry. Now, according to the College Board, planar geometry falls under *additional topics* and only accounts for 6 problems on the exam. However, based on my experience perusing the new SAT material and taking the test at Gunn¹, I do not believe this is true. In fact, a lot of the planar geometry falls under the study of lines, which accounts for 32% of the test! Even the Khan Academy tests each contain at least 3-4 trig questions.

Hence, as a compromise, we will cover planar geometry and trigonometry for the next 2 lectures, and then the *extremely* rare topics in one.

7.2 Life Begins on a Line

The simplest geometric object you will see on the SAT is that of a line. Now, we've already covered a *ton* of line facts in the first two chapters, but let us finish the rest here, starting with the number line:



Here are two of the most important number line formulas you need to know:

The **distance** from A to B is

$$|B - A|$$

and the **midpoint** of $\overline{AB}$ is

$$\frac{A + B}{2}$$

¹In fact, on May 2016, the exam referenced one of the more advanced circle theorems.

Be careful,



The midpoint formula is NOT $\frac{B-A}{2}$!

$\frac{B-A}{2}$ is the formula for half the distance, not midpoint!

If you want to remember the midpoint formula, just think of it as the average of A and B (which is indeed the definition of a midpoint).

If the tick marks below are evenly spaced, which number is closest to the value of x ?



- A) 17.3
- B) 17.6
- C) 18.3
- D) 18.6

Solution: The segment is being partitioned into 6 intervals, hence we can calculate the length of each interval as

$$\frac{\text{Distance}}{\text{Number of Intervals}} = \frac{2}{6} = \frac{1}{3}$$

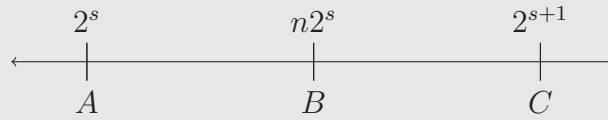
Therefore, x is located at

$$17 + 5 \times \frac{1}{3} \approx 18.6$$

So **(D)** is the answer.

Here is one of my favorite problems from the October 2008 SAT that is still valid for the new SAT:

If B is the midpoint of A and C , then what is the value of n ?



- A) 1
- B) 1.5
- C) 2
- D) 2.5

Solution: By midpoint formula, we have

$$n2^s = \frac{2^s + 2^{s+1}}{2}$$

Applying exponent properties, we can pull out the 2^s from the right hand side:

$$\frac{2^s + 2^{s+1}}{2} = \frac{2^s + 2^s \cdot 2}{2} = \frac{2^s(1 + 2)}{2}$$

Hence,

$$n2^s = \frac{2^s(1 + 2)}{2}$$

Dividing out the 2^s yields

$$n = \frac{3}{2}$$

So the answer is **(B)**.

Suppose you did not see that you can manipulate exponent properties to eliminate the s variable. Alternatively, you can notice that the answer is independent of s , so just plug in your own!



If your answer is independent of a variable, try plugging in your own value for that variable.

Solution: As an alternate solution, plug in $s = 1$, we then have

$$n2^1 = \frac{2^1(1 + 2)}{2}$$

which yields,

$$n = \frac{3}{2}$$

Moving on to two dimensions, the extension of distance and midpoint is pretty simple:

The **distance** from (x_1, y_1) to (x_2, y_2) is

$$\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

and the **midpoint** of $\overline{AB}$ is

$$\left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right)$$

If $(4, 6)$ is the midpoint of $(1, 2)$ and (x, y) , then what is (x, y) ?

- A) $(6, 4)$
- B) $(6, 10)$
- C) $(7, 4)$
- D) $(7, 10)$

Solution: By the midpoint formula, we know that the x coordinate satisfies

$$\frac{1 + x}{2} = 4$$

and y coordinate satisfies

$$\frac{2 + y}{2} = 6$$

Solving these equations, we see that $(7, 10)$ is the midpoint **(D)**.

Finally, we repeat some facts about lines in Chapter 2, namely,

The general formula for a line is

$$y = mx + b$$

where b is the y -intercept and slope m can be calculated as

$$m = \frac{\Delta y}{\Delta x}$$

We also need to know that

- Two lines are *parallel* if their slopes are the *same*.
- Two lines are *perpendicular* if their slopes are the *negative reciprocal*.

Suppose

$$\begin{aligned} 3x + 5y &= 2 \\ mx + 6y &= 4 \end{aligned}$$

Describes a system of two perpendicular lines. Then what is m ?

- A) 3
- B) -3
- C) 10
- D) -10

Solution: Putting each equation in standard form we get

$$\begin{aligned} y &= -\frac{3}{5}x + \frac{2}{5} \\ y &= -\frac{m}{6}x + \frac{4}{6} \end{aligned}$$

Since the lines are perpendicular,

$$-\frac{m}{6} = \frac{5}{3}$$

so

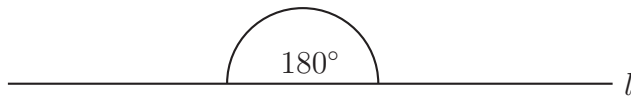
$$m = -10$$

giving us the answer **(D)**.

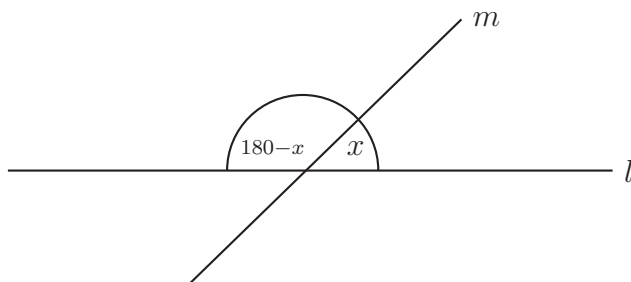
7.3 All About Angles

Now that we've finished looking at lengths and distances, let's review angles.

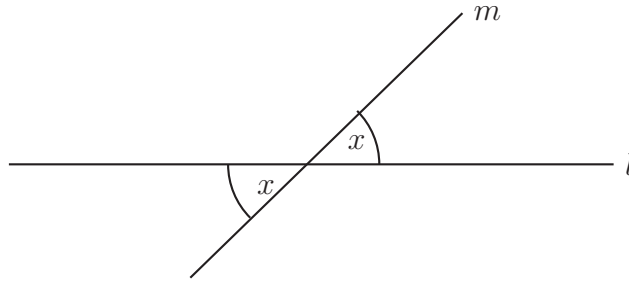
We associate a line with 180° :



By splitting the line, we can create *supplementary angles*, or angle pairs whose degree measure add to 180° :

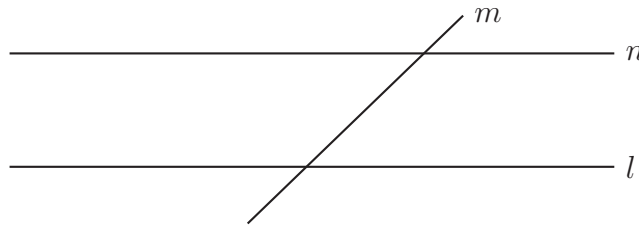


But notice that m is also a line, so it too has an angle associated with 180° . This means that the angle beneath $180^\circ - x$ must be x :



Hence the angles in an intersection that are opposite of each other are congruent. These are known as *vertical angles*.

Now suppose we add a third line l . The figure we create is called a *transversal*.



When it comes to transversals, you need to know that

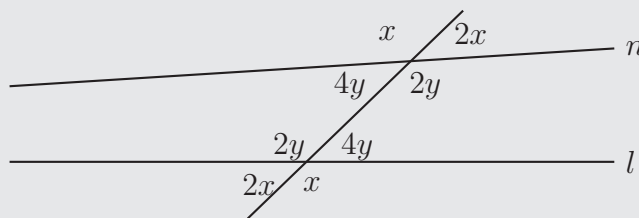
If two lines are *parallel* in a transversal, the corresponding angles are *congruent*:

The converse is also true:

If the corresponding angles are *congruent*, then the two lines in a transversal are *parallel*.

Consider the figure below:

Note: Figure not drawn to scale

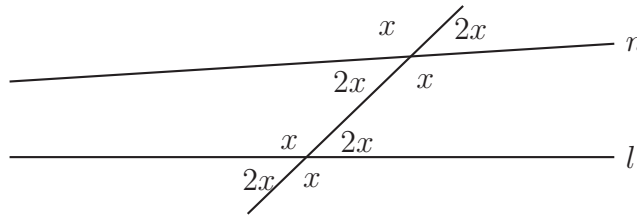


Which of the following must be true?

- A. $x = y$
- B. $x + y = 180^\circ$.
- C. $l \parallel n$
- D. None of the above

Solution: From vertical angles, we see that $x = 2y$. Rewriting each angle in terms of x , we have

Note: Figure not drawn to scale



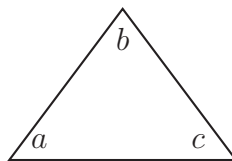
Since the corresponding angles are congruent, the two lines are parallel(C).

7.4 All About Triangles

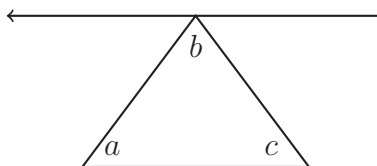
First, you will need to know the following facts about general triangles:

- Area = Base $\times$ Height
- The sum of 2 sides is bigger than the third
- The difference of 2 sides is less than the third
- The biggest side is opposite the biggest angle
- The smallest side is opposite the smallest angle
- The sum of the angles of a triangle is 180° .

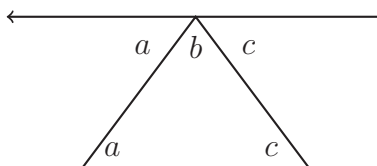
These should all be familiar, in particular, you should remember the last fact is just the result of our theory of transversals:



Just draw a line parallel to the base through the top vertex,

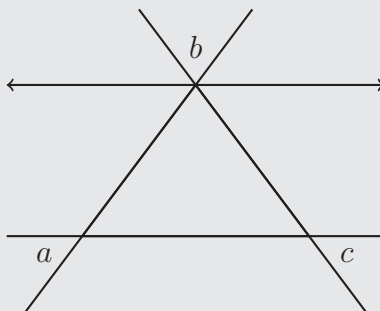


apply our transversal facts:



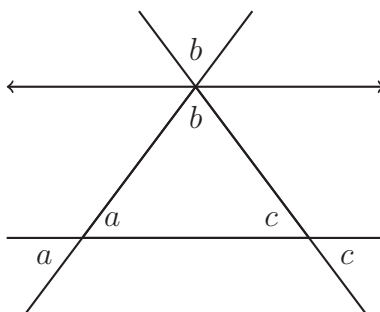
Then $a + b + c = 180^\circ$ by supplementary angles.

In the diagram below, what is $a + b + c$?



- A. 90°
- B. 180°
- C. 360°
- D. Cannot be determined

Solution: By vertical angles, we know the angles are just the angles of a triangle:



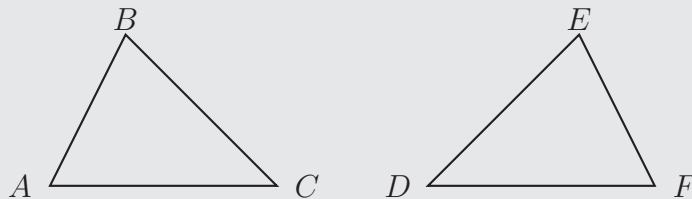
Hence, $a + b + c = 180^\circ$ **(B)**.

The new SAT will also test you on *congruence* and *similarity*.

As you may recall from Geometry, two triangles are congruent (or the same) if they satisfy one of the following conditions

SSS
 ASA
 SAS
 AAS

Consider triangles $\triangle ABC$ and $\triangle DEF$:



If $\overline{AB} = \overline{EF}$, $\overline{AC} = \overline{DF}$ and $m\angle A = m\angle F$, then what must be true?

- A. $\overline{AB} = \overline{DF}$
- B. $m\angle C = m\angle D$
- C. $\overline{BC} = \overline{EF}$
- D. None of the above

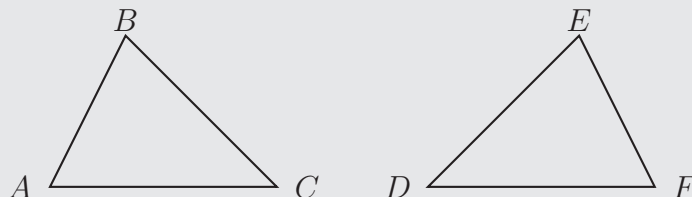
Solution: These two triangles are congruent by *SAS*. Hence $m\angle C = m\angle D$ **(B)**.
Careful though,



Two triangles satisfying *ASS* need not be congruent!

You can easily have two triangles satisfy *ASS* yet not be congruent.

Consider triangles $\triangle ABC$ and $\triangle DEF$:



If $\overline{AB} = \overline{EF}$, $\overline{AC} = \overline{DF}$ and $m\angle B = m\angle E$, then what must be true?

- A. $\overline{AB} = \overline{DF}$
- B. $m\angle C = m\angle D$
- C. $\overline{BC} = \overline{EF}$
- D. None of the above

Solution: Since only *ASS* is satisfied, the answer is none of the above **(D)**.

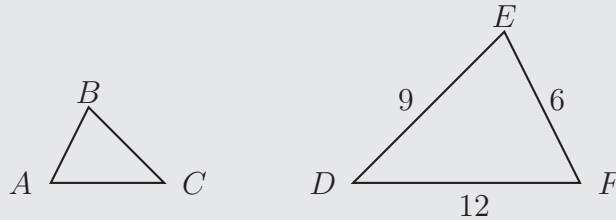
Two triangles are *similar* if their corresponding sides are in *ratio*. Moreover,

*If two triangles are **similar**, then their corresponding **angles are the same**.*

and

*If their corresponding **angles are the same**, the two triangles are **similar**.*

Consider triangles $\triangle ABC$ and $\triangle DEF$:



Suppose we are given that

$$\begin{aligned} m\angle A &= m\angle F \\ m\angle D &= m\angle C \\ m\angle B &= m\angle E \end{aligned}$$

Moreover, $BC = 3$. What is the perimeter of $\triangle ABC$?

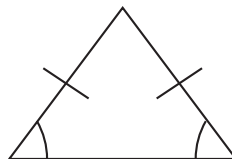
- A. 6.25
- B. 9
- C. 18
- D. Cannot be determined

Solution: Since the two triangles have the same angles, they are similar. And because $\overline{ED}$ is triple $\overline{BC}$, each side of $\triangle ABC$ is a third in length to its corresponding side in $\triangle DEF$. The perimeter is then $3 + 2 + 4 = 9$ **(B)**.

There are three main classes of triangle you should know for the exam:

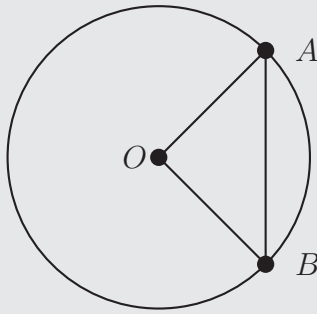
Isoceles Triangle:

An isoceles triangle is one that has at least **two equal sides**. The angles opposite the equal sides are also equal.



Isoceles triangles can show up in the strangest of places:

Consider the circle centered at O and $m\angle AOB = 50^\circ$. What is $m\angle A$?



- A. 60°
- B. 65°
- C. 130°
- D. Cannot be determined

Since $\overline{OA}$ and $\overline{OB}$ are radii, $\triangle OAB$ is in fact an isosceles triangle. This implies that $m\angle A = m\angle B$. Since the sum of the angles of a triangle is 180°

$$180 = m\angle A + m\angle A + 50$$

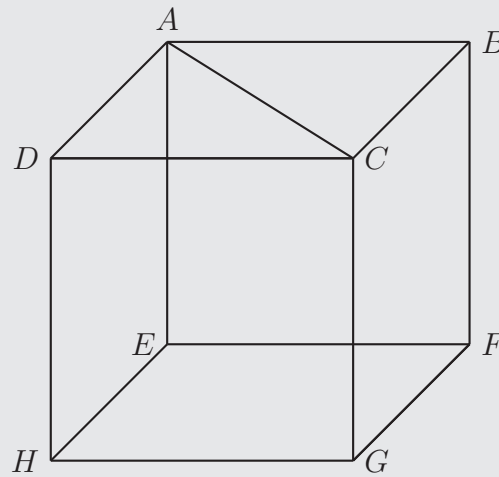
and hence $m\angle A = 65^\circ$ **(B)**.

Equilateral Triangle

An equilateral triangle is the extreme case of an isosceles triangle. All the sides are equal and all the angles are equal.

Like equilateral triangles, they show up in the strangest of places. Consider the classic SAT problem from January 2010:

The length of $\overline{AC}$ in the cube below is 2.



What is the perimeter of the triangle formed by connecting vertices A , C and H ?

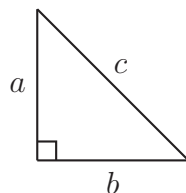
- A) 6
- B) 8
- C) 10
- D) 12

Solution: Notice that each side of the triangle is a diagonal of a face of the cube, hence, the triangle is equilateral. The answer is therefore **(A)**.

Right Triangles

The most important facts you need to remember about right triangles are

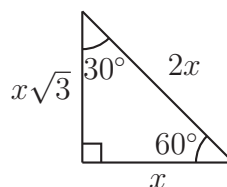
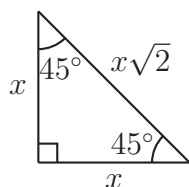
Pythagorean Theorem: For any right triangle,



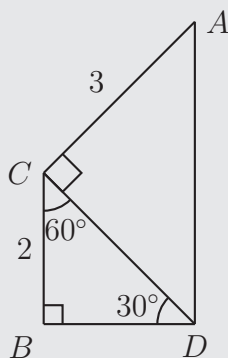
$$a^2 + b^2 = c^2$$

as well as the two classic right triangles:

$45^\circ - 45^\circ - 90^\circ$ and $30^\circ - 60^\circ - 90^\circ$: The following ratios always hold:



In the diagram below, what is the length of $\overline{AD}$?



- A) 5
- B) $5\sqrt{2}$
- C) 10
- D) $10\sqrt{2}$

Solution: Since $\triangle BCD$ is a $30^\circ - 60^\circ - 90^\circ$ triangle, $\overline{CD}$ is 4. But now we have a $3 - 4 - 5$ triangle, so the answer is 5 (**A**).

7.5 A Touch of Trigonometry

One of the major differences of the new SAT from the old SAT is the inclusion of trigonometry. But don't worry if you never took trig. This is because you will only be tested on the

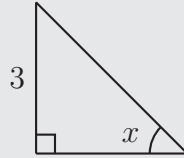
- The *meaning* of sine, cosine, tangent
- A few simple trigonometric *identities*.

This means no double angle formulas, no law of cosines, and no inverse trig functions.

By definition, sine, cosine, and tangent are *ratios*² of the right triangle:

$$\begin{aligned}\sin(\theta) &= \frac{\text{Opposite}}{\text{Hypotenuse}} \\ \cos(\theta) &= \frac{\text{Adjacent}}{\text{Hypotenuse}} \\ \tan(\theta) &= \frac{\text{Opposite}}{\text{Adjacent}}\end{aligned}$$

In the diagram below, if $\tan x = \frac{3}{4}$, then what is the hypotenuse of the triangle?



- A) 3
- B) 4
- C) 5
- D) $5\sqrt{2}$

Solution: By definition of $\tan(x)$,

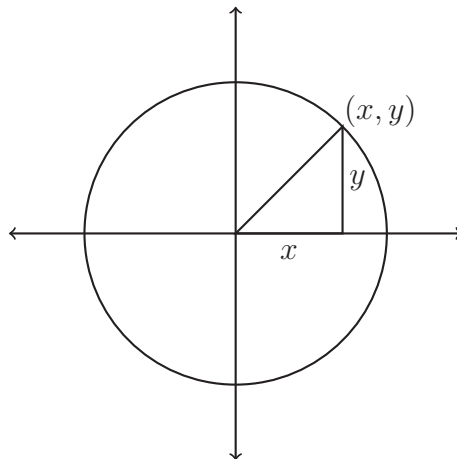
$$\tan(x) = \frac{\text{opposite}}{\text{adjacent}} = \frac{3}{4}$$

Hence the base is 4. But now we have a 3-4-5 triangle **(C)**.

As for the trigonometric identities, you only need to know three. The simplest being

$$\sin^2(\theta) + \cos^2(\theta) = 1.$$

This identity comes from inscribing a triangle on the unit circle:



²One way to remember these ratios is the mnemonic SOHCAHTOA:

Using SOHCAHTOA, we see that

$$\sin(\theta) = \frac{y}{1}$$

$$\cos(\theta) = \frac{x}{1}$$

and hence

$$x = \cos(\theta)$$

$$y = \sin(\theta)$$

Applying Pythagorean Theorem yields

$$\sin^2(\theta) + \cos^2(\theta) = 1$$

Find all θ such that

$$\sin^2(\theta) = 2 - \cos^2(\theta)$$

A) $\theta = 0$

B) $\theta = 90^\circ$

C) All real θ .

D) There is no such θ .

Solution: Moving the $\sin(\theta)$ and $\cos(\theta)$ terms to one side yields

$$\sin^2(\theta) + \cos^2(\theta) = 2$$

or

$$1 = 2$$

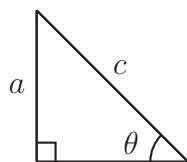
which is never true **(D)**.

The next two identities are *extremely* important. They have appeared on almost every one of the College Board Practice exams and in fact, the College Board states them by name in the appendix. Namely, they are the identities that allows one to rewrite $\sin(x)$ in terms of $\cos(x)$ and vice versa:

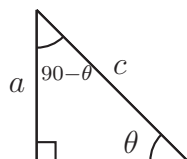
$$\sin(90^\circ - \theta) = \cos(\theta)$$

$$\cos(90^\circ - \theta) = \sin(\theta)$$

Even though these identities look intimidating, they are very simple: it is just an application of the fact that the sum of the angles in a triangle is 180° . Consider a right triangle with angle θ and a leg of length a and hypotenuse c :



Since the sum of the angles of a triangle add up to 180° , the remaining angle must be $90 - \theta$:



Using the triangle, we can calculate,

$$\begin{aligned}\sin(\theta) &= \frac{a}{c} \\ \cos(90^\circ - \theta) &= \frac{a}{c}\end{aligned}$$

Hence

$$\cos(90^\circ - \theta) = \sin(\theta)$$

Voila!

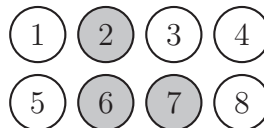
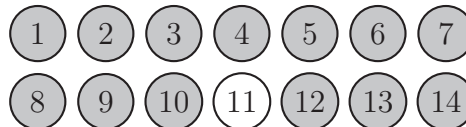
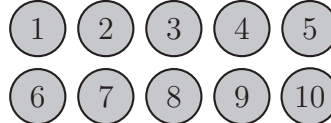
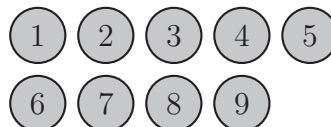
Which of the following is the same as $\cos(63^\circ)$

- A) $\cos(27^\circ)$
- B) $\sin(27^\circ)$
- C) $\cos(37^\circ)$.
- D) $\sin(37^\circ)$

Solution: $90 - 63 = 27$, so by our identity on complimentary angles, the answer is **(B)**.

Commandments Covered

- A2. Use trigonometric ratios and the Pythagorean theorem to solve applied problems involving right triangles.
- A6. Use concepts and theorems about congruence and similarity to solve problems about lines, angles, and triangles.
- A7. Use the relationship between similarity, right triangles, and trigonometric ratios; use the relationship between sine and cosine of complementary angles.



Chapter 8

Plane and Simple: Circles and Beyond

If there is one positive thing I can say about circles on the SAT, it is that they got more well rounded

- Anonymous SAT Math Teacher

8.1 Arcs and Areas

The simplest circle questions you may encounter are those that ask you about area and circumference. Just in case you forgot,

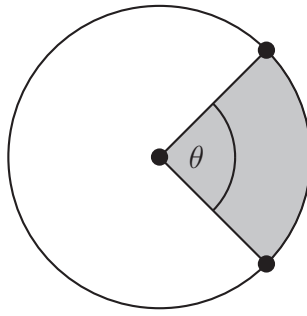
- The **area** of a circle is πr^2
- The **circumference** of a circle is $2\pi r$

If the area of a circle is 25π , then what is the circle's circumference?

- A) 5π
- B) 10π
- C) 20π .
- D) 25π

Solution: Setting $25\pi = \pi r^2$, we find that $r = 5$. Using the circumference formula, the answer is then **(B)**.

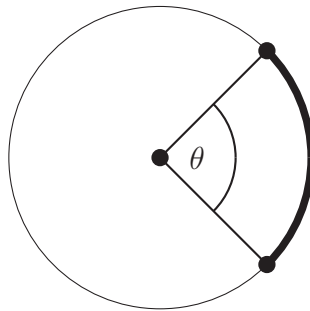
You also need to know how to calculate a *part* of the circle. In particular, you may be asked to find the area of a *sector*:



The area of a sector is just a *proportion* of the full area. And since the full degree measure of a circle is 360° , the area of a sector is just

$$\frac{\theta}{360} \cdot \pi r^2$$

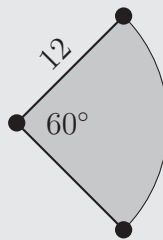
You also need to know how to compute the arc-length of a sector:



The trick is the same: the arc-length formula is just a *proportion* of the circumference formula:

$$\frac{\theta}{360} \cdot 2\pi r$$

What is the area of the sector below?



- A) 20π
- B) 24π
- C) 28π .
- D) 32π

Solution: Since the circle has a radius of 12, the area is 144π . $1/6$ of 144π is 24π **(B)**.

Note that you will also need to know *radians* for the new SAT:

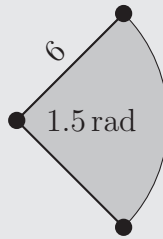
$$\text{Degrees} \times \frac{\pi}{180} = \text{Radians}$$

$$\text{Radians} \times \frac{180}{\pi} = \text{Degrees}$$

You especially need to know that radians *simplifies* the arc-length formula as

$$r\theta$$

What is the perimeter of the sector below?



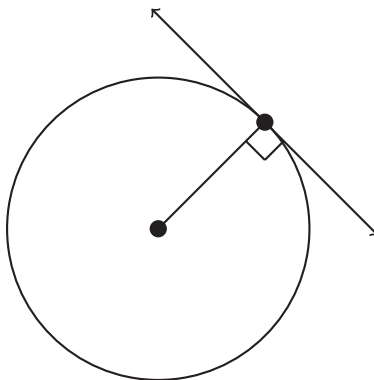
- A) $6 + 9\pi$
- B) $12 + 9\pi$
- C) 21.
- D) 25

Solution: The length of the arc is $1.5 \cdot 6 = 9$. Hence, the full perimeter is $9 + 6 + 6 = 21$. **(C)**.

8.2 Going on a Tangent

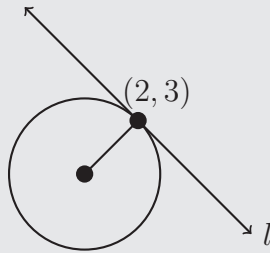
The circumference and area formulas are need to know circle properties for the SAT. But the most forgettable circle property that I've seen tested on both the old and new SAT is the following fact:

*Given a circle and a line tangent to the circle, the angle formed from connected a radius to the point of tangency is always a **right angle**.*



Let's do my favorite example from the old SAT:

Suppose a circle centered at the origin intersects a tangent line l at $(2, 3)$:

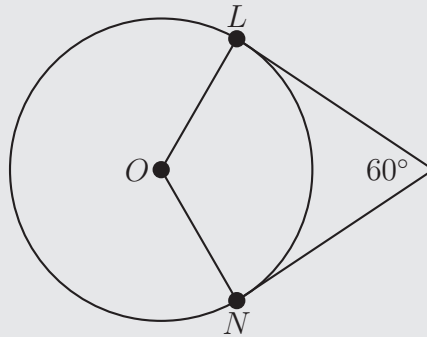


What is the slope of l ?

- A) $\frac{2}{3}$
- B) $\frac{3}{2}$
- C) $-\frac{2}{3}$
- D) $-\frac{3}{2}$

Solution: Since the circle is centered at the origin, the slope of the radius connecting the origin and point of tangency is $\frac{3}{2}$. But because the tangent line is perpendicular, its slope is $-\frac{2}{3}$ (**C**).

The next problem is the very last question from the second College Board released exam:



In the figure above, point O is the center of the circle, line segment LM and MN are tangent to the circle at points L and N , respectively, and the segments intersect at point M as shown. If the circumference of the circle is 96, what is the length of minor arc $\widehat{LN}$?

Solution: By tangent property of circles, we know that

$$\begin{aligned} m\angle L &= 90 \\ m\angle N &= 90 \end{aligned}$$

Since the sum of the interior angles of a quadrilateral add up to 360° , we can compute $m\angle O$ as

$$360 - 90 - 90 - 60 = 120.$$

Moreover, because the circumference of the circle is 96, the radius is

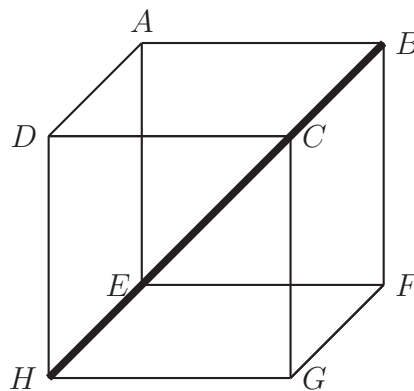
$$r = \frac{96}{2\pi}.$$

Using our trusty arc-length formula,

$$\frac{120}{360} \cdot 2\pi \cdot \frac{96}{2\pi} = \mathbf{32}.$$

8.3 Adding Another Dimension

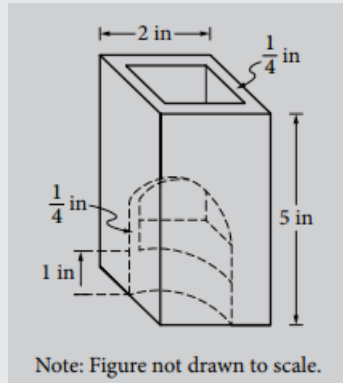
In the past, the SAT has given a potpourri of fantastic 3D questions. And most of these questions were a variation of finding the distance between two opposite vertices of the cube.



Technically, these questions are still new SAT valid since they can be solved by applying Pythagorean Theorem twice. However, the only type of 3D problems that I have seen on the new SAT so far is questions¹ that involve applying *volume* to the *real world*. And there are two variations of the volume question that you should know.

The first involves adding and subtracting volumes to form a more complicated shape. Here is the example from the *Additional Topics* pdf:

¹In fact, every question on the 6 released exams involved finding the volume of a silo of grain. I guess the College Board really likes farmers :P



A glass vase is in the shape of a rectangular prism with a square base. The figure above shows the vase with a portion cut out. The external dimensions of the vase are height 5 inches (in), with a square base of side length 2 inches. The vase has a solid base of height 1 inch, and the sides are each $\frac{1}{4}$ inch thick. Which of the 4 following is the volume, in cubic inches, of the glass used in the vase?

- A) 6
- B) 8
- C) 9
- D) 11

Solution: To calculate the volume of the vase, we subtract the inner volume from the outer volume. Since the base is a square, the outer volume is just

$$\text{Outer Volume} = 2 \cdot 2 \cdot 5 = 20$$

From the diagram, we see that the height is reduced by 1 and the side of a base is reduced by $\frac{1}{2}$. Hence the volume of the inner rectangular prism is

$$\text{Inner Volume} = 1.5 \cdot 1.5 \cdot 4 = 9$$

The volume of the vase is then $20 - 9 = 11$ **(D)**.

The second type of volume question involves transferring volume from one shape to another. And the key here is that **volume is preserved**:

Suppose Philip the farmer decides to transfer grain from one cylindrical silo to another. The first silo has a height of 5 meters and a base of radius 2 meters whereas the second silo has a height of 10 meters and a base of radius 4 meters. If the height of the grain in the first silo is 4 meters and the second silo is empty, what percent of the second silo is filled after all the grain of the first is transferred to the second?

- A) .1
- B) 10
- C) .4
- D) 40

Solution: Just in case you forgot, to calculate the volume of a cylinder, just think of it as the area of a circle raised by a height of h :

$$\underbrace{\pi r^2}_{\text{Area}} \cdot h$$

But if you did forget, remember:



The Surface Area and Volume Formulas are on the first page!

The volume of the grain in the first container is

$$\pi \cdot 2^2 \cdot 4 = 16\pi$$

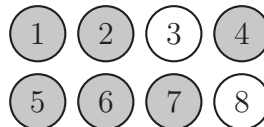
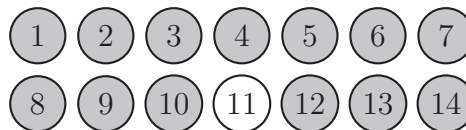
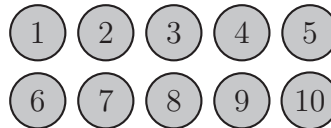
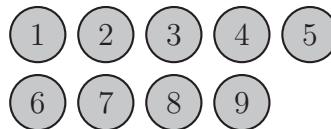
When we move the grain to the second silo, we can solve for the height of the grain using

$$\pi \cdot 4^2 \cdot h = 16\pi$$

which yields $h = 1$. Hence, the second silo is only 10% full **(B)**.

Commandments Covered

- A1. Solve problems using volume formulas.
- A4. Convert between degrees and radians and use radians to determine arc lengths.
- A5. Apply theorems about circles to find arc lengths, angle measures, chord lengths, and areas of sectors.
- A6. Use concepts and theorems about congruence and similarity to solve problems about lines, angles, and triangles.
- A7. Use the relationship between similarity, right triangles, and trigonometric ratios; use the relationship between sine and cosine of complementary angles.



Chapter 9

Once in a Blue Moon

*Have you ever heard the wolf cry to the blue corn moon?
Well I haven't seen a Rational Roots Theorem SAT problem either.*

- Sohcahtoa

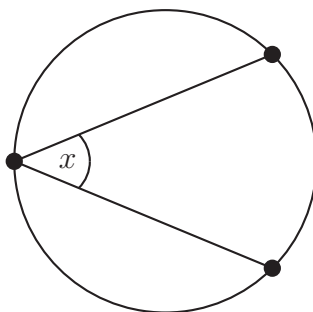
Deep down, I dislike the fact that the SAT goes so deep for only 6 questions. And to quote my friend Elsa, I would personally

Let It Go.

Just focus on the other chapters and break a 700. But if you really want to get that 800, here are the remaining fair game material. And to prove to any disbelievers that these are actually¹ tested on the SAT, I have included several real SAT problems in this chapter.

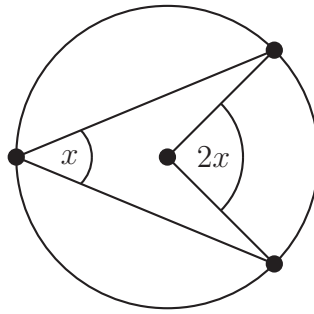
9.1 More Advanced Circle Properties

Other than the circle properties from the preceding chapter, you will also be tested on *inscribed angles*, or angles formed from 3 points chosen on the circumference:



It turns out that the measure of the inscribed angle is always *half* the measure of the corresponding central angle.

¹I don't blame you: these questions are rarer than Mewtwo and Articuno combined.



Be careful, here's a common mistake I've seen a thousand times in my life:



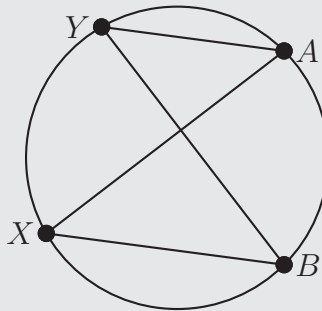
The inscribed angle is half the central angle and NOT the other way around!

Just look at the diagram: the central angle is obviously the bigger one!

Because of this result, it turns out that

Inscribed angles corresponding to the same arc are the same.

Here's a question from the College Board's *Additional Topics in Mathematics* pdf that tests this concept:



In the figure above, $\angle AXB$ and $\angle AYB$ are inscribed in the circle. Which of the following statements is true?

- A) The measure of $\angle AXB$ is greater than the measure of $\angle AYB$.
- B) The measure of $\angle AXB$ is less than the measure of $\angle AYB$.
- C) The measure of $\angle AXB$ is equal to the measure of $\angle AYB$.
- D) There is not enough information to determine the relationship between the measure of $\angle AXB$ and the measure of $\angle AYB$.

Solution: Both angles correspond to the same central angle, hence the answer is (C).

You will also need to know the conic section²:

$$(x - h)^2 + (y - k)^2 = r^2$$

where (h, k) is the center and r is the radius.

You will be tested on the circle formula in two ways:

- Understanding what the given formula *represents* and knowing how to manipulate it.
- **Completing the square** to generate the circle formula.

And just in case you forgot,

To complete the square of $x^2 + bx = d$ just add $\left(\frac{b}{2}\right)^2$ to both sides!

Let's do an example of each circle formula problem you can see on the exam.

Suppose the circle described by

$$(x - 3)^2 + (y - 1)^2 = 4$$

has its radius increased by 1 and is translated 1 to the right and 1 up. What is the resulting circle equation?

- A. $(x - 2)^2 + y^2 = 5$
- B. $(x - 4)^2 + (y - 2)^2 = 5$
- C. $(x - 2)^2 + y^2 = 9$
- D. $(x - 4)^2 + (y - 2)^2 = 9$

Solution: From a center of $(3, 1)$, the new center is $(4, 2)$. Moreover, the radius of 2 is now increased to 3. Hence the answer is $(x - 4)^2 + (y - 2)^2 = 9$ (D).

Describe the graph of the equation

$$x^2 - 4x + y^2 - 6y + 9 = 0$$

- A. A circle centered at $(-2, -3)$ with radius 2
- B. A circle centered at $(-2, -3)$ with radius 4
- C. A circle centered at $(2, 3)$ with radius 2
- D. A circle centered at $(2, 3)$ with radius 4

²This is the *only* conic section formula you will need to know. Moreover, out of all the theorems in this chapter, this is perhaps the most frequent, about one in every other test. So instead of being once in a blue moon, it is more like once every other moon.

Solution: Completing the square for the x terms, we add $(\frac{4}{2})^2$ to both sides:

$$x^2 - 4x + 4 + y^2 - 6y + 9 = 0 + 4$$

which factors as

$$(x - 2)^2 + y^2 - 6y + 9 = 4$$

We then complete the square for the y terms by adding $(\frac{6}{2})^2$ to both sides

$$(x - 2)^2 + y^2 - 6y + 9 + 9 = 4 + 9$$

which factors as

$$(x - 2)^2 + (y - 3)^2 = 4$$

Comparing our equation to the circle formula, we see that we have a circle of radius 2 centered at (2, 3) (C).

9.2 More Advanced Root Theorems

If you can manipulate polynomials algebraically and apply quadratic equation, you are likely to break a 700. But if you want that 800, here are a few root theorems that have shown up on previous exams:

Sum of Roots and Product of Roots Theorem (Simple Case³): *For a quadratic $x^2 + bx + c$, b is the negative sum of roots and c is the product of the roots.*

Here is a question from the non-calculator portion of the second SAT exam ever released:

What is the sum of all values of m that satisfy $2m^2 - 16m + 8 = 0$?

- A. -8
- B. $-4\sqrt{3}$
- C. $4\sqrt{3}$
- D. 8

Solution: Dividing the equation by 2 yields

$$m^2 - 8m + 4 = 0$$

So by the preceding theorem, the answer is 8 (D).

Note that even though the College Board explicitly lists this theorem (in the non-simple case) on its *Passport to Advanced Math* pdf, you honestly don't need to worry much about it. **In the worst case scenario, just apply quadratic equation.**

³You can always divide out by the leading coefficient a

Remainder Theorem: For a polynomial $p(x)$, if the value of $p(a) = k$, then the remainder when $p(x)$ is divided by $x - a$ is $p(k)$. In particular, if $k = 0$, then $x - a$ is a factor of $p(x)$.

Here is a problem from the very first SAT exam ever released.

For a polynomial $p(x)$, the value of $p(3)$ is -2. Which of the following must be true about $p(x)$?

- A. $x - 5$ is a factor of $p(x)$
- B. $x - 2$ is a factor of $p(x)$
- C. $x + 2$ is a factor of $p(x)$
- D. The remainder when $p(x)$ is divided by $x - 3$ is -2.

Solution: This is *exactly* the Remainder Theorem (D).

Rational Roots Theorem (Simple Case³): If $x^n + a_{n-1}x^{n-1} + a_{n-2}x^{n-2} + \dots + a_0$ has a rational root, that root must be a factor of a_0 .

I've only seen this theorem come up once: the fourth released SAT test, on a non-calculator grid-in:

$$x^3 - 5x^2 + 2x - 10$$

For what real value of x is the equation above true?

Solution: The factorization of this polynomial is not obvious and the formula for solving for the roots of a cubic equation is horrendous. Instead, we can apply rational root theorem and guess that the root is either $\pm 1, \pm 2, \pm 5, \pm 10$. But we know that the SAT does not allow negative grid-in's, hence it is either 1, 2, 5, or 10. Plugging in 5 yields $5^3 - 5 \cdot 5^2 + 2 \cdot 5 - 10 = 0$. So the answer is **5**.

Alternatively, if you did not know rational roots theorem, you can also do some fancy factoring:

Solution: Pulling out an x^2 from the first two terms yields

$$x^2(x - 5) + 2x - 10$$

Now pull out an $x - 5$ from the first term and $2x - 10$:

$$(x - 5)(x^2 + 2)$$

Therefore, **5** is a factor.

³I have yet to see the case where $a \neq 1$

9.3 Not so Complex Numbers

Even though Complex numbers seem harrowing, you won't need to know anything as scary as De Moirve's Theorem. According to the SAT, all you need to know is how to

Add, subtract, multiply, divide, and simplify complex numbers.

Let's start!

To add and subtract complex numbers, we just treat them as the same as normal polynomials. Group the i terms together and real number terms together and add: it's as simple as π .

Compute $(3 - 2i) - (-2 + 3i)$:

- A. $5 - 3i$
- B. $5 - 5i$
- C. $1 - 3i$
- D. $1 - 5i$

Solution: Distributing, we get

$$3 - 2i + 2 - 3i$$

which simplifies to $5 - 5i$ (**B**).

To multiply two complex numbers, again we treat them like polynomials: just FOIL:

$$(a + bi)(c + di)$$

and remember to use the fact that $i^2 = -1$.

Compute $(4 - 3i)(4 + 3i)$:

- A. 25
- B. $16 - 9i$
- C. $16 + 9i$
- D. $9 - 16i$

Solution: Applying FOIL, we get

$$(4 - 3i)(4 + 3i) = \underbrace{16}_{\text{First}} + \underbrace{12i}_{\text{outer}} - \underbrace{12i}_{\text{inner}} + \underbrace{9i^2}_{\text{last}}$$

Using the fact that $i^2 = -1$, this simplifies to 25(A).

Addition, subtraction, and multiplication are easy. But division takes a little work. Suppose you want to divide one complex number by another, say

$$(11 + 10i) \div (4 - i)$$

The key is to represent the division as a fraction:

$$\frac{11 + 10i}{4 - i}$$

And then multiply by 1 using the conjugate of the denominator (the conjugate is just a complex number with the imaginary part negated):

$$\frac{11 + 10i}{4 - i} \cdot \frac{4 + i}{4 + i}$$

Multiply via FOIL-ing yields

$$\frac{34 + 51i}{17}$$

which simplifies⁴ to

$$2 + 3i.$$

Compute $(14 + 8i) \div (2 + 4i)$:

- A. $2 + 3i$
- B. $3 + 2i$
- C. $3 - 2i$
- D. $2 - 3i$

Solution: Representing the division as a fraction, we multiply by 1:

$$\frac{14 + 8i}{2 + 4i} \cdot \frac{2 - 4i}{2 - 4i}$$

FOIL-ing yields:

$$\frac{60 - 40i}{20}$$

But this is just $3 - 2i$ (C).

⁴In a normal course, you split into $a + bi$ form and your answer will likely have fractional parts. But this is the SAT: the answer will look nice!

As for the last topic, simplifying complex numbers, other than the steps we took in simplifying the complex numbers involved in products and quotients, you also need to know how to simplify powers of i . But that's simple, just look at the pattern:

$$\begin{aligned} i^0 &= 1 \\ i^1 &= i \\ i^2 &= -1 \\ i^3 &= -i \\ i^4 &= 1 \\ i^5 &= i \\ i^6 &= -1 \\ i^7 &= -i \\ i^8 &= 1 \\ i^9 &= i \\ &\vdots \end{aligned}$$

The pattern⁵ repeats every fourth power. Hence, all you need to remember is



If the remainder of p when dividing by 4 is r , then

$$\begin{aligned} r = 0 &\rightarrow i^p = 1 \\ r = 1 &\rightarrow i^p = i \\ r = 2 &\rightarrow i^p = -1 \\ r = 3 &\rightarrow i^p = -i \end{aligned}$$

Simplify:

$$i^{12} + i^{13} + i^{14} + i^{15}$$

- A. i
- B. $-i$
- C. 0
- D. 1

Solution: We look at the remainders when we divide the powers by 4:

$$\begin{aligned} 12 &\rightarrow 0 \\ 13 &\rightarrow 1 \\ 14 &\rightarrow 2 \\ 15 &\rightarrow 3 \end{aligned}$$

Hence, the expression we are trying to evaluate is the same as

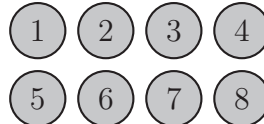
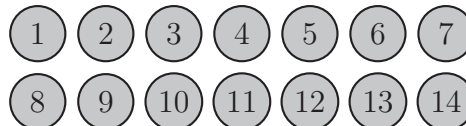
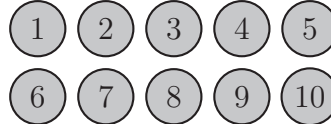
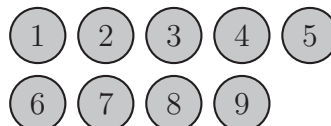
$$1 + i + -1 + -i$$

which is just 0(C).

⁵If you want to be uber formal, you can use exponent properties to get $i^q = i^{4n+r} = (i^4)^n i^r = (1^n)(i^r) = i^r$. But come on, this is SAT math, not rocket science :P.

Commandments Covered

- A3. Add, subtract, multiply, divide, and simplify complex numbers.
- A5. Apply theorems about circles to find arc lengths, angle measures, chord lengths, and areas of sectors.
- A8. Create or use an equation in two variables to solve a problem about a circle in the coordinate plane.
- P11. Understand the relationship between zeros and factors of polynomials, and use that knowledge to sketch graphs and determine whether an expression is a factor of a polynomial.



Chapter 10

Appendix: The SAT Math Commandments

10.1 Heart of Algebra

- ♡1 Create, solve, or interpret a linear expression or equation in one variable that represents a context.
- ♡2 Create, solve, or interpret linear inequalities in one variable that represent a context.
- ♡3 Build a linear function that models a linear relationship between two quantities.
- ♡5 Describe a linear relationship that models a context using either an equation in two variables or function notation.
- ♡6 Create, solve, and interpret systems of linear inequalities in two variables.
- ♡7 Create, solve, and interpret systems of two linear equations in two variables.
- ♡8 Algebraically solve linear equations (or inequalities) in one variable.
- ♡9 Algebraically solve systems of two linear equations in two variables.
- ♡10 Understand connections between algebraic and graphical representations.

10.2 Problem Solving and Data Analysis

- D1. Use ratios, rates, proportional relationships, and scale drawings to solve single- and multistep problems.
- D2. Solve single- and multistep problems involving percentages.
- D3. Solve single- and multistep problems involving measurement quantities, units, and unit conversion. Calculate density; or use the concept of density to solve a multistep problem.
- D4. Given a scatterplot, use linear, quadratic, or exponential models to describe how the variables are related. Select the equation of a line or curve of best fit; interpret the line in the context of the situation; or use the line or curve of best fit to make a prediction.

- D5. Use the relationship between two variables to investigate key features of the graph.
- D6. Compare linear growth with exponential growth.
- D7. Use two-way tables to summarize categorical data and relative frequencies, and calculate conditional probability, association of variables, or independence of events.
- D8. Make inferences about population parameters based on sample data. The sample statistics may mention confidence intervals and measurement error that the student should understand and make use of, but need not calculate.
- D9. Use statistics to investigate measures of center of data and analyze shape, center, and spread. The measures of center include mean, median, and mode, and the measures of spread range (may be calculated) and standard deviation (non-calculated).
- D10. Evaluate reports to make inferences, justify conclusions, and determine appropriateness of data collection methods. The reports may consist of tables, graphs, or text summaries.

10.3 Passport to Advanced Math

- P1. Create a quadratic or exponential function or equation that models a context.
- P2. Determine the most suitable form of an expression or equation to reveal a particular trait.
- P3. Create equivalent expressions involving rational exponents and radicals, including simplifying or rewriting in other forms.
- P4. Create an equivalent form of an algebraic expression by using structure and fluency with operations.
- P5. Solve a quadratic equation having rational coefficients.
- P6. Add, subtract, and multiply polynomial expressions and simplify the result.
- P7. Solve an equation in one variable that contains radicals or contains the variable in the denominator of a fraction. The student may be required to identify when a resulting solution is extraneous.
- P8. Solve a system of one linear equation and one quadratic equation.
- P9. Rewrite simple rational expressions. Students will add, subtract, multiply, or divide two rational expressions or divide two polynomial expressions and simplify the result.
- P10. Interpret parts of nonlinear expressions in terms of their context. Identify or describe the real-life meaning of a constant term, a variable, or a feature of the given equation.
- P11. Understand the relationship between zeros and factors of polynomials, and use that knowledge to sketch graphs and determine whether an expression is a factor of a polynomial.

- P12. Understand a nonlinear relationship between two variables by making connections between their algebraic and graphical representations. Determine the equation of a curve given a verbal description of a graph; determine key features of the graph of a linear function from its equation; or determine the impact on a graph of a change in the defining equation.
- P13. Use function notation, and interpret statements using function notation. Use function notation to solve conceptual problems related to transformations and compositions of functions.
- P14. Use structure to isolate or identify a quantity of interest in an expression. Rearrange an equation or formula to isolate a single variable.

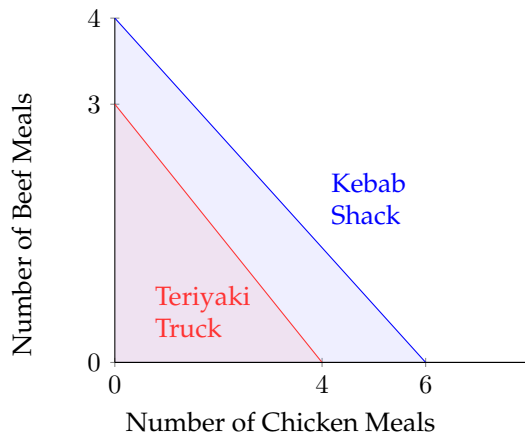
10.4 Additional Topics

- A1. Solve problems using volume formulas.
- A2. Use trigonometric ratios and the Pythagorean theorem to solve applied problems involving right triangles.
- A3. Add, subtract, multiply, divide, and simplify complex numbers.
- A4. Convert between degrees and radians and use radians to determine arc lengths.
- A5. Apply theorems about circles to find arc lengths, angle measures, chord lengths, and areas of sectors.
- A6. Use concepts and theorems about congruence and similarity to solve problems about lines, angles, and triangles.
- A7. Use the relationship between similarity, right triangles, and trigonometric ratios; use the relationship between sine and cosine of complementary angles.
- A8. Create or use an equation in two variables to solve a problem about a circle in the coordinate plane.



1

The graph below shows two consumer options for spending \$30 on beef and chicken entrees with sides at two different food trucks.



Which of the following statements must be true about the consumer options?

- A) At the Teriyaki Truck the chicken meal and the beef meal are closer in price than at the Kebab Shack.
- B) Chicken is more expensive than beef at either location.
- C) A customer always gets more food at the Teriyaki Truck than at the Kebab Shack.
- D) The Kebab Shack sells more food than the Teriyaki Truck.

2

Consider the fraction

$$n = \frac{4}{2 + \frac{1}{x}}$$

where x is a positive integer. As x gets bigger and bigger, what happens to the value of n ?

- A) n remains constant
- B) n approaches 0
- C) n approaches 1
- D) n approaches 2



3

Gender	Eye Color	
	Brown or Hazel	Blue or Green
Female		
Male		
Total	606	70

The incomplete table above summarizes the number of students with brown or hazel eyes and the number of students with green or blue eyes by gender for both grades at Jefferson Junior High School. There are 11 times as many female students with brown or hazel eyes as female students with blue or green eyes. There are 7 times as many male students with brown or hazel eyes as male students with blue or green eyes. None of the students identify as belonging to more than one of the four category boxes in the table. Which of the following is closest to the probability that a brown or hazel eyed student is a brown or hazel eyed male?

- A) 0.875
- B) 0.586
- C) 0.526
- D) 0.474

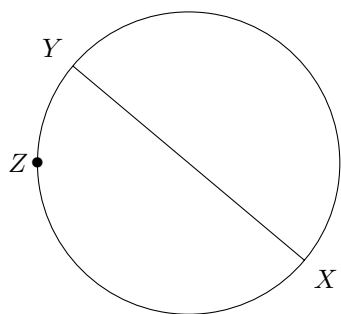
4

If $\sin(x) = \frac{1}{2}$, then what is $\cos(x)$?

- A) $\frac{1}{5}$
- B) $\frac{2}{5}$
- C) $\frac{3}{5}$
- D) Cannot be determined



5



In the diagram above, $\overline{XY}$ passes through the center of the circle. If the length of $\widehat{XZY}$ is 24π , what is the length of the radius of the circle?

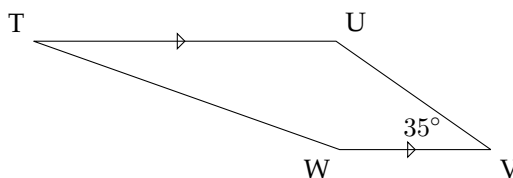
- A) 6
- B) 12
- C) 24
- D) 48

6

What is 12% of 50?

- A) 6
- B) 12
- C) 24
- D) 36

7

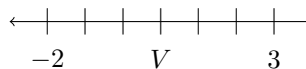


Note: diagram is **NOT** to scale.

In quadrilateral TUVW, $\overline{TU} \parallel \overline{VW}$. If $m\angle V$ measures 35 degrees, what is $m\angle T - m\angle U + m\angle V + m\angle W$?

- A) 35°
- B) 70°
- C) 145°
- D) 325°

8



In the number line above, V is the midpoint between -2 and 3 . What is $V - 1$?

- A) $-\frac{1}{2}$
- B) $\frac{1}{2}$
- C) 1
- D) $1\frac{1}{2}$



9

Simplify:

$$\frac{3}{w-3} - \frac{4}{w-1}$$

- A) $\frac{w-9}{(w-3)(w-1)}$
- B) $\frac{9-w}{(w-3)(w-1)}$
- C) $\frac{-w-9}{(w-3)(w-1)}$
- D) $\frac{-3}{w-1}$

10

The city library donated some children's books to Mr. Clark's first-grade class. If each student takes 4 books, there will be 20 books left. If 3 students do not take a book and the rest of the students take 5 books each, there will be no books left. How many books were donated to the class?

- A) 120
- B) 140
- C) 160
- D) 185

Answer Key

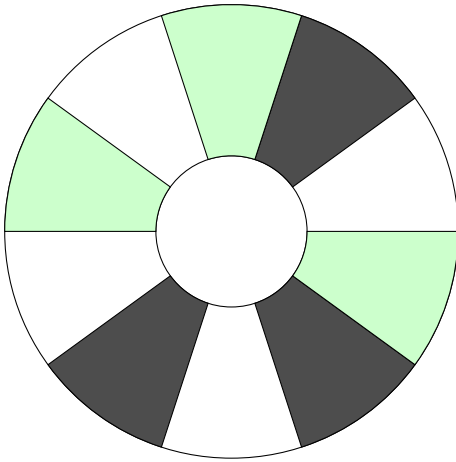
Calculator Off	
1	B
2	D
3	D
4	D

⋮

Calculator On	
5	C
6	A
7	B
8	A
9	B
10	C



1



Note: diagram is **NOT** to scale.

In the diagram above, the shaded regions are congruent sectors of a donut shape that comes from removing a circle of radius 1 from a concentric circle of radius 3. If a point is selected at random within this diagram, what is the probability it lands in a shaded region?

- A) $\frac{8}{15}$
- B) $\frac{6}{11}$
- C) $\frac{5}{9}$
- D) $\frac{3}{5}$

2

In a geometric series consisting of only positive terms, the third term is 12 and the seventh term is 192. What is the first term?

- A) 2
- B) 3
- C) 4
- D) 6

3

If $f(3x - 4) = 4x - 10$ and $f(E) = 2$ then E must equal?

- A) 2
- B) 3
- C) 5
- D) 9



4

Consider the following system of equations:

$$\begin{aligned}6x - 3y &= -1 \\ ax + 2y &= -4\end{aligned}$$

For what value of a does the above system have no solutions?

- A) 6
- B) 8
- C) 10
- D) -4



5

Suppose $|b - a| = 0$. Which of the following must be true?

- A) $b > 0$
- B) $b < a$
- C) $b = a$
- D) None of the above

6

In a certain sequence, other than the first two terms, each subsequent term is the sum of the two preceding terms. What is the eighth term of this sequence?

- A) 5
- B) 7
- C) 9
- D) Cannot be determined

7

$$\sqrt{j - k} = j + 1$$

If $k = -21$, what is the solution set for j in the equation above?

- A) $j \in \emptyset$
- B) $j \in \{-5, 4\}$
- C) $j \in \{-5\}$
- D) $j \in \{-1\}$

8

Solve for y :

$$\sqrt[3]{y} + \sqrt[3]{-27y} = -8$$

- A) $y = \frac{64+96\sqrt{3}}{23}$
- B) $y = 64$
- C) $y = -8$
- D) $y = \frac{-32}{117}$



9

$$\begin{array}{r} A \quad B \\ - \quad B \quad A \\ \hline C \end{array}$$

A , B , and C represent digits in the correctly worked out subtraction problem above. Which of the following could be the value of C ?

- i. 0
- ii. 1
- iii. 9

- A) i only
 B) ii only
 C) i and iii only
 D) i, ii and iii

10

The parabola with equation $y = ax^2 + bx + c$ and vertex (h, k) is reflected about the line $y = k$. This results in the parabola with equation $y = dx^2 + ex + f$. Which of the following equals $a + b + c + d + e + f$?

- A) $2b$
 B) $2c$
 C) $2a + 2b$
 D) $2k$

Answer Key

Calculator Off	
1	A
2	B
3	C
4	D

⋮

Calculator On	
5	C
6	D
7	A
8	B
9	C
10	D



1

In a bag containing red marbles and blue marbles, after adding n red marbles to a bag, the probability of choosing a red marble is 50%. If there were originally 8 red marbles and 20 blue marbles, then what is the value of n ?

- A) 12
- B) 14
- C) 20
- D) 22

2

A farmer uses 40 meters of fencing to enclose a rectangular area of length x and width y . Which of the following functions below represents the area in terms of x ?

- A) $f(x) = x(20 - x)$
- B) $f(x) = x(30 - x)$
- C) $f(x) = x(40 - x)$
- D) $f(x) = x(80 - x)$

3

If $0 < x < 90^\circ$ and $\tan(x) = \frac{1}{u}$, then what is $\sin(x)$?

- A) u
- B) $\frac{1}{\sqrt{u^2+1}}$
- C) $\frac{u}{\sqrt{u^2+1}}$
- D) Cannot be determined

4

Which of the following is the same as $\sin^2(x) + \cos^2(x)$?

- A) $\sin(x)$
- B) 0
- C) $\tan(x)$
- D) 1



5

Suppose a is irrational. Which must always be rational?

- A) a^2
- B) $\sqrt{a}$
- C) $a + a$
- D) None of the above

6

x	$j(x)$
-2	$\frac{5}{16}$
0	5
1	20

Above are 3 values for the function j . If $j(k) = ab^k$ for some constants a and b , what is the value of b ?

- A) $\frac{1}{4}$
- B) $\frac{1}{2}$
- C) 4
- D) 8

7

$$\begin{array}{r} \text{A} \quad \text{B} \\ + \quad \text{B} \quad \text{A} \\ \hline 1 \quad \text{A} \quad 7 \end{array}$$
 A and B represent different digits in the correctly worked out addition problem above. What is $A+B$?

- A) 6
- B) 12
- C) 14
- D) 17

8

Let $f(z)$ be defined by the formula

$$f(z) = \frac{1 - z^{2015}}{1 - z}.$$

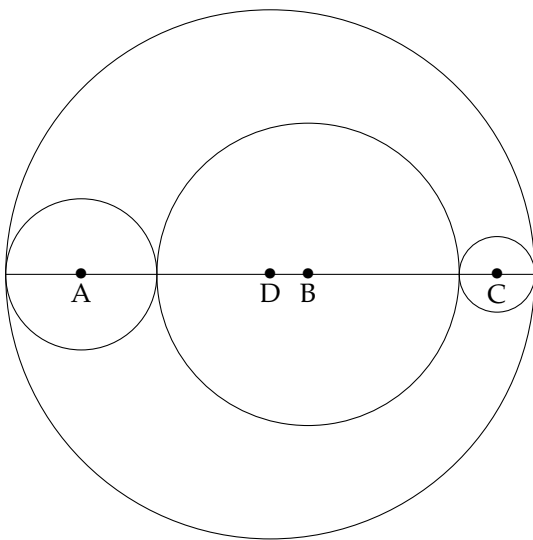
Evaluate

$$\lim_{z \rightarrow 1} f(z)$$

- A) 0
- B) 1
- C) 2015
- D) The result cannot be obtained from the information given.



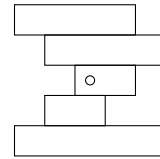
9



Circle A has a radius of 2. Circle B has a radius of 8. Circle C has a radius of 4. All the circles are tangent to each other as shown. What is the radius of circle D?

- A) 7
- B) 8
- C) 14
- D) 16

10



If the figure above were rotated clockwise 90 degrees about O, which of the following would be the result?

- A)
- B)
- C)
- D)

Answer Key

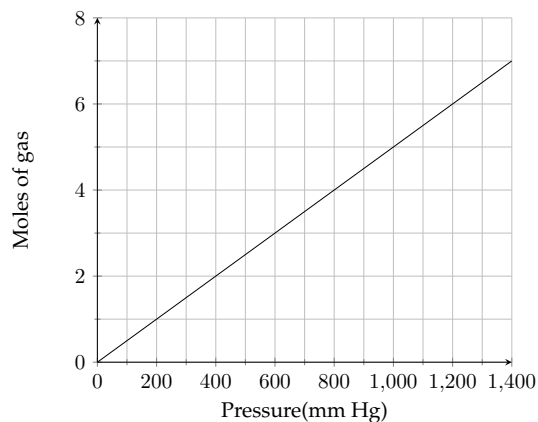
Calculator Off	
1	A
2	A
3	B
4	D

⋮

Calculator On	
5	D
6	C
7	D
8	C
9	C
10	D



1



A particular sample of gas behaves in accordance with ideal gas principles under its controlled volume and temperature. Which of the following functions most accurately predicts the pressure (P) of the gas sample given the sample's moles (M) for a point on this graph?

- A) $P = \frac{1}{200} M$
- B) $P - 3 = 200(M - 600)$
- C) $P - 600 = \frac{1}{200} (M - 3)$
- D) $P - 600 = 200(M - 3)$

2

If $x^2 = y^2$, then which of the following must be true?

- A) $x = -y$
- B) $x = y$ or $x = -y$
- C) $x = y$ and $x = -y$
- D) None of the above

3

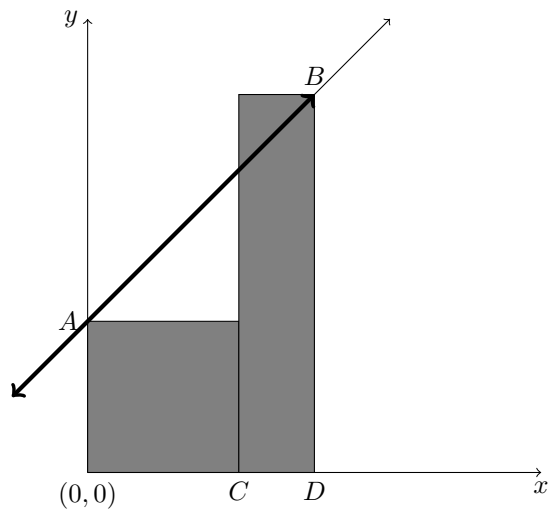
When 9 times the number k is subtracted from -3 , the result is 15. What number results when k is tripled and added to 1?

- A) 7
- B) -5
- C) -13
- D) -125



4

The line $y = x + 2$ intersects the two rectangles below at A and B . Moreover, C is located at $(2,0)$ and D is at $(3,0)$, what is the total area of the two rectangles?



- A) 4
- B) 5
- C) 9
- D) 10



5

Duyen bought a patio set at a store that gave a 10% discount off its original price. In addition, Duyen had a manufacturer's coupon for 15% off the store's selling price. The total amount Duyen paid was c dollars, including a 9% sales tax on the final price after both discounts. Which of the following represents the original price of the patio set in terms of c ?

- A) $\frac{c}{(0.9)(0.85)(1.09)}$
- B) $(0.9)(0.85)(1.09)c$
- C) $\frac{c}{0.84}$
- D) $0.84c$

6

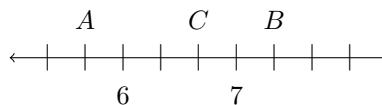
Consider the set of positive integers $\{a, b, c, d, e\}$ where

$$a < b < c < d < e$$

If d is replaced by $2d$ which of the following must be true?

- A) The mean is decreased
- B) The median is increased.
- C) The median is unchanged.
- D) There is no mode.

7



In the number line diagrammed above, what is the difference between the length of AB and the length of BC ?

- A) $\frac{-22}{3}$
- B) 4
- C) $\frac{5}{2}$
- D) 1

8

Translate the following verbal mathematical expression into their equivalent symbolic counterparts. The product of three numbers is five times their sum.

- A) $xy = 2(x - y)$
- B) $xyz = 5(x + y + z)$
- C) None of these.
- D) The answer cannot be determined from the information given.



9

When is $\frac{\sqrt{4x-4}}{x-1}$ undefined?

- A) $x < 4$
- B) $x \leq 1$
- C) $x \geq 4$
- D) $x = 0$

10

If x , y , and z are positive numbers satisfying

$$x + 1/y = 4, \quad y + 1/z = 1, \quad \text{and} \quad z + 1/x = 7/3,$$

then $xyz =$

- A) $\frac{2}{3}$
- B) 1
- C) $\frac{4}{3}$
- D) $\frac{7}{3}$

Answer Key

Calculator Off	
1	D
2	C
3	B
4	C

⋮

Calculator On	
5	A
6	C
7	D
8	B
9	B
10	B



1

If $xy = 0$, then which of the following must be true?

- A) $x = 0$
- B) $x = 0$ or $y = 0$
- C) $x = 0$ and $y = 0$
- D) None of the above

2

Which of the following polar coordinates represents the same Cartesian coordinate as $(2, 2\sqrt{3})$?

- A) $(2, 30^\circ)$
- B) $(4, 30^\circ)$
- C) $(4, 60^\circ)$
- D) $(8, 30^\circ)$

3

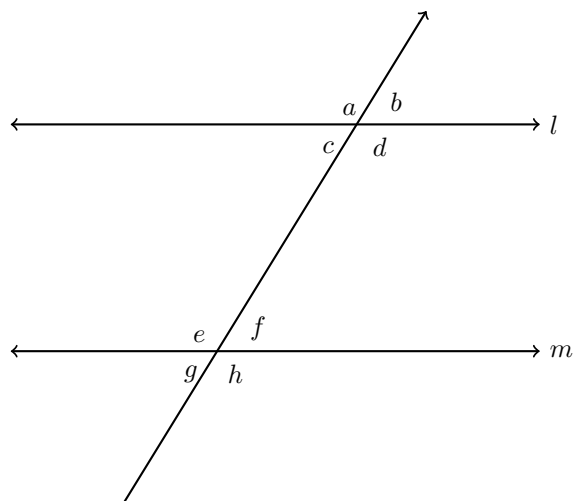
Simplify:

$$\frac{x+7}{x+1} - \frac{x-1}{x+7}$$

- A) $\frac{48}{x^2+7}$
- B) -1
- C) $\frac{2(7x+24)}{(x+1)(x+7)}$
- D) $\frac{2(7x+25)}{(x+1)(x+7)}$



4



Which of the following lists the exterior angles where $l \parallel m$?

- A) a, b, g, h
- B) e, f, g, h
- C) b, d, f, h
- D) c, d, e, f



5

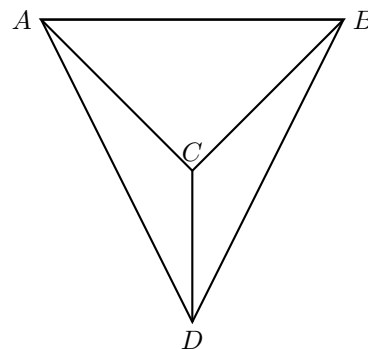
$$\begin{aligned}v(x) &= 2x^2 + x - 6 \\w(x) &= 6x^3 + 3x^2 - 18x\end{aligned}$$

The polynomials $v(x)$ and $w(x)$ are defined above. Which of the following polynomials is divisible by $3x - 2$?

- A) $a(x) = w(x) - 2v(x)$
- B) $b(x) = w(x) + 2v(x)$
- C) $c(x) = 3w(x) - v(x)$
- D) $d(x) = 3w(x) - 2v(x)$

6

Suppose $m\angle ACB = 90^\circ$ and $m\angle ACD = m\angle BCD$. If $AB = 10$ and $AC = CB$, then which of the following cannot be a value for BD ?



- A) 7
- B) 9
- C) 10
- D) 11



7

Six different points J, K, L, M, N, and O are on a number line in that order. JM is 10.1 and KO is 9.7. If LM is 8.4, what is one possible value for KL?

- A) -3
- B) 0.49
- C) $\frac{1}{4}$
- D) 8

8

If $\frac{2a+5}{3} = b$ and $b = 23$, what is the value of a ?

- A) 64
- B) 32
- C) 27
- D) $\frac{32}{3}$

9

If $(3x + 10)(3x - 10) = 300$, what is the value of $9x^2$?

- A) 100
- B) 200
- C) 400
- D) 1200

10

If $(ux + 2)(vx - 5) = 15x^2 + tx - 10$ for all values of x , and $u + v = 8$, what are the two possible values for t ?

- A) $-\frac{2}{5}$ and $\frac{5}{3}$
- B) -5 and -19
- C) 3 and 5
- D) 8 and 15

Answer Key

Calculator Off	
1	B
2	C
3	D
4	A

⋮

Calculator On	
5	A
6	A
7	B
8	B
9	C
10	B

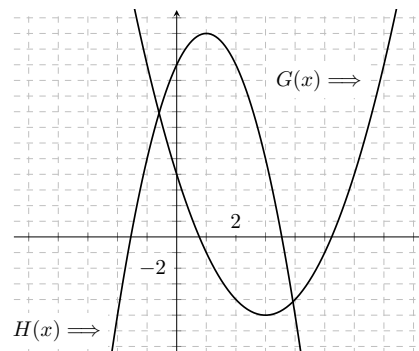


1

Which of the following polynomials have no roots?

- A) $x^2 + 1$
- B) $x^2 - 1$
- C) $x^2 - 4$
- D) $x^2 + 6x + 5$

2



Graphs of the functions $G(x)$ and $H(x)$ are shown in the xy -plane above. For which of the following values of x does $G(x) + H(x) = 0$?

- A) 4
- B) 3
- C) 2
- D) 1



3

Suppose that N is the cost of k CD's at jTones web-store. Which of the following expressions best describes the price of x CD's?

- A) N/x
- B) N/k
- C) kN
- D) None of the above

4

Which of the following is not equivalent to $6^{\frac{3}{2}}$?

- A) $2\sqrt{54}$
- B) $6\sqrt{6}$
- C) $9\sqrt{8}$
- D) $\sqrt{216}$



5

In an election, 3.6 million votes were cast. Everyone voted for either Xavier, Javier, or Zavier. Xavier received 72,000 fewer votes than Javier and Zavier combined. What percent of the 3.6 million votes were cast for Xavier?

- A) 48%
- B) 49%
- C) 49.5%
- D) 50.5%

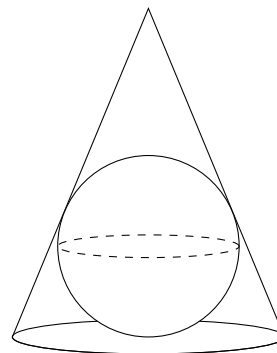
6

A policy analyst is attempting to determine a reasonable amount of available parking spaces per household to advise a city to require for new developments. Which of the following makes the best sense for an outlier resistant approach to looking at the center and spread of the city's current demands for parking spaces per household?

- A) Geometric mean and 90th percentile
- B) Median and interquartile range
- C) Mode and range
- D) None of the above combinations are consistently outlier resistant.

7

A sphere of radius 4 inches is inscribed in a cone with a base of radius 6 inches. In inches, what is the height of the cone? Express your answer as a decimal to the nearest tenth.

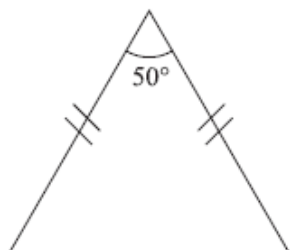


- A) 13.8
- B) 14.1
- C) 14.2
- D) 14.4



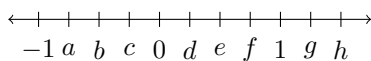
8

The isosceles triangle below has one angle measure as shown. What is the measure of each of the other angles?



- A) 30°
- B) 45°
- C) 50°
- D) 65°

9



The letters $a, b, c, d, e, f, g,$ and h above represent numbers as shown on the number line above. Which of the following expressions has the greatest value?

- A) $d - a$
- B) $f + b$
- C) $g - e$
- D) $h - f$

10

$$h(t) = -16t^2 + 208t + k$$

The equation above models the height in feet of a cannonball t seconds after firing from a location atop a cliff that is k feet above the valley below. After how many seconds will the cannonball return to its starting height?

- A) 13
- B) 15
- C) 16
- D) It cannot be determined from the information given.

Answer Key

Calculator Off	
1	A
2	B
3	D
4	C

⋮

Calculator On	
5	B
6	C
7	D
8	D
9	A
10	A



1

For integers x and y , we say (x, y) is an integer pair solution to an equation if it makes that equation true. How many integer pair solutions does

$$4x + 6y = 71$$

have?

- A) 0
- B) 1
- C) 4
- D) More than 2

2

An elevator has a maximum capacity of 15 persons or 4000 pounds. A football team's linemen and backs are completely worn out from their second workout session of the day and want to get up to their room without taking the stairs. The linemen weigh about 300 pounds each and the backs weigh about 180 pounds each. Let l represent the number of linemen and b represent the number of backs. Which of the following sets of inequalities represents the ways the players could get into the elevator without exceeding the limits?

- A) $\begin{cases} 300l + 180b \leq 4000 \\ l + b \leq 15 \end{cases}$
- B) $\begin{cases} \frac{l}{300} + \frac{b}{180} \leq 15 \\ l + b \leq 4000 \end{cases}$
- C) $\begin{cases} 300l + 180b \leq 15 \\ l + b \leq 4000 \end{cases}$
- D) $\begin{cases} 300l + 180b \leq 4000 \\ l + b \leq 4000 \end{cases}$



3

If $q - r > q$, which of the following must be true?

- A) $q < 0$
- B) $q > 0$
- C) $r = 0$
- D) $r < 0$

4

What is the square of $9x$?

- A) $3x$
- B) $9x$
- C) $3x^2$
- D) $81x^2$



5

$$\begin{aligned}y &= -6 \\ y &= qx^p + q\end{aligned}$$

In the system of equations above, p and q are constants. For which of the following values of p and q does the system of equations have no real solutions?

- A) $p = 1, q = -10$
- B) $p = 2, q = -6$
- C) $p = 3, q = -2$
- D) $p = 4, q = 2$

6

$$\begin{aligned}y &= 2x^2 - x - 6 \\ y &= -x^2 + 2x + 12\end{aligned}$$

What is/are the solution(s) to the system above?

- A) $(x, y) \in \{(-2, 3)\}$
- B) $(x, y) \in \{(-2, 4)\}$
- C) $(x, y) \in \{(3, 9)\}$
- D) $(x, y) \in \{(-2, 3), (3, 9)\}$

7

The function $f(t)$ satisfies the following equation

$$f(3t - 1) = \frac{1}{2t + 1}$$

whenever $t \neq -1/2$. Find $f(f(-2))$.

- A) $\frac{1}{2}$
- B) $\frac{5}{6}$
- C) $\frac{3}{10}$
- D) $\frac{3}{11}$

8

Let M and N be integers. If

$$N = |5M^2 - 2|$$

which of the following is NOT a possible value of N ?

- A) -2
- B) 1
- C) 2
- D) 7



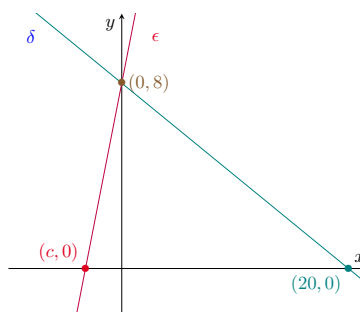
9

$$f_h = \left(\frac{c}{c-v} \right) f_e$$

The formula above gives the frequency heard (f_h) at a stationary point from a moving object given the velocity of sound in the pertinent medium (c), the speed the object is moving towards the listener relative to the medium (v), and the actual frequency of the sound at the source (f_e). Which of the following is a correct formula for v ?

- A) $v = c \left(1 - \frac{f_e}{f_h} \right)$
- B) $v = c(f_h - f_e)$
- C) $v = f_e - f_h$
- D) $v = 1 - \frac{f_e}{f_h}$

10



In the xy -plane above, line δ is perpendicular to line ϵ when the x and y axes are using the same scale. In the diagram the x and y axes are scaled differently. What is the value of c ?

- A) $-\frac{2}{5}$
- B) $-\frac{16}{5}$
- C) $-\frac{5}{2}$
- D) $\frac{5}{2}$

Answer Key

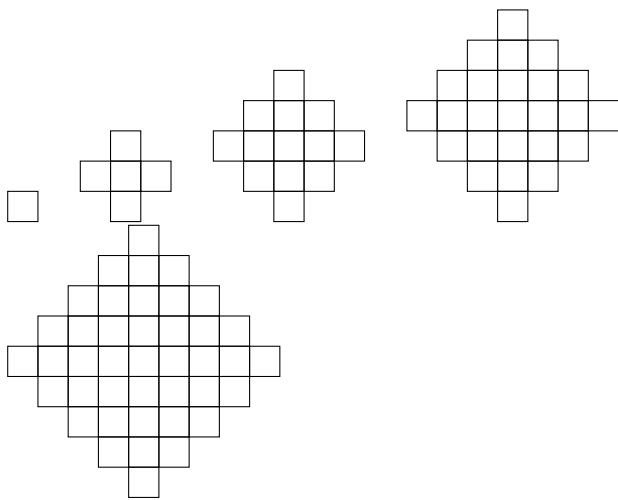
Calculator Off	
1	A
2	A
3	D
4	D

⋮

Calculator On	
5	D
6	D
7	D
8	A
9	A
10	B



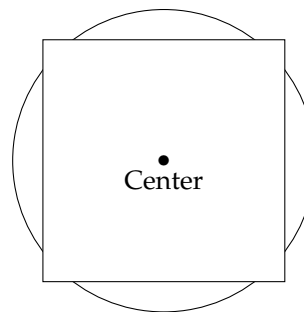
1



The first five pictures in a series of compositions made with unit squares are shown above. If the series were to continue to the 20th picture, how many unit squares would be in that 20th picture?

- A) 400
- B) 761
- C) 2,870
- D) 1,048,573

2



A space figure is formed by the union of a cube with a side of 8 with the same center as a sphere with a radius of 5. The figure is then cut in half through its center, and the cut side is shown above. When the figure is put back together, approximately what percent of the cube's surface area is still visible?

- A) 2%
- B) 22%
- C) 25%
- D) 50%



3

If $\frac{-5}{9}\Delta = \frac{2}{3}$, what is the value of Δ ?

- A) $\frac{-6}{5}$
- B) $\frac{-10}{27}$
- C) $\frac{1}{9}$
- D) $\frac{11}{9}$

4

What is the smallest positive number divisible by 3, 5, and 9?

- A) 30
- B) 45
- C) 90
- D) 135



5

What is the simplified form of $5\sqrt{12} - \sqrt{75}$?

- A) $4\sqrt{3}$
- B) $5\sqrt{3}$
- C) $9\sqrt{3}$
- D) $10\sqrt{3}$

6

Super thin basalt fiber wool can be manufactured to have a density of $2.5 \frac{\text{kg}}{\text{m}^2}$. A metric ton is 1000 kg. How many metric tons of super thin basalt fiber wool would be needed to cover one square kilometer?

- A) 0.0025
- B) 2.5
- C) 2,500
- D) 2,500,000

7

$$M = \frac{5}{9}(I + 460)$$

The equation above shows how a temperature I , measured in degrees Fahrenheit, relates to a temperature M , measured in degrees Kelvin. Based on the equation, which of the following must be true?

- I. A temperature increase of 1 degree Kelvin is equivalent to a temperature increase of $\frac{5}{9}$ degree Fahrenheit.
 - II. A temperature decrease of 1 degree Fahrenheit is equivalent to a temperature decrease of 1.8 degree Kelvin.
 - III. A temperature increase of $\frac{5}{9}$ degree Kelvin is equivalent to a temperature increase of 1 degree Fahrenheit.
- A) I only.
 - B) III only.
 - C) I and II only.
 - D) I and II and III.



8

If $\Phi(x) = -5x + 11$, what is $\Phi(-3y)$ equal to?

- A) $-15y + 11$
- B) $15y + 11$
- C) $-15y - 11$
- D) $15y - 33$

9

If $\frac{4}{a} = \frac{8}{3a-12}$, what is the value of $3a - 12$?

- A) -3
- B) 0
- C) 3
- D) 12

10

A remodeling company estimates the price of a job, in dollars, using the equation $P = F + 45wt$, where P is the estimated price of the job, F is the estimated fixed cost for materials, w is the number of employees devoted to the job, and t is the number of hours spent completing the job. Which of the following is the best interpretation of the number 45 in the equation?

- A) Each construction worker works 45 hours a week.
- B) The price of a job increases by \$45 every hour.
- C) There will be 45 construction workers on the job.
- D) The company charges \$45 per hour for each worker on the job.

Answer Key

Calculator Off	
1	B
2	D
3	A
4	B

⋮

Calculator On	
5	B
6	C
7	B
8	B
9	C
10	D



1

In a class, every student scored either an 85, 90, or 95. The frequency chart is recorded below:

Grade	Number of students
85	10
90	7
95	3

What is the class average?

- A) 87.25
- B) 88.25
- C) 90
- D) 90.25

2

Consider the quadratic $x^2 - 2kx + k^2$. What must be true about this quadratic?

- A) It has 0 real roots
- B) It has a double root
- C) It is always positive
- D) None of the above

3

Which of the following describes the set of points that lie strictly above the line $y = 2x + 1$?

- A) $y < 2x + 1$
- B) $y > 2x + 1$
- C) $y \leq 2x + 1$
- D) $y \geq 2x + 1$

4

Keiko tosses one penny and Ephraim tosses two pennies. The probability that Ephraim gets the same number of heads that Keiko gets is

- A) $\frac{1}{4}$
- B) $\frac{3}{8}$
- C) $\frac{1}{2}$
- D) $\frac{3}{4}$



5

Consider a set of 10 consecutive integers. Which of the following has the greatest probability?

- A) Choosing a number from this set that is less than the median.
- B) Choosing a number from this set that is even.
- C) Choosing a number from this set that is odd.
- D) All of the above have equal probability.

6

What is the smaller root of $4x^2 + 13x + 3$?

- A) -3
- B) $.25$
- C) 3
- D) 4

7

A line in the xy -plane passes through $(4, 5)$ and has a slope of $-\frac{1}{3}$. Which of the following points is on the line?

- A) $(1, 2)$
- B) $(3, 8)$
- C) $(7, 4)$
- D) $(7, 6)$

8

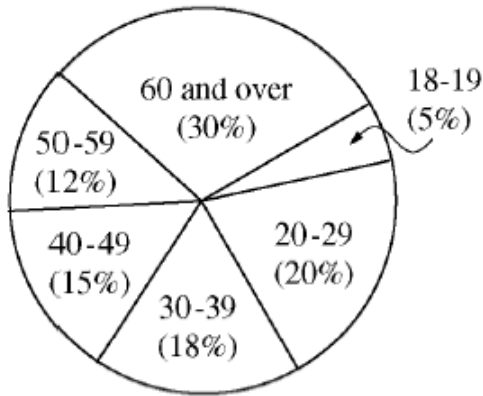
If $3x + y = 6$, what is the value of $2^y 8^x$?

- A) 2^4
- B) 4^3
- C) 8^0
- D) The value cannot be determined from the information given.



9

AGE DISTRIBUTION OF REGISTERED VOTERS IN WASHINGTON COUNTY



According to the graph above, if there are 6,000 registered voters aged 60 and over in Washington County, how many registered voters are under the age of 30 ?

- A) 1,000
- B) 2,000
- C) 4,000
- D) 5,000

10

If

$$\frac{3}{2x-3} = \frac{6}{x}$$

then what is the value of x ?

- A) -2
- B) -3
- C) 3
- D) 2

Answer Key

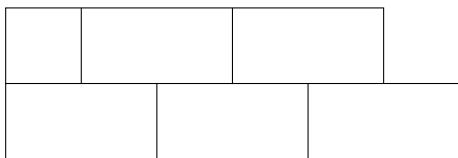
Calculator Off	
1	B
2	B
3	B
4	B

⋮

Calculator On	
5	D
6	A
7	C
8	B
9	D
10	D



1



A block wall 100 feet long and 7 feet high will be constructed using blocks that are 1 foot high and either 2 feet long or 1 foot long (no blocks may be cut). The vertical joins in the blocks must be staggered as shown, and the wall must be even on the ends. What is the smallest number of blocks needed to build this wall?

- A) 344
- B) 347
- C) 350
- D) 353

2

For any integer n , which of the following is always 1 more than double an even number?

- A) $n + 1$
- B) $2n + 1$
- C) $4n + 1$
- D) $4n + 3$

3

Let z be a complex number. Define a transformation of the plane by the function $f(z)$ using the following procedure: Let $w_0 = f(0)$, and define a sequence recursively by $w_k = f(w_{k-1})$. This sequence is periodic, which means that for some $p > 0$, for each $k \geq 0$, we have that $w_{k+p} = w_k$. Find the smallest such value for p .

- A) 8
- B) 10
- C) 12
- D) More than 12.

4

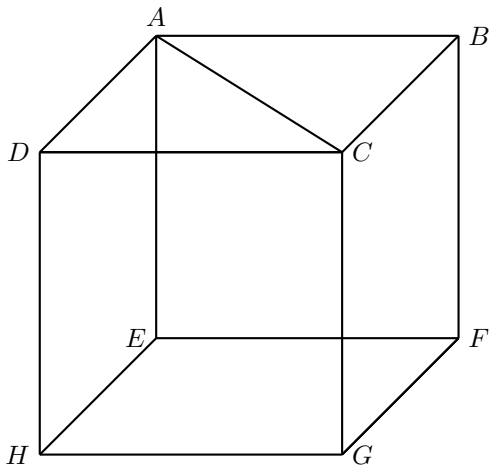
Which of the numbers below is the largest?

- A) $\log(50)$
- B) $3\log(3)$
- C) $5\log(2)$
- D) $3\log(4)$



5

The length of $\overline{AC}$ in the cube below is 2.



What is the perimeter of the triangle formed by connecting vertices A , C and H ?

- A) 6
- B) 8
- C) 10
- D) 12

6

What is $\frac{5-12i}{3-2i}$?

- A) $3 - 2i$
- B) $2 - 2i$
- C) $2 + 2i$
- D) $4 - 2i$

7

$$\begin{aligned} 2x - 4y &= -1 \\ \frac{1}{4}(x + 2) &= y - 2 \end{aligned}$$

What is the solution to the above system of equations in the xy plane?

- A) $(3, -1.75)$
- B) $(8.5, 4.5)$
- C) $(9, 4.75)$
- D) $(15, 7.75)$



8

Chandra pays an on-line service provider a fixed monthly fee plus an hourly charge for connect time. Her December bill was \$12.48, but in January her bill was \$17.54 because she used twice as much connect time as in December. What is the fixed monthly fee?

- A) \$2.53
- B) \$5.06
- C) \$6.24
- D) \$7.42

9

$$\begin{aligned} Bx - 5y &= -3 \\ 3x + 4y &= 10 \end{aligned}$$

In the system of equations above, B is a constant and x and y are variables. For what value of B will the system of equations have no solution?

- A) 25
- B) $\frac{15}{4}$
- C) 0
- D) $-\frac{15}{4}$

10

$$\begin{aligned} g &= 3.99 - 0.20w \\ b &= 2.39 + 0.20w \end{aligned}$$

In the equations above, g and b represent the price of a gallon of premium gas and the price of a pint of blueberries w weeks after the peak of blueberry season a previous year at a particular store. What was the price per gallon of premium gasoline when it matched the price of a pint of blueberries?

- A) \$3.79
- B) \$3.19
- C) \$2.79
- D) \$1.60

Answer Key

Calculator Off	
1	D
2	C
3	A
4	D

⋮

Calculator On	
5	A
6	A
7	C
8	D
9	D
10	B



1



Which of the following letters satisfy $x < \sqrt{x} < \sqrt[3]{x}$?

- A) a
- B) b
- C) c
- D) None of the above

2

A pet hotel advertises that they have enough space and supplies to regularly house a ton of cats and dogs (literally, 2,000 pounds of cats and dogs) on a daily basis. At present the pet hotel is housing a sled dog team of 18 Siberian Huskies, and the average weight of the Siberian Huskies is 40 pounds. They are also housing 3 cats with an average weight of 10 pounds each. If the hotel currently has no other tenants, how many more pounds of dogs and cats could the pet hotel take in if they are true to their advertisement?

- A) 1930
- B) 1280
- C) 1250
- D) 750

3

Which one of the numbers below is the same as double its square?

- A) .25
- B) .5
- C) 1
- D) 4



4

Define $f(x) = x^a$ where a is a positive number.
Which of the following is the same as $f(x + y)$?

- A) $f(x) + f(y)$
- B) $f(x) - f(y)$
- C) $f(x)f(y)$
- D) None of the above.



5

Gender	Age		Total
	0 to 24	25 to 40	
Male	28	17	45
Female	15	3	18
Total	43	20	63

The table above shows the distribution of age and gender for an amateur athletic event. Due to the number of contestants, somebody in the first round must receive a bye. If the recipient of the bye is selected at random, what is the probability that the recipient will either be a female under the age of 25 or a male over the age of 24?

- A) $\frac{15}{63}$
- B) $\frac{17}{63}$
- C) $\frac{31}{63}$
- D) $\frac{32}{63}$

6

The ratio of purple marbles to blue marbles in a jar is 5 : 2. If there are 6 blue marbles, then how many marbles are in the jar?

- A) 15
- B) 20
- C) 21
- D) 35

7

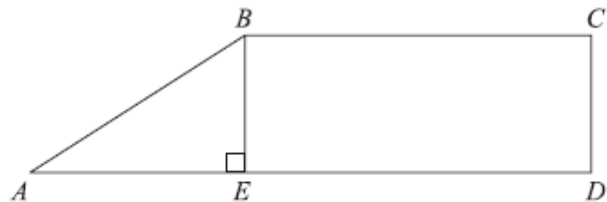
The expression $\frac{16p-7}{4p+3}$ is equivalent to which of the following?

- A) $4 - \frac{7}{3}$
- B) $\frac{4+7}{-3}$
- C) $4 - \frac{19}{4p+3}$
- D) $4 + \frac{5}{4p+3}$



8

In the figure below $AD = 16$, $ED = 11$, and AE is congruent to CD . What is the length of AB ?



- A) $5\sqrt{2}$
- B) 6
- C) $11\sqrt{2}$
- D) 25

9

25% of k is equal to $\frac{8}{9}$ of r . What is $\frac{r}{k}$?

- A) $\frac{225}{8}$
- B) $\frac{32}{9}$
- C) $\frac{9}{32}$
- D) $\frac{2}{9}$

10

In a mathematics contest with ten problems, a student gains 5 points for a correct answer and loses 2 points for an incorrect answer. If Olivia answered every problem and her score was 29, how many correct answers did she have?

- A) 5
- B) 7
- C) 8
- D) 9

Answer Key

Calculator Off	
1	C
2	D
3	B
4	D

⋮

Calculator On	
5	D
6	C
7	C
8	A
9	C
10	B



1

$$\begin{array}{r} A \quad B \\ + \quad B \quad A \\ \hline 1 \quad D \quad C \end{array}$$

A , B , C and D are all digits in the correctly worked out problem above. What is the value of $D - C$?

- A) 0
- B) 1
- C) 4
- D) 9

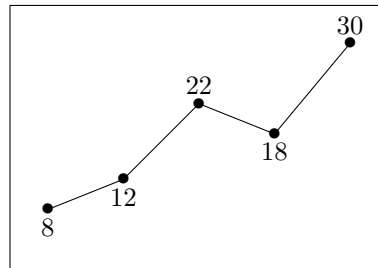
2

What is $\log_{12}(8) \log_8(144)$?

- A) 1
- B) 2
- C) 4
- D) 8

3

Blue Jade Shoes sold per month



April May June July August

The line graph above shows the number of blue jade shoes sold by Tian Jewelers from April to August of 1961.

Tian Jewelers sold how many more shoes in August than in April and July combined?

- A) 22
- B) 12
- C) 4
- D) 2



4

Find all x such that $0 < x < 360^\circ$ and $2 \tan^2(x) = \frac{2}{3}$.

- A) $30^\circ, 60^\circ$
- B) $30^\circ, 210^\circ$
- C) $30^\circ, 150^\circ, 210^\circ, 330^\circ$
- D) 330°



5

Consider the following system of equations:

$$\begin{aligned}2x + 3y &= -2 \\ 6x + 9y &= -4\end{aligned}$$

How many solutions does this system have?

- A) 0
- B) 1
- C) 2
- D) Infinitely many

6

Which of the following expresses the relationship, x is the square of y ?

- A) $x = \sqrt{y}$
- B) $x = y$
- C) $x = y^2$
- D) $x = y^5$

7

If $x > 6$, which of the following is equivalent to

$$\frac{1}{\frac{2}{x-6} + \frac{4}{x-3}}?$$

- A) $\frac{6x-30}{x^2-9x+18}$
- B) $\frac{x^2-9x+18}{6x-30}$
- C) $x^2 - 9x + 18$
- D) $\frac{2x-9}{6}$

8

What is the sum of the digits of the decimal form of the product $2^{1999} \cdot 5^{2001}$?

- A) 2
- B) 4
- C) 7
- D) 10



9

A lucky year is one in which at least one date, when written in the form month/day/year, has the following property: The product of the month times the day equals the last two digits of the year. For example, 1956 is a lucky year because it has the date 7/8/56 and $7 \times 8 = 56$. Which of the following is NOT a lucky year?

- A) 1990
- B) 1991
- C) 1993
- D) 1994

10

Which of the following expressions is equal to 5 for some real value of z ?

- A) $2 + \sqrt{z}$
- B) $-2 - 3\sqrt{z}$
- C) $4 - \sqrt{z}$
- D) $-1 + \sqrt{16 - z^2}$

Answer Key

Calculator Off	
1	B
2	B
3	C
4	C

⋮

Calculator On	
5	A
6	C
7	B
8	C
9	D
10	A



1

Suppose $(x - 3)$ is the only factor of the quadratic $P(x)$. Then what is $P(x)$?

- A) $x^2 + 9$
- B) $x^2 - 6x + 9$
- C) $x^2 + 6x + 9$
- D) $x^2 + 6x - 9$

2

Each of five friends take either a car or a bus to get to a party. If at least one friend needs to be in each vehicle, then how many ways can the friends be assigned to a bus or a car?

- A) 28
- B) 29
- C) 30
- D) 32

3

$\frac{u^3}{b}$ is an integer, but $\frac{u^2}{b}$ is **NOT** an integer. Which of the following could be the values of u and b ?

- A) $u = 1, b = 1$
- B) $u = 3, b = 81$
- C) $u = 4, b = 32$
- D) $u = 5, b = 25$

4

If $\frac{x}{y} = \frac{x-2}{y+3}$ and $x > \frac{3}{2}$, what is the restriction on the range of y ?

- A) $y < \frac{-9}{4}$
- B) $y < \frac{3}{2}$
- C) $y > \frac{3}{2}$
- D) $y > \frac{-9}{4}$



5

Which point does not lie on the unit circle?

- A) $(-1, 0)$
- B) $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$
- C) $(0, -1)$
- D) $(1, 1)$

6

The cost to rent a computer with the fastest internet connection speed possible at an internet cafe is \$.001 per second. Which of the following equations represents the total rent r , in dollars, for h hours of computer rental?

- A) $r = \frac{0.001}{3600h}$
- B) $r = \frac{60h}{0.001}$
- C) $r = \frac{3600h}{0.001}$
- D) $r = 0.001(3600h)$

7

Six trees are equally spaced along one side of a straight road. The distance from the first tree to the fourth is 60 feet. What is the distance in feet between the first and last trees?

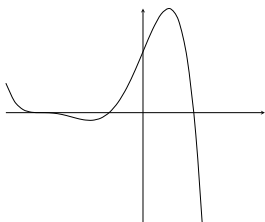
- A) 90
- B) 100
- C) 105
- D) 120



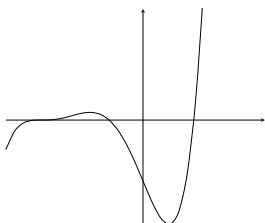
8

Which of the following graphs could be the graph of $y = 4(x - \frac{1}{6})^3(x + \frac{1}{9})(x + \frac{1}{3})$?

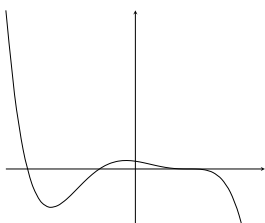
A)



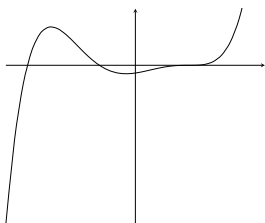
B)



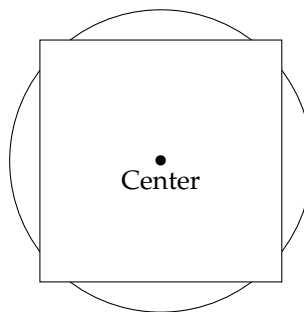
C)



D)



9



A space figure is formed by the union of a cube with a side of 8 with the same center as a sphere with a radius of 5. The figure is then cut in half through its center, and the cut side is shown above. When the figure is put back together, approximately what percent of the cube's surface area is still visible?

A) 19%

B) 22%

C) 25%

D) 50%



10

If $\log_2(13) = x$ and $\log_2(26) = y$, what is $x - y$?

- A) -1
- B) 2
- C) 2.5
- D) 5.9

Answer Key

Calculator Off	
1	B
2	C
3	C
4	A

⋮

Calculator On	
5	D
6	D
7	B
8	D
9	D
10	A



1

The sum of all the integers from 1 to 50 (inclusive) is 1275. What is the sum from 51 to 100?

- A) 3775
- B) 3825
- C) 3950
- D) 4000

2

For any real numbers a, b , if the point (a, b) is reflected across the line $y = x$, what is the resulting point?

- A) $(a, -b)$
- B) (b, a)
- C) $(-b, a)$
- D) $(-b, -a)$

3

A researcher conducted a survey to determine whether voters in a certain large city prefer private sector union protections or right-to-work laws. The researcher asked 2409 voting-age people at bus stops near the industrial zones, and 352 refused to respond. Which of the following factors makes it least likely that a reliable conclusion can be drawn about the political opinions of all the voters in the city?

- A) The place the survey was given
- B) The large percentage that refused to respond
- C) The sample size
- D) The population size

4

The terms of a sequence are defined by $a_n = a_{n-2} + 5a_{n-1}$ for $n > 2$. What is the value of a_4 if $a_1 = 2$ and $a_2 = 4$?

- A) 100
- B) 114
- C) 49
- D) 56



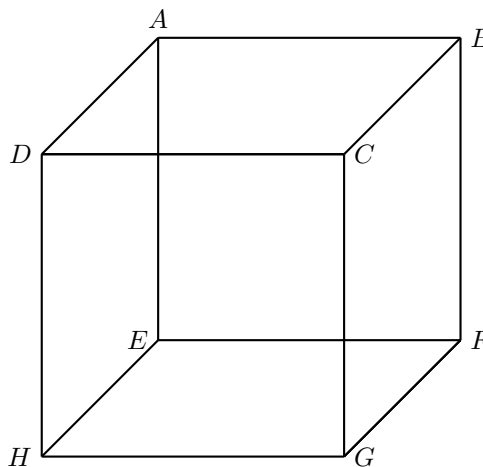
5

Which of the following has only complex roots?

- A) $x^2 - 1$
- B) $x^2 + 16$
- C) $x^2 - 2x + 1$
- D) $x^2 - 3x + 2$

6

The length of each edge in the cube below is 2.



What is the length of $\overline{AG}$ (not shown)?

- A) 2
- B) $2\sqrt{3}$
- C) $2\sqrt{5}$
- D) 4



7

Which of the following represents a way of rewriting $y = 2x^2 - x - 10$ such that the roots of the equation are explicitly stated as values inside the equation?

- A) $y = 2(x - \frac{1}{4})^2 - \frac{81}{8}$
- B) $y = (2x - 5)(x + 2)$
- C) $y = 2(x + \frac{5}{2})(x - 2)$
- D) $y = 2(x - \frac{5}{2})(x + 2)$

8

Twice as many countries adopted the Euro from 1999 to 2001 as had adopted the currency from 2002 to 2014, with no countries deciding to leave the currency. If n countries adopted the currency from 1999 to 2001, which of the following expressions represents the total number of countries that were using the Euro in 2014?

- A) $\frac{1}{2}n$
- B) $\frac{3}{2}n$
- C) $2n$
- D) $3n$

9

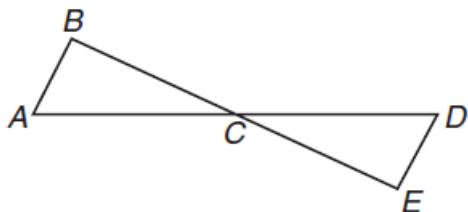
For positive integers x , let $\text{r}_3 x$ be defined as the remainder when x is divided by 3. Which of the following is equal to $\text{r}_3 11$?

- A) $\text{r}_3 15 - \text{r}_3 4$
- B) $\text{r}_3 14 - \text{r}_3 3$
- C) $\text{r}_3 13 - \text{r}_3 2$
- D) $\text{r}_3 12 - \text{r}_3 1$



10

The following is a partially completed correct proof of a fact from plane geometry.



GIVEN: Point C is the midpoint of $\overline{AD}$ and $\overline{BE}$.

PROVE: $\triangle ABC \cong \triangle DEC$

Statement	Justification
1. C is the midpoint of $\overline{AD}$ and $\overline{BE}$.	Given
2. $\overline{AC} \cong \overline{DC}$ and $\overline{BC} \cong \overline{EC}$	Definition of midpoint
3. $\angle ACB \cong \angle DCE$	?
4. $\triangle ABC \cong \triangle DEC$	SAS

Which of the following best completes the proof?

- A) The distributive property of polygons.
- B) The angle equivalence principle.
- C) The congruence property of vertical angles.
- D) The inverse property of translations.

Answer Key

Calculator Off	
1	A
2	B
3	A
4	B

⋮

Calculator On	
5	B
6	B
7	D
8	C
9	B
10	C



1

When 9.479 is rounded to the nearest whole number, the result is how much less than when 9.479 is rounded to the nearest hundredth?

- A) 9
- B) 0.48
- C) 1.0
- D) 25

2

If

$$\frac{x}{x+2} = \frac{1}{3}$$

what is x ?

- A) 1
- B) 2
- C) 3
- D) 5

3

Define $f(x) = a^x$ where a is a positive number. Which of the following is the same as $f(x+y)$?

- A) $f(x) + f(y)$
- B) $f(x) - f(y)$
- C) $f(x)f(y)$
- D) None of the above.

4

Which of the following polynomials have only positive roots?

- A) $x^2 - 5x + 6$
- B) x^2
- C) $x^2 - 7x + 6$
- D) $x^2 + 2x + 1$



5

Suppose

$$(4x - 5)(4x + 5) = 75$$

Then what is $4x^2$?

- A) 100
- B) 75
- C) 50
- D) 20

6

For which of the following years is the predicted GDP per capita for the United Kingdom less than the actual GDP per capita for the United Kingdom?

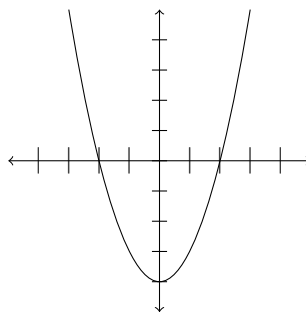
- A) 2003
- B) 2005
- C) 2007
- D) 2009

7

If $\frac{c+d}{c} = \frac{6}{7}$, which of the following must also be true?

- A) $\frac{2c+d}{c} = 1$
- B) $\frac{d-c}{c} = \frac{-1}{7}$
- C) $\frac{d}{c} = \frac{-1}{7}$
- D) $\frac{d}{c} = \frac{13}{7}$

8



$g(x)$ is one result when $f(x) = x^2 - 4$, shown above, is horizontally shifted. If $g(7) = 0$, which of the following could be the equation of $g(x)$?

- A) I only
- B) II only
- C) I and III only
- D) I, II, and III



9

$$\begin{aligned}x &= \frac{1}{2}(y - 4) \\ y &= (-2x + 7)(x - 1)\end{aligned}$$

How many ordered pairs of real numbers (x, y) satisfy the system of equations shown above?

- A) 0
- B) 2
- C) 3
- D) Infinitely many.

10

2, 3, 6, n , 42, ...

The first term of the sequence above is 2, and each term after the first is 3 less than 3 times the previous term. What is the value of n ?

- A) 9
- B) 12
- C) 15
- D) 21

Answer Key

Calculator Off	
1	B
2	A
3	C
4	A

⋮

Calculator On	
5	A
6	C
7	C
8	C
9	A
10	C



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

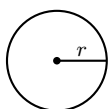
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

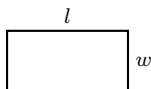
1. The use of a calculator **is not permitted**.
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3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

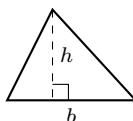


$$A = \pi r^2$$

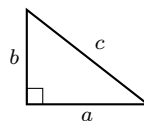
$$C = 2\pi r$$



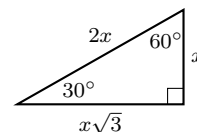
$$A = lw$$



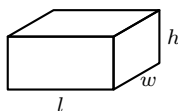
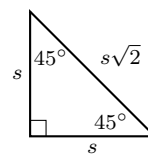
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



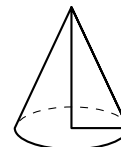
$$V = lwh$$



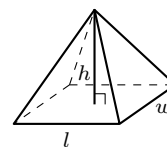
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

If $f(x) = x^2$ and $g(x) = x + 1$, then what is $f(g(x))$?

- A) $x^2 + 1$
- B) $x^2 + x + 1$
- C) $x^2 + 2x + 1$
- D) $x^2 + 2x + 2$

2

Philip buy 2 hats for \$3 each and 6 socks for \$1 each. Which of the following shows the correct matrix multiplication to compute the amount of money Philip spends?

- A) $\begin{bmatrix} 2 & 3 \end{bmatrix} \begin{bmatrix} 6 & 1 \end{bmatrix}$
- B) $\begin{bmatrix} 2 & 6 \end{bmatrix} \begin{bmatrix} 3 & 1 \end{bmatrix}$
- C) $\begin{bmatrix} 2 \\ 6 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$
- D) $\begin{bmatrix} 2 & 6 \end{bmatrix} \begin{bmatrix} 3 \\ 1 \end{bmatrix}$

3

Let a, b, c, d, e , and f be distinct lines. If

$$\begin{array}{ll} a & \parallel b, \\ b & \perp c, \\ c & \parallel d, \\ d & \perp e, \\ e & \parallel f \end{array}$$

then how many of the aforementioned lines are perpendicular to a ?

- A) 2
- B) 3
- C) 4
- D) 6

4

If the difference of two numbers is 7 and their sum is 31, then what is the larger of these two numbers?

- A) 12
- B) 14
- C) 18
- D) 19



5

For points A, B, C , $\overline{AB} = \overline{BC}$. Then which of the following must be true?

- A) B bisects $\overline{AC}$
- B) B is the midpoint of $\overline{AC}$
- C) $\overline{AB}$ is parallel to $\overline{BC}$
- D) None of the above

6

Suppose $4x^2 + 1 < 17$. What is the set of all x that satisfies this inequality?

- A) $x > 4$
- B) $x < 4$
- C) $-4 < x < 4$
- D) None of the above

7

In a basketball tournament there are 8 teams. If each team plays 4 matches with every other team, then how many matches are there?

- A) 32
- B) 56
- C) 110
- D) 112

8

Which of the following expressions equals $(x+5)(x-3)$?

- A) $x^2 - 2x - 15$
- B) $x^2 + 2x - 15$
- C) $x^2 - 2x + 15$
- D) $x^2 + 2x + 15$



9

Suppose x satisfies $3^x = 7$. Then which of the following is equal to x ?

- A) $\log_3(7)$
- B) $\log_7(3)$
- C) $\log_7(10)$
- D) None of the above

10

If a is 50% of 150% of b , then what is a in terms of b ?

- A) $\frac{3}{4}b$
- B) b
- C) $\frac{3}{2}b$
- D) $2b$

11

What is

$$\frac{10 \cdot 8 \cdot 6 \cdot 4 \cdot 2}{5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}?$$

- A) 5
- B) 8
- C) 16
- D) 32

12

Which of the following best describes the set of all points satisfying $x^2 + y^2 > 1$?

- A) A circle
- B) The set of all points outside a circle
- C) The set of all points between two circles
- D) The empty set



13

Which equation represents the following relationship: *The sum of two numbers is double the difference.*

- A) $x + y = x - y$
- B) $x + y = 2(x - y)$
- C) $2(x + y) = x - y$
- D) $x = 2y$

14

What is not in the range of the real valued function $f(x) = x^2 + 2$?

- A) 1
- B) 2
- C) 4
- D) All the values above are in the range.

15

Suppose parallelogram $ABCD$ has two congruent adjacent angles. Which of the following must be true?

- A) All sides are equal.
- B) All angles are equal.
- C) $ABCD$ is a rhombus.
- D) None of the above.

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

5. **Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If  is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

6. **Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write answer in boxes. →

Grid in result. →

← Fraction line

← Decimal point

Answer: 2.5

	7	/	1	2		2	.	5	
	☐	☒	☐	☐		☐	☐	☒	☐
	☐	☐	☐	☐		☐	☐	☐	☐
1	☐	☐	☐	☐	1	☐	☐	☐	☐
2	☐	☐	☐	☐	2	☒	☐	☐	☐
3	☐	☐	☐	☐	3	☐	☐	☐	☐
4	☐	☐	☐	☐	4	☐	☐	☐	☐
5	☐	☐	☐	☐	5	☐	☐	☐	☒
6	☐	☐	☐	☐	6	☐	☐	☐	☐
7	☒	☐	☐	☐	7	☐	☐	☐	☐
8	☐	☐	☐	☐	8	☐	☐	☐	☐
9	☐	☐	☐	☐	9	☐	☐	☐	☐

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3		.	6	6	6		.	6	6	7
	☐	☒	☐		☒	☐	☐	☐		☒	☐	☐	☐
	☐	☐	☐		☐	☐	☐	☐		☐	☐	☐	☐
1	☐	☐	☐	1	☐	☐	☐	☐	1	☐	☐	☐	☐
2	☐	☐	☐	2	☐	☐	☐	☐	2	☐	☐	☐	☐
3	☐	☐	☐	3	☐	☐	☐	☐	3	☐	☐	☐	☐
4	☐	☐	☐	4	☐	☐	☐	☐	4	☐	☐	☐	☐
5	☐	☐	☐	5	☐	☐	☐	☐	5	☐	☐	☐	☐
6	☐	☐	☐	6	☒	☒	☒		6	☒	☒	☒	
7	☐	☐	☐	7	☐	☐	☐	☐	7	☐	☐	☐	☒
8	☐	☐	☐	8	☐	☐	☐	☐	8	☐	☐	☐	☐

Answer: 201 – either position is correct

	2	0	1		2	0	1
	☐	☐	☐		☐	☐	☐
	☐	☐	☐		☐	☐	☐
1	☐	☐	☐	1	☐	☐	☐
2	☒	☐	☐	2	☒	☐	☐
3	☐	☐	☐	3	☐	☐	☐

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16

The sum of two numbers is 8 and their product is 12. What is the larger of these two numbers?

17

If

$$5^x = \sqrt{125}$$

then what is the value of x ?

18

If $7x - 8 = 12$, what is $7x + 8$?

19

Evaluate $(3^2)^3 - (3^3)^2$

20

Bob is currently 5 years younger than Ana. Five years from now, Ana will be 1.5 times Bob's age. How old is Ana currently?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

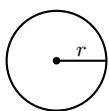
DIRECTIONS

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NOTES

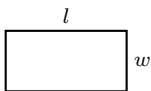
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REFERENCE

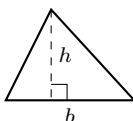


$$A = \pi r^2$$

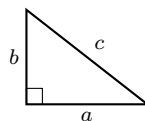
$$C = 2\pi r$$



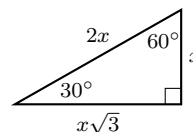
$$A = lw$$



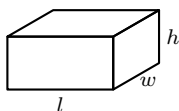
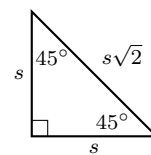
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



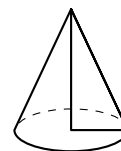
$$V = lwh$$



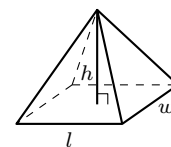
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

Suppose the point (a, b) satisfies $a \cdot b = 1$. Then what quadrants can (a, b) be in?

- A) I, II only
- B) I, III only
- C) I, II, IV only
- D) Any quadrant

2

Which of the following expresses the relationship, x is the square of y ?

- A) $x = \sqrt{y}$
- B) $x = y$
- C) $x = y^2$
- D) $x = y^3$

3

Let a, b , and c be integers. If $2a = 4b$ and $2b = 4c$, then what is a in terms of c ?

- A) c
- B) $3c$
- C) $4c$
- D) $6c$

4

If

$$|a + b| = |a - b|$$

then which of the following must be true?

- A) $a = 0$
- B) $a = 0$ or $b = 0$
- C) $a = 0$ and $b = 0$
- D) None of the above



5

How many pairs (a, b) satisfy

$$a + b = 1$$

$$a + b = 0$$

- A) 0
- B) 2
- C) 4
- D) Infinitely many

6

Which of the following has the same value as $\sin(140^\circ)$?

- A) $\sin(40^\circ)$
- B) $\sin(150^\circ)$
- C) $\sin(220^\circ)$
- D) $\sin(230^\circ)$

7

Which of the following values is not in the range of $f(x) = \frac{\sqrt{x+3}}{2}$?

- A) 1
- B) 2
- C) 3
- D) $\sqrt{8}$

8

Which of the following has the same value as $\tan(15^\circ)$?

- A) $\tan(-15^\circ)$
- B) $\tan(105^\circ)$
- C) $\tan(195^\circ)$
- D) $\tan(360^\circ)$



9

Consider the isosceles triangle $\triangle ABC$. If one pair of angles has a measure whose sum is 160° , then what is the largest possible angle that $\triangle ABC$ can have?

- A) 60°
- B) 100°
- C) 140°
- D) 150°

10

A spherical container of radius r that is completely filled with water is poured into a cylinder of height h and radius r . If, after pouring the water, the cylinder is three quarters full, then what is h in terms of r ? (Note: the volume of a sphere is $\frac{4}{3}\pi r^3$.)

- A) $\frac{4}{3}r$
- B) $\frac{5}{3}r$
- C) $\frac{14}{9}r$
- D) $\frac{16}{9}r$

11

Which of the following best describes the set of all points satisfying $3 > x^2 + y^2 > 1$

- A) A circle
- B) The set of all points outside a circle
- C) The set of all points between two circles
- D) The empty set

12

Suppose the sum of the measures of two adjacent angles of a quadrilateral is 270° . Which of the following cannot be true?

- A) The quadrilateral is a parallelogram.
- B) The remaining two angles are complementary.
- C) The sum of the measures of the angles of this quadrilateral is 360° .
- D) Two sides of the quadrilateral are equal in length.



13

What is $\log_{12}(36) + \log_{12}(4)$?

- A) 2
- B) 3
- C) 4
- D) 5

14

What are all values of x such that $|5x - 1| = 4$?

- A) 0
- B) 1
- C) .6, 1
- D) $-.6, 1$

15

What is $\frac{4\pi}{5}$ radians in degrees?

- A) 120°
- B) 124°
- C) 144°
- D) 150°

16

Suppose the quadratic $x^2 - 2bx + c$ has a double root where b and c are both positive. What is b in terms of c ?

- A) $\sqrt{c}$
- B) c
- C) $2c$
- D) None of the above



17

Suppose a and b are numbers such that $a + b$ is rational. What must be true about a ?

- A) a is rational
- B) a is an integer
- C) $a = 0$
- D) None of the above

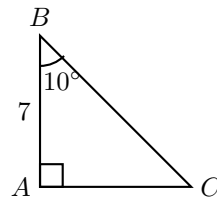
18

Expand $(x + y)(x - y)$

- A) $x^2 + 2x + y^2$
- B) $x^2 + y^2$
- C) $x^2 - 2x + y^2$
- D) $x^2 - y^2$

19

In the following diagram, what is the length of $\overline{AC}$?



- A) $\frac{\tan(10^\circ)}{7}$
- B) $7 \tan(10^\circ)$
- C) $\frac{7}{\sin(10^\circ)}$
- D) $7 \cos(10^\circ)$

20

For integers x and y , we say (x, y) is an integer pair solution to an equation if it makes that equation true. How many integer pair solutions does

$$4x + 6y = 71$$

have?

- A) 0
- B) 1
- C) 2
- D) 4



21

Suppose $f(x) = f(y)$ for every x and y . Which of the following can be f ?

- A) $f(x) = x$
- B) $f(x) = x - 2$
- C) $f(x) = 5$
- D) None of the above.

22

The solution set of $y > x^2 + 1$ lie in which quadrant(s)?

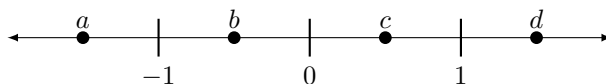
- A) *I* only
- B) *I* and *II* only
- C) *I*, *II*, *IV* only
- D) *I*, *II*, *III*, *IV*

23

Suppose $\sin(x + a) + b$ has a range of $[1, 3]$. Then which of the following must be true?

- A) $a = 2$
- B) $a = -2$
- C) $b = 2$
- D) $b = -2$

24



Which of the following has the greatest value?

- A) $a + b$
- B) $b + c$
- C) $a - d$
- D) $d - a$



25

Which of the following Cartesian coordinates, when represented as (r, θ) , has the greatest value of r ?

- A) $(3, 4)$
- B) $(3, 3)$
- C) $(4, 4)$
- D) $(5, 0)$

26

The solution set of $-3y - 6 > -3x$ lie in which quadrant(s)?

- A) *I* only
- B) *I, II, III* only
- C) *I, III, IV* only
- D) *I, II, III, IV*

27

The number 386231444 is closest to which of the following numbers?

- A) 3862×10^4
- B) 3.86×10^9
- C) 3.86×10^8
- D) 386×10^8

28

Which equation can be used to solve for the smallest number in the following relationship: *The sum of three consecutive odd numbers is 15.*

- A) $x + (x + 1) + (x + 2) = 15$
- B) $x + (x + 1) + (x + 3) = 15$
- C) $x + (x + 2) + (x + 4) = 15$
- D) $x + (3x) + (5x) = 15$



29

What is the value of a such that the system below has no solutions?

$$\begin{aligned}x - 2y &= -2 \\ -4x + 8y &= a\end{aligned}$$

- A) 4
- B) 6
- C) 8
- D) 10

30

Which of the following can be the first three terms of a geometric sequence?

- A) 1, 5, 9
- B) 2, 4, 6
- C) -2, 4, -8
- D) -2, 5, 9

**DIRECTIONS**

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- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

7	/	1	2
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 2.5

← Decimal point

2	.	5
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

2	/	3
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

2	0	1
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

2	0	1
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

In a certain recipe, 12 eggs are used to create 16 cakes. How many eggs are required to make 88 cakes?

32

If

$$y = x + 12$$

and

$$3y = -2y + 5$$

what is x ?

33

What is $\log_{12}(8) \log_8(144)$?

34

A dumbbell rack holds a collection of weights. There are 5 two pound weights, 4 three pound weights, 6 five pound weights, x ten pound weights, and 5 twenty pound weights. If the median weight on the rack is 10 pounds, what is the smallest number that x can be?

35

In a class, everyone is in swim club, ski club, or both. If there are 24 students in class, 17 students are in ski club, and 12 students are in swim club, how many are in both clubs?

36

In a bag containing red marbles and blue marbles, after adding n red marbles to a bag, the probability of choosing a red marble is 50%. If there were originally 8 red marbles and 20 blue marbles, then what is the value of n ?



37

If 12% of 35 is $x\%$ of 12, what is the value of x ?

38

What is $\log_4(1024)$?

Answer Key

Calculator Off	
1	C
2	D
3	A
4	D
5	D
6	C
7	D
8	A
9	A
10	A
11	D
12	B
13	B
14	A
15	B
16	6
17	1.5
18	28
19	0
20	10

.....

Calculator On	
1	B
2	C
3	C
4	B
5	A
6	A
7	A
8	C
9	C
10	D
11	C
12	A
13	A
14	D
15	C
16	A
17	A
18	D
19	B
20	A
21	C
22	B
23	C
24	D
25	C
26	C
27	C
28	C
29	C
30	C
31	66
32	-11
33	2
34	11
35	5
36	12
37	35
38	5



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

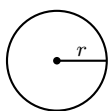
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

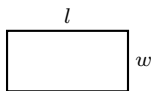
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

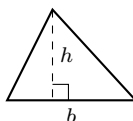


$$A = \pi r^2$$

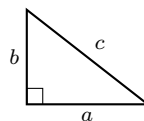
$$C = 2\pi r$$



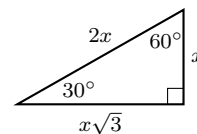
$$A = lw$$



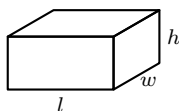
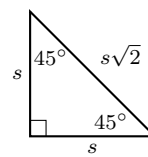
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



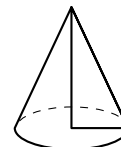
$$V = lwh$$



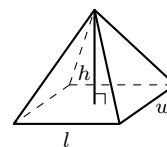
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

What is the distance between the x -intercept and y -intercept of the function $f(x) = \frac{4}{9}(x - 3)^2$?

- A) 3
- B) $3\sqrt{3}$
- C) $4\sqrt{2}$
- D) 5

2

Suppose the sum of ten distinct positive numbers is 57. What is the largest possible number of these ten numbers?

- A) 10
- B) 11
- C) 12
- D) 13

3

Find all x such that $x^2 + x - 6 < 0$.

- A) $x < -3$ or $x > 2$
- B) $-3 < x < 2$
- C) $-2 < x < 3$
- D) $x < -2$ or $x > 3$

4

Which of the following can be the first three terms of an arithmetic sequence?

- A) 2, 4, 8
- B) $-1, 1, -1$
- C) $-2, 1, 3$
- D) $-2, 1, 4$



5

Consider the sequence of 1's:

$$1, 1, 1, 1, \dots$$

What is the sum of the first 19 terms of this sequence?

- A) 19
- B) 100
- C) 190
- D) 200

6

Which of the following numbers below is the largest?

- A) $\frac{1}{2 \cdot 3 \cdot 5 \cdot 7}$
- B) $\frac{1}{2 \cdot 3 \cdot 5 \cdot 13}$
- C) $\frac{1}{2 \cdot 3 \cdot 7 \cdot 13}$
- D) $\frac{1}{3 \cdot 5 \cdot 7 \cdot 13}$

7

Suppose

$$6 < 3(x - 5) < 9$$

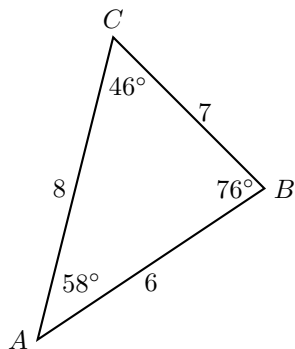
Then which of the following describe all possible x ?

- A) $2 < x < 8$
- B) $3 < x < 7$
- C) $3 < x < 8$
- D) $7 < x < 8$



8

Consider the figure below:



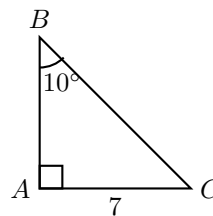
Which equation holds true?

(Note: For any triangle with sides a, b, c that are opposite angles A, B, C respectively, $c^2 = a^2 + b^2 - 2ab \cos(C)$)

- A) $7^2 = 6^2 + 8^2 - 2 \cdot 6 \cdot 8 \cdot \cos(58^\circ)$
- B) $6^2 = 7^2 + 8^2 - 2 \cdot 7 \cdot 8 \cdot \cos(58^\circ)$
- C) $8^2 = 6^2 + 7^2 + 2 \cdot 6 \cdot 7 \cdot \cos(58^\circ)$
- D) $6^2 = 7^2 + 8^2 + 2 \cdot 7 \cdot 8 \cdot \cos(58^\circ)$

9

In the following diagram, what is the length of $\overline{BC}$?



- A) $\frac{\cos(10^\circ)}{7}$
- B) $\frac{\tan(10^\circ)}{7}$
- C) $\frac{7}{\sin(10^\circ)}$
- D) $\frac{7}{\cos(10^\circ)}$

10

Expand $(a - 2b + c) + (-2a + b - 3c)$:

- A) $-a - b - 2c$
- B) $a + 2b - c$
- C) $-a + b - c$
- D) $a - b - c$



11

If $\sin(x) = \frac{1}{2}$, then what is $\cos(x)$?

- A) $\frac{1}{5}$
- B) $\frac{2}{5}$
- C) $\frac{3}{5}$
- D) Cannot be determined

12

Which of the following is the same as $\sin^2(x) + \cos^2(x)$?

- A) $\cos(x)$
- B) $\sin(x)$
- C) 0
- D) 1

13

If $x = 2y + 1$, what is $4y$ in terms of x ?

- A) $2x - 1$
- B) $2x + 2$
- C) $2x - 2$
- D) $2x - 4$

14

Which of the following functions satisfy $f(-x) = f(x)$ for all real numbers x ?

- A) $x^2 + x + 1$
- B) $x^2 + 2|x|$
- C) $|x - 1|$
- D) $x^3 + 2$



15

Suppose A is in the fourth quadrant, B at the origin, and C on the negative x -axis. Moreover, $\angle ABC = 150^\circ$ and the length of AB is 1. Then where is A located?

- A) $(-\frac{1}{2}, \frac{\sqrt{3}}{2})$
- B) $(-\frac{\sqrt{3}}{2}, \frac{1}{2})$
- C) $(\frac{1}{2}, -\frac{\sqrt{3}}{2})$
- D) $(\frac{\sqrt{3}}{2}, -\frac{1}{2})$

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

- Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If  is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

7	/	1	2
☐	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 2.5

← Decimal point

	2	.	5
☐	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
☐	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	6
☐	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
☐	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

	2	0	1
☐	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

	2	0	1
☐	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.

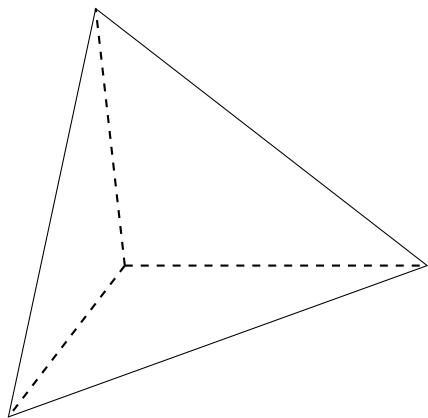


16

Evaluate $2^{-2} + (-2)^2$

17

The figure below is a tetrahedron: a pyramid where each side is an equilateral triangle. By connecting edges together, how many 60° angles does a tetrahedron have?



18

What is the 100th term of the following sequence?

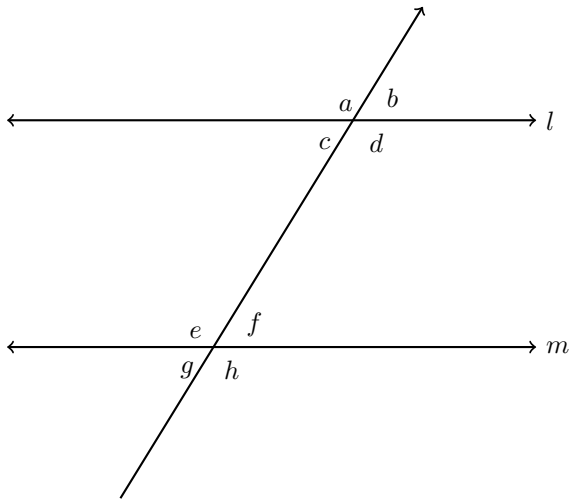
$$\begin{aligned} a_1 &= 4 \\ a_n &= \begin{cases} \frac{a_{n-1}}{-2} & \text{if } n \text{ is even} \\ 2a_{n-1} & \text{if } n \text{ is odd and } n > 1 \end{cases} \end{aligned}$$

19

What is the degree of the minimum non-zero polynomial with real coefficients that contains $1 - i$ and 3 as roots?



20



If $m\angle b = 4x + 2$ and $m\angle g = 2x + 26$, then what is x if $l \parallel m$?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

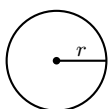
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

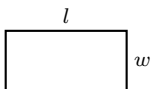
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
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4. All figures lie in a plane unless otherwise indicated.
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REFERENCE

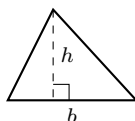


$$A = \pi r^2$$

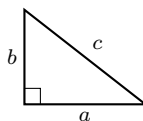
$$C = 2\pi r$$



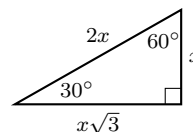
$$A = lw$$



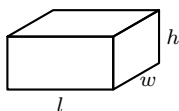
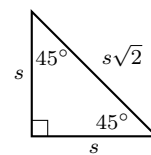
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



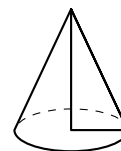
$$V = lwh$$



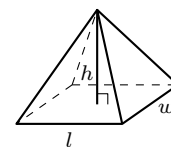
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

Suppose $m\angle ABD = 120^\circ$ and $m\angle ABC = 30^\circ$. If C is in the interior of $\angle ABD$, then what is $m\angle CBD$?

- A) 30°
- B) 90°
- C) 100°
- D) Cannot be determined

2

Consider the fraction

$$n = \frac{4}{2 + \frac{1}{x}}$$

where x is a positive integer. As x gets bigger and bigger, what happens to the value of n ?

- A) n remains constant
- B) n approaches 0
- C) n approaches 1
- D) n approaches 2

3

Which of the following, when defined, is equivalent to

$$\frac{\cos(2\theta)}{\cos(\theta) + \sin(\theta)}?$$

(Note: For any angle θ , $\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$)

- A) $\cos(2\theta)$
- B) $\sin(2\theta)$
- C) $\cos(\theta) - \sin(\theta)$
- D) 1

4

Which point does not lie on the unit circle?

- A) $(0, 1)$
- B) $(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2})$
- C) $(0, -1)$
- D) $(1, 1)$



5

Suppose $f(x)$ is non-negative for all real numbers x . Which of the following can be f ?

- A) $f(x) = 1/x$
- B) $f(x) = 1 - x$
- C) $f(x) = x^2$
- D) None of the above.

6

Consider a set of 10 consecutive integers. Which of the following has the greatest probability?

- A) Choosing a number from this set that is less than the mean.
- B) Choosing a number from this set that is less than the median.
- C) Choosing a number from this set that is even.
- D) All of the above have equal probability.

7

Suppose A is in the first quadrant, B at the origin, and C on the negative x -axis. Moreover, $\angle ABC = 120^\circ$ and the length of AB is 1. Then where is A located?

- A) $(\frac{1}{2}, \frac{\sqrt{3}}{2})$
- B) $(-\frac{1}{2}, \frac{\sqrt{3}}{2})$
- C) $(-\frac{\sqrt{3}}{2}, \frac{1}{2})$
- D) $(\frac{1}{2}, -\frac{\sqrt{3}}{2})$

8

The product of three numbers is odd. Which of the following must be true?

- A) The sum of these three numbers is even.
- B) The sum of these three numbers is odd.
- C) The product is also divisible by 2.
- D) None of the above



9

The set of points satisfying $y < 2x - 1$ are contained and only contained in which quadrants?

- A) I, II, III
- B) I, II, III, IV
- C) I, II
- D) I, III, IV

10

If $\triangle ABC$ has sides of integer length, and $AB = BC = 12$, then what is the smallest length $\overline{CD}$ can have?

- A) 0
- B) 1
- C) 12
- D) 24

11

In an election, everyone voted for Klaus, Philip, or both candidates. If x people voted for Klaus, y people voted for Philip, and b people voted for both, then how many people voted for Philip or Klaus, but not both?

- A) $x + y - b$
- B) $x + y - 2b$
- C) $x - y + b$
- D) $x - y + 2b$

12

Which of the following best describes the set of all points satisfying $x^2 + y^2 < -1$

- A) A circle
- B) The set of all points outside a circle
- C) The set of all points between two circles
- D) The empty set



13

The solution set of $y - x < 1$ lie in which quadrant(s)?

- A) *I* only
- B) *I, II, III* only
- C) *I, II, IV* only
- D) *I, II, III, IV*

14

In right triangle $\triangle ABC$, $\cos(C) = \frac{5}{13}$. If the side opposite C is 24, then what is the length of the hypotenuse?

- A) 13
- B) 25
- C) 26
- D) Cannot be determined

15

What is 12% of 50?

- A) 6
- B) 12
- C) 24
- D) 50

16

Suppose $x^2 = y^2$. Then which of the following must be true?

- A) $|x| = |y|$
- B) $x = y$
- C) $xy \geq 0$
- D) None of the above



17

For which of the following x is $\sin(x) < 0$ and $\tan(x) > 0$?

- A) $0 < x < 90^\circ$
- B) $180^\circ < x < 270^\circ$
- C) $180^\circ < x < 360^\circ$
- D) None of the above.

18

Which of the following, when defined, is equivalent to

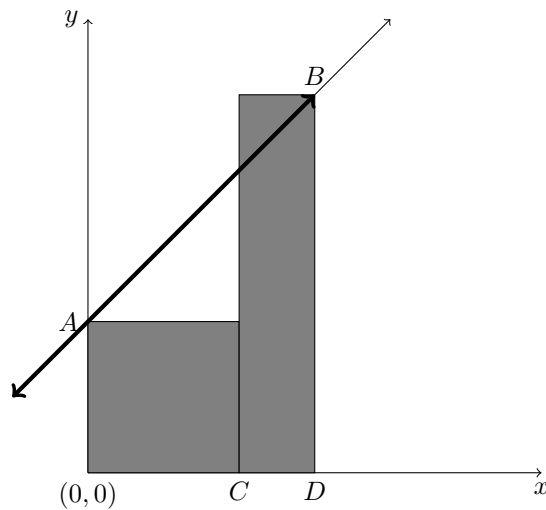
$$\sin(2\theta) \cdot \cos(\theta)$$

(Note: For any angle θ , $\sin(2\theta) = 2\sin(\theta)\cos(\theta)$)

- A) $2\sin^2(\theta)$
- B) $2\sin^3(\theta)$
- C) $2 - 2\sin^2(\theta)$
- D) $2\sin(\theta) - 2\sin^3(\theta)$

19

The line $y = x + 2$ intersects the two rectangles below at A and B . Moreover, C is located at $(2,0)$ and D is at $(3,0)$, what is the total area of the two rectangles?



- A) 5
- B) 9
- C) 10
- D) 12



20

Suppose the parabola

$$y = ax^2 + bx + c$$

intersects the line

$$y = mx + b$$

at $(1, 0)$. Then m equals which of the following?

- A) $c + a$
- B) $c + 2a$
- C) $a - c$
- D) $c - a$

21

If

$$2^a 3^a = 6^b,$$

then what is b in terms of a ?

- A) a
- B) $2a$
- C) $3a$
- D) Cannot be determined

22

Suppose that k is a positive number that has 2 as the hundredth digit in its decimal representation. Then what is the biggest possible value of

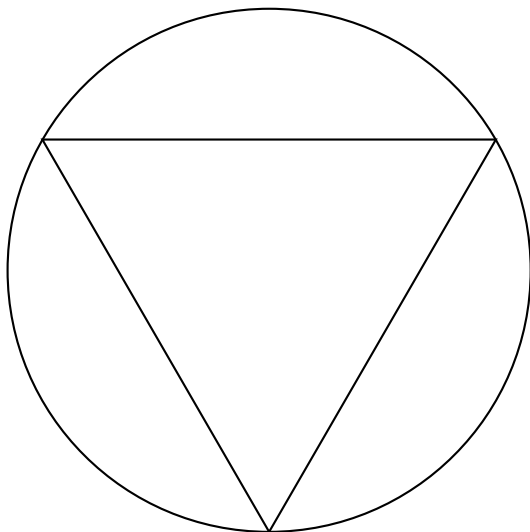
$$\frac{k+1}{k}$$

- A) 1
- B) 2
- C) 51
- D) 52



23

The following is an equilateral triangle of perimeter 12 inscribed inside a circle:



How many equilateral triangles of perimeter 12 can you inscribe within this circle?

- A) 2
- B) 4
- C) 8
- D) More than 4

24

For what A and B is

$$\frac{1}{x(x+1)} = \frac{A}{x} + \frac{B}{x+1}$$

- A) $A = 1, B = -1$
- B) $A = -1, B = 1$
- C) $A = -1, B = -1$
- D) $A = 2, B = 1$

25

Consider the following system of equations:

$$\begin{aligned} 6x + 3y &= -1 \\ mx + 9y &= -4 \end{aligned}$$

For what value of m does the above system have no solutions?

- A) 6
- B) 8
- C) 16
- D) 18



26

Let $f(x) = mx + b$ for some m and b . If $f(3) = 10$, which of the following cannot be true?

- A) $f(1) = 12$ and $f(5) = 8$
- B) $f(2) = 12$ and $f(4) = 10$
- C) $b = 0$
- D) All of the above must be true.

27

A wheel makes 4 revolutions in 3 minutes. How many revolutions does the wheel make in 270 seconds?

- A) 6
- B) 8
- C) 10
- D) 14

28

If $2 \sin(a) = 0$ and $2 \cos(\frac{\pi}{2} - b) = 0$, then which cannot be a possible value of $a + b$?

- A) 0
- B) $\frac{\pi}{2}$
- C) 2π
- D) 8π

29

Consider the following system of equations:

$$\begin{aligned} 2x + 3y &= -2 \\ 4x + 6y &= -4 \end{aligned}$$

How many solutions does this system have?

- A) 0
- B) 2
- C) 4
- D) Infinitely many



30

Which of the following has the same graph as $\cos(x)$?

- A) $\sin(x - 90^\circ)$
- B) $\cos(x - 90^\circ)$
- C) $\cos(x - 270^\circ)$
- D) $\cos(x - 360^\circ)$

.....

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

5. **Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If $\begin{array}{|c|c|c|c|} \hline 3 & 1 & / & 2 \\ \hline \bullet & \bullet & \bullet & \bullet \\ \hline \end{array}$ is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

6. **Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write answer in boxes. →

7	/	1	2
•	•	•	•
0	0	0	0
1	1	•	1
2	2	2	•
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
•	7	7	7
8	8	8	8
9	9	9	9

← Fraction line

Grid in result.

Answer: 2.5

	2	.	5
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	3
4	4	4	4
5	5	5	•
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

← Decimal point

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	•
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	•
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

	2	0	1
•	•	•	•
0	•	0	0
1	1	1	•
2	•	2	2
3	3	3	3

2	0	1	
•	•	•	•
•	0	0	0
1	1	•	1
•	2	2	2
3	3	3	3

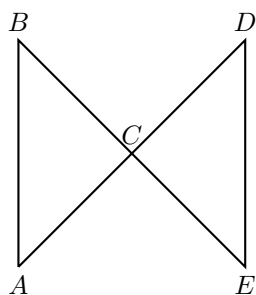
NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

Suppose John can travel 2 miles in 40 minutes and Jill can travel 5 miles in one hour. If Jill and John start at the location and walk in the same direction, what is the distance separating Jack and Jill in 4 hours?

32



In the figure above, $m\angle ACB = 3x - 10$ and $m\angle ECD = 2x + 15$. What is the value of x ?

33

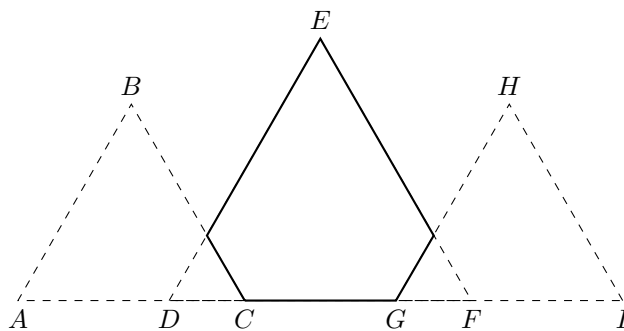
If

$$f(x) = ax^2 + bx + c$$

and the graph of y has a vertex when at $(-1, -1)$ and has a point $(-3, 15)$, what is $f(1)$?

34

$\triangle ABC$ and $\triangle GHI$ are equilateral triangles with perimeter 9 and triangle $\triangle DEF$ is an equilateral triangle with perimeter 12. If $AI = 8$ and $DC = GF$, then what is the perimeter of the bolded region?



35

The ratio of purple marbles to blue marbles in a jar is $5 : 2$. If there are 6 blue marbles, then how many marbles are in the jar?

36

The ratio of boys to girls in a class is $2 : 5$. If there are 35 students in class, then how many boys are there?



37

On three exams, John scores a 70, 80, and 98. If John has a score of S on his next exam, and the average of the four exams is a 85, then what is the value of S ?

38

In a class, every student scored either an 85, 90, or 95. The frequency chart is recorded below:

Grade	Number of students
85	10
90	7
95	3

What is the class average?

Answer Key

Calculator Off	
1	D
2	C
3	B
4	D
5	A
6	A
7	D
8	A
9	C
10	A
11	D
12	D
13	C
14	B
15	D
16	4.25
17	12
18	2
19	3
20	12

Calculator On	
1	B
2	D
3	C
4	D
5	C
6	D
7	A
8	B
9	D
10	B
11	A
12	D
13	D
14	C
15	A
16	A
17	B
18	D
19	B
20	A
21	A
22	C
23	D
24	A
25	D
26	B
27	A
28	B
29	D
30	D
31	8
32	25°
33	15
34	10
35	21
36	10
37	92
38	88.25



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

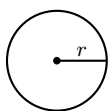
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

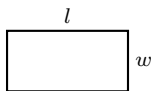
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

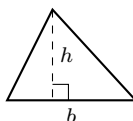


$$A = \pi r^2$$

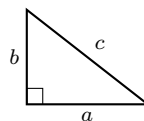
$$C = 2\pi r$$



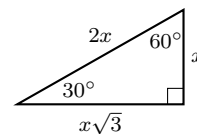
$$A = lw$$



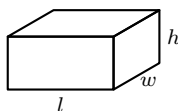
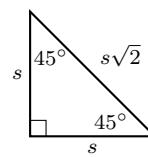
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



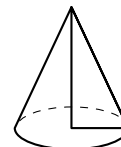
$$V = lwh$$



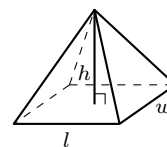
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

Suppose A is in the third quadrant, B at the origin, and C on the positive x -axis. Moreover, $\angle ABC = 150^\circ$ and the length of AB is 1. Then where is A located?

- A) $(\frac{1}{2}, \frac{\sqrt{3}}{2})$
- B) $(-\frac{1}{2}, \frac{\sqrt{3}}{2})$
- C) $(-\frac{\sqrt{3}}{2}, \frac{1}{2})$
- D) $(\frac{1}{2}, -\frac{\sqrt{3}}{2})$

2

Consider the following system of equations:

$$\begin{aligned}2x + 3y &= -2 \\6x + 9y &= -4\end{aligned}$$

How many solutions does this system have?

- A) 0
- B) 1
- C) 2
- D) 4

3

What values for x would make $\frac{1}{\sqrt{x-1}}$ undefined?

- A) 1 only
- B) -1 only
- C) All reals less than 1
- D) All reals less than or equal to 1

4

Which of the following Cartesian coordinates represents the same polar coordinate as $(3, 45^\circ)$?

- A) $(3, 3)$
- B) $(3, 3\sqrt{2})$
- C) $(3, 3\sqrt{3})$
- D) $(\frac{3\sqrt{2}}{2}, \frac{3\sqrt{2}}{2})$



5

A set consists of all the positive integers less than 21. What is the probability that the integer is divisible by 4?

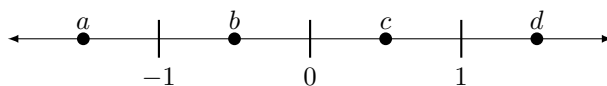
- A) $\frac{1}{4}$
- B) $\frac{1}{5}$
- C) $\frac{1}{2}$
- D) $\frac{2}{5}$

6

Which of the following numbers is irrational?

- A) $\sqrt[3]{8}$
- B) $\sqrt[4]{16}$
- C) $\sqrt[3]{25}$
- D) $\sqrt[3]{27}$

7



Which of the following has the smallest value?

- A) ab
- B) bc
- C) cd
- D) $c - b$

8

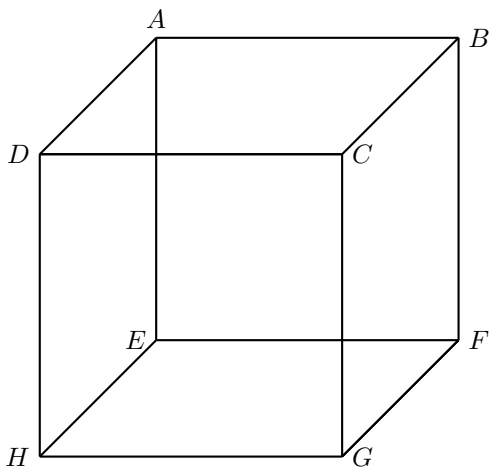
Which of the following polar coordinates represents the same Cartesian coordinate as $(2, 2\sqrt{3})$?

- A) $(2, 60^\circ)$
- B) $(4, 30^\circ)$
- C) $(4, 60^\circ)$
- D) $(8, 30^\circ)$



9

The length of each edge in the cube below is 2.



What is the length of $\overline{AG}$ (not shown)?

- A) 2
- B) $2\sqrt{2}$
- C) $2\sqrt{3}$
- D) $2\sqrt{5}$

10

How many lines go through the points $(a, 0)$ and $(0, b)$ if a and b are fixed numbers greater than 0?

- A) 0
- B) 1
- C) Infinitely many
- D) Cannot be determined

11

Suppose $(2, 3)$ and $(5, 6)$ lie on a circle. Which of the following could be a third point on the circle?

- A) $(6, 2)$
- B) $(1, 3)$
- C) $(5, 3)$
- D) $(8, 3)$



12

A cylinder has its height increased by 50% and its radius decreased by 50%. Which of the following must be true?

- A) The cylinder's volume is unchanged.
- B) The cylinder's surface area is unchanged.
- C) The volume is increased by more than 25%
- D) The volume is decreased by more than 25%

13

If $5^{2x}2^{2x} = 10000$, what is x ?

- A) 1
- B) 2
- C) 3
- D) 5

14

What is the graph of the intersection of a plane and a right circular cone if the plane bisects the height and is parallel to the base?

- A) Circle
- B) Ellipse
- C) Quadrilateral
- D) Point

15

For positive reals x and y , suppose $y^2 = 9x^4$. What is y in terms of x ?

- A) $3x$
- B) $3x^2$
- C) $9x$
- D) $9x^2$

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

5. **Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If  is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

6. **Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write answer in boxes. →

← Fraction line

Grid in result.

	7	/	1	2
	☐	☒	☐	☐
	☐	☐	☐	☐
	0	0	0	
1	1	☒	1	
2	2	2	☒	
3	3	3	3	
4	4	4	4	
5	5	5	5	
6	6	6	6	
☒	7	7	7	
8	8	8	8	
9	9	9	9	

Answer: 2.5

← Decimal point

	2	.	5
	☐	☐	☒
	☐	☐	☐
	0	0	0
1	1	1	1
2	☒	2	2
3	3	3	3
4	4	4	4
5	5	5	☒
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
	☐	☒	☐
	☐	☐	☐
	0	0	0
1	1	1	1
2	☒	2	2
3	3	3	☒
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
☒	☐	☐	☐
	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	☒	☒	☒
7	7	7	7
8	8	8	8

.	6	6	7
☒	☐	☐	☐
	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	☒	☒	6
7	7	7	☒
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
	☐	☐	☐
	☐	☐	☐
	0	☒	0
1	1	1	☒
2	☒	2	2
3	3	3	3

2	0	1	
☐	☐	☐	☐
	☐	☐	☐
	☒	0	0
1	1	☒	1
☒	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16

If the measure of angle A is double the measure of angle B and angle A and angle B are complementary, then what is the measure of angle A ?

17

What is $0^1 + 1^0$?

18

In a sequence of 4 positive numbers, the first three terms are an arithmetic sequence and the last three terms are a geometric sequence. If the second term is 2 and the last term is 8, then what is the first term?

19

There are two containers each containing a mixture composed of alcohol and water. We also know that the volume of the mixture in the second container is double that of the volume of the first. If the first container's mixture is composed of three fourths alcohol and the second container mixture is composed of one third water, then what is the ratio of alcohol to water when both mixtures are combined?

20

Distinct lines a, b, c are parallel to each other and distinct lines d, e, f are parallel to each other. Moreover, line $a \perp b$. How many intersections are there among a, b, c, d, e and f ?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

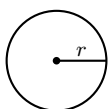
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

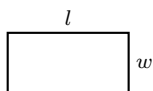
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

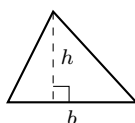


$$A = \pi r^2$$

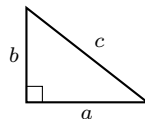
$$C = 2\pi r$$



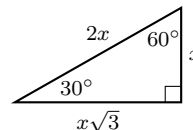
$$A = lw$$



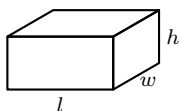
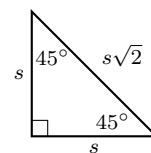
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



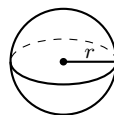
Special Right Triangles



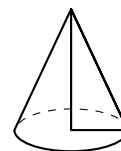
$$V = lwh$$



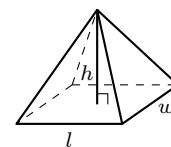
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

Which of the following graphs can never be described by a single function?

- A) A circle
- B) A parabola
- C) A polynomial
- D) None of the above

2

What is the smallest non-zero polynomial with real coefficients that contains $1 - i$ as a root?

- A) $x^2 - 1$
- B) $x^2 + 1$
- C) $x^2 - 2x + 2$
- D) $x^2 + 2x + 2$

3

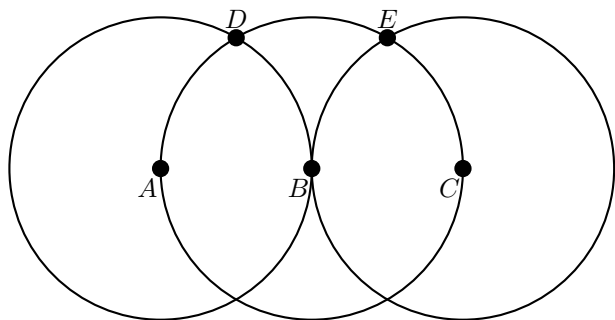
Find all x such that $0 < x < 360^\circ$ and $(2 \cos(x) - 1)(2 \sin(x) - 1) = 0$.

- A) $30^\circ, 60^\circ, 120^\circ, 150^\circ$,
- B) $30^\circ, 60^\circ, 150^\circ, 300^\circ$
- C) $30^\circ, 60^\circ, 210^\circ, 300^\circ$
- D) $30^\circ, 60^\circ, 210^\circ, 240^\circ$



4

The three circles below are congruent and have radius 3:



If A , B and C are the centers of the three circles, then what is the perimeter of the trapezoid $ABCED$ (not drawn)?

- A) $9\sqrt{3}$
- B) 12
- C) 15
- D) Cannot be determined

5

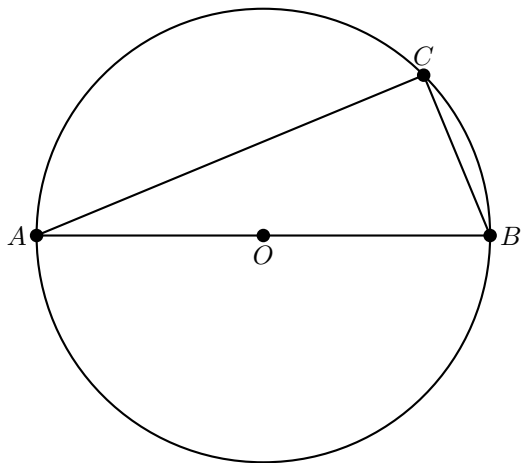
Which of the coordinate pairs below is on the circle that is centered at the origin with radius 5?

- A) $(-3, 4)$
- B) $(6, 1)$
- C) $(-1, 5)$
- D) $(4, 1)$



6

If the radius of the circle centered at O is r , $\overline{AOB}$ is a diameter, $AC = 2r - 4$, $BC = 2r - 2$ then what is r ?



- A) 5
- B) 10
- C) 12
- D) Cannot be determined.

7

What values of x would make $\frac{x}{x^2 - x}$ undefined?

- A) 0 only
- B) 1 only
- C) 0, 1
- D) 0, 1, -1

8

How many 3 digit combinations are there such that:

1. Each digit is an odd number
2. If 3 is used, 9 cannot be used.
3. The digits, when read left to right, is strictly increasing.

- A) 6
- B) 7
- C) 8
- D) 9



9

Klaus has a collection of books. Suppose 20% of his books are French. Moreover, 10% of these French books were written by Victor Hugo. If there are n books written by Victor Hugo, what is one possible size of Klaus' original book collection?

- A) 150
- B) 160
- C) 170
- D) 190

10

What is 150° in radians?

- A) $\frac{2\pi}{3}$
- B) $\frac{4\pi}{5}$
- C) $\frac{5\pi}{6}$
- D) $\frac{6\pi}{7}$

11

Which of the following is the same as $\sin(90^\circ - x) + \cos(x)$?

- A) $2 \cos(x)$
- B) 0
- C) $\tan(x)$
- D) 1

12

If $\log_x(2) = 4$, then x is approximately

- A) 2
- B) 4
- C) 1.19
- D) 3.2



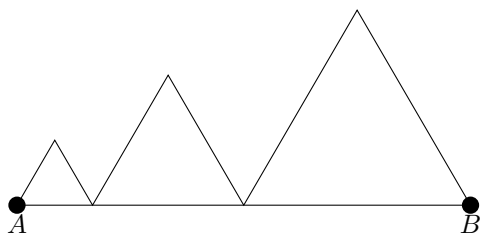
13

How many four digit numbers begin and end with an even digit?

- A) 250
- B) 500
- C) 1600
- D) 2000

14

Suppose $AB = 20$ and the three triangles that lie on $\overline{AB}$ are all equilateral. What is the sum of the perimeters of these three triangles?



- A) 30
- B) 40
- C) 60
- D) Cannot be determined

15

In $\triangle ABC$,

$$\begin{aligned} AB &= 5 \\ BC &= 6 \\ AC &= 5 \end{aligned}$$

What must be true about the measure of $\angle BAC$?

- A) $180^\circ > m\angle BAC > 90^\circ$
- B) $90^\circ > m\angle BAC > 60^\circ$
- C) $m\angle BAC = m\angle ABC$
- D) None of the above

16

Suppose the product of a and b is equal to the quotient of a and b . Then which of the following must be true?

- A) $a = 0$
- B) $b = 1$
- C) $a = 0$ or $b = 1$ or $b = -1$
- D) None of the above



17

In a chain, every third link is made of gold and every fifth link is painted blue. If there are 2000 links, how many of these links are gold links that are painted blue?

- A) 132
- B) 133
- C) 134
- D) 136

18

As x gets larger and larger, what does the function $\frac{x^2+2x^3}{x^2+x^3}$ approach?

- A) 0
- B) 1
- C) 2
- D) Infinity

19

In a geometric series consisting of only positive terms, the third term is 12 and the seventh term is 192. What is the first term?

- A) 2
- B) 3
- C) 5
- D) 6

20

What is the area of the quadrilateral formed from the intersections and vertices of the parabolas $y = x^2$ and $y = -x^2 + 10$?

- A) $5\sqrt{5}$
- B) $10\sqrt{5}$
- C) $10\sqrt{10}$
- D) $12\sqrt{7}$



21

If $y = x^3$ and $z^3 = y^2$, what is z in terms of x ?

- A) $\sqrt[3]{x}$
- B) x^2
- C) x^3
- D) $2x$

22

Which of the following equations is false for some x , y and z ?

- A) $(x \div y) \div z = x \div (y \div z)$
- B) $2(x + y) = 2x + 2y$
- C) $xy + xz = x \cdot (y + z)$
- D) $x \cdot yz = z \cdot yx$

23

Suppose p and q are prime numbers such that $p + q = 43$. What is pq ?

- A) 41
- B) 82
- C) 84
- D) 43

24

What is the range of $3 \cos(x) - 2$?

- A) $[-3, 3]$
- B) $[-3, 1]$
- C) $[-5, 1]$
- D) $[-5, -3]$



25

If $0 < x < 90^\circ$ and $\cos(x) = \frac{4}{5}$, then what is $\sin(x)$?

- A) $\frac{1}{5}$
- B) $\frac{3}{5}$
- C) $\frac{4}{5}$
- D) Cannot be determined

26

Suppose a and b are integers and $a^2 - b^2$ is odd. What must be true?

- A) $a + b$ is even
- B) $a - b$ is even
- C) ab is even
- D) None of the above

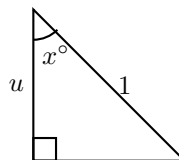
27

Suppose $f(x, y) = xy + x + y$. Which of the following is the same as $f(3, 6)$?

- A) $f(3, -6)$
- B) $f(-6, 3)$
- C) $f(6, 3)$
- D) None of the above.

28

In the following diagram, what is x ?



- A) $\arctan\left(\frac{u}{1-u^2}\right)$
- B) $\arctan\left(\frac{u}{\sqrt{1-u^2}}\right)$
- C) $\arctan\left(\frac{\sqrt{1-u^2}}{u}\right)$
- D) $\arctan\left(\frac{\sqrt{u^2-1}}{u}\right)$



29

At how many points do the graphs described by $2x - y = 2$ and $6x - 3y = 6$ intersect?

- A) 0
- B) 1
- C) 2
- D) Infinitely many

30

If $x = 2y + 1$ and $y = 2x - 1$, what is the value of $x + y$?

- A) -1
- B) 0
- C) 1
- D) Cannot be determined.

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

- Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If  is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write answer in boxes. →

7	/	1	2
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

← Fraction line

Grid in result.

Answer: 2.5

2	.	5	
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

← Decimal point

Acceptable ways to grid $\frac{2}{3}$ are:

2	/	3	
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

2	0	1	
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

2	0	1	
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



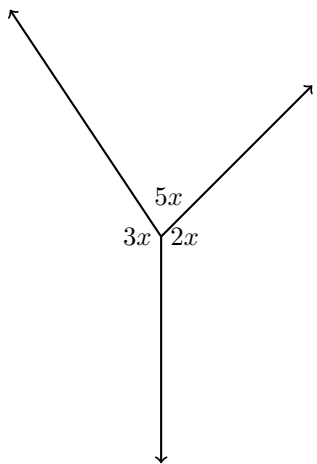
31

What is the square root of the cube of 25?

32

In parallelogram $ABCD$, the measure of angle A is $3x + 10$ and the measure of angle B is $10x - 25$. What is the value of x ?

33



The degree measures are listed in the figure above. What is the smallest angle in this figure?

34

If

$$(3^x 2^{2x})^6 = 144$$

then what is x ?

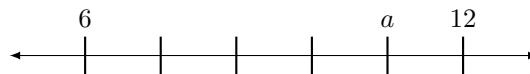
35

Simplify: $\left(\frac{1}{16}\right)^{-\frac{3}{4}}$

36

What is the slope of the line that goes through $(-1, 3)$ and $(1, 5)$?

37



Suppose each of the tick marks are equally spaced on the above number line. What is the value of a ?



38

What is the amplitude of $2 \cos(3x - 4) + 1$?

.....

Answer Key

Calculator Off	
1	C
2	A
3	D
4	D
5	A
6	C
7	B
8	C
9	C
10	B
11	D
12	D
13	B
14	A
15	B
16	60°
17	1
18	0
19	$\frac{25}{11}$
20	9

Calculator On	
1	A
2	C
3	B
4	C
5	A
6	A
7	C
8	B
9	A
10	C
11	A
12	C
13	D
14	C
15	B
16	C
17	B
18	C
19	B
20	B
21	A
22	A
23	B
24	C
25	B
26	C
27	C
28	C
29	D
30	B
31	125
32	15°
33	72°
34	$\frac{1}{3}$
35	8
36	1
37	10.8
38	2



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

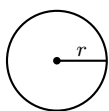
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

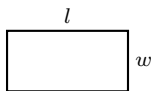
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

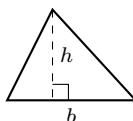


$$A = \pi r^2$$

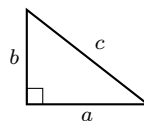
$$C = 2\pi r$$



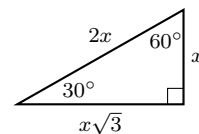
$$A = lw$$



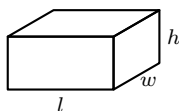
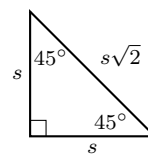
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



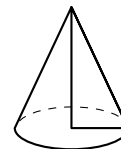
$$V = lwh$$



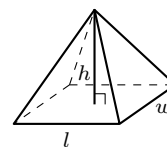
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

If $2^{2x+1} = 512$, then what is x ?

- A) 1
- B) 2
- C) 4
- D) 5

2

Which of the following inequalities yields all b such that $2x^2 + b + 18$ has no real roots?

- A) $b > 6$
- B) $b < 6$
- C) $b > 12$
- D) $-12 < b < 12$

3

If

$$2^{x+1} \cdot 4^{x+1} \cdot 8^{x+1} = 2^y$$

what is y in terms of x ?

- A) $3x + 3$
- B) $3x + 4$
- C) $6x + 2$
- D) $6x + 6$

4

The set of points satisfying $y > -1$ are contained and only contained in which quadrants?

- A) *III, IV*
- B) *I, II, III*
- C) *I, II, III, IV*
- D) *I, II*



5

If $\log_b(a) > 0$ and

$$\frac{\log_b(a)}{\log_a(b)} = 16$$

then what is a in terms of b ?

- A) b^2
- B) b^3
- C) b^4
- D) None of the above.

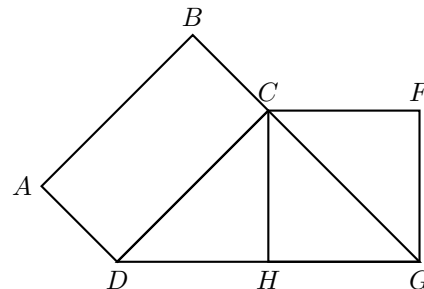
6

Which one of the numbers below is the same as double its square?

- A) .25
- B) .5
- C) 1
- D) 4

7

Suppose $ABCD$ and $CFGH$ are rectangles.



Which of the following must be true?

- A) $m\angle CDH = m\angle FCG$
- B) $m\angle DCH = m\angle HCG$
- C) $m\angle CDH = m\angle CGF$
- D) None of the above

8

Which of the following describes the set of points that lie strictly above the line $y = 2x + 1$?

- A) $y > 2x + 1$
- B) $y \leq 2x + 1$
- C) $y \geq 2x + 1$
- D) $y = 2x + 1$



9

Which of the following polynomials have only positive roots?

- A) $x^2 - 5x + 6$
- B) x^2
- C) $x^2 - 7x + 6$
- D) $x^2 + 2x + 1$

10

If the diameter of a circle is 6, then what is the area of the circle in terms of π ?

- A) 3π
- B) 6π
- C) 9π
- D) 36π

11

A meal consists of an appetizer, a dessert, and a main course. If there are a appetizers, b desserts, and c main courses, and 200 different meals can be made, then which of the following cannot be the value of a ?

- A) 1
- B) 20
- C) 50
- D) 60

12

Which of the following polynomials has $x + 2$ as a factor?

- A) $x^2 + 1$
- B) $x^2 - 1$
- C) $3x^2 + 2x - 1$
- D) $3x^2 + 2x - 8$



13

What is i^{77} ?

- A) 1
- B) i
- C) $-i$
- D) -1

14

Find all x such that $0 < x < 360^\circ$ and $\sin(x) = \sqrt{3}\cos(x)$.

- A) 30°
- B) 60°
- C) $30^\circ, 210^\circ$
- D) $60^\circ, 240^\circ$

15

Which of the decimal expansions below represents an irrational number?

- A) $.\bar{3}$
- B) $.\bar{6}$
- C) $.\bar{9}$
- D) None of the above

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
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as 3.5 or $7/2$. (If  is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

6. **Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write answer in boxes. →

Grid in result. →

	7	/	1	2
	☐	☒	☐	☐
	☐	☐	☐	☐
	0	0	0	0
1	1	☒	1	1
2	2	2	☒	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
☒	7	7	7	7
8	8	8	8	8
9	9	9	9	9

← Fraction line

Answer: 2.5

	2	.	5
	☐	☐	☒
	☐	☐	☐
	0	0	0
1	1	1	1
2	☒	2	2
3	3	3	3
4	4	4	4
5	5	5	☒
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

← Decimal point

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
	☐	☒	☐
	☐	☐	☐
	0	0	0
1	1	1	1
2	☒	2	2
3	3	3	☒
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
☒	☐	☐	☐
☐	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	☒	☒	☒
7	7	7	7
8	8	8	8

.	6	6	7
☒	☐	☐	☐
☐	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	☒	☒	6
7	7	7	☒
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
	☐	☐	☐
	☐	☐	☐
	0	☒	0
1	1	1	☒
2	☒	2	2
3	3	3	3

2	0	1	
☐	☐	☐	☐
☐	☐	☐	☐
☒	0	0	0
1	1	☒	1
☒	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16

How many ways can one rearrange the letters A, C, T ?

17

How many positive factors does 49 have?

18

Each marble in a jar of marbles is colored precisely one color. If the probability of drawing a red marble is 20% and the probability of drawing a green marble is 12%, then what is the probability of drawing neither a red nor green marbles?

19

Evaluate $\sqrt{\sqrt{9} - \sqrt{4}}$

20

What is $e^{\ln(5)}$?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

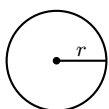
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

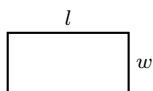
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

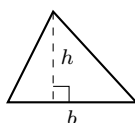


$$A = \pi r^2$$

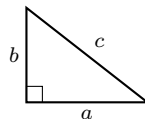
$$C = 2\pi r$$



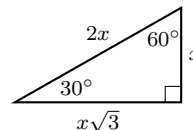
$$A = lw$$



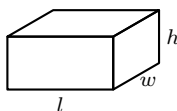
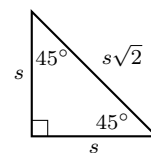
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



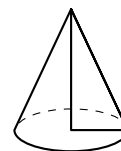
$$V = lwh$$



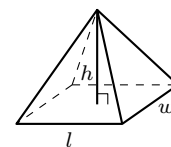
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

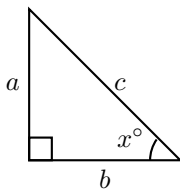
The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

In the right triangle below, what is $\tan(x)$?



- A) $\frac{a}{b}$
- B) $\frac{b}{a}$
- C) $\frac{b}{\sqrt{a^2 + b^2}}$
- D) Cannot be determined

2

Suppose a phone company charges a fixed charge of F dollars for making the call and charges C dollars for each K minute. If we represent the cost in dollars as a function y of minutes x , what is the coordinate that represents a call of $\frac{3K}{2}$ minutes?

- A) $(\frac{3K}{2}, \frac{3C}{2})$
- B) $(\frac{3K}{2}, \frac{3C}{2} + F)$
- C) $(\frac{3K}{2}, \frac{3C}{2K} + F)$
- D) $(\frac{3K}{2}, \frac{C}{K} + F)$

3

Consider the set of positive integers $\{a, b, c, d, e\}$ where

$$a < b < c < d < e$$

If d is replaced by $2d$ which of the following must be true?

- A) The mean is decreased
- B) The range is decreased
- C) The median is unchanged.
- D) There is no mode.

4

$\triangle ABC$ has the following angle measures:

$$\begin{aligned} m\angle ABC &= 40^\circ \\ m\angle ACB &= 80^\circ \\ m\angle BAC &= 60^\circ \end{aligned}$$

Which of the following must be true?

- A) $\overline{BC}$ is the longest side
- B) $\overline{BC}$ is the shortest side
- C) $\overline{AB}$ is the shortest side
- D) $\overline{AB}$ is the longest side



5

Suppose $|b - a| = 0$. Which of the following must be true?

- A) $b < a$
- B) $b = a$
- C) a and b are both 0.
- D) None of the above

6

What is the center and radius of the circle $(x - 3)^2 + (y + 4)^2 = 4$?

- A) center: $(3, 4)$ radius: 4
- B) center: $(-3, 4)$ radius: 4
- C) center: $(-3, 4)$ radius: 2
- D) center: $(3, -4)$ radius: 2

7

If $983210 + 12310 = a \times 10^b$ where $0 < a < 10$ and b is an integer, then what is b ?

- A) 4
- B) 5
- C) 7
- D) 8

8

What is the square of $9x$?

- A) $3x$
- B) $9x$
- C) $3x^2$
- D) $81x^2$



9

If $x^2 - 2x + c$ intersects the x -axis only once, then what must be the value of c ?

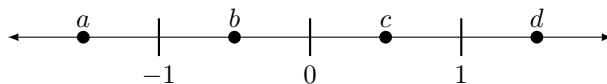
- A) -1
- B) 1
- C) 3
- D) 4

10

What are all values of x such that the inequality $x^2 < 6 - x$ holds?

- A) $-2 < x < 3$
- B) $-3 < x < 2$
- C) $x > 3$ or $x < -2$
- D) $x > 2$ or $x < -3$

11



Which of the following has the greatest value?

- A) c^a
- B) c^c
- C) c^d
- D) d^a

12

In a certain right triangle with angles A , B and C , $\cos(C) = \frac{4}{5}$. What is the length of the hypotenuse of this right triangle?

- A) 3
- B) 5
- C) 6
- D) Cannot be determined



13

The set of all real number x satisfying $-\sqrt{x^2} = -x$ is

- A) All real numbers
- B) All positive reals
- C) All negative reals
- D) All non-negative reals

14

A printer prints 36 pages per minute. How many seconds does it take to print 120 pages?

- A) 160
- B) 180
- C) 200
- D) 240

15

What is $\frac{5-12i}{3-2i}$?

- A) $3 - 2i$
- B) $3 + 2i$
- C) $2 + 2i$
- D) $4 - 2i$

16

Which of the following cannot be the greatest common factor of 44 and $11x^2$ where x is a positive integer?

- A) 1
- B) 4
- C) 22
- D) All of the above are possible.



17

Which of the following is equivalent to $\sqrt[3]{4}$?

- A) $2^{\frac{1}{3}}$
- B) $2^{-\frac{1}{3}}$
- C) $2^{\frac{2}{3}}$
- D) $2^{\frac{3}{2}}$

18

If

$$\sqrt{x^2} = -x,$$

then x must be

- A) 0
- B) negative
- C) positive
- D) non-positive

19

Which of the following functions intersect both axes?

- A) $f(x) = 1$
- B) $f(x) = \frac{1}{x}$
- C) $f(x) = x^2 + 1$
- D) $f(x) = x^2 + 5x + 4$

20

What does the solution set of $-3x^2 - 3y^2 + 12 < 0$ look like?

- A) The empty set
- B) A point
- C) A half plane
- D) The outside of a circle



21

Which of the following is the same as $2^{10} \cdot 4^5$?

- A) 2^{15}
- B) 4^{15}
- C) 4^{20}
- D) $(2^{10})^2$

22

Suppose $(x - 3)$ is the only factor of the quadratic $P(x)$. Then what is $P(x)$?

- A) $x^2 - 9$
- B) $x^2 - 6x + 9$
- C) $x^2 + 6x + 9$
- D) $x^2 + 6x - 9$

23

If $\frac{5}{a} = \frac{4}{b}$, what is $\frac{b}{a}$?

- A) .8
- B) 1.2
- C) 9
- D) 20

24

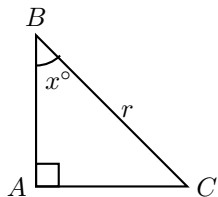
A right circular cone is intersected by a plane. Which of the following cannot be the shape of the intersection?

- A) Ellipse
- B) Quadrilateral
- C) Triangle
- D) Point



25

In the following diagram, what is the length of $\overline{AB}$?



- A) r
- B) $r \cos(x)$
- C) $r \sin(x)$
- D) r^2

26

For any numbers a and b , which of the following is always the same as

$$\frac{|b - a| + |a - b|}{2}$$

- A) 0
- B) $b - a$
- C) $|a - b|$
- D) None of the above

27

What is $a + b + c + d$ if

$$\begin{bmatrix} 3 & 0 \\ 2 & 1 \end{bmatrix} + \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}?$$

- A) 10
- B) 13
- C) 15
- D) 16

28

i is a root of which of the following polynomials?

- A) $1 + x$
- B) $1 + x + x^2 + x^3$
- C) $1 + x + x^2 + x^3 + x^4$
- D) $1 + x + x^2 + x^3 + x^4 + x^5$



29

Which of the numbers below is the largest?

- A) $2 \log(7)$
- B) $3 \log(3)$
- C) $5 \log(2)$
- D) $3 \log(4)$

30

What is the least common multiple of 12 and 20?

- A) 20
- B) 40
- C) 60
- D) None of the above.

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

	7	/	1	2
.	.	.	.	.
	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

Answer: 2.5

← Decimal point

	2	.	5
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	7
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

	2	0	1
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

Consider a group of n students in a room. At noon, a third of the students leave. At 1pm, a third of the number of students who left at noon leave. If there are now 25 students, then what is the value of n ?

32

As x gets larger and larger, what does $\frac{3x^2+2x+7}{x^3+1}$ approach?

33

The terms of a sequence are defined by $a_n = 2a_{n-2} - a_{n-1}$ for $n > 2$. What is the value of a_5 if $a_1 = 2$ and $a_2 = 4$?

34

In a jar, there are 30 red jelly beans, 20 white jelly beans, and 10 green jelly beans. If we are to add x green jelly beans, then how many do we need to add so the probability of randomly drawing a green jelly bean is $\frac{1}{2}$?

35

If

$$3^x + 3^x + 3^x = 3^7$$

then what is x ?

36

What is $\log_4(2) + \log_2(4)$?

37

If two points $(3, -4)$ and $(-2, x)$ are a distance 13 from each other, what is a possible value of x ?



38

What is the value of m such that the system below has no solutions?

$$2x + 4y = 12$$

$$mx + 14y = 35$$

.....

Answer Key

Calculator Off	
1	C
2	D
3	D
4	C
5	C
6	B
7	C
8	A
9	A
10	C
11	D
12	D
13	B
14	D
15	D
16	6
17	3
18	68%
19	1
20	5

:

Calculator On	
1	A
2	B
3	C
4	D
5	B
6	D
7	B
8	D
9	B
10	B
11	A
12	D
13	D
14	C
15	A
16	C
17	C
18	D
19	D
20	D
21	D
22	B
23	A
24	B
25	B
26	C
27	D
28	B
29	D
30	C
31	45
32	0
33	-8
34	40
35	6
36	2.5
37	-16
38	7



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

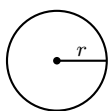
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

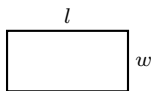
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

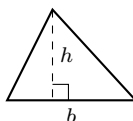


$$A = \pi r^2$$

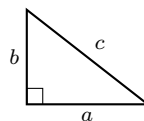
$$C = 2\pi r$$



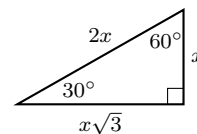
$$A = lw$$



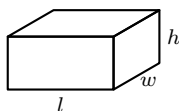
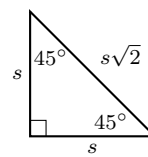
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



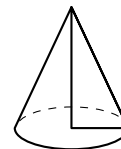
$$V = lwh$$



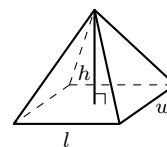
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

Suppose $|a - b| = |a|$. Which of the following must be true?

- A) $b = 0$
- B) $a = 0$ and $b = 0$
- C) $a = 0$ or $b = 0$
- D) None of the above

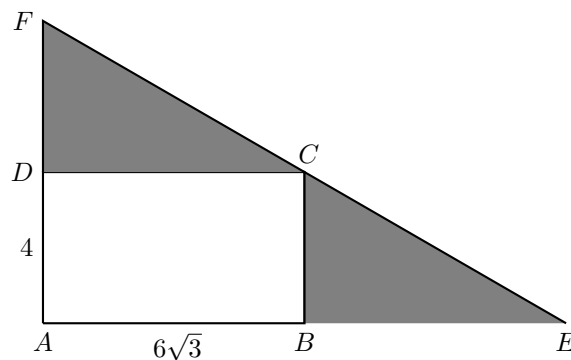
2

In a class, everyone is either in math club or science club. There are 17 students in math club and 13 students in science club. If there are 5 students in both clubs, then how many students are in the class?

- A) 20
- B) 25
- C) 26
- D) 30

3

Let $ABCD$ be a rectangle with $AB = 6\sqrt{3}$ and $AD = 4$. Moreover, $m\angle BEC = 30^\circ$. What is the total area of the shaded regions?



- A) $23\sqrt{3}$
- B) $24\sqrt{3}$
- C) $25\sqrt{3}$
- D) $26\sqrt{3}$



4

Suppose the Mexican peso is directly proportional to the American dollar. If 20 American dollars is 400 pesos, then what is 900 pesos in American dollars?

- A) 42
- B) 45
- C) 50
- D) 55

5

Suppose $m\angle ABD = 120^\circ$ and $m\angle ABC = 30^\circ$. What is $m\angle CBD$?

- A) 90°
- B) 100°
- C) 105°
- D) Cannot be determined.

6

In scientific notation, what is $.000123 + .00002$?

- A) 1.431×10^{-3}
- B) 1.431×10^{-4}
- C) 1.431×10^{-6}
- D) 14.31×10^{-2}

7

If

$$\frac{1}{x} = 25$$

what is $\sqrt{x}$?

- A) 5
- B) 10
- C) .2
- D) .04

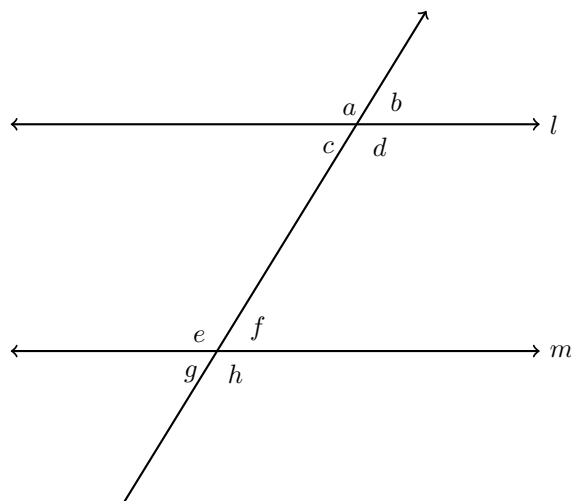


8

Which of the complex numbers below is equal to $\frac{2 - 10i}{3 - 2i}$?

- A) $1 - i$
- B) $2 + i$
- C) $2 + 2i$
- D) $2 - 2i$

9



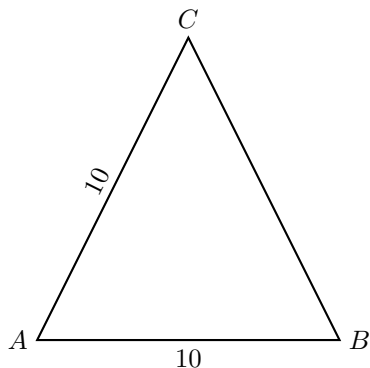
Which of the following lists the exterior angles where $l \parallel m$?

- A) a, b, e, f
- B) a, b, g, h
- C) e, f, g, h
- D) c, d, e, f



10

Consider the following triangle:



If $m\angle A = 60^\circ$, then what is the perimeter of this triangle?

- A) 30
- B) 50
- C) 75
- D) Cannot be determined

11

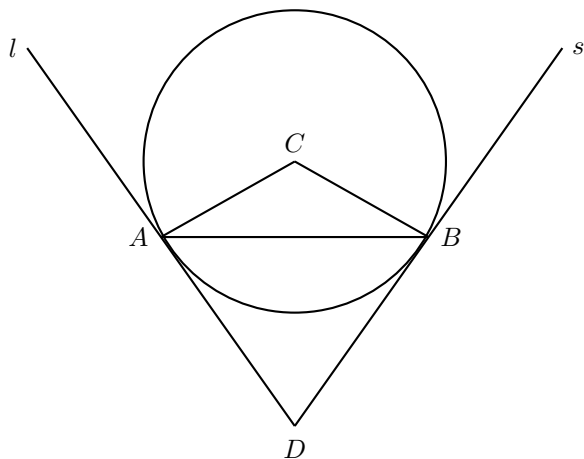
Define $f(x) = x^a$ where a is a positive number. Which of the following is the same as $f(f(x))$ for every possible a ?

- A) $f(a)$
- B) x^{2a}
- C) x^{a^2}
- D) None of the above.



12

Lines l and s are tangent to the circle (centered at C) at points A and B .



Which of the following must be true?

- I. $m\angle CAB = m\angle ABD$
 - II. $m\angle BAD = m\angle ABD$
 - III. If $m\angle ACB = 90^\circ$, then quadrilateral $ACBD$ is a rectangle.
- A) I only
B) II only
C) II and III only
D) I, II and III

13

Which of the following expressions is the same as $\frac{n+1!}{n!}$ for any integer n ?

- A) 1
B) $n!$
C) $n + 1$
D) $n + 1!$

14

Which of the following numbers is not a factor of 60?

- A) 2
B) 3
C) 5
D) 7



15

Consider the data points

$(1, 11.5), (2, 13), (3, 16.2), (4, 17.5), (5, 21)$

Which of the following lines describe this collection of points the best?

- A) $y = x + 10$
- B) $y = -x + 12$
- C) $y = 2x$
- D) $y = 2x + 10$

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
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- Some problems may have more than one correct answer. In such cases, grid only one answer.

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as 3.5 or $7/2$. (If  is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

6. **Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write answer in boxes. →

← Fraction line

Grid in result.

7	/	1	2
.	.	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 2.5

← Decimal point

2	.	5	
.	.	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

2	/	3	
.	.	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	6
.	.	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
.	.	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

2	0	1	
.	.	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

2	0	1	
.	.	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16

What is the distance between the points (3,4) and (8, 16)?

17

If a line $y = mx + b$ goes through the points (1,2) and (3,5), what is $m + b$?

18

What is $\sqrt[3]{-125}$?

19

What is $(\frac{1}{2} + \frac{1}{3})^{-1}$?

20

If $16^{12} = 2^y$, what is y ?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

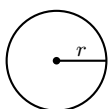
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

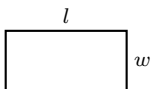
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
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REFERENCE

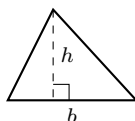


$$A = \pi r^2$$

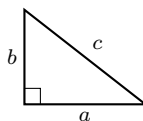
$$C = 2\pi r$$



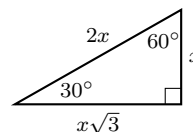
$$A = lw$$



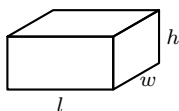
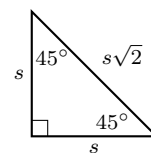
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



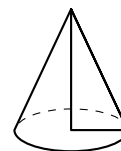
$$V = lwh$$



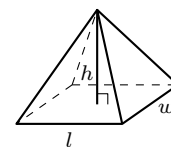
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

What is $a + b + c + d$ if

$$\begin{bmatrix} 4 & 4 \\ 1 & -2 \end{bmatrix} - \begin{bmatrix} -4 & 1 \\ 1 & 3 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}?$$

- A) 6
- B) 8
- C) 11
- D) 16

2

What is $\log_{17}(\sqrt{17})$?

- A) $\frac{1}{2}$
- B) 1
- C) $\frac{3}{2}$
- D) 2

3

What is the sum of the lengths of the major and minor axis of the ellipse $\frac{(x-2)^2}{16} + \frac{(y-4)^2}{9} = 1$?

- A) 10
- B) 14
- C) 25
- D) 50

4

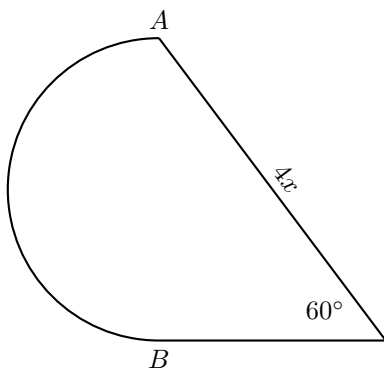
Which of the following has the same value as $\cos(15^\circ)$?

- A) $\cos(195^\circ)$
- B) $\cos(215^\circ)$
- C) $\cos(315^\circ)$
- D) $\cos(345^\circ)$



5

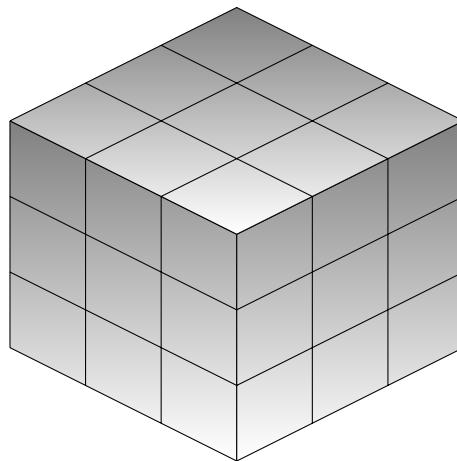
If $\overline{AB}$ (not drawn) is a diameter of a semicircle, then what is the area of the region below in terms of x ?



- A) $1.5x^2\pi + 2x^2\sqrt{3}$
- B) $3x^2\pi + 2x^2\sqrt{3}$
- C) $3x^2\pi + 4x^2\sqrt{3}$
- D) $3x^2\pi + 8x^2\sqrt{3}$

6

A cube of surface area 54 in^2 is composed of 27 smaller identical cubes. If a corner cube is removed, what is the new surface area (in square inches)?



- A) 50
- B) 54
- C) 62
- D) 65



7

Which of the circles below is tangent to both the x and y axes?

- A) $(x - 1)^2 + (y + 2)^2 = 1$
- B) $(x + 1)^2 + (y - 1)^2 = 1$
- C) $(x + 2)^2 + (y - 2)^2 = 2$
- D) $(x + 3)^2 + (y + 4)^2 = 5$

8

Suppose

$$(4x - 5)(4x + 5) = 75$$

Then what is $4x^2$?

- A) 100
- B) 50
- C) 25
- D) 20

9

Which of the following is the same as $\log_b(b^a)$

- A) a
- B) b
- C) 2
- D) b^2

10

Which of the following graphs intersect both axes?

- A) $x = 1$
- B) $y = -1$
- C) $y = 2x + 1$
- D) $(x - 3)^2 + (y - 4)^2 = 1$



11

Suppose $(3 - 2i)x$ is a real number. Then what is a possible value of x ?

- A) $2 + 3i$
- B) $3 - 2i$
- C) $3 + 2i$
- D) 11

12

What is the y -intercept of the function

$$f(x) = 2 \sin(x) + 3 \cos(x)$$

- A) 0
- B) 1
- C) 3
- D) 5

13

Which of the following is the same as $2^3 + 2^3 + 2^3 + 2^3$?

- A) 4^3
- B) 2^5
- C) 2^{10}
- D) 2^{12}

14

Which of the following is the same as $(2x + y - z) - (x - y + z)$?

- A) $x + 2y + 2z$
- B) $x + 2y - 2z$
- C) $x - 2y + 2z$
- D) $x + y - 2z$



15

Consider a fixed point and a line such that the point lies outside this line. The set of all points that are equidistant from this point and this line forms which shape?

- A) Circle
- B) Hyperbola
- C) Parabola
- D) None of the above

16

Suppose for some value a , $f(a) = 11$. Which of the following functions can be $f(x)$?

- A) $\cos(x) + 9$
- B) $2\sin(x) + 8$
- C) $3\cos(x) - 11$
- D) $\tan(x)$

17

Which of the following is a true statement?

- A) All rectangles are parallelograms and all parallelograms are squares.
- B) All rectangles are parallelograms and all squares are rectangles.
- C) All rectangles are squares and all squares are parallelograms.
- D) All parallelograms are squares and all squares are rectangles.

18

At how many points do the graphs described by $2x - y = 7$ and $4x - 2y = 3$ intersect?

- A) 0
- B) 2
- C) 3
- D) Infinitely many



19

Suppose two sides of a triangle have lengths 12 and 13. What is the biggest possible area of this triangle?

- A) 12
- B) 13
- C) 78
- D) Cannot be determined

20



Suppose x is located on the number line as shown above. Which of the following inequalities describe all possible x ?

- A) $|x - 7| < 6$
- B) $|x - 13| < 6$
- C) $|x - 10| < 6$
- D) $|x - 10| < 3$

21

Which of the following expressions have more than one root?

- A) $x^2 + 6x + 9$
- B) $x^2 - 6x + 9$
- C) $x^2 - 10x + 25$
- D) $x^2 - 25$

22

A 13 feet ladder is leaning against a wall that is perpendicular to the ground. After the ladder slips, the top of the ladder falls 7 feet along the wall. If the base of the ladder is currently 12 feet from the wall, how far (in feet) did the base of the ladder move away from the wall after it slipped?

- A) 4
- B) 5
- C) 7
- D) 8



23

Find all x such that $x^2 + 2x - 15 > 0$.

- A) $x < -5$ or $x > 3$
- B) $-5 < x < 3$
- C) $-3 < x < 5$
- D) $x < -3$ or $x > 5$

24

In a jar, there are 4 red marbles and 6 blue marbles. How many red marbles need to be added so that there is a 70% chance of drawing a red marble?

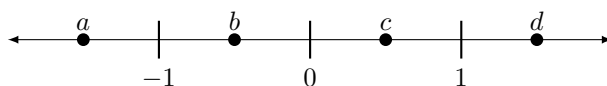
- A) 5
- B) 10
- C) 15
- D) 30

25

Given $7^x 3^y = 63^x$, what is y in terms of x ?

- A) $2x$
- B) $3x$
- C) $4x$
- D) $8x$

26



Which of the following letters satisfy $x < \sqrt{x} < \sqrt[3]{x}$?

- A) a
- B) b
- C) c
- D) None of the above



27

Which of the following is the same as $x^2 - 8x + 15$?

- A) $(x - 2)(x - 3)$
- B) $(x - 3)(x - 1)$
- C) $(x - 3)(x - 5)$
- D) $(x - 6)(x - 1)$

28

Let $f(x) = a^x$ where a is some positive real number. Which of the following must be true?

- I. f has a y -intercept of 1
- II. f has no x -intercept
- III. f is always positive

- A) I only
- B) I,III only
- C) II,III only
- D) I,II,III

29

Which of the following is equivalent to i^{923} ?

- A) i
- B) -1
- C) $-i$
- D) 0

30

Which of the following polynomials have a vertex that lies on the y -axis?

- A) $x^2 + 6x + 5$
- B) $x^2 - 4$
- C) $x^2 - 8x - 9$
- D) $x^2 + 8x - 9$

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

	7	/	1	2
.	.	.	.	.
	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

Answer: 2.5

← Decimal point

	2	.	5
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	7
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

	2	0	1
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

If $ax = 5$ and $ya = 7$, what is $a(x + y)$?

32

If the circumference of a circle is 36π , then what is the radius?

33

A meal consists of an appetizer, a dessert, and a main course. If there are 3 appetizers, 2 desserts, and 5 main courses, then how many different meals can be made?

34

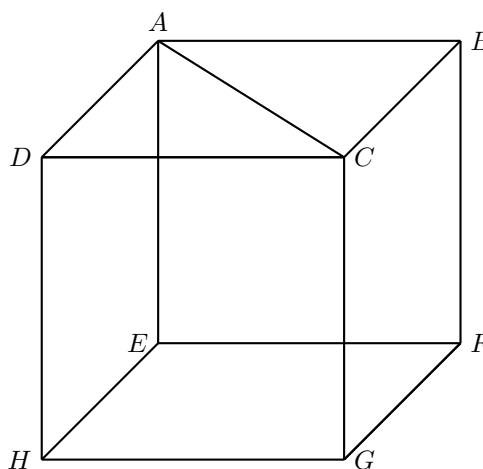
Suppose $x^2 - c^2$ has 7 as a root. Then what is the other root?

35

What is 25% of 50% of 96?

36

The length of $\overline{AC}$ in the cube below is 2.



What is the perimeter of the triangle formed by connecting vertices A , C and H ?

37

If $-4 \leq x \leq 4$, the smallest value of $f(x) = x^2 + 3$ is



38

If

$$\frac{x}{x+2} = \frac{1}{3}$$

what is x ?

.....

Answer Key

Calculator Off	
1	D
2	B
3	D
4	B
5	D
6	B
7	C
8	D
9	B
10	A
11	C
12	C
13	C
14	D
15	D
16	13
17	2
18	-5
19	1.2
20	48

:

Calculator On	
1	A
2	A
3	B
4	D
5	A
6	B
7	B
8	A
9	A
10	C
11	C
12	C
13	B
14	B
15	C
16	D
17	B
18	A
19	C
20	D
21	D
22	C
23	B
24	B
25	A
26	C
27	C
28	D
29	C
30	B
31	12
32	18
33	30
34	-7
35	12
36	6
37	3
38	1



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

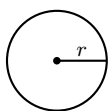
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

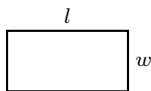
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

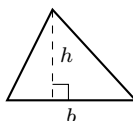


$$A = \pi r^2$$

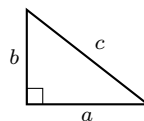
$$C = 2\pi r$$



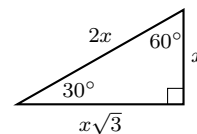
$$A = lw$$



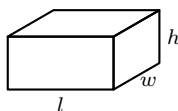
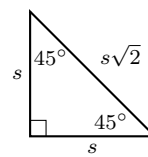
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



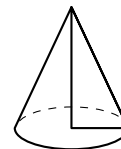
$$V = lwh$$



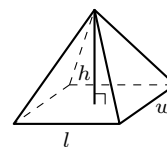
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

Find all x such that $0 < x < 360^\circ$ and $2 \tan^2(x) = \frac{2}{3}$.

- A) $30^\circ, 60^\circ$
- B) $30^\circ, 210^\circ$
- C) $30^\circ, 150^\circ$
- D) $30^\circ, 150^\circ, 210^\circ, 330^\circ$

2

If $\angle ABC$ is supplementary to $\angle CBD$ and $\angle DBE$ is supplementary to $\angle CBD$, then what must be true?

- A) $\angle ABC = 90^\circ$
- B) $m\angle DBE = m\angle CBD$
- C) $m\angle ABC = m\angle CBD$
- D) $m\angle ABC = m\angle DBE$

3

Suppose points A, B, C and D lie on a line in that order. If $AC = 12$, $BD = 11$, and $AD = 16$, then what is the length of $\overline{BC}$?

- A) 7
- B) 5
- C) 4
- D) Cannot be determined

4

For positive numbers $a < b < c < d < e$, we have

$$a + b + c + d + e = 100$$

What is the largest possible value of a ?

- A) 17
- B) 18
- C) 19
- D) 21



5

If $a^2b - ab^2 = 12$ and $ab = 6$, then what is $a - b$?

- A) 1
- B) 2
- C) 4
- D) 6

6

If $k^2 = x$, what is $5x^{-3}$?

- A) $5k^5$
- B) $\frac{5}{k^5}$
- C) $\frac{5}{k^6}$
- D) $\frac{6}{k^6}$

7

If r is the hypotenuse of a right triangle and θ is an angle of this triangle such that $0 < \theta < 90^\circ$, then what is the area of this triangle in terms of θ ?

- A) r
- B) $r \cos(\theta)$
- C) $\frac{r^2 \sin(\theta) \cos(\theta)}{2}$
- D) Cannot be determined

8

If $a < b$, then which of the following must be true?

- A) $\sqrt{a} < \sqrt{b}$
- B) $ab < a$
- C) $ab > a$
- D) $-b + 1 < -a + 1$



9

What is

$$\frac{10}{9^{22}} - \frac{1}{9^{21}}$$

- A) $\frac{1}{9^{21}}$
- B) $\frac{1}{9^{22}}$
- C) $\frac{10}{9^{21}}$
- D) $\frac{1}{9^{23}}$

10

Where does the line $y = -6x - 5$ intersect the parabola $y = x^2$?

- A) $(-1, 1)$
- B) $(-2, 4)$ and $(-4, 16)$
- C) $(1, 1)$ and $(-4, 16)$
- D) $(-1, 1)$ and $(-5, 25)$

11

For some number M , a polynomial $f(x)$ satisfies $f(x) \leq M$ for every x . Which of the following can be that polynomial?

- A) $x^2 - 9$
- B) $-x^2$
- C) $x + 1$
- D) $1 - x$

12

Suppose A is an $a \times b$ matrix and B is a $c \times d$ matrix and the product $A+B$ exists. Which of the following must be true?

- A) $a = b$ and $c = d$
- B) $a = c$ and $b = d$
- C) $c = d$
- D) $b = d$



13

If $\log_{27}(x) = -\frac{1}{3}$, then what is x ?

- A) $\frac{1}{3}$
- B) $\frac{2}{3}$
- C) $-\frac{1}{3}$
- D) $-\frac{2}{3}$

14

Which of the following is irrational?

- A) $\sqrt{25}$
- B) $\sqrt{9}$
- C) $\sqrt{0}$
- D) $\sqrt{62}$

15

Suppose an amusement park ride only allows people who are between 60 and 72 inches tall. Which equation represents the height (in inches) of all people who are forbidden to ride?

- A) $|x - 66| > 6$
- B) $|x - 66| < 6$
- C) $|x - 72| < 12$
- D) $|x - 72| > 24$

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

5. **Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If  is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

6. **Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write answer in boxes. →

Grid in result. →

	7	/	1	2
	☐	☒	☐	☐
	☐	☐	☐	☐
	0	0	0	0
1	1	☒	1	1
2	2	2	☒	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
☒	7	7	7	7
8	8	8	8	8
9	9	9	9	9

← Fraction line

Answer: 2.5

	2	.	5
	☐	☐	☒
	☐	☐	☐
	0	0	0
1	1	1	1
2	☒	2	2
3	3	3	3
4	4	4	4
5	5	5	☒
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

← Decimal point

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
	☐	☒	☐
	☐	☐	☐
	0	0	0
1	1	1	1
2	☒	2	2
3	3	3	☒
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
☒	☐	☐	☐
☐	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	☒	☒	☒
7	7	7	7
8	8	8	8

.	6	6	7
☒	☐	☐	☐
☐	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	☒	☒	6
7	7	7	☒
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
	☐	☐	☐
	☐	☐	☐
	0	☒	0
1	1	1	☒
2	☒	2	2
3	3	3	3

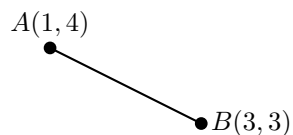
2	0	1	
☐	☐	☐	☐
☐	☐	☐	☐
☒	0	0	0
1	1	☒	1
☒	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16

Consider the following segment $\overline{AB}$:



If $\overline{AB}$ is part of square $ABCD$, then what is the slope of $\overline{BC}$?

17

The time for Bob to finish a project takes 2 hours. Together with Alice, the project takes 40 minutes. If Alice worked on the project alone, how many minutes would it take her?

18

Suppose $f(x, y) = \sqrt{x^2 + y^2}$. What is $f(5, -12) - f(-3, 4)$?

19

If the average of 3 consecutive numbers is 27, what is the larger of these three numbers?

20

On a number line, suppose A, B, C, D lie on the line in this order. If the length of $\overline{CD}$ is 3, and B is the midpoint of $\overline{AD}$ and C is the midpoint of $\overline{BD}$, then what is the length of $\overline{AC}$?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

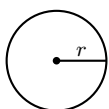
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

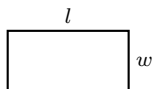
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

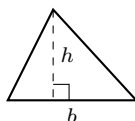


$$A = \pi r^2$$

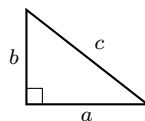
$$C = 2\pi r$$



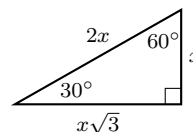
$$A = lw$$



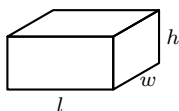
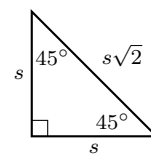
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



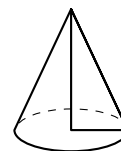
$$V = lwh$$



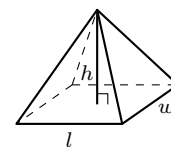
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

Which function satisfies $f(-x) = f(x)$ for any x ?

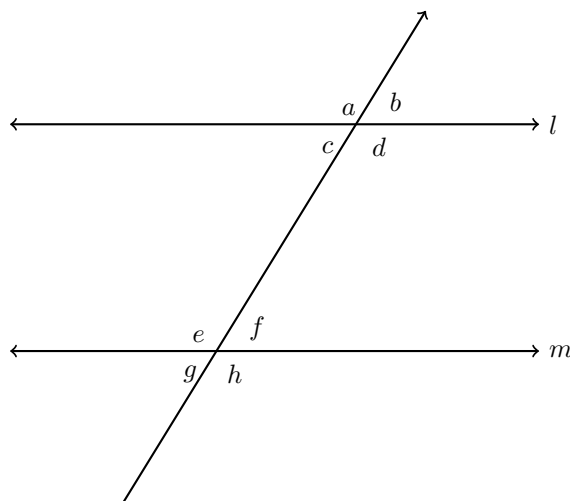
- A) $f(x) = x^2$
- B) $f(x) = x^3$
- C) $f(x) = x + 1$
- D) $f(x) = 2x$

2

If $0 < a < b < c < d$, then which of the following fractions is the smallest?

- A) $\frac{b}{c}$
- B) $\frac{a}{d}$
- C) $\frac{b}{d}$
- D) $\frac{c}{d}$

3



If $m\angle a = 3x + 20$ and $m\angle g = 2x + 10$, then what is x if $l \parallel m$?

- A) 14
- B) 12
- C) 20
- D) 30



4

The slope of the line that goes through $(-1, -4)$ and (a, b) is less than 0. Which of the following can be (a, b) ?

- A) $(-2, -5)$
- B) $(-1, -5)$
- C) $(1, -5)$
- D) $(2, 5)$

5

Suppose printer A prints 2 pages per minute and printer B prints 30 pages per hour. How many pages do they print together in 3 hours?

- A) 150
- B) 450
- C) 500
- D) 600

6

If the only prime factors of x are 2, 5, and 7, then which of the following cannot be a factor of x ?

- A) 10
- B) 14
- C) 35
- D) 36

7

Which of the following lines is parallel to the line

$$y = -\frac{5}{2}x + 3?$$

and has y -intercept 5?

- A) $y = \frac{2}{5}x$
- B) $y = \frac{2}{5}x + 5$
- C) $y = -\frac{5}{2}x + 2$
- D) $y = -\frac{5}{2}x + 5$



8

In a certain sequence, other than the first two terms, each subsequent term is the sum of the two preceding terms. What is the eighth term of this sequence?

- A) 5
- B) 7
- C) 11
- D) Cannot be determined

9

Which of the following has no real roots?

- A) $x^2 - 1$
- B) $x^2 + 1$
- C) $x^2 + x - 12$
- D) $x^2 - 3x + 2$

10

Which of the following polynomials have only one root?

- A) $x^2 - 5x + 6$
- B) $x^2 - 2x + 1$
- C) $x^2 + 6x + 8$
- D) $x^2 - 9x + 20$



11

In a restaurant, each table either sits 3 people or sits 5 people. If there are 22 tables in the restaurant, and the restaurant can seat 86 people, then which equation can be used to find the number of 3-seat tables and 5-seat tables?

A)

$$\begin{aligned}x + y &= 22 \\ 3x + 5y &= 86\end{aligned}$$

B)

$$\begin{aligned}x + 3y &= 22 \\ x + 5y &= 86\end{aligned}$$

C)

$$\begin{aligned}3x + y &= 22 \\ x + 5y &= 86\end{aligned}$$

D)

$$\begin{aligned}3x + y &= 86 \\ x + 5y &= 22\end{aligned}$$

12

For any real numbers a, b , if the point (a, b) is reflected across the line $y = x$, what is the resulting point?

A) $(-a, b)$ B) $(a, -b)$ C) (b, a) D) $(-b, a)$

13

Suppose the fifth term of a geometric series is 48 and the eighth term is 384. What is the first term?

A) 2

B) 3

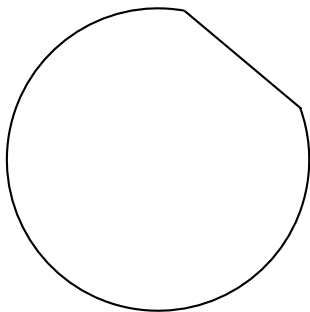
C) 4

D) 6



14

Suppose a circle has a 60° arc removed. Then, the endpoints of the remaining arc are connected with a line segment to create the following shape:



If the original circle had a radius of 6, then what is the perimeter of this shape?

- A) 10π
- B) $10\pi + 6$
- C) 12π
- D) $12\pi + 12$

15

Which of the following polynomials is always non-negative?

- A) $(x - 1)^3(x - 2)^3(x - 3)^3$
- B) $(x - 1)(x - 2)^2(x - 3)^3$
- C) $(x - 1)^2(x - 2)^4(x - 3)^6$
- D) None of the above

16

In scientific notation, what is $1100000 + 910000$?

- A) 201×10^4
- B) 20.1×10^5
- C) 2.01×10^5
- D) 2.01×10^6



17

Fiona has a colored egg collection. Suppose three quarters of her eggs are blue. Moreover, a sixth of her blue eggs are not ostrich eggs. If she has ten blue ostrich eggs, then how many eggs does she have?

- A) 12
- B) 14
- C) 16
- D) 18

18

If for some integer $k \geq 3$,

$$3^{k+1} + 3^k = 2^s$$

then what is 3^{k+2} in terms of s ?

- A) 2^s
- B) $9 \cdot 2^s$
- C) $6 \cdot 2^{s-2}$
- D) $9 \cdot 2^{s-2}$

19

Which of the following shifts of the graph of $y = x^2$ would result in the graph of $y = x^2 + 4x + c$ where c is a positive constant greater than 4?

- A) Left 1 units and up $c - 4$ units
- B) Right 1 units and up $c + 4$ units
- C) Right 2 units and up $c - 4$ units
- D) Left 2 units and up $c - 4$ units

20

Consider the equation,

$$f(x) = |7 - x|$$

Which of the following is the same as $f(15)$?

- A) $f(-8)$
- B) $f(-1)$
- C) $f(1)$
- D) $f(7)$



21

If $\sqrt{3} = x - 1$, what is $2(x - 1)^2$?

- A) 36
- B) 30
- C) 15
- D) 6

22

Simplify: $\frac{\sqrt{600}}{\sqrt{200}}$

- A) 2
- B) $\sqrt{3}$
- C) 4
- D) $\sqrt{30}$

23

Which of the following is not the same as $x^{\frac{3}{2}}$?

- A) $x \cdot \sqrt{x}$
- B) $(\sqrt{x^2})^3$
- C) $\sqrt{x^3}$
- D) $\frac{x^2}{\sqrt{x}}$

24

For any real numbers a, b , if the point (a, b) is reflected across the line $y = -x$, what is the resulting point?

- A) $(-a, b)$
- B) $(a, -b)$
- C) (b, a)
- D) $(-b, -a)$



25

Which of the following polynomials have no roots?

- A) $x^2 + 1$
- B) x^2
- C) $x^2 - 1$
- D) $x^2 + 6x + 5$

26

If $y = 3x + 1$ and $y = 3x - 5$, what is the value of x ?

- A) 12
- B) 13
- C) 10
- D) No such x exist.

27

Which of the following is not equivalent to $6^{\frac{3}{2}}$?

- A) $2\sqrt{54}$
- B) $6\sqrt{6}$
- C) $9\sqrt{8}$
- D) All of the above are equivalent

28

Suppose (x, y) lies on the unit circle. Then which of the following must hold true?

- A) $x > 0, y > 0$
- B) $xy > 0$
- C) $x < 1$
- D) $x^2 + y^2 = 1$



29

Simplify:

$$(-2x^2y^3)^3$$

- A) $-2x^6y^9$
- B) $-8x^5y^6$
- C) $-8x^6y^6$
- D) $-8x^6y^9$

30

From the same starting point, John walks 4 mph north and Jim walks 3 mph south. By the time they finish walking, 35 miles separates them. How many hours elapsed from when they first started walking?

- A) 1
- B) 2
- C) 3
- D) 5

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

5. **Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If $\begin{array}{|c|c|c|c|} \hline 3 & 1 & / & 2 \\ \hline \bullet & \bullet & \bullet & \bullet \\ \hline \end{array}$ is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

6. **Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write answer in boxes. →

7	/	1	2
•	•	•	•
0	0	0	0
1	1	•	1
2	2	2	•
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
•	7	7	7
8	8	8	8
9	9	9	9

← Fraction line

Grid in result.

Answer: 2.5

	2	.	5
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	3
4	4	4	4
5	5	5	•
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

← Decimal point

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	•
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	•
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

	2	0	1
•	•	•	•
0	•	0	0
1	1	1	•
2	•	2	2
3	3	3	3

2	0	1	
•	•	•	•
•	0	0	0
1	1	•	1
•	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

What is the smaller root of $4x^2 - 9$?

32

Evaluate $|4 - 5| + |3 - 4| + |3 - 2| + |1 - 2|$

33

Given that a and b are positive integers and

$$2^{ab} = 512,$$

what is the minimum possible value of $a + b$?

34

If A is the smallest prime number greater than 50 and B is the largest prime number less than 50, then what is $A - B$?

35

What is $3a + 2b$ if

$$\begin{bmatrix} a & b \\ b & a \end{bmatrix} - \begin{bmatrix} -b & a \\ a & -b \end{bmatrix} = \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix}?$$

36

What is the smallest positive number divisible by 3, 5, and 9?

37

Which of the following sets contain a number that is smaller than any other number in that set?

38

Suppose the greatest common divisor of 2000 and integer n is 1. Then the minimum number of digits to the right of the decimal point in the decimal expansion of $\frac{n}{2000}$ is

Answer Key

Calculator Off	
1	D
2	D
3	A
4	B
5	B
6	C
7	C
8	B
9	B
10	D
11	B
12	B
13	A
14	D
15	A
16	2
17	60
18	8
19	28
20	9

:

Calculator On	
1	A
2	B
3	D
4	C
5	B
6	D
7	D
8	D
9	B
10	B
11	A
12	C
13	B
14	B
15	C
16	D
17	C
18	D
19	D
20	B
21	D
22	B
23	B
24	D
25	A
26	D
27	C
28	D
29	D
30	D
31	-1.5
32	4
33	6
34	6
35	8
36	45
37	None of the above
38	4



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

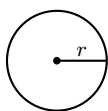
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

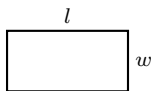
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REFERENCE

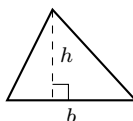


$$A = \pi r^2$$

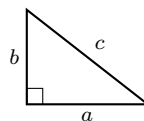
$$C = 2\pi r$$



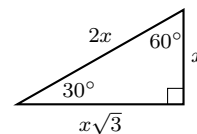
$$A = lw$$



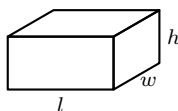
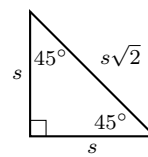
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



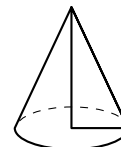
$$V = lwh$$



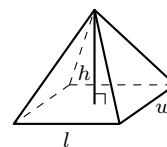
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

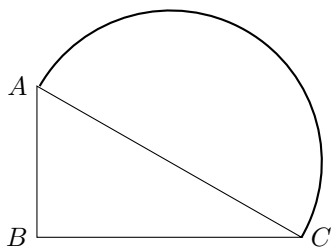
The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1



The area of the semicircle below is 36π . If $BC = AB$, then what is the area of triangle $\triangle ABC$?

- A) 18
- B) 36
- C) 50
- D) Cannot be determined

2

Let $f(x) = x^b$ where b is some positive integer. Which of the following must be true?

- I. f has a y -intercept of 1
 - II. f can have more than one x -intercept
 - III. f is always non-negative
- A) I,III only
 - B) I,II only
 - C) I,II,III
 - D) None of the above

3

Which of the following, when defined, is equivalent to

$$\cos^4(\theta) - \sin^4(\theta)?$$

(Note: For any angle θ , $\cos(2\theta) = \cos^2(\theta) - \sin^2(\theta)$)

- A) $\cos(2\theta)$
- B) $\sin(2\theta)$
- C) $\cos(\theta) + \sin(\theta)$
- D) $\cos(\theta) - \sin(\theta)$



4

What is the period(in radians) of $5 \cos(2x - 4) + 3$?

- A) $\frac{\pi}{2}$
- B) π
- C) 2π
- D) 4π

5

Two weeks ago, a price of a shirt increased by 10% and then decreased by 10% the following week. Assuming the shirt wasn't free, which of the following must be true?

- A) The price of the shirt now is cheaper than two weeks ago.
- B) The price of the shirt now is more expensive than two weeks ago.
- C) The price of the shirt is \$11.
- D) Cannot be determined without knowing the original price

6

Let x and y be integers. If

$$y = |3x - 2|$$

which of the following is NOT a possible value of y ?

- A) -2
- B) 2
- C) 5
- D) 7

7

If there are j jars, and each of these jars contain b jellybeans, and there are 40 jellybeans, then which is a possible value for j ?

- A) 6
- B) 8
- C) 12
- D) 22



8

Suppose $\tan(x)$ is undefined. Then what is one possible value of x ?

- A) 0
- B) $\frac{\pi}{4}$
- C) $\frac{\pi}{2}$
- D) 2π

9

What number is exactly halfway between $\frac{2}{5}$ and $\frac{4}{7}$?

- A) $\frac{17}{35}$
- B) $\frac{3}{6}$
- C) $\frac{2}{3}$
- D) $\frac{1}{35}$

10

In a clothing store, the cost of three shirts and two ties is \$27 and the cost of two shirts and three ties is \$23, what is the cost of one shirt and one tie?

- A) 8
- B) 9
- C) 10
- D) 11

11

Suppose the quotient of two positive real numbers is equal to their product. Then what must be true?

- A) At least one of the numbers must be an integer
- B) At least one of the numbers must be 1.
- C) Both of the numbers must be 1.
- D) None of the above



12

What is the greatest common factor of 36 and 84?

- A) 6
- B) 9
- C) 12
- D) None of the above.

13

If $0 < x < 90^\circ$ and $\tan(x) = \frac{1}{u}$, then what is $\sin(x)$?

- A) u
- B) $\frac{1}{u^2+1}$
- C) $\frac{1}{\sqrt{u^2+1}}$
- D) $\frac{u}{\sqrt{u^2+1}}$

14

A circle of radius 3 is centered at $(-1, 2)$. Which of the following lines is not tangent to this circle?

- A) $x = 3$
- B) $y = 5$
- C) $y = -1$
- D) $x = 2$

15

What is the smaller root of $4x^2 + 13x + 3$?

- A) -3
- B) $.25$
- C) 3
- D) 4

DIRECTIONS

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
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- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or $7/2$. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write $\rightarrow$
answer
in boxes.

Grid in
result.

	7	/	1	2
		0	0	0
1	1	1		1
2	2	2	2	
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7		7	7	7
8	8	8	8	8
9	9	9	9	9

Answer: 2.5

← Fraction line

← Decimal point

	2	.	5

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
			
			
			
①	①	①	①
②		②	②
③	③	③	
④	④	④	④
⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧

.	6	6	6
			
			
	0	0	0
①	①	①	①
②	②	②	②
③	③	③	③
④	④	④	④
⑤	⑤	⑤	⑤
⑥			
⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧

.	6	6	7
①	①	①	①
②	②	②	②
③	③	③	③
④	④	④	④
⑤	⑤	⑤	⑤
⑥			⑥
⑦	⑦	⑦	
⑧	⑧	⑧	⑧

Answer: 201 – either position is correct

	2	0	1
	/	/	
.	.	.	.
	0	●	0
①	①	①	●
②	●	②	②
③	③	③	③

2	0	1	

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16

In an aquarium containing 52 fish, each fish has either blue fins, a green tail, or both. If there are 37 fishes with blue fins and 22 fishes with a green tail, how many fishes have both blue fins and a green tail?

17

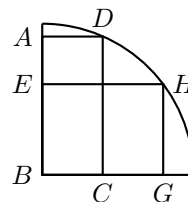
If there are 10 jars, and each of these jars contain 3 jellybeans, then how many jellybeans are there?

18

A triangle has a height of length 20 and a base of length 5. What is the area of this triangle?

19

Rectangles $ABCD$ and $EBGH$ are inscribed in the quarter circle centered at B with radius 5.



What is the sum of the lengths of the segments (not drawn) $\overline{AC}$ and $\overline{EG}$?

20

Let x be a positive integer such that

$$|5x - 16| > 20$$

What is the smallest possible value of x ?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

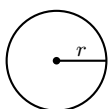
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

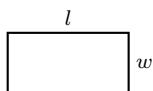
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

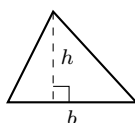


$$A = \pi r^2$$

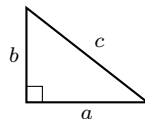
$$C = 2\pi r$$



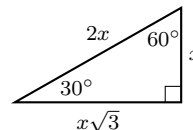
$$A = lw$$



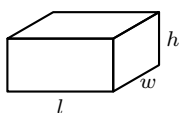
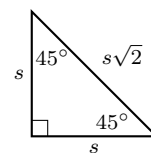
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



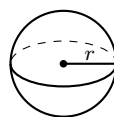
Special Right Triangles



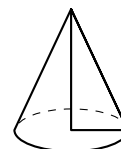
$$V = lwh$$



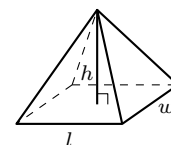
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



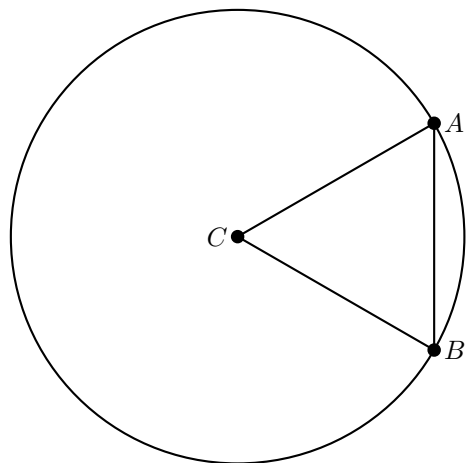
1

Which of the following is the greatest?

- A) $(\frac{1}{2})^{-2}$
- B) $(\frac{1}{2})^0$
- C) 2^{-2}
- D) $(-2)^3$

2

The circle below is centered at C and has radius 3. If $m\angle C = 60^\circ$, then what is the perimeter of $\triangle ABC$?

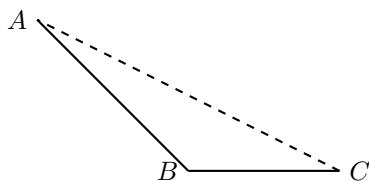


- A) 3
- B) 6
- C) 9
- D) Cannot be determined



3

Consider line segments $\overline{AB}$ and $\overline{BC}$ with lengths 5 and 6, respectively.

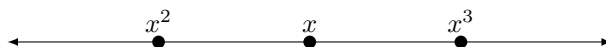


If $\angle ABC$ is chosen such that $\triangle ABC$ has the greatest possible area, which of the following describes $\angle ABC$?

- A) $m\angle ABC = 90^\circ$
- B) $m\angle ABC > 90^\circ$
- C) $m\angle ABC < 90^\circ$
- D) Cannot be determined

4

Consider a standard number line with points located at x , x^2 , x^3 as shown below.



Which of the following describe all possible x ?

- A) $0 < x < 1$
- B) $-2 < x$
- C) $-1 < x < 0$
- D) No such x exists.

5

The sum of integers from 1 to 30 (inclusive) is 465. What is the sum of integers from 31 to 60 (inclusive)?

- A) 465
- B) 930
- C) 1365
- D) 1830



6

What does the solution set of $x^2 + y^2 + 1 < 0$ look like?

- A) The empty set
- B) A point
- C) The inside of a circle
- D) The outside of a circle

7

Suppose for some value a , $f(a) = 3$. Which of the following functions can be $f(x)$?

- A) $\cos(x)$
- B) $\sin(x)$
- C) $3\cos(x) - 1$
- D) $3\sin(x) + 1$

8

Which of the following numbers is rational?

- A) $\sqrt{13}$
- B) $\sqrt{14}$
- C) $\sqrt{15}$
- D) $\sqrt{16}$

9

Which of the following inequalities is true?

- A) $\frac{2}{3} < \frac{3}{5} < \frac{4}{7}$
- B) $\frac{2}{3} < \frac{4}{7} < \frac{3}{5}$
- C) $\frac{4}{7} < \frac{3}{5} < \frac{2}{3}$
- D) $\frac{4}{7} < \frac{2}{3} < \frac{3}{5}$



10

What is the sum of the first 12 terms of an arithmetic sequence with 7 as the first term, and 4 as the second?

- A) -114
- B) -57
- C) 112
- D) 230

11

For integers n and m where $m > n$, suppose an m degree polynomial is multiplied with an n degree polynomial. What is the greatest number of non-zero terms that the resulting polynomial can have?

- A) $m + n$
- B) $m + n + 1$
- C) mn
- D) $mn + 2$

12

How many ways can one choose 2 flavors out of 5 possible flavors of ice cream?

- A) 5
- B) 10
- C) 15
- D) 20

13

For all $x \neq -1, -2$, the rational function $\frac{x^2+x}{x^2+3x+2}$ has the same output as which function

- A) $\frac{x}{x+2}$
- B) $\frac{x}{x-2}$
- C) $\frac{1}{x+2}$
- D) $\frac{1}{x-2}$

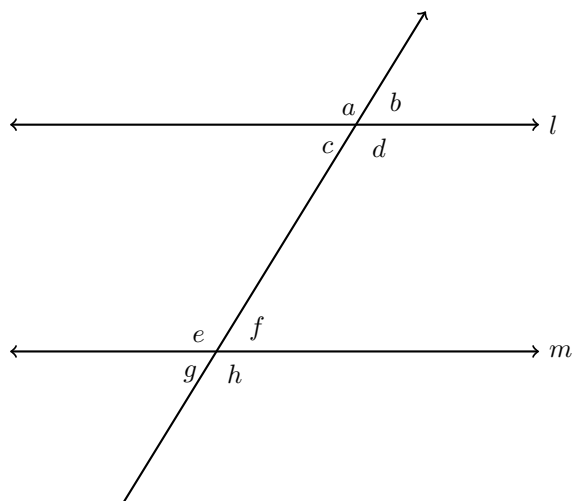


14

What is $(2 + 3i)(6 - i)$?

- A) $14 - 12i$
- B) $15 + 16i$
- C) $13 - 13i$
- D) $16 + 15i$

15



Which of the following lists all angles that must be congruent to angle a ($l \parallel m$)?

- A) $a, d, f, e,$
- B) a, d, f, g
- C) a, d, e, g
- D) a, d, e, h



16

How many vertical asymptotes does the function $\frac{x-2}{x^3-4x}$ have?

- A) 1
- B) 2
- C) 3
- D) 4

17

For integers n and m where $m > n$, suppose an m degree polynomial is added to a n degree polynomial. What is the greatest number of non-zero terms that the resulting polynomial can have?

- A) m
- B) n
- C) $m + 1$
- D) mn

18

Let n be an integer greater than 6. Which of the following inequalities is true?

- A) $2^n < n! < n^n$
- B) $n! < 2^n < n^n$
- C) $2^n < n^n < n!$
- D) $n! < n^n < 2^n$

19

The cost of a taxi ride is \$5 for the first mile and \$2 for each additional mile. Which of the following functions give the total cost, in dollars, for n miles?

- A) $C(n) = 2n + 5$
- B) $C(n) = 2n - 5$
- C) $C(n) = 2n + 3$
- D) $C(n) = 2n - 3$



20

If a , b , and c are negative numbers such that

$$\frac{1}{3}a = 2b$$

$$\frac{1}{4}b = \frac{1}{5}c$$

then which of the following orders the numbers from least to greatest?

- A) $a < b < c$
- B) $a < c < b$
- C) $b < c < a$
- D) $c < a < b$

21

Which of the following lists all the asymptotes of the graph

$$y = \frac{2x^2 - 2}{x^2 + 2x + 1}?$$

- A) $y = 2$ and $x = 1$
- B) $y = 2$ and $x = -1$
- C) $y = 2$
- D) $y = -2$ and $x = 1$

22

In which quadrants is $\sin(x) \cos(x) > 1$

- A) II only
- B) II and III only
- C) II and IV only
- D) None of the above

23

Let x be directly proportional to y . Then $\frac{1}{y^2}$ is inversely proportional to which of the following?

- A) x
- B) x^2
- C) $\frac{1}{x^2}$
- D) None of the above



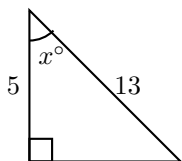
24

What does the solution set of $x + 3y < 10$ look like?

- A) The empty set
- B) A point
- C) A half plane
- D) The inside of a circle

25

In the following diagram, what is x ?



- A) $\arctan\left(\frac{5}{13}\right)$
- B) $\arccos\left(\frac{13}{5}\right)$
- C) $\arccos\left(\frac{12}{13}\right)$
- D) $\arcsin\left(\frac{12}{13}\right)$

26

Which of the following quadratics has 5 and -5 as roots?

- A) $x^2 + 5x + 10$
- B) $x^2 - 10$
- C) $x^2 - 25$
- D) $x^2 + 25$

27

Each of five friends take either a car or a bus to get to a party. If at least one friend needs to be in each vehicle, then how many ways can the friends be assigned to a bus or a car?

- A) 28
- B) 29
- C) 30
- D) 31



28

Suppose $(-4, -3)$ and $(-2, -5)$ lie on a circle. Which of the following could be the center of the circle?

- A) $(6, 2)$
- B) $(-2, -3)$
- C) $(3, 5)$
- D) $(1, 1)$

29

In which quadrants is $\sin(x) \cos(x)$ negative?

- A) *II* only
- B) *II* and *III* only
- C) *II* and *IV* only
- D) *II*, *III*, and *IV* only.

30

If

$$\begin{aligned} n^{2x} \cdot n^3 &= n^{13} \\ (n^{2y})^3 &= n^{18} \end{aligned}$$

then what is $x + y$?

- A) 8
- B) 9
- C) 10
- D) 11


DIRECTIONS

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

- Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If $\begin{array}{|c|c|c|c|} \hline 3 & 1 & / & 2 \\ \hline \bullet & \bullet & \bullet & \bullet \\ \hline \end{array}$ is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

	7	/	1	2
•	•	•	•	•
	0	0	0	
1	1	•	1	
2	2	2	•	
3	3	3	3	
4	4	4	4	
5	5	5	5	
6	6	6	6	
•	7	7	7	
8	8	8	8	
9	9	9	9	

Answer: 2.5

← Decimal point

	2	.	5
•	•	•	•
	0	0	0
1	1	1	1
2	•	2	2
3	3	3	3
4	4	4	4
5	5	5	•
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
•	•	•	•
	0	0	0
1	1	1	1
2	•	2	2
3	3	3	•
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
•	•	•	•
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	7
8	8	8	8

.	6	6	7
•	•	•	•
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	6
7	7	7	•
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
•	•	•	•
	0	•	0
1	1	1	•
2	•	2	2
3	3	3	3

2	0	1	
•	•	•	•
•	•	0	0
1	1	•	1
•	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

If $x = 4$, what is $x^{-\frac{1}{2}}$?

32

What is $64^{2/3} - 4^{3/2}$?

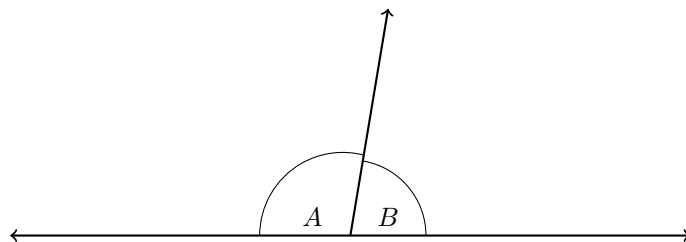
33

If
$$\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 4 & 3 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix}?$$
 then what is the product of a, b, c, d ?

34

If
$$x^{\frac{3}{4}} = 8$$
 what is x ?

35



In the figure above, $m\angle A = 3x + 5$ and $m\angle B = 20x - 55$. What is the measure of the smaller angle?

36

Consider the quadratic $x^2 - bx + 5$. If the vertex lies on the line $x = 3$, then what is b ?

37

If $f(x) = 3x + 7$, what is $f^{-1}(10)$?



38

Suppose the sum of seven consecutive integers is 56.
What is the difference between the mean and median
of these seven numbers?

.....

Answer Key

Calculator Off	
1	A
2	D
3	A
4	A
5	A
6	A
7	B
8	C
9	A
10	C
11	B
12	C
13	C
14	A
15	A
16	7
17	30
18	50
19	10
20	8

Calculator On	
1	A
2	C
3	A
4	D
5	C
6	A
7	D
8	D
9	C
10	A
11	B
12	B
13	A
14	B
15	D
16	B
17	C
18	A
19	C
20	C
21	B
22	D
23	B
24	C
25	D
26	C
27	C
28	B
29	C
30	A
31	.5
32	8
33	9
34	16
35	35°
36	6
37	1
38	0



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

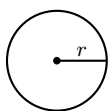
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

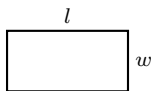
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

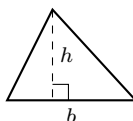


$$A = \pi r^2$$

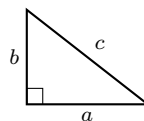
$$C = 2\pi r$$



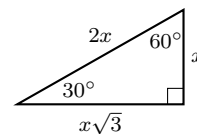
$$A = lw$$



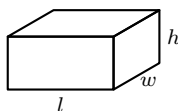
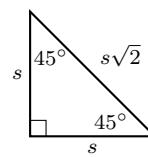
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



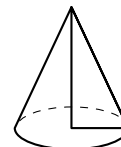
$$V = lwh$$



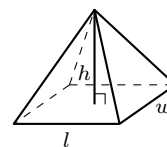
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

Which of the following polynomials have only negative roots?

- A) $x^2 + 7x + 10$
- B) $x^2 - 4x + 3$
- C) $x^2 - 4x - 5$
- D) $x^2 - 2x - 8$

2

Which of the following binomials is a factor of $x^2 - 2x - 24$?

- A) $x - 4$
- B) $x + 4$
- C) $x - 12$
- D) $x + 12$

3

If $F = \frac{kq^2}{r}$, what happens to the value of F when k , q , and r are all doubled?

- A) F remains the same.
- B) F is doubled.
- C) F is multiplied by $\frac{1}{4}$.
- D) F is quadrupled.

4

Consider the vertical line that goes through $(3, 4)$. Which of the points below lie on this vertical line?

- A) $(3, 2)$
- B) $(2, 4)$
- C) $(3, 0)$
- D) $(0, 3)$



5

What is the domain of $7 \sin(x) + 1$?

- A) All real numbers
- B) $[-3, 1]$
- C) $[-1, 1]$
- D) $[-6, 8]$

6

A farmer uses 40 meters of fencing to enclose a rectangular area of length x and width y . Which of the following functions below represents the area in terms of x ?

- A) $f(x) = x(10 - x)$
- B) $f(x) = x(20 - x)$
- C) $f(x) = x(40 - x)$
- D) $f(x) = x(80 - x)$

7

Which of the following polar coordinates represents the same polar coordinate as $(2, 15^\circ)$?

- A) $(2, -335^\circ)$
- B) $(2, -315^\circ)$
- C) $(2, 105^\circ)$
- D) $(2, 365^\circ)$

8

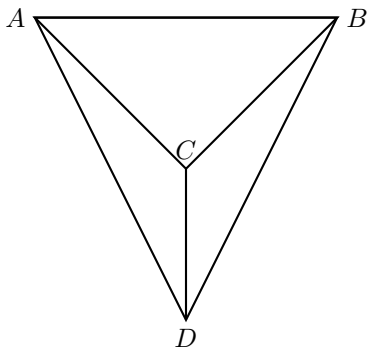
Suppose that, on an exam, the class average of n students is 90. However, 6 students were caught cheating and their scores were replaced with 0's. If the class average dropped to 50, which of the following gives the average of the 6 students who cheated prior to getting caught?

- A) $6n$
- B) $\frac{19n}{3}$
- C) $\frac{20n}{3}$
- D) $8n$



9

Suppose $m\angle ACB = 90^\circ$ and $m\angle ACD = m\angle BCD$. If $AB = 10$ and $AC = CB$, then which of the following cannot be a value for BD ?



- A) 7
- B) 9
- C) 10
- D) 11

10

What is the domain of the real valued function $f(x) = \sqrt{x}$?

- A) All real numbers.
- B) All non-positive numbers.
- C) All non-negative numbers.
- D) All numbers except 0.

11

Evaluate:

$$\frac{1 + \frac{1}{2}}{1 + \frac{1}{1 + \frac{1}{2}}}$$

- A) $\frac{1}{5}$
- B) $\frac{5}{2}$
- C) $\frac{9}{10}$
- D) $\frac{10}{9}$



12

Simplify:

$$\frac{a^2 b^{-\frac{1}{2}} c^{-1}}{a^{-1} b^2 c^{-\frac{1}{2}}}$$

- A) $a^3 b^{-2.5} c^{-1.5}$
 B) $a^3 b^{-2.5} c^{-0.5}$
 C) $a^3 b^{2.5} c^{-0.5}$
 D) $a^3 b^{2.5} c^{0.5}$

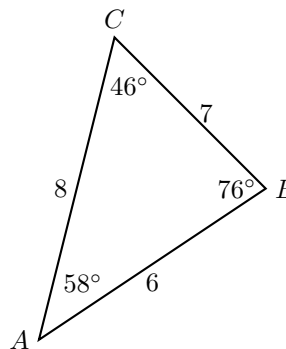
13

Which of the following expressions has only one root?

- A) $x^2 - 9$
 B) $x^2 - 7x + 10$
 C) $x^2 - 8x + 16$
 D) $x^2 - 25$

14

Consider the figure below:



Which equation holds true?

(Note: For any triangle with sides a, b, c that are opposite angles A, B, C respectively, $\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$.)

- A) $\frac{\sin(58^\circ)}{8} = \frac{\sin(76^\circ)}{7}$
 B) $\frac{\sin(76^\circ)}{8} = \frac{\sin(58^\circ)}{7}$
 C) $\frac{\sin(46^\circ)}{6} = \frac{\sin(58^\circ)}{8}$
 D) $\frac{\sin(76^\circ)}{8} = \frac{\sin(46^\circ)}{8}$



15

When is $\frac{\sqrt{x-4}}{x-4}$ undefined?

- A) $x = 4$
- B) $x < 4$
- C) $x \leq 4$
- D) $x \geq 4$

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

7	/	1	2
☐	☒	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 2.5

← Decimal point

	2	.	5
☐	☐	☒	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
☐	☐	☒	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
☒	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	7
☒	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
☐	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

2	0	1	
☐	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16

What is the remainder when 289 is divided by 9?

17

A jar contains red, blue, and green balls. The number of blue balls is double the number of red balls and the number of green balls is triple the number of blue balls. What is the probability of pulling a red ball?

18

In a certain sequence, every term beyond the second is the product of the two preceding terms. If the first two terms of this sequence is 1 and 2, then what is the sixth term of this sequence?

19

How many real numbers satisfy the property that its square is equal to itself?

20

The high school senate is composed of one junior and two seniors. If there are 3 juniors and 5 seniors to choose from, how many different senates can be formed?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

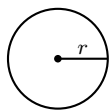
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

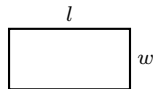
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

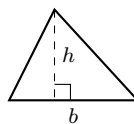


$$A = \pi r^2$$

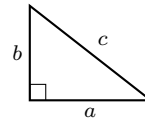
$$C = 2\pi r$$



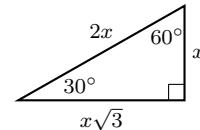
$$A = lw$$



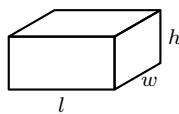
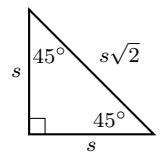
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



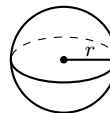
Special Right Triangles



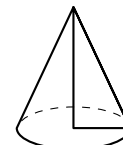
$$V = lwh$$



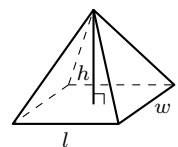
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

Suppose two sides of a triangle has lengths 5 and 12. If the third side has integer length, what is the largest possible length of the third side?

- A) 13
- B) 16
- C) 17
- D) 18

2

For what values of x does the graph of $y = x^2 - 9x + 14$ intercept the x -axis?

- A) 7 and 2
- B) -7 and 2
- C) 7 and -2
- D) -7 and -2

3

If for some integer $n > 2$,

$$2^{n+1} - 2^n = k - 1$$

then what is 2^{n-1} in terms of k ?

- A) $\frac{k-1}{2}$
- B) $k - 1$
- C) k
- D) $2k$

4

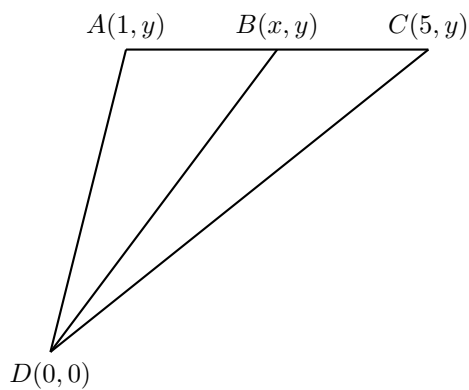
Which of the following inequalities is true?

- A) $4^5 < 2^7 < 8^3$
- B) $2^7 < 4^5 < 8^3$
- C) $2^7 < 8^3 < 4^5$
- D) $8^3 < 2^7 < 4^5$



5

If the area of $\triangle ACD$ is 100 and B is the midpoint of $\overline{AC}$, then what is the (x, y) coordinate of B ?



- A) (2.5, 25)
- B) (3, 50)
- C) (3, 25)
- D) (3, 30)

6

Define $f(x) = x^a$ where a is a positive number. Which of the following is the same as $f(x + y)$?

- A) $f(x) + f(y)$
- B) $f(x) - f(y)$
- C) $x + y$
- D) None of the above.

7

Suppose we know

$$7a - 6b - 5c = 4$$

$$2a - 2b - 2c = 8$$

Then what is $a + c$?

- A) -20
- B) 0
- C) 12
- D) 28



8

Suppose $(x - 3)$ is a factor of $x^2 + x - c$. Then what is c ?

- A) -6
- B) 6
- C) 5
- D) 12

9

Suppose for integers a and b ,

$$\frac{a + b}{2}$$

is also an integer. Then which of the following must be true?

- A) a and b are both zero.
- B) a and b are both even.
- C) a and b are both odd.
- D) The sum of a and b is even.

10

Define $f(x) = a^x$ where a is a positive number. Which of the following is the same as $f(x + y)$?

- A) $f(x) + f(y)$
- B) $f(x) - f(y)$
- C) $f(x)f(y)$
- D) None of the above.

11

Suppose for some positive reals a and b , $\log_b(a) = \frac{1}{\log_b(a)}$, then which can be the value of a in terms of b ?

- A) b^2
- B) b^3
- C) $\frac{1}{b}$
- D) $\frac{1}{b^2}$



12

Expand $(x + y)^2$

- A) $x^2 + 2x + y^2$
- B) $x^2 - y^2$
- C) $x^2 - 2x + y^2$
- D) $x^2 - y^2$

13

Which of the following best describes the set of all points that are closer to $(3, 4)$ than $(1, 2)$?

- A) A line
- B) A half plane
- C) A circle
- D) The empty set

14

Suppose $(-3, 7)$ and $(11, -5)$ lie on the circumference of a circle such that the line formed from these two points goes through the circle's center. What is the radius of the circle?

- A) $\sqrt{85}$
- B) $2\sqrt{85}$
- C) $\sqrt{17}$
- D) $5\sqrt{2}$

15

Suppose a is irrational. Which must always be rational?

- A) a^2
- B) a^3
- C) $\sqrt{a}$
- D) None of the above



16

Let A, B, C be $m \times m$ matrices. Which of the following need not be true?

- A) $A + B = B + A$
- B) $A(BC) = (AB)C$
- C) $AB = BA$
- D) $A(B + C) = AB + AC$

17

Which of the following expressions is never negative for any value of x ?

- A) $x^2 - 1$
- B) $x^2 - 2x + 1$
- C) $\sqrt[3]{x}$
- D) None of the above

18

If $\frac{2}{7}$ of a number is 10, what is $\frac{1}{7}$ of that number?

- A) 2
- B) 5
- C) 10
- D) 20

19

What is $\log(\ln(e^{10}))$?

- A) 0
- B) 1
- C) 2.17
- D) 10



20

What is

$$|3 - 7| - |7 - 3| + |6 - 5| - |5 - 6|?$$

- A) -8
- B) -4
- C) 0
- D) 2

21

What is the simplified form of $5\sqrt{12} - \sqrt{75}$

- A) $4\sqrt{3}$
- B) $5\sqrt{3}$
- C) $9\sqrt{3}$
- D) $10\sqrt{3}$

22

The sum of all the integers from 1 to 50 (inclusive) is 1275. What is the sum from 51 to 100?

- A) 2550
- B) 3775
- C) 3950
- D) 4000

23

How many positive factors does 5^3 have?

- A) 1
- B) 2
- C) 4
- D) 5



24

For what values of a is

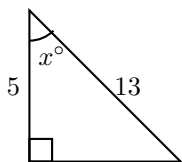
$$\frac{|a| - a}{2}$$

a positive number?

- A) All real a .
- B) No real a .
- C) All positive a .
- D) All non-zero a .

25

In the following diagram, what is $\sin(x) \cos(x)$?



- A) $\frac{5}{13}$
- B) $\frac{12}{13}$
- C) $\frac{60}{13}$
- D) $\frac{60}{169}$

26

Consider the quadratic $x^2 - 2kx + k^2$. What must be true about this quadratic?

- A) It has a double root
- B) It has 2 distinct real roots
- C) It is always positive
- D) None of the above

27

Which of the following is always positive for all real x ?

- A) x^2
- B) $\sqrt{x}$
- C) $\sqrt{x^2}$
- D) $\sqrt{x^2 + 1}$



28

What is the y -intercept of $\frac{x^2+4x}{x^2+2x}$?

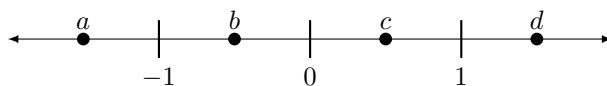
- A) 1
- B) 2
- C) 4
- D) Does not exist.

29

Which of the following expressions below represents x percent of y percent of 100000?

- A) xy
- B) $10xy$
- C) $100xy$
- D) $1000xy$

30



Which of the following letters satisfy $x < x^2$?

- A) d only
- B) a, b, d only
- C) a, c, d only
- D) a, b, c, d

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

5. **Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If  is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

6. **Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write answer in boxes. →

7	/	1	2
•	•	•	•
0	0	0	0
1	1	•	1
2	2	2	•
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
•	7	7	7
8	8	8	8
9	9	9	9

← Fraction line

Grid in result.

Answer: 2.5

	2	.	5
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	3
4	4	4	4
5	5	5	•
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

← Decimal point

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	•
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	•
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

	2	0	1
•	•	•	•
0	•	0	0
1	1	1	•
2	•	2	2
3	3	3	3

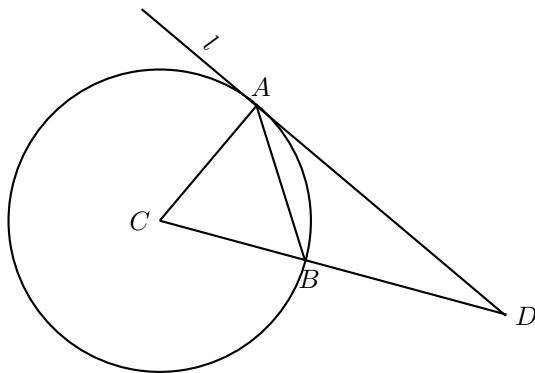
2	0	1	
•	•	•	•
•	0	0	0
1	1	•	1
•	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

Line l is tangent to the circle at point A . If point C is the center of the circle and $m\angle ADC = 30^\circ$, then what is the measure of $\angle ABD$?

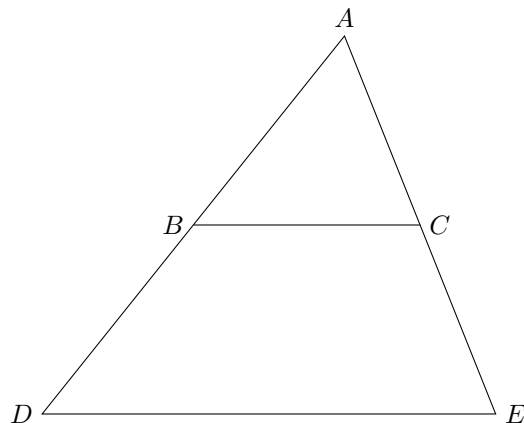


32

On a Cartesian Plane, if $\overline{AB}$ has midpoint C and the slope of $\overline{BC}$ is 12, then what is the slope of $\overline{AB}$?

33

In the figure below, $\overline{BC}$ is parallel to $\overline{DE}$. If $AC = 6$, $DE = 18$, and $CE = 12$, then what is the length of $\overline{BC}$?



34

On a number line, suppose the distance between points A and C is 3 and the distance from points A to B is the same as the distance from points B to C . If C is located at 1, where is B located?



35

For what value of c does the equation $|2x - 3| + 3 = c$ have exactly one solution?

36

Evaluate $2^1 \cdot 2^2 \cdot 2^3$

37

Consider the graph of $f(x) = x^2 + ax + b$ with vertex $(0, -3)$. If the graph of $g(x) = f(x + h) + k$ has vertex $(4, 2)$, what is hk ?

38

Suppose the coordinate $(3, 0)$ is shifted four units to the right and two units up, and then rotated 90° counterclockwise about the point $(5, 2)$. If the new point is (x, y) , then what is $x + y$?

Answer Key

Calculator Off

1	A
2	B
3	D
4	B
5	A
6	B
7	A
8	C
9	A
10	C
11	C
12	B
13	C
14	B
15	C

16	1
17	$\frac{1}{9}$
18	32
19	1
20	30

.....

Calculator On

1	B
2	A
3	A
4	C
5	B
6	D
7	A
8	D
9	D
10	C
11	C
12	A
13	B
14	A
15	D
16	C
17	B
18	B
19	B
20	C
21	B
22	B
23	C
24	C
25	D
26	A
27	D
28	D
29	B
30	B

31	120°
32	12
33	6
34	−.5
35	3
36	64
37	−20
38	9



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

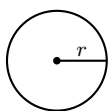
DIRECTIONS

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NOTES

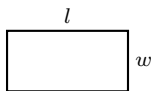
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

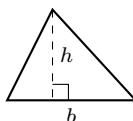


$$A = \pi r^2$$

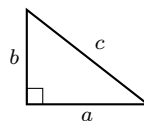
$$C = 2\pi r$$



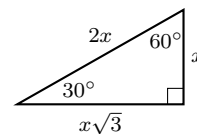
$$A = lw$$



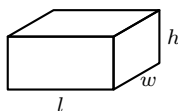
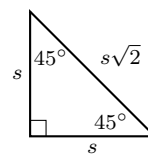
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



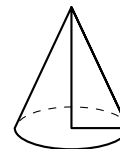
$$V = lwh$$



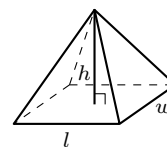
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

If $x > 6$, which of the following is equivalent to

$$\frac{1}{\frac{2}{x-6} + \frac{4}{x-3}}?$$

- A) $\frac{6x-30}{x^2-9x+18}$
- B) $\frac{x^2-9x+18}{6x-30}$
- C) $x^2 - 9x + 18$
- D) $\frac{2x-9}{6}$

2

If

$$\frac{3}{2x-3} = \frac{6}{x}$$

then what is the value of x ?

- A) -2
- B) -3
- C) 3
- D) 2

3

Two figures have the same area. One is triangle with a base $4(m-3)$ meters and a corresponding height of $(m+1)$ meters. The other is a parallelogram with a $50m$ cm base and a corresponding height of $240(m-1)$ cm. Find m .

- A) 5
- B) 3
- C) $\frac{3}{2}$
- D) 2

4

For $i = \sqrt{-1}$, what is the difference $(3 - 8i) - (-3 + 5i)$?

- A) $-3i$
- B) $-16i$
- C) $6 - 3i$
- D) $6 - 13i$



5

Lara is a logistics specialist for a shipping company. Each week, she receives a shipment that needs to be distributed throughout the entire country. The number of kilometers left to travel for the last part of the shipment to be offloaded can be estimated with the equation $d = 6000 - 80t$, where d is the distance left to be traveled and t is the number of hours the shipment has been in transit. What is the meaning of the value 80 in this equation?

- A) Lara's shipment moves about 80 kilometers per hour.
- B) Lara starts the week with 80 shipments.
- C) Lara should expect the shipment to take 80 hours to finish.
- D) Lara splits the shipment into 80 kilometer segments.

6

Which of the following equations has a graph in the xy -plane for which y is always less than -2 ?

- A) $y = \sqrt[3]{-2 - x}$
- B) $y = -2 - |4 - x|$
- C) $y = \frac{-2}{x}$
- D) $y = -2 - \sqrt{x^2 + 1}$

7

$$h(t) = -\frac{1}{2}gt^2 + v_0t + h_0$$

Vertical motion for a launched object without its own source of propulsion is modeled by the equation above, where g is the gravitational constant for the planet in question, v_0 is the starting vertical velocity, and h_0 is the starting height of the object. For this equation, time is typically measured in seconds and altitude is measured in meters. Gravity on Mars is approximately $\frac{15\text{m}}{4\text{s}^2}$.

According to the model, if a volcanic rock is launched into the Martian atmosphere from ground level at $\frac{45\text{m}}{\text{s}}$ and the rock lands at ground level, about how many seconds is the rock in midair?

- A) 0s
- B) $2\sqrt{6}\text{s}$
- C) 12s
- D) 24s



8

If $\frac{2a+5}{3} = b$ and $b = 23$, what is the value of a ?

- A) 64
- B) 32
- C) 27
- D) $\frac{32}{3}$

9

If $x^2 = y^2$, then which of the following must be true?

- A) $x = -y$
- B) $x = y$ or $x = -y$
- C) $x = y$ and $x = -y$
- D) None of the above

10

If $xy = 0$, then which of the following must be true?

- A) $x = 0$
- B) $x = 0$ or $y = 0$
- C) $x = 0$ and $y = 0$
- D) None of the above

11

$$g = 3.99 - 0.20w$$

$$b = 2.39 + 0.20w$$

In the equations above, g and b represent the price of a gallon of premium gas and the price of a pint of blueberries w weeks after the peak of blueberry season a previous year at a particular store. What was the price per gallon of premium gasoline when it matched the price of a pint of blueberries?

- A) \$3.79
- B) \$3.19
- C) \$2.79
- D) \$1.60



12

$$y = -3.64x + 18$$

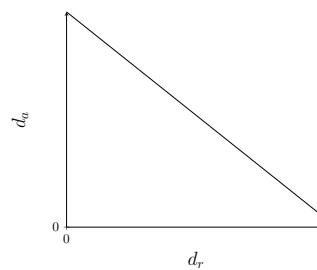
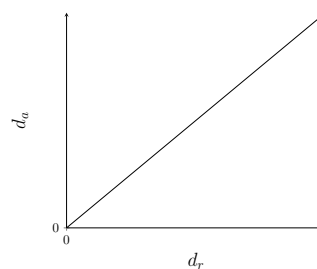
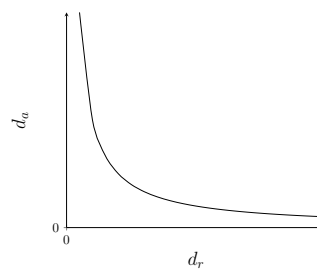
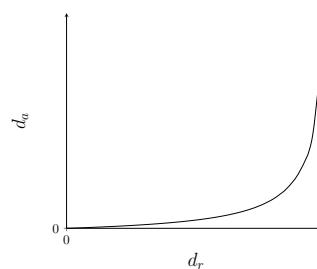
An economist comes up with the linear equation above as an approximation of the United States demand curve for butter, where y is the quantity demanded in pounds per person per year and x is the average price per pound of butter. He figures the linear model is appropriate for prices between \$3 and \$4 per pound. Which of the following statements correctly describes the role of the slope in the model?

- A) The slope indicates that an increase of one pound in butter demanded per person per year will increase the average price per pound of butter by \$3.64.
- B) The slope indicates that an increase of one pound in butter demanded per person per year will decrease the average price per pound of butter by \$3.64.
- C) The slope indicates that butter never costs more than \$18 per pound.
- D) The slope indicates that an average price increase of \$1 per pound would be expected to cause consumers to desire on average 3.64 fewer pounds of butter per year.

13

$$\frac{1}{f} = \frac{1}{d_a} + \frac{1}{d_r}$$

The equation above is used for describing the apparent distance of an image in a concave mirror (d_a) in relation to the real distance of the object (d_r) and the focal length of the mirror (f). If the focal length of the mirror is constant, and the object is closer than the focal length, which of the following graphs accurately represents the apparent distance in the mirror as a function of the actual distance of the object?





14

$$M = \frac{5}{9}(I + 460)$$

The equation above shows how a temperature I , measured in degrees Fahrenheit, relates to a temperature M , measured in degrees Kelvin. Based on the equation, which of the following must be true?

- I. A temperature increase of 1 degree Kelvin is equivalent to a temperature increase of $\frac{5}{9}$ degree Fahrenheit.
 - II. A temperature decrease of 1 degree Fahrenheit is equivalent to a temperature decrease of 1.8 degree Kelvin.
 - III. A temperature increase of $\frac{5}{9}$ degree Kelvin is equivalent to a temperature increase of 1 degree Fahrenheit.
- A) I only.
B) III only.
C) I and II only.
D) I and II and III.

15

If $x = 8^2$, which of the following is 2^{10} ?

- A) $2x$
B) 2^3x
C) 2^4x
D) 2^5x

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

7	/	1	2
•	•	•	•
0	0	0	0
1	1	•	1
2	2	2	•
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
•	7	7	7
8	8	8	8
9	9	9	9

Answer: 2.5

← Decimal point

	2	.	5
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	3
4	4	4	4
5	5	5	•
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	•
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	7
8	8	8	8

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	•
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
•	•	•	•
0	•	0	0
1	1	1	•
2	•	2	2
3	3	3	3

	2	0	1
•	•	•	•
0	•	0	0
1	1	•	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16

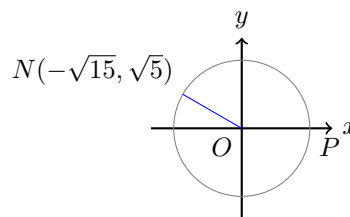
Three vases are filled with flowers. If the ratio of flowers in the three vases is $1 : 2 : 3$, then how many flowers can there be in total?

17

$$\begin{aligned}\alpha x + \beta y &= 2 \\ 3x + y &= -12\end{aligned}$$

In the system of equations above, α and β are constants. If the system has no solutions, what is the value of $\frac{\alpha}{\beta}$?

18



In the xy -plane above, O is the center of the circle, and the measure of $\angle NOP$ is $\theta\pi$ radians. What is the value of θ ?

19

What is the value of the x intercept of $y = 6x - 2$?

20

Evaluate $\frac{1+\frac{1}{2}}{1-\frac{1}{2}}$



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

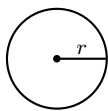
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

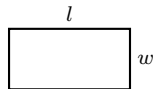
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
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REFERENCE

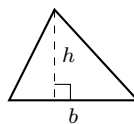


$$A = \pi r^2$$

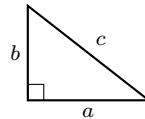
$$C = 2\pi r$$



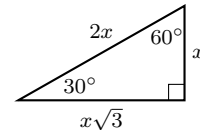
$$A = lw$$



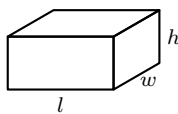
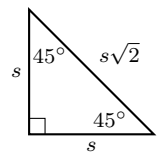
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



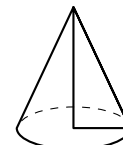
$$V = lwh$$



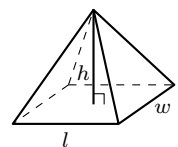
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

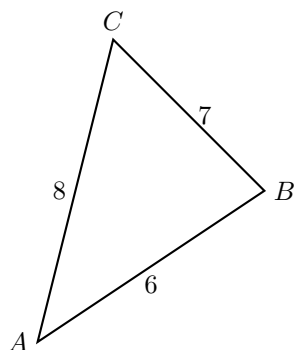
The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

In the figure below, which of the following equations can be used to find angle C ?



(Note: For any triangle with sides a, b, c that are opposite angles A, B, C respectively, $c^2 = a^2 + b^2 - 2ab \cos(C)$)

- A) $8^2 = 6^2 + 7^2 - 2 \cdot 6 \cdot 7 \cdot \cos(C)$
- B) $6^2 = 7^2 + 8^2 - 2 \cdot 7 \cdot 8 \cdot \cos(C)$
- C) $8^2 = 6^2 + 7^2 + 2 \cdot 6 \cdot 7 \cdot \cos(C)$
- D) $6^2 = 7^2 + 8^2 + 2 \cdot 7 \cdot 8 \cdot \cos(C)$

2

If $|3 - 6x| \geq 2$, what is the domain of x ?

- A) $x \in (-\infty, \frac{1}{6}]$
- B) $x \in (-\infty, \frac{1}{6}] \cup [\frac{5}{6}, \infty)$
- C) $x \in \mathbb{R}$
- D) $x \in \emptyset$

3

A pet hotel advertises that they have enough space and supplies to regularly house a ton of cats and dogs (literally, 2,000 pounds of cats and dogs) on a daily basis. At present the pet hotel is housing a sled dog team of 18 Siberian Huskies, and the average weight of the Siberian Huskies is 40 pounds. They are also housing 3 cats with an average weight of 10 pounds each. If the hotel currently has no other tenants, how many more pounds of dogs and cats could the pet hotel take in if they are true to their advertisement?

- A) 1930
- B) 1280
- C) 1250
- D) 750



4

Let A and B represent $n \times n$ square matrices. If $\det(A) = 3$ and $\det(B) = 5$ what is the value of $\det(AB)$?

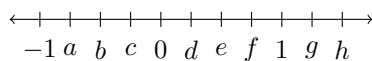
- A) $3/5$
- B) $5/3$
- C) 15
- D) 1

5

The amount of money Maude the magician earns is a fixed commission per member of her audience, but she has to pay \$200 to reserve the stage for each performance. Based upon prior reputation, Maude is allowed to have the stage fee taken from her payout rather than required up front. During one performance on a slow night, Maude is paid \$500 based upon an audience of 140. In her budgeting, Maude reserves 18% of her pay for her insurance bills and materials for her tricks that don't count towards her taxable profits. How much taxable profit will Maude report to the IRS for this performance?

- A) \$90
- B) \$114.80
- C) \$410
- D) \$482

6



The letters a, b, c, d, e, f, g , and h above represent numbers as shown on the number line above. Which of the following expressions has the greatest value?

- A) $d - a$
- B) $f + b$
- C) $g - e$
- D) $h - f$

7

Find all x such that $0 < x < 360^\circ$ and $\sin^2 + \sin(x) + \frac{1}{4} = 0$.

- A) $30^\circ, 210^\circ$
- B) $210^\circ, 240^\circ$
- C) $210^\circ, 330^\circ$
- D) 330°

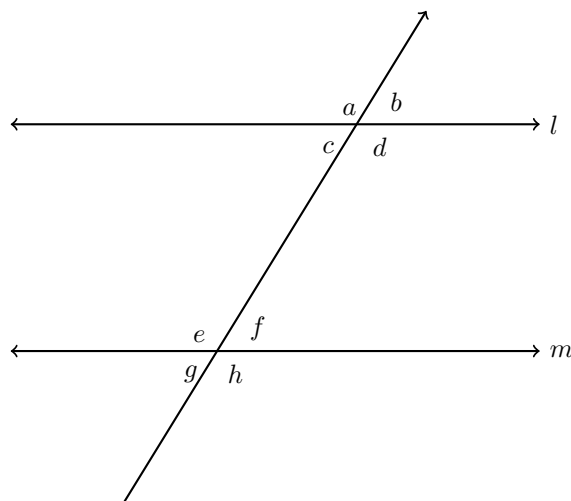


8

What values for x would make $\frac{1+\sqrt{x}}{1-\sqrt{x}}$ undefined?

- A) 1 only
- B) $(-1, \infty)$
- C) $(-\infty, 0) \cup \{x = 1\}$
- D) All reals less than -1

9



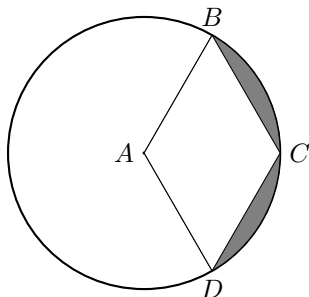
Which of the following lists all angles that must be supplementary to c ($l \parallel m$)?

- A) $a, d, f, e,$
- B) a, b, c, d
- C) a, d, f, g
- D) a, d, e, h



10

In the circle below, the center is at A and the radius is 6. If the measures of arcs BC and CD are equal and $m\angle BAD = 120^\circ$, then what is the perimeter of the shaded region?



- A) $10 + 2\pi$
- B) $12 + 2\pi$
- C) $12 + 4\pi$
- D) $12 + 8\pi$

11

Which of the following shifts of the graph of $y = x^2$ would result in the graph of $y = x^2 - 4x + c$ where c is a positive constant greater than 4?

- A) Left 1 units and up $c - 4$ units
- B) Right 2 units and up $c - 4$ unit
- C) Left 2 units and up $c - 4$ units
- D) Left 2 units and up $c + 4$ units

12

$$7(x - 4) + 4 = 5(x - 2)$$

If the statement above is true, what is the value of x ?

- A) -5
- B) -1
- C) 7
- D) 11



13

Foul Shots Made by College Basketball Centers by Situation in Sample Games

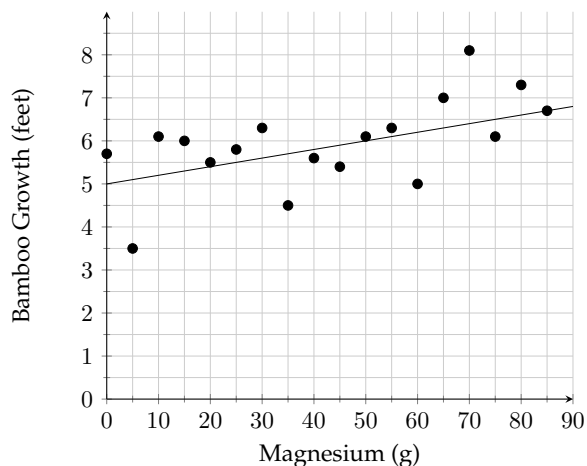
	Made 0	Made 1	Made 2	Total
1-and-1	19	63	17	99
Basket-and-1	24	49	0	73
2 shots	12	58	20	90
Total	55	170	37	262

The data in the table above were produced by a sports researcher examining the potential value of using the intentional fouling strategy on centers in college basketball. If a center made at least 1 foul shot, what is the probability the center made 2 foul shots on the play?

- A) $\frac{37}{207}$
- B) $\frac{37}{170}$
- C) $\frac{78}{207}$
- D) $\frac{90}{207}$

14

Bamboo Growth versus Amount of Magnesium Fertilizer



Gao Cao is examining whether or not providing magnesium fertilizer to young bamboo plants will cause them to grow taller even faster. He applies fertilizer to 17 bamboo plants and keeps track of an 18th bamboo plant for two weeks. The line of best fit for the data Gao Cao gathered is shown. For the plant that received 75 grams of Magnesium fertilizer, what was the difference between the bamboo's actual growth and the bamboo's predicted growth?

- A) -0.9 feet
- B) -0.4 feet
- C) 0.7 feet
- D) 1.7 feet



15

The mass of an object is equal to the weight of the object divided by the gravity acting upon the object. In SI units, weight is measured by Newtons, where $1 \text{ N} = 1 \text{ kg} \cdot \frac{\text{m}}{\text{sec}^2}$. What is the mass, in kilograms, of an object with a weight of 196 N on a planet with a gravity of $9.8 \frac{\text{m}}{\text{sec}^2}$?

- A) 20,000
- B) 1920.8
- C) 186.2
- D) 20

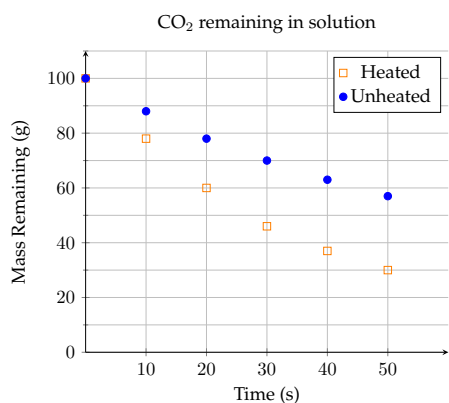
16

Nora has a series of molds she uses to generate prosthetic horns in the shape of right circular cones. Each of the cones generated by these molds have radii equal to their heights. She pours a liquid polymer from a two-gallon container into each mold until each mold is filled to a depth of one inch. If she uses the entire two-gallon container on making these prosthetic horns, how many prosthetic horns will she have made? (Note: There are 231 cubic inches in one gallon.)

- A) 441
- B) 220
- C) 154
- D) 149



17



Two carbonic acid solutions each have 100 grams of dissolved CO₂ and then sealed. One of them is placed in a heated chamber and the other is placed in an unheated chamber. The containers are opened simultaneously and left inside their respective chambers for 50 minutes. The graph above shows the amount of CO₂ still left in each container at 10 minute intervals. Which of the following statements correctly compares the average rates at which the two samples lose CO₂?

- A) In every 10-minute interval, the magnitude of the rate of change for the solution in the heated chamber is clearly greater than that of the solution in the unheated chamber.
- B) In every 10-minute interval, the magnitude of the rate of change for the solution in the unheated chamber is clearly greater than that of the solution in the heated chamber.
- C) At first the rate of change is the same between the two chambers, but the magnitude of the rate of change for the heated chamber quickly becomes greater than that of the unheated chamber.
- D) The magnitude of the rate of change for the heated chamber is clearly greater during the first 30 minutes of the experiment, but the rate of change appears nearly identical during the last 20 minutes.

18

A triangle was altered by increasing its base by 40 percent and decreasing its height by d percent. If these alterations increased the area of the triangle by 12 percent, what is the value of d ?

- A) 14
- B) 20
- C) 28
- D) 30

19

Suppose for some value a , $f(a) = 0$. Which of the following functions can be $f(x)$?

- A) $\cos(x) - 2$
- B) $2\sin(x) - 3$
- C) $3\cos(x) - 4$
- D) None of the above



20

Consider the following system of equations:

$$\begin{aligned}2x - 3y &= -2 \\4x - 6y &= -4\end{aligned}$$

How many solutions does this system have?

- A) 0
- B) 1
- C) 2
- D) Infinitely many

21

At Krabby Kravings, customers can choose from a crab cake dish or a crab legs dish with some degree of customization. Regardless of how they customize their orders, all crab cake dishes cost the same and all crab legs dishes cost the same. However, prices vary seasonally. One day, Alicia came in to treat her department to lunch. They ordered 7 leg dishes and 4 cake dishes for \$247. The same day Bob came in with Regina and her son Phil. They ordered 2 leg dishes and 1 cake dish for \$68. On that day, how much did a crab cake dish cost at Krabby Kravings?

- A) \$18
- B) \$20
- C) \$22
- D) \$25

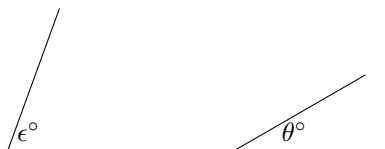
22

A positive integer is subtracted from its square. The units digit of the result could be which of the following?

- A) 4
- B) 5
- C) 6
- D) 7



23



Note: Figures not drawn to scale.

The angles drawn above are acute and $\cos(\theta^\circ) = \sin(\epsilon^\circ)$. If $\theta = 4\nu - 6$ and $\epsilon = 8\nu + 6$, what is the value of ν ?

- A) 7.5
- B) 15
- C) 24
- D) 66

24

The amount of money Maude the magician earns is a fixed commission per member of her audience, but she has to pay \$200 to reserve the stage for each performance. Based upon prior reputation, Maude is allowed to have the stage fee taken from her payout rather than required up front. During one performance on a slow night, Maude is paid \$500 based upon an audience of 140. On a much better night, Maude has 600 audience members. How much money does Maude get paid on the better night?

- A) \$3000
- B) \$2800
- C) \$2142.86
- D) \$208

25

For what values of q is $|5 - q^2| - 16 = 9$?

- A) $q \in \{\pm\sqrt{30}\}$
- B) $q \in \{\pm 2i\sqrt{5}\}$
- C) $q \in \{\pm\sqrt{30}, \pm 2i\sqrt{5}\}$
- D) There is no such value of q .



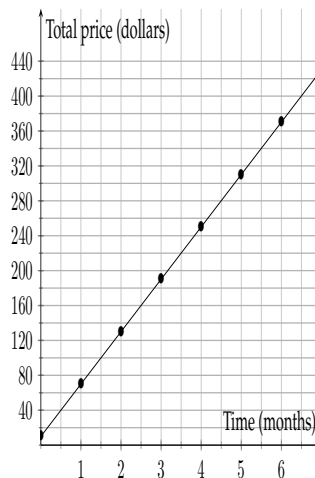
26

A store has to charge two sales taxes. One is a state sales tax for 8.5% and the other is a municipal sales tax for 1.25%. All the items this store sells are subject to the municipal sales tax, but the state sales tax is not applied to items deemed to be necessities. How much sales tax does the store have to collect if it sells m of merchandise, n of which are necessities?

- A) $0.0125n + 0.085m$
- B) $0.0125m + 0.085n$
- C) $0.0975m + 0.0125n$
- D) $0.0975m - 0.085n$

27

Total Cost of Belonging to an Organic Cooperative by the Month



The graph above displays the total price p paid by a customer to belong to an organic cooperative for m months. What does the slope represent in the graph?

- A) The initial cost of joining the cooperative.
- B) The total number of months spent with the cooperative.
- C) The total number of organic products purchased via the cooperative.
- D) The increase in cost to stay with the cooperative for each additional month.



28

The amount of coal left in a seam being mined is estimated over the course of three years, as shown in the table.

Time (months)	Tons
0	1,450,000
6	1,325,000
12	1,200,000
18	1,075,000
24	950,000
30	825,000
36	700,000

Which of the following best describes the relationship between time and the estimated amount of coal remaining in the seam?

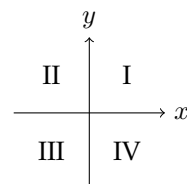
- A) Exponential growth
- B) Exponential decay
- C) Increasing linear
- D) Decreasing linear

29

Which of the following formulas defines an odd function of the variable x ?

- A) $x + x^3$
- B) $x^2 + x + 1$
- C) $x^2 + 2|x|$
- D) $x^3 + 2$

30



If the system of inequalities $y > x^2 - 4$ and $y \leq 2x - 1$, which quadrant contains no solutions to the system?

- A) Quadrant IV
- B) Quadrant II
- C) Quadrant I
- D) There are solutions in all four quadrants.

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

7	/	1	2
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 2.5

← Decimal point

2	.	5
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

2	/	3
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

2	0	1
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

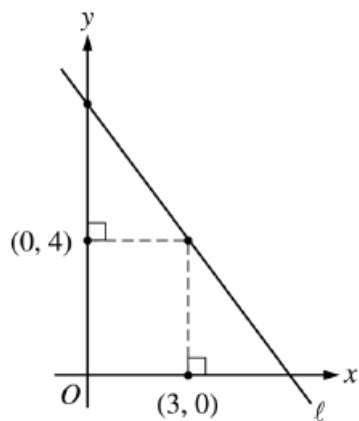
2	0	1
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

Line ℓ is shown in the graph below. What is the y intercept of line ℓ ?



Note: Figure not drawn to scale.

32

Suppose we know

$$7a^2 - 6b - 5c^{-1} = 4$$

$$2a^2 - 2b - 2c^{-1} = 8$$

Then what is $a^2 + c^{-1}$?

33

If $f(x) = 3x^{1/3}$, what is $f^{-1}(3)$?

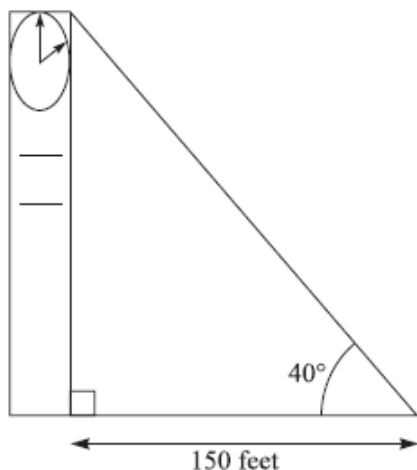
34

The city library donated some children's books to Mr. Clark's first-grade class. If each student takes 4 books, there will be 20 books left. If 3 students do not take a book and the rest of the students take 5 books each, there will be no books left. How many books were donated to the class?



35

A clock tower casts a 150-foot shadow on level ground, as shown below. The angle of elevation from the tip of the shadow to the top of the tower is 40° . To the nearest tenth of a foot, what is the height of the clock tower?



36

In the social sciences and in biology, game theory is used to predict the behavior of actors that have to interact with one another. Ordered pairs are used to communicate how much a particular actor prefers or dislikes given potential outcomes. The game theory table below is used to reflect the possible outcomes of an evening where Vi and Sheila have three different group activities they could participate in but are asked independently and simultaneously which they would prefer. In the ordered pairs below, Vi's satisfactions with each particular outcome are the y coordinates and Sheila's satisfactions are the x coordinates.

		Vi		
		Go to Club	Play Frisbee	See A Play
Sheila	Go to Club	(8, 9)	(6, 6)	(-6, -1)
	Play Frisbee	(5, 7)	(9, 8)	(-6, -1)
	See a Play	(-6, 7)	(-6, 6)	(-1, 3)

If Vi reasons that they both won't go to see the play, what would be her average satisfaction with choosing by flipping a fair coin to decide if she will go to the club or play frisbee?

37

If $-1 \leq x \leq 100$, the smallest value of $f(x) = x^2 + 3$ is



38

Before Change	
Team	Tackles per Game
Binobos	26.3
Tamarinds	25.4
Apes	24.6
Orcas	23.9
Wyverns	22.1
Urchins	21.3
Molerats	20.4

After Change	
Team	Tackles per Game
Binobos	27.5
Tamarinds	29.7
Apes	30.8
Orcas	28.9
Wyverns	22.9
Urchins	22.5
Molerats	25.7

An amateur soccer league has decided to make tackles worth 0.02 points each to boost entertainment value. The tables above indicate the tackles per game for the various teams the season before and the season after the rule change. According to the data collected above, what was the average increase in team tackles per game, rounded to the nearest tenth?

Answer Key

Calculator Off

1	B
2	D
3	A
4	D
5	A
6	D
7	D
8	B
9	C
10	B
11	B
12	D
13	A
14	B
15	C
16	180
17	3
18	$\frac{5}{6}$ or .833
19	$\frac{1}{3}$ or .333
20	3

Calculator On

1	B
2	C
3	D
4	C
5	C
6	A
7	C
8	C
9	D
10	C
11	B
12	D
13	A
14	B
15	D
16	A
17	D
18	B
19	D
20	D
21	A
22	C
23	A
24	B
25	C
26	D
27	D
28	D
29	A
30	B
31	10
32	-20
33	1
34	160
35	126.0
36	7.5 or $\frac{15}{2}$
37	3
38	3.4



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

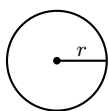
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

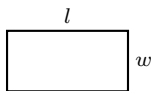
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

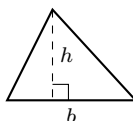


$$A = \pi r^2$$

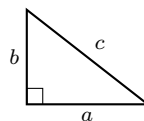
$$C = 2\pi r$$



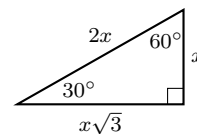
$$A = lw$$



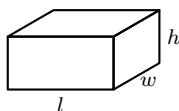
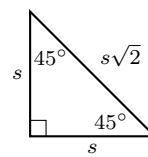
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



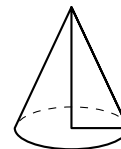
$$V = lwh$$



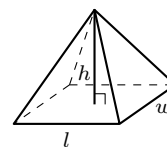
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

Suppose a and b are numbers such that ab is rational. What must be true about a ?

- A) a is rational
- B) a is irrational
- C) a is an integer
- D) None of the above

2

$$d = 1 + \frac{p}{100} \cdot 4.8$$

A sports scientist uses the model above to estimate the average distance d run by a defender per game, in miles, in terms of a player's percentage readiness for professional soccer, p , for any player that is 40% to 100% ready for NCAA soccer. Based upon the model, what is the estimated increase, in miles per game run, for a 25% increase in readiness to play NCAA soccer?

- A) 5.20625
- B) 5
- C) 2.2
- D) 1.2

3

$$\begin{aligned} x^2 + y^2 &= 1 \\ y &= ax^2 + 1 \end{aligned}$$

If a is a real constant, which of the following statements must be false about the system above in the xy plane?

- A) The system has three solutions.
- B) The system cannot have any solutions.
- C) The system has only one solution.
- D) $x = 0$ for a solution to the equation.

4

If $3v = -25$, what is the value of $12v + 100$?

- A) -200
- B) $-8.\bar{3}$
- C) 0
- D) 100



5

$$A = \frac{F}{100}(20 - \sqrt[3]{V})$$

An automobile's acceleration when the gas pedal is completely pressed, A , is determined by the formula above. V is the speed of the car, and F is the fuel grade of the gas in the tank. Which of the following equations represents the fuel grade of the gas in the tank in terms of the other variables?

- A) $F = \frac{100A(400+40\sqrt[3]{V}+\sqrt[3]{V^2})}{8000-V}$
- B) $F = \frac{100A(400-40\sqrt[3]{V}+\sqrt[3]{V^2})}{8000+V}$
- C) $F = \frac{100A(20+\sqrt[3]{V})}{400-V}$
- D) $F = \frac{100A(400+\sqrt[3]{V})}{8000-V}$

6

$$P = 0.211G + 1.444$$

The equation above describes a linear correlation between individual European country's military expenditures and their GDPs in 2012. In the equation, P is the percentage of the country's 2012 GDP devoted to military expenditures and G is the GDP of the country in 2012 measured in trillions of U.S. dollars. Which of the following statements is true about the model?

- A) The model predicts that military expenditures increase by 1 percent of GDP per 1.444 trillion dollars increase in GDP.
- B) The slope of the line represents the predicted increase in GDP per percent increase in military spending.
- C) The P -intercept has no meaning.
- D) The model indicates an additional million dollars of GDP has a very small effect on the portion of a European country's budget devoted to military expenditures.



7

$$\begin{aligned}x &= \frac{1}{2}(y - 4) \\y &= (-2x + 7)(x - 1)\end{aligned}$$

How many ordered pairs of real numbers (x, y) satisfy the system of equations shown above?

- A) 0
- B) 2
- C) 3
- D) Infinitely many.

8

ONLINE USED BOOK SALE

Paperback Novels \$2	Shipping and handling \$5
Hardcover Novels \$5	
Nonfiction \$4	Free shipping on orders of \$30 or more.

Martha wants to buy used hardcover fiction from an online bookstore. Which of the following correctly gives $P(n)$, the price P Martha will pay to buy n hardcover novels?

- A) $P(n) = \begin{cases} n < 6 : 5n + 5 \\ n \geq 6 : 5n \end{cases}$
- B) $P(n) = \begin{cases} n < 15 : 2n + 5 \\ n \geq 15 : 2n \end{cases}$
- C) $P(n) = \begin{cases} n < 30 : 2n + 5 \\ n \geq 30 : 2n \end{cases}$
- D) $P(n) = \begin{cases} n < 30 : 5n + 5 \\ n \geq 30 : 5n \end{cases}$



9

If $\overline{AB}$ has midpoint $(4, -1)$ and B is located at $(3, 2)$, then what is the coordinate of point A ?

- A) $(5, -4)$
- B) $(4, -5)$
- C) $(4, 5)$
- D) $(3, -4)$

10

Which of the following expressions is equal to 5 for some real value of z ?

- A) $2 + \sqrt{z}$
- B) $-2 - 3\sqrt{z}$
- C) $4 - \sqrt{z}$
- D) $-1 + \sqrt{16 - z^2}$

11

A radioactive substance decays at an hourly rate of 5%. If the initial amount of the substance is $12.8\mu\text{g}$ (micrograms), which of the following functions l models the remaining amount of the substance, in μg , t hours later?

- A) $l(t) = 0.05(12.8)^t$
- B) $l(t) = 0.95(12.8)^t$
- C) $l(t) = 12.8(0.05)^t$
- D) $l(t) = 12.8(0.95)^t$

12

$$\sqrt{j - k} = j + 1$$

If $k = -21$, what is the solution set for j in the equation above?

- A) $j \in \emptyset$
- B) $j \in \{-5, 4\}$
- C) $j \in \{-5\}$
- D) $j \in \{-1\}$



13

If $3x + 11 = -2$, what is the value of $6x + 7$?

- A) $-\frac{13}{3}$
- B) -19
- C) -11
- D) 25

14

Which of the following complex numbers is equivalent to $\frac{4-3i}{2+9i}$? (Note: $i = \sqrt{-1}$)

- A) $\frac{19+42i}{77}$
- B) $\frac{-19-42i}{85}$
- C) $\frac{-35+42i}{77}$
- D) $\frac{35-42i}{85}$

15

$$\frac{8 - 15i}{3 + 4i}$$

If the expression above is rewritten in the form $a + bi$, where a and b are real numbers, what is the value of a ? (Note: $i = \sqrt{-1}$)

- A) $-\frac{36}{25}$
- B) $-\frac{36}{7}$
- C) $\frac{24}{25}$
- D) -3

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

	7	/	1	2
	☐	☒	☐	☐
	☐	☐	☐	☐
	0	0	0	0
1	1	☒	1	1
2	2	2	☒	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
☒	7	7	7	7
8	8	8	8	8
9	9	9	9	9

Answer: 2.5

← Decimal point

	2	.	5
	☐	☐	☒
	☐	☐	☐
	0	0	0
1	1	1	1
2	☒	2	2
3	3	3	3
4	4	4	4
5	5	5	☒
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
	☐	☒	☐
	☐	☐	☐
	0	0	0
1	1	1	1
2	☒	2	2
3	3	3	☒
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
☒	☐	☐	☐
☐	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	☒	☒	☒
7	7	7	7
8	8	8	8

.	6	6	7
☒	☐	☐	☐
☐	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	☒	☒	6
7	7	7	☒
8	8	8	8

Answer: 201 – either position is correct

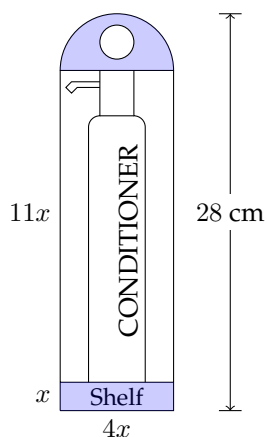
	2	0	1
	☐	☐	☐
	☐	☐	☐
	0	☒	0
1	1	1	☒
2	☒	2	2
3	3	3	3

	2	0	1
	☐	☐	☐
	☐	☐	☐
	☒	0	0
1	1	☒	1
☒	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16



Lena has a storage device that hangs from her showerhead. The top of the storage device is the apparatus that secures the device to the showerhead and has no storage space. What is the maximum height, in centimeters, of a bottle that can stand upright on the middle shelf?

17

Evaluate: $1^0 + 2^1 + 3^2$?

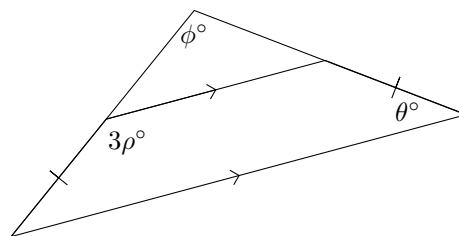
18

If

$$\frac{x}{x+2} = \frac{3}{5}$$

what is x ?

19

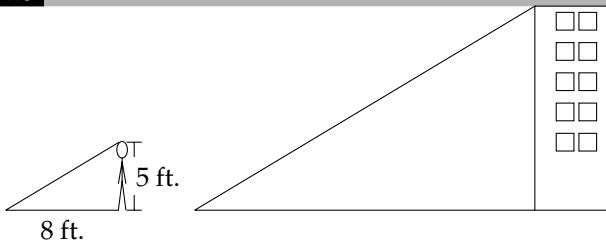


Note: diagram is NOT to scale.

If $\phi = \theta$, what is the value of ρ ?



20



The diagrams above, drawn to different scales, represent the measurement of two shadows. One of the shadows belongs to a teenager standing orthogonal to the ground, and the other is of an edifice that is likewise at a right angle to the ground. As the shadows are measured at the same time there is no observable difference between the corresponding angles on the ground. If the building's shadow is measured to be 280 feet, what is the building's height? Grid your answer in terms of feet.



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

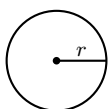
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

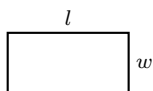
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

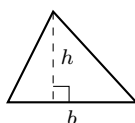


$$A = \pi r^2$$

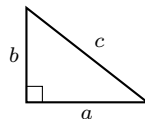
$$C = 2\pi r$$



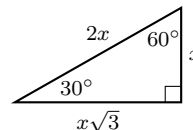
$$A = lw$$



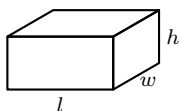
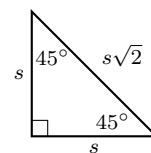
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



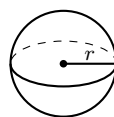
Special Right Triangles



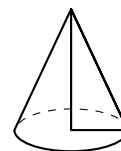
$$V = lwh$$



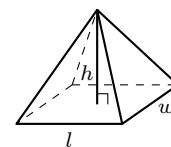
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.

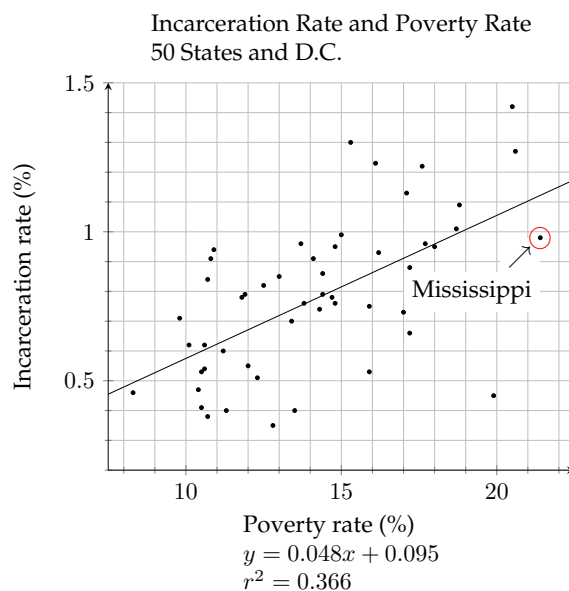


1

Duyen bought a patio set at a store that gave a 10% discount off its original price. In addition, Duyen had a manufacturer's coupon for 15% off the store's selling price. The total amount Duyen paid was c dollars, including a 9% sales tax on the final price after both discounts. Which of the following represents the original price of the patio set in terms of c ?

- A) $\frac{c}{(0.9)(0.85)(1.09)}$
- B) $(0.9)(0.85)(1.09)c$
- C) $\frac{c}{0.84}$
- D) $0.84c$

2



The graph above shows a line of best fit through data points comparing incarceration rates to poverty rates. Which of the following statements best describes the data point that represents the state of Mississippi?

- A) Mississippi's poverty rate is about 2.2% higher than the prediction.
- B) Mississippi's poverty rate is about 0.15% lower than the prediction.
- C) Mississippi's incarceration rate is about 0.15% lower than the prediction.
- D) Mississippi's incarceration rate is about 0.15% higher than the prediction.

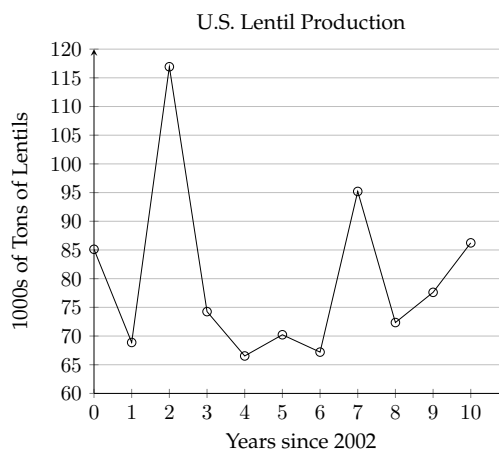


3

Which of the following is the same as $ab + a + b + 1$?

- A) $(a + b)^2$
- B) $(a + b + 1)^2$
- C) $(a + 1)(b + 1)$
- D) $(a - 1)(b - 1)$

4

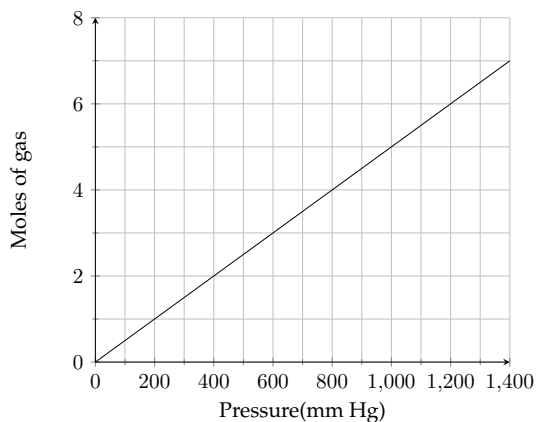


According to the graph above, which of the following statements best describes the general trend in lentil production for the United States from 2006 to 2012?

- A) The production of lentils fluctuates wildly from 2006 to 2012.
- B) There is a large decline in the production of lentils from 2006 to 2012.
- C) If the 2009 burst in production is removed, the trend is for lentil production to gradually increase from 2006 to 2012.
- D) The graph provides incomplete data for the years 2006 to 2012, so the question cannot be answered with the information provided.



5



A particular sample of gas behaves in accordance with ideal gas principles under its controlled volume and temperature. Which of the following functions most accurately predicts the pressure (P) of the gas sample given the sample's moles (M) for a point on this graph?

- A) $P = \frac{1}{200} M$
- B) $P - 3 = 200(M - 600)$
- C) $P - 600 = \frac{1}{200} (M - 3)$
- D) $P - 600 = 200(M - 3)$

6

The molecular weight of an unknown compound, in atomic mass units (amu), is approximately 10% less than the atomic weight of gold. The atomic weight of gold 200 amu. Which of the following best approximates the molecular weight, in amu, of the unknown element?

- A) 220
- B) 20
- C) 190
- D) 180

7

$$f(x) = \frac{-1}{2}(x - 4)^2 + 1$$

Which of the following represents the y -intercept of f ?

- A) $(0, -8)$
- B) $(0, -7)$
- C) $(4 + \sqrt{2}, 0)$
- D) $(4 - \sqrt{2}, 0)$



8

For which of the following years is the predicted GDP per capita for the United Kingdom less than the actual GDP per capita for the United Kingdom?

- A) 2003
- B) 2005
- C) 2007
- D) 2009

9

The number of full-time employments (FTEs) assigned to art teachers employed by a particular school district from 2005 to 2012 can be modeled by the following equation: $y = -0.35x + 8.75$. In this equation, x represents the number of years since 2005 and y represents the number of full-time employments. Which of the following best describes the meaning of the number 8.75 in the equation?

- A) The estimated increase in art teacher FTEs per year
- B) The difference between the number of FTEs in 2012 and the number of FTEs in 2005
- C) The number of full-time employments assigned by the district in 2005
- D) The total number of art teachers employed by the district in 2005

10

What is $(4, 45^\circ)$ in Cartesian coordinates?

- A) $(\sqrt{2}, \sqrt{2})$
- B) $(2, 2)$
- C) $(2\sqrt{2}, 2\sqrt{2})$
- D) $(4, 4)$

11

When the positive integer k is divided by 9, the remainder is 4. What is the remainder when k^2 is divided by 9?

- A) 4
- B) 5
- C) 7
- D) 8



12

An elevator has a posted weight limit of 3000 pounds. A crew is moving several pieces of industrial equipment with this elevator and realizes they can't put more than 7 of their lathes in the elevator for any given trip, even though 12 of their lathes would fit based upon volume. Based upon this information, which of the following can be determined about the weight of the lathes?

- I. The lathes must weigh more than 350 pounds each.
 - II. Filling the elevator with all the lathes that would fit would go beyond the weight limit by at least 4800 pounds.
 - III. The speakers can't weigh 450 pounds or more.
- A) III only.
 - B) I and II only.
 - C) I and III only.
 - D) I, II and III.

13

$$C = K - 273.15$$

$$C = \frac{5}{9}(F - 32)$$

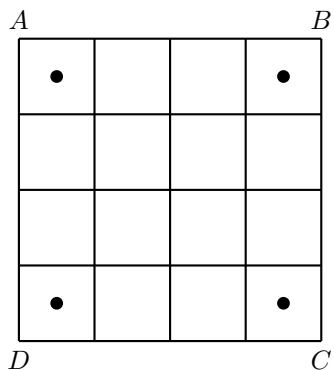
Ferris Cantor needs to convert 68°F into degrees Kelvin. However, he is only able to find formulas to convert either measurement (F for Fahrenheit and K for Kelvin) into degrees Celsius (C in the equations above). Which of the following is closest to Ferris's temperature in degrees Kelvin?

- A) 5.78
- B) 278.93
- C) 293.15
- D) 341.15



14

Square $ABCD$ is composed of 16 identical squares. Moreover, a point is placed in the center of each corner square:



If the area of $ABCD$ is 64, then what is the area of the circle (not shown) that goes through the four points?

- A) 16π
- B) 18π
- C) 22π
- D) 25π

15

Let p and q be prime numbers such that $p + q = 48$. What is the biggest possible value of p ?

- A) 10
- B) 41
- C) 43
- D) 47

16

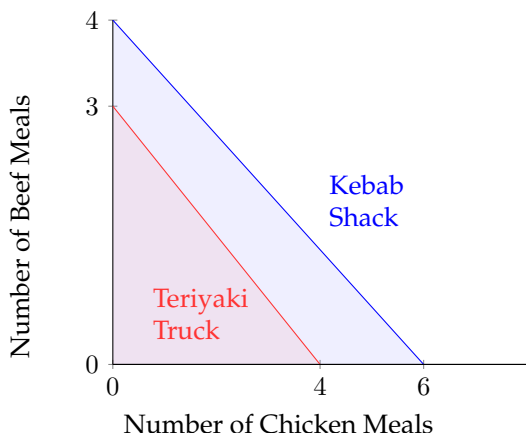
The cost to rent a computer with the fastest internet connection speed possible at an internet cafe is \$.001 per second. Which of the following equations represents the total rent r , in dollars, for h hours of computer rental?

- A) $r = \frac{0.001}{3600h}$
- B) $r = \frac{60h}{0.001}$
- C) $r = \frac{3600h}{0.001}$
- D) $r = 0.001(3600h)$



17

The graph below shows two consumer options for spending \$30 on beef and chicken entrees with sides at two different food trucks.



Which of the following statements must be true about the consumer options?

- A) At the Teriyaki Truck the chicken meal and the beef meal are closer in price than at the Kebab Shack.
- B) Chicken is more expensive than beef at either location.
- C) A customer always gets more food at the Teriyaki Truck than at the Kebab Shack.
- D) The Kebab Shack sells more food than the Teriyaki Truck.

18

A researcher conducted a survey to determine whether voters in a certain large city prefer private sector union protections or right-to-work laws. The researcher asked 2409 voting-age people at bus stops near the industrial zones, and 352 refused to respond. Which of the following factors makes it least likely that a reliable conclusion can be drawn about the political opinions of all the voters in the city?

- A) The place the survey was given
- B) The large percentage that refused to respond
- C) The sample size
- D) The population size

19

$$y = x^2 - 4x - 21$$

The equation above represents a parabola in the xy -plane. Which of the following equivalent forms of the equation displays the x -intercepts of the parabola as constants or coefficients?

- A) $y + 21 = x(x - 4)$
- B) $y = x^2 - 4(x + \frac{5}{4})$
- C) $y = (x - 7)(x + 3)$
- D) $y + 25 = (x - 2)^2$



20

If $\sqrt{xy^2} = 21$, which of the following ordered pairs is a solution to the equation?

- A) $(-1, 21)$
- B) $(3, 7)$
- C) $(9, -7)$
- D) $(42, 1)$

21

In a particular pickleball tournament, each player starts the game with p points. Players gain 11 points for victories and lose 11 points for losses, and there are no ties. In this tournament, a player won 3 games, lost 7 games, and had a final score of 66 points. What was p ?

- A) 22
- B) 33
- C) 110
- D) 143

22

Translate the following verbal mathematical expression into their equivalent symbolic counterparts. The sum of two numbers is half the difference between them.

- A) $x + y = (1/2)(x - y)$
- B) $xy = 2(x - y)$
- C) $xyz = 5(x + y + z)$
- D) None of these.

23

If a is the average (arithmetic mean) of $2x$ and 5, b is the average of $3x$ and 8, and c is the average of $7x$ and 17, what is the average of a , b , and c in terms of x ?

- A) $x + 6$
- B) $2x + 5$
- C) $4x + 10$
- D) $6x + 15$



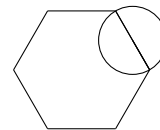
24

Gender	Eye Color	
	Brown or Hazel	Blue or Green
Female		
Male		
Total	606	70

The incomplete table above summarizes the number of students with brown or hazel eyes and the number of students with green or blue eyes by gender for both grades at Jefferson Junior High School. There are 11 times as many female students with brown or hazel eyes as female students with blue or green eyes. There are 7 times as many male students with brown or hazel eyes as male students with blue or green eyes. None of the students identify as belonging to more than one of the four category boxes in the table. Which of the following is closest to the probability that a brown or hazel eyed student is a brown or hazel eyed male?

- A) 0.875
- B) 0.586
- C) 0.526
- D) 0.474

25



In the diagram above, the diameter of a circle with an area of $64\pi\text{cm}^2$ is also the side of a regular hexagon. What is the area, in square centimeters, of the hexagon?

- A) $96\sqrt{3}$
- B) $384\sqrt{3}$
- C) 288
- D) 1152

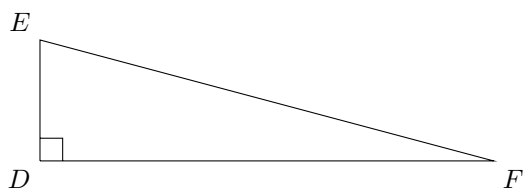
26

When 9 times the number k is subtracted from -3 , the result is 15. What number results when k is tripled and added to 1?

- A) 7
- B) -5
- C) -13
- D) -125



27



In the diagram above, $DE = 4$ and the area of $\triangle JKL$ is 30 square units. Given those two facts and the diagram above, which of the following trigonometric statements must be true?

- A) $\sin \angle E = 3.75$
- B) $\cos \angle E = 3.75$
- C) $\tan \angle E = 3.75$
- D) $\cot \angle E = 3.75$

28

A baker is giving away mini-cupcakes as samples to all the small businesses within 3 miles of her bakery. If she gives 5 mini-cupcakes to each small business, she will have 11 left over for samples in her shop. If she wants to give away 6 mini-cupcakes to each small business, she will need to bake at least 9 more mini-cupcakes. How many mini-cupcakes did the baker make for samples?

- A) 21
- B) 41
- C) 71
- D) 111

29

If $\log_{81}(x) = -\frac{1}{4}$, then what is x ?

- A) $\frac{1}{3}$
- B) $\frac{2}{3}$
- C) $-\frac{1}{3}$
- D) $-\frac{3}{2}$



30

A policy analyst is attempting to determine a reasonable estimate for the additional sales tax revenue to be generated via the legalization of marijuana for recreational use in his state. Which of the following would be most helpful for the policy analyst's estimate?

- A) An estimate of the average marijuana product price and of price's standard deviation.
- B) An estimate of the range of marijuana product prices and of the estimated number of products to be marketed.
- C) An estimate of the average marijuana product price and of the estimated number of products that would be sold.
- D) An estimate of how much black market sales of marijuana make in the state currently.

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

- Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If  is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write answer in boxes. →

7	/	1	2
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

← Fraction line

Grid in result.

Answer: 2.5

2	.	5	
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

← Decimal point

Acceptable ways to grid $\frac{2}{3}$ are:

2	/	3	
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

2	0	1	
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

2	0	1	
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

A rectangle in the xy coordinate plane has one side that is a segment of $2x + 8y = -1$. What is the sum of the slopes of the sides of the rectangle?

32

Yessenia opened a bank account that earns $100 \cdot \ln(1.04)$ percent annual interest compounded continually. Her initial deposit was \$5000, and she uses the expression $5000(m)^t$ to find the value of the account after t years. What is the value of m in the expression?

33

The towing limit for a particular pickup truck is 4,600 pounds. A company wants to carry as many 11.2 pound parcels as possible in a trailer that weighs 1,300 pounds when empty that's attached to this pickup truck. What is the maximum number of parcels the company could place in the trailer without exceeding the towing limit?

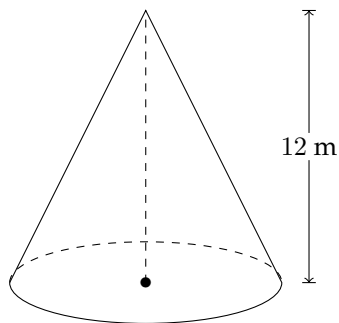
34

In ancient Rome, a cubit was equal to 6 palm widths. In modern units, a cubit is estimated to be equal to 17.5 inches. Based on these relationships, how many *yards* are in 432 palm widths? (1 yard = 36 inches)

35

What is $\log_{\pi}(\sqrt{\pi})$?

36



An ice cream company plans on using a freezer with a storage space of a right cone as drawn above for keeping inventory. If the volume is supposed to be $256\pi \text{ m}^3$, what is the diameter of the base of the cone, in meters?



37

Consider the following sequence:

$$1, 0, -1, 0, 1, 0, \dots$$

What is the sum of the first 100 terms of this sequence?

38

Consider the following system of equations:

$$\begin{aligned} 6x - 3y &= -1 \\ ax + 2y &= -4 \end{aligned}$$

For what value of a does the above system have no solutions?

Answer Key

Calculator Off	
1	D
2	D
3	B
4	D
5	A
6	D
7	A
8	A
9	A
10	A
11	D
12	A
13	B
14	B
15	A
16	22
17	12
18	3
19	40
20	175 feet

Calculator On	
1	A
2	D
3	C
4	C
5	D
6	D
7	C
8	C
9	C
10	C
11	C
12	D
13	C
14	B
15	C
16	D
17	B
18	A
19	C
20	C
21	C
22	A
23	B
24	D
25	B
26	B
27	C
28	D
29	A
30	C
31	7.5 or $\frac{15}{2}$
32	1.04
33	294
34	35
35	$\frac{1}{2}$
36	16
37	0
38	-4



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

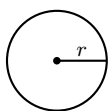
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

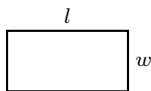
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

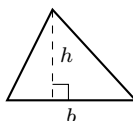


$$A = \pi r^2$$

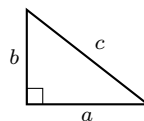
$$C = 2\pi r$$



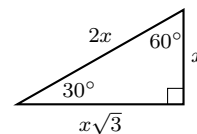
$$A = lw$$



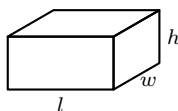
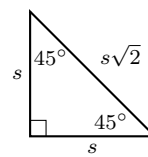
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



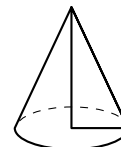
$$V = lwh$$



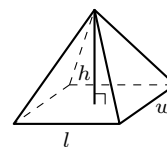
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

If

$$\frac{4}{x-2} \div \frac{(x+3)^2}{3x-6} = 6$$

what are the possible values of x ?

- A) $x \in \{2\}$
- B) $x \in \{-3 \pm \sqrt{2}\}$
- C) $x \in \{1, 2, -7\}$
- D) $x \in \emptyset$

2

For a small population of a particular invasive species released with no natural predator extant in the local habitat, Maren uses the following exponential model to describe its expected population prior to reaching 60% of the carrying capacity of the habitat.

$$N = Be^{0.1m}$$

In this equation, N represents the population after m months, B represents the initial population, and e is the natural exponential base. If Maren takes interest in having an inverse function $m(N)$ that gives the number of months the invasive species has been present based upon its current population, which of the following equations should she use?

- A) $m(N) = \frac{\ln N - \ln B}{0.1}$
- B) $m(N) = 0.1(\ln N - \ln B)$
- C) $m(N) = \frac{N}{0.1e^B}$
- D) $m(N) = \frac{1}{3} \left(\frac{N}{B} \right)^{10}$



3

Suppose f is an increasing function. Which of the following can be f ?

- A) $f(x) = 1/x$
- B) $f(x) = 1 - x$
- C) $f(x) = x^2$
- D) None of the above.

4

Simplify:

$$\frac{3}{w-3} - \frac{4}{w-1}$$

- A) $\frac{w-9}{(x-3)(w-1)}$
- B) $\frac{9-w}{(w-3)(w-1)}$
- C) $\frac{-w-9}{(w-3)(w-1)}$
- D) $\frac{-3}{w-1}$

5

$$625g^4 - h^4$$

Which of the following is equivalent to the expression shown above? (Note: $i = \sqrt{-1}$)

- A) $(5g - h)^4$
- B) $(5g^2 + ih^2)^2$
- C) $(5g - h)^2(5g + h)^2$
- D) $(5g + ih)(5g - ih)(5g + h)(5g - h)$

6

$$\begin{aligned} Bx - 5y &= -3 \\ 3x + 4y &= 10 \end{aligned}$$

In the system of equations above, B is a constant and x and y are variables. For what value of B will the system of equations have no solution?

- A) 25
- B) $\frac{15}{4}$
- C) 0
- D) $-\frac{15}{4}$



7

$$y = a(x - 5)(x + 3)$$

In the quadratic equation above, a is a nonzero constant. The graph of the equation in the xy -plane is a parabola with vertex (a, m) . Which of the following is equal to m ?

- A) -12
- B) -1
- C) 16
- D) It cannot be determined from the information provided.

8

x	$j(x)$
0	8
1	21
4	60
8	0

The function j is defined by a polynomial. Some values of x and $j(x)$ are shown in the table above. Which of the following must be a factor of $j(x)$?

- A) $x - 21$
- B) $x - 8$
- C) $x - 4$
- D) $x - 1$



9

$$\begin{aligned} 2x - 4y &= -1 \\ \frac{1}{4}(x + 2) &= y - 2 \end{aligned}$$

What is the solution to the above system of equations in the xy plane?

- A) $(3, -1.75)$
- B) $(8.5, 4.5)$
- C) $(9, 4.75)$
- D) $(15, 7.75)$

10

$$\begin{aligned} \frac{y}{x} &= -8 \\ -2(x + 1) &= y \end{aligned}$$

If (x, y) is the solution to the system of equations above, what is the value of x ?

- A) $-\frac{1}{3}$
- B) $\frac{1}{3}$
- C) $-\frac{1}{5}$
- D) $\frac{1}{5}$

11

Which of the following represents a way of rewriting $y = 2x^2 - x - 10$ such that the roots of the equation are explicitly stated as values inside the equation?

- A) $y = 2(x - \frac{1}{4})^2 - \frac{81}{8}$
- B) $y = (2x - 5)(x + 2)$
- C) $y = 2(x + \frac{5}{2})(x - 2)$
- D) $y = 2(x - \frac{5}{2})(x + 2)$

12

Which of the following is not a polynomial?

- A) 0
- B) $\sqrt{x}$
- C) x^2
- D) $x^2 - x^3$

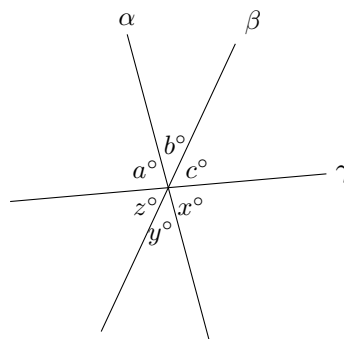


13

Which is the same as $\sqrt{50}$

- A) $2\sqrt{5}$
- B) $\sqrt{25}$
- C) $2\sqrt{25}$
- D) None of the above.

14



Note: Figure is not drawn to scale.

In the figure above lines α , β , and γ intersect at a point. If $b - c = x - z$, which of the following must be true?

I. $a = y$ II. $b = x$ III. $c = z$

- A) I and II only.
- B) II and III only.
- C) I, II, and III.
- D) None of the above.



15

Which of the following is not the same as 8^2 ?

- A) 2^6
- B) 4^3
- C) $16^{\frac{2}{3}}$
- D) $16^{\frac{3}{2}}$

.....

DIRECTIONS

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

1. Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
2. Mark no more than one circle in any column.
3. No question has a negative answer.
4. Some problems may have more than one correct answer. In such cases, grid only one answer.
5. **Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or $7/2$. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
6. **Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write $\rightarrow$
answer
in boxes.

Grid in
result.

	7	/	1	2
		0	0	0
1	1	1		1
2	2	2	2	
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7		7	7	7
8	8	8	8	8
9	9	9	9	9

Answer: 2.5

← Fraction line

← Decimal point

	2	.	5
	0	0	0
1	1	1	1
2		2	2
3	3	3	3
4	4	4	4
5	5	5	
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
	/	●	
●	●	●	●
	0	0	0
1	1	1	1
2	●	2	2
3	3	3	●
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
			
			
	0	0	0
①	①	①	①
②	②	②	②
③	③	③	③
④	④	④	④
⑤	⑤	⑤	⑤
⑥			
⑦	⑦	⑦	⑦
	8	8	8

.	6	6	7
			
			
			
			
			
			
			
			
			
			
			

Answer: 201 – either position is correct

	2	0	1
	/	/	
.	.	.	.
	0	●	0
1	1	1	●
2	●	2	2
3	3	3	3

2	0	1	

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



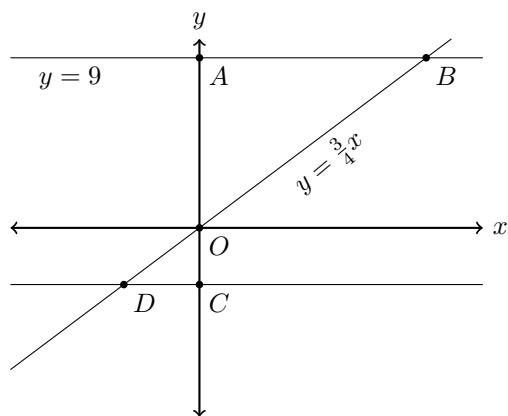
16

Consider the following collection of data:

2, 3, 5, 1, 1, 7, 8, 8, 10

If a is the median and b is the mean, then what is $a - b$?

17



In the diagram above, $\frac{AO}{OC} = \frac{BO}{OD}$. What is $\sin \angle D$?

18

$$-12x + y = -6$$

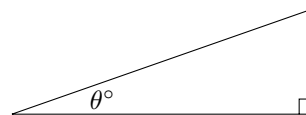
$$6x + 5y = 25$$

If (x, y) is the solution to the system of equations above, what is the value of x ?

19

If $\frac{2}{3}k - \frac{3}{5}k = \frac{3}{8} - \frac{1}{24}$, what is the value of k ?

20



In the triangle above, $\tan \theta^\circ = \frac{7}{\sqrt{51}}$. What is the value of $\sin \theta^\circ$?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

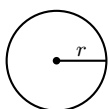
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

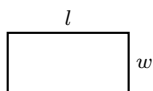
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

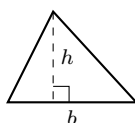


$$A = \pi r^2$$

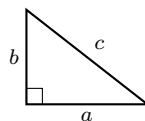
$$C = 2\pi r$$



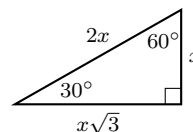
$$A = lw$$



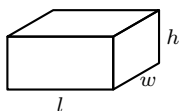
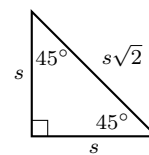
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



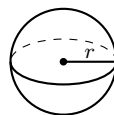
Special Right Triangles



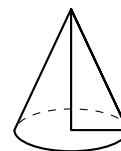
$$V = lwh$$



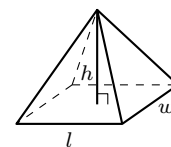
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

Let A represent a 4×4 matrix with $\det(A) = 2$. What is the value of $\det(2A)$?

- A) 4
- B) 8
- C) 16
- D) 32

2

Let $a = \frac{1}{2}$. Which of the following is true?

- A) $a < a^{\frac{1}{2}} < a^{-1}$
- B) $a^{\frac{1}{2}} < a < a^{-1}$
- C) $a^{\frac{1}{2}} < a^{-1} < a$
- D) $a < a^{-1} < a^{\frac{1}{2}}$

3

At certain universities, one of the requirements for becoming a varsity policy debater is the ability to “spread”, which means to verbally read 250 words per minute. Sumaya can currently verbally read 130 words per minute, and believes that with practice she can increase her speed by 12 words per minute per week. Which of the following represents the number of words per minute Sumaya believes she will be able to read w weeks from now?

- A) $12 + 130w$
- B) $130 + 12w$
- C) $130 - 12w$
- D) $250 - 12w$

4

When 9.479 is rounded to the nearest whole number, the result is how much less than when 9.479 is rounded to the nearest hundredth?

- A) 9
- B) 0.48
- C) 1.0
- D) 25



5

If $\sqrt{\Omega} + 3 \leq -5$, what is the greatest possible value of $\sqrt{\Omega} + 5$?

- A) -10
- B) -3
- C) -2
- D) 0

6

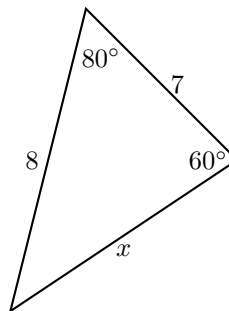
$$m(x) = \frac{-1}{4}(x+3)^2 + 1$$

Which of the following represents the solutions to m ?

- A) $x \in \{-5, -1\}$
- B) $x \in \{1, 5\}$
- C) $x \in \{-4\}$
- D) $x \in \{\pm 2\sqrt{10}\}$

7

Which of the following equations can be used to solve for x ?



(Note: For any triangle with sides a, b, c that are opposite angles A, B, C respectively, $c^2 = a^2 + b^2 - 2ab \cos(C)$)

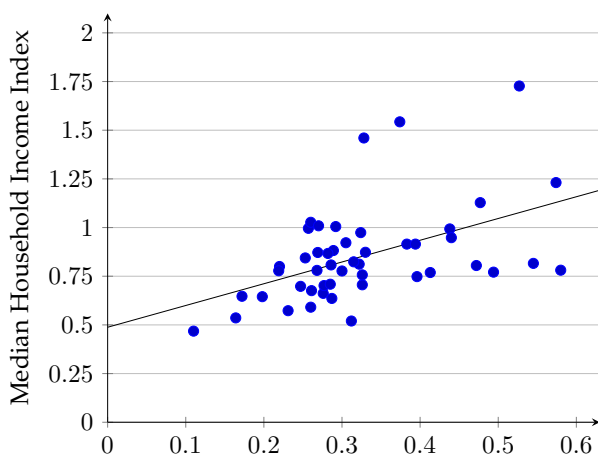
- A) $x^2 = 7^2 + 8^2 - 2 \cdot 7 \cdot 8 \cdot \cos(80^\circ)$
- B) $x^2 = 7^2 + 8^2 - 2 \cdot 7 \cdot 8 \cdot \cos(60^\circ)$
- C) $x^2 = 7^2 + 8^2 - 2 \cdot 7 \cdot 8 \cdot \cos(40^\circ)$
- D) $8^2 = x^2 + 7^2 + 2 \cdot 7 \cdot 8 \cdot \cos(40^\circ)$



8

The median household income index for a US city or town is defined to be the ratio

$$\frac{\text{median household income for the city or town}}{\text{median household income for the United States}}$$



Proportion of 25+ year olds with Bachelor's Degrees or Higher

The scatterplot above shows the median household income index relative to the proportion of the city or town population that has earned a minimum of a bachelor's degree using data on 50 cities and towns throughout the United States from 2013. The line of best fit is also shown and has the equation $y = 1.117x + 0.488$. Which of the following best explains how the number 1.117 relates to the scatterplot?

- A) On average, towns and cities have median household incomes 1.117 times the median household income of the United States.
- B) Higher median incomes indicate the presence of social pressure for more people to go to college, and an increase of 1% on the median household income index would be expected to raise the proportion of people in the city or town who are college educated by 1.117%.
- C) The median income for a household in any city or town won't exceed 1.117 times the national median household income, except in outlier data points.
- D) On average, having a more college educated city or town brings up the median household income, and having 10% more college educated people in a city or town would be expected to raise the median household income there by about 0.117 times the median household income in the United States.

9

Best Documentaries According to Rotten Tomatoes By MPAA Rating

October, 2015, unrated movies not included

MPAA rating	Year of U.S. Release				Total
	1991-2000	2001-2005	2006-2010	2011-2015	
G	0	5	0	2	7
PG	2	0	5	5	12
PG-13	1	2	5	15	23
R	0	4	5	8	17
Total	3	11	15	30	59

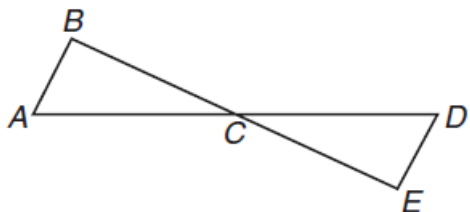
The table above represents the 59 movies that had MPAA ratings from the 100 top documentaries according to Rotten Tomatoes. They are categorized by those ratings and their United States release dates. What proportion of the documentaries were released in the United States between 2001 and 2010 with a PG-13 rating?

- A) $\frac{23}{59}$
- B) $\frac{15}{59}$
- C) $\frac{7}{59}$
- D) $\frac{5}{59}$



10

The following is a partially completed correct proof of a fact from plane geometry.



GIVEN: Point C is the midpoint of $\overline{AD}$ and $\overline{BE}$.

PROVE: $\triangle ABC \cong \triangle DEC$

Statement	Justification
1. C is the midpoint of $\overline{AD}$ and $\overline{BE}$.	Given
2. $\overline{AC} \cong \overline{DC}$ and $\overline{BC} \cong \overline{EC}$	Definition of midpoint
3. $\angle ACB \cong \angle DCE$	?
4. $\triangle ABC \cong \triangle DEC$	SAS

Which of the following best completes the proof?

- A) The distributive property of polygons.
- B) The angle equivalence principle.
- C) The congruence property of vertical angles.
- D) The inverse property of translations.

11

The median sale price of a home in a given zip code is \$1,800,000. The mean sale price of a home in the same zip code is \$4,393,449. Which of the following situations could explain the difference between the mean and median home sales prices?

- A) Many of the homes have sale prices between \$1,800,000 and \$4,393,449.
- B) Most of the homes have sale prices that are close to each other.
- C) There are a few homes that are selling for much less than the rest.
- D) There are two homes in the zip code on sale for more than 20,000,000.

12

At the Slightly Fluffy Cats and Dogs pet store, 40% of the 30 cats have brown eyes. The fraction of dogs at the store with green eyes is equal to the fraction of cats at the store with brown eyes. How many of the 40 dogs at the store have green eyes?

- A) 8
- B) 10
- C) 14
- D) 16



13

The radius of a cylinder is double its height. In terms of radius r , what is the surface area of the cylinder?

- A) $\frac{\pi r^3}{2}$
- B) $3\pi r^2$
- C) $6\pi r^2$
- D) $8\pi r^2$

14

For a polynomial $p(x)$, the value of $p(1) > p(2)$ and $|p(2)| = p(1)$. Which of the following must be true about $p(x)$?

- A) $x - 1.5$ is a factor of $p(x)$.
- B) $p(x)$ must be symmetrical.
- C) The remainder when $p(x)$ is divided by $x - 2$ is $p(1)$.
- D) The remainder when $p(x)$ is divided by $x - 1$ is $-p(2)$.

15

$$\begin{aligned}y &= -6 \\ y &= qx^p + q\end{aligned}$$

In the system of equations above, p and q are constants. For which of the following values of p and q does the system of equations have no real solutions?

- A) $p = 1, q = -10$
- B) $p = 2, q = -6$
- C) $p = 3, q = -2$
- D) $p = 4, q = 2$

16

If x and y are integers such that $x^2 = 64$ and $y^3 = 64$, which of the following could be true?

- I. $x = 8$
- II. $y = -4$
- III. $x + y = -4$

- A) I only
- B) I and III
- C) II and III
- D) I, II, and III



17

$$P = 0.75(W + 16)$$

The equation above represents the purchase price, P , in dollars, of buying W pounds of fresh local organic vegetables and having them delivered to a personal residence. The company only sells to local residents out of concerns for product quality. Which of the following best explains the real world implication for the number 0.75 in the equation above?

- A) The smallest possible delivery costs \$0.75.
- B) This company charges \$0.75 per pound for its organic vegetables.
- C) \$0.75 is the amount of money needed to pay postage and get an order through customs.
- D) \$0.75 is the company's profit margin per pound of produce sold.

18

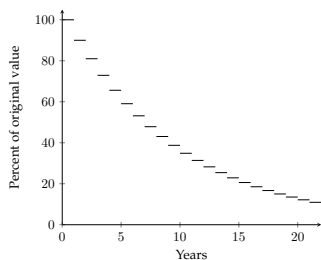
Suppose $\sin(x) < \cos(x)$. Which of the following is a possible value of x ?

- A) 60°
- B) 120°
- C) 210°
- D) 330°

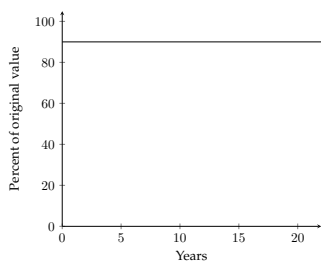


19

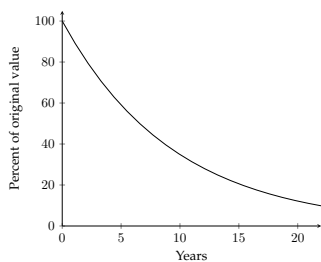
The depreciated value of an item refers to the assessed value of an item that has lost value due to wear and tear. If an item is losing about 10% of its prior year's value each year, which of the following graphs could model the depreciated value of an item as a function of time?



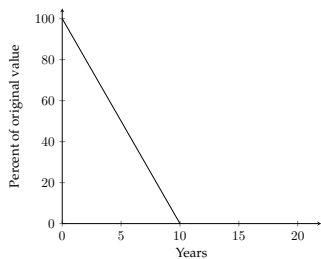
A)



B)



C)



D)

20

A *perfect* number is an integer that is a sum of its positive divisors, excluding the number itself. Which of the numbers below is *perfect*?

- A) 10
- B) 15
- C) 20
- D) 28

21

A policy analyst is attempting to determine a reasonable amount of available parking spaces per household to advise a city to require for new developments. Which of the following makes the best sense for an outlier resistant approach to looking at the center and spread of the city's current demands for parking spaces per household?

- A) Geometric mean and 90th percentile
- B) Median and interquartile range
- C) Mode and range
- D) None of the above combinations are consistently outlier resistant.



22

Suppose A is an $a \times b$ matrix and B is a $c \times d$ matrix and the product AB exists. Which of the following must be true?

- A) $a = b$
- B) $b = c$
- C) $c = d$
- D) $b = d$

23

Which of the following is the greatest?

- A) 30% of 5
- B) 50% of 10
- C) 16% of 25
- D) 29% of 50

24

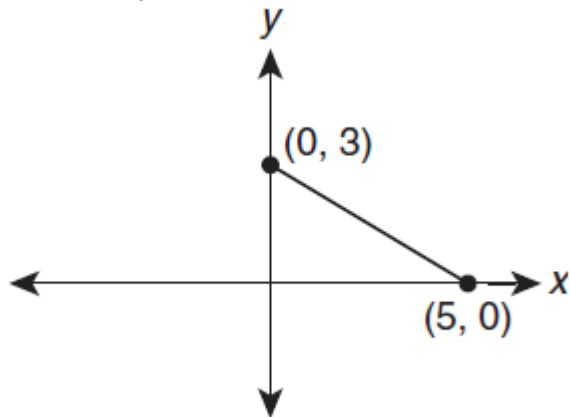
In a certain sequence, other than the first two terms, each subsequent term is the difference of the two preceeding terms. What is the eighth term of this sequence if the difference between the second and first term is zero?

- A) 7
- B) 9
- C) 11
- D) Cannot be determined



25

In the diagram below, the line segment is rotated 180° about the y axis.



Which of the following correctly describes the volume of the resulting solid?

- A) 15π
- B) 25π
- C) The value cannot be determined from the information given.
- D) None of these.

26

An economist who makes his living advising commodity traders about agricultural futures has derived the following two equations with regards to California nectarine production and California nectarine prices.

$$H = 48,400R + 2900$$

$$P = -0.000008H + 65$$

In the equations, H represents the number of hundredweights of California nectarines that will be produced and utilized that year, R represents the average rainfall in the state of California that year in inches, and P represents the price per hundredweight nectarine producers should be able to get from food shipping companies that year. If the economist believes the average rainfall in California will be 17 inches next year, which of the following is closest to the price per pound for nectarines he would be predicting? A hundredweight is 112 pounds.

- A) \$58.39
- B) \$52.14
- C) \$0.58
- D) \$0.57



27

If $(3x + 10)(3x - 10) = 300$, what is the value of $9x^2$?

- A) 100
- B) 200
- C) 400
- D) 1200

28

$$\sqrt{5p} = \sqrt{11}$$

What is the value of p ?

- A) $\frac{13}{2}$
- B) $\frac{15}{2}$
- C) $\frac{9}{2}$
- D) $\frac{11}{2}$

29

$$I = \frac{S}{4\pi d^2}$$

The intensity of a sound wave at a given location is approximated by the formula above, where I is the intensity, S is the loudness of the sound at its source, and d is the distance from the source of the sound. Which of the following expresses the loudness of the sound at its source in terms of its density and the distance from the source?

- A) $S = \frac{I}{4\pi d^2}$
- B) $S = 4\pi d^2 I$
- C) $S = \frac{4\pi d^2}{I}$
- D) $S = 2d\sqrt{I\pi}$

30

Let M and N be integers. If

$$N = |5M^2 - 2|$$

which of the following is NOT a possible value of N ?

- A) -2
- B) 1
- C) 2
- D) 7

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

7	/	1	2
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 2.5

← Decimal point

2	.	5
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

2	/	3
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

2	0	1
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

2	0	1
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

Claude purchased a bond for \$4000 that pays 3% annual interest compounded continually over the course of ten years upon maturity. Alfred found a better deal and purchased a bond for \$4000 that pays 3.4% annual interest compounded continually over the course of ten years upon maturity. When the bonds mature, how much more money will Alfred receive than Claude? (Round the answer to the nearest dollar when gridding your response, and don't grid in the dollar sign.)

32

How many pairs (a, b) satisfy

$$a + b = 1$$

$$a - b = 0$$

33

If $f(x) = \frac{1}{\sqrt{x}}$ and $g(x) = 2x - x^2$, then what is $g(f(4))$?

34

A radio station sells advertising slots in 10 hour packages for any given broadcasting month. The station operates every day, 24 hours per day, and advertises for 20 minutes per hour on average. How many packages of advertising slots does the radio station need to sell to fill all of its slots in a 30 day month?

35

Athletic pursuits that require foot grip are safer with higher friction coefficients. The typical basketball shoe with a rubber sole generates about 60% more friction than a leather-soled shoe when interacting with most hard surfaces. Dan's footing on an asphalt basketball court is secured by 810N when standing still in rubber shoes. What should be his estimate for the friction coefficient of leather on asphalt?



36

Ages of the First 12 Class 1 Senators from Nevada at the Beginning of Their First Terms in Office

Senator	Age (years)	Senator	Age (years)
Stewart	37	Bunker	34
Sharon	54	Scrugham	62
Fair	49	Carville	60
Nixon	44	Malone	56
Massey	55	Cannon	46
Pittman	40	Hecht	54

The table above lists the ages of the first 12 Class 1 Nevada Senators when they began their first terms in office. According to the table, what was the mean age, in years, of these senators at the beginning of their terms? (Round your answer to the nearest tenth.)

37

What is the 100th term of the following sequence?

$$\begin{aligned}
 a_1 &= 6 \\
 a_n &= \begin{cases} \frac{a_{n-1}}{2} & \text{if } n \text{ is even} \\ 2a_{n-1} & \text{if } n \text{ is odd and } n > 1 \end{cases}
 \end{aligned}$$

38

If $(3x + 5)(4x - 1) = ax^2 + bx + c$, find $a + b + c$.

Answer Key

Calculator Off	
1	B
2	A
3	D
4	B
5	D
6	D
7	D
8	B
9	C
10	B
11	D
12	B
13	A
14	D
15	C
16	0
17	$\frac{3}{5}$ or 0.6
18	$\frac{5}{6}$ or .833
19	$\frac{5}{6}$
20	0.7 or $\frac{7}{10}$

.....

Calculator On	
1	D
2	A
3	B
4	B
5	B
6	A
7	A
8	D
9	C
10	C
11	D
12	D
13	B
14	D
15	D
16	B
17	C
18	D
19	C
20	D
21	C
22	B
23	D
24	D
25	D
26	C
27	C
28	D
29	B
30	A
31	220
32	1
33	$\frac{3}{4}$
34	24
35	0.506
36	49.3
37	3
38	24



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

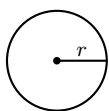
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

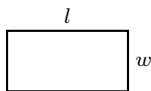
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

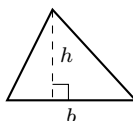


$$A = \pi r^2$$

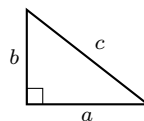
$$C = 2\pi r$$



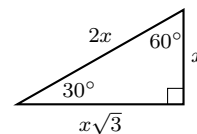
$$A = lw$$



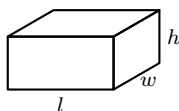
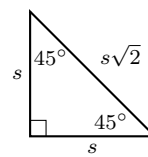
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



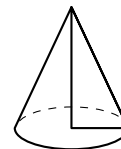
$$V = lwh$$



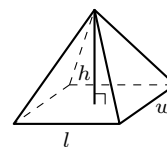
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



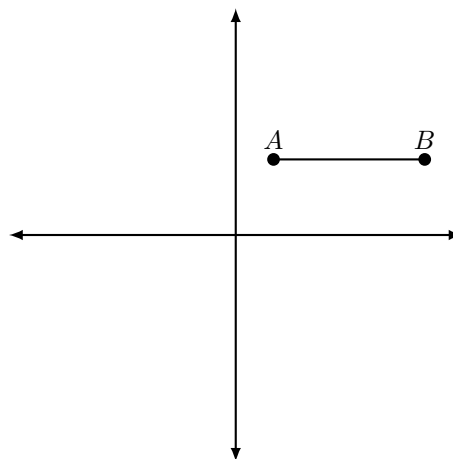
1

Which of the following is a perfect square for every integer value of x ?

- A) $x^2 - 5x + 6$
- B) $x^2 + 1$
- C) $x^2 - 1$
- D) $x^2 + 6x + 9$

2

On the Cartesian plane below, consider line segment $\overline{AB}$:



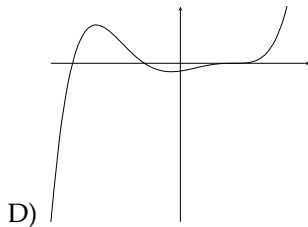
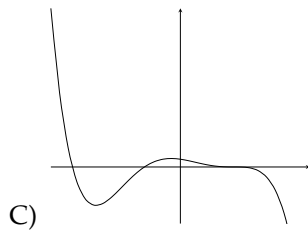
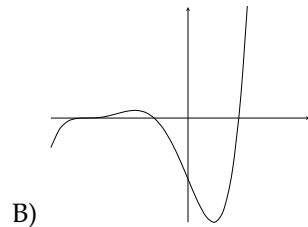
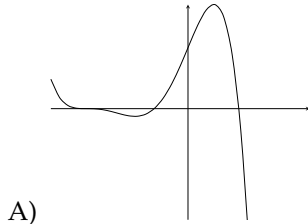
How many points C are on the plane such that $AC = CB$.

- A) 2
- B) 4
- C) More than 4
- D) 0



3

Which of the following graphs could be the graph of $y = 4(x - \frac{1}{6})^3(x + \frac{1}{9})(x + \frac{1}{3})$?



4

Which of the following algebraic operations is neither associative nor commutative?

- A) Addition
- B) Multiplication
- C) Subtraction
- D) All of the above

5

A cafeteria's industrial size dishwasher is down so they are paying willing employees Sri and Mikaela overtime on a day off to wash dishes at home. When the dishwasher broke down the cafeteria needed 37 loads of dishes washed. Sri is able to get her home dishwasher to get through 5 loads of dishes every 4 hours. Virgil is able to get his home dishwasher to get through 6 loads of dishes every 5 hours. What is the least amount of time, rounded to the nearest hour, it could take Brianna and Virgil to finish washing all 37 loads of dishes?

- A) 15 hours
- B) 16 hours
- C) 17 hours
- D) 18 hours



6

$$\frac{b^{x^2}}{b^{y^2}} = b^{30}$$

Given the equation above is true, if $b > 1$, and $x + y = 3$, what is the value of $x - y$?

- A) 10
- B) $\sqrt{30}$
- C) 15
- D) 30

7

Which of the following shifts of the graph of $y = x^2$ would result in the graph of $y = x^2 + 4x + c$ where c is a positive constant greater than 4?

- A) Left 1 units and up $c - 4$ units
- B) Right 2 units and up $c - 4$ unit
- C) Left 2 units and up $c - 4$ units
- D) Left 2 units and up $c + 4$ units

8

What are the solutions to $3x^2 - 9x = 3$?

- A) $x = \frac{3}{2} \pm \frac{\sqrt{3}}{2}$
- B) $x = \frac{3}{2} \pm \frac{\sqrt{13}}{2}$
- C) $x = \frac{9}{2} \pm \frac{\sqrt{77}}{4}$
- D) $x = -1$

9

Holly and Salman each order an organic smoothie and summer rawsagna at a restaurant in Portland, Oregon, where there is no sales tax. The price of Holly's meal was d dollars, and the price of Salman's meal was 2 dollars less than Holly's. If Holly and Salman agree to split the bill and a 15% gratuity evenly, which of the following represents the amount, in dollars, each of them paid?

- A) $1.15d - 1.15$
- B) $2.3d - 2.3$
- C) $d + 14$
- D) $2d + 13$



10

Simplify: $(x + y)^2 - (x - y)^2$

- A) $x^2 + y^2$
- B) xy
- C) $2xy$
- D) $4xy$

11

The graph of a line in the xy -plane has a slope of $\frac{1}{4}$ and contains the point $(-2, -9)$. The graph of a second line passes through the points $(8, -8)$ and $(13, -3)$. If the two lines intersect at the point (f, g) , what is the value of $f - g$?

- A) -16
- B) 4
- C) 16
- D) $-\frac{5}{3}$

12

Vinny may be hired out to dust a rectangular field with one pesticide. His fee on any particular rectangular field can be calculated by the expression $clw + kd$, where c is the cost per square meter, l is the length of the field in meters, w is the width of the field in meters, k is a cost per kilometer Vinny has to fly to get to the field, and d is the distance Vinny has to fly to get to the field. As long as the plane can hold enough pesticide, Vinny will be willing to dust more than one field at a time, and a lot of his customers take him up on this offer. Which factor in the expression likely decreases (on average) by placing orders for multiple fields at once?

- A) c
- B) k
- C) d
- D) w

13

What is the sum of all values of w that satisfy the following?

$$2w^2 - 8w - 11 = 0$$

- A) 4
- B) 0
- C) $-\sqrt{38}$
- D) -4



14

If $12^x 2^y = 6^x$, what is y in terms of x ?

- A) x
- B) $-x$
- C) $2x$
- D) $-2x$

15

Which of the following functions is positive for all real x ?

- I. $3x^2 - 1$
- II. $x^2 + 3$
- III. $3^{-x-5} + 1$

- A) II only
- B) III only
- C) II and III only
- D) I, II, and III

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

- Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If  is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

	7	/	1	2
	☐	☒	☐	☐
	☐	☐	☐	☐
	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

Answer: 2.5

← Decimal point

	2	.	5
	☐	☐	☒
	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
	☐	☒	☐
	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
☒	☐	☐	☐
☐	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	7
☒	☐	☐	☐
☐	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
	☐	☐	☐
	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

	2	0	1
	☐	☐	☐
	☐	☐	☐
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16

Kova's GPS in her van says that she can reach her destination in 20 minutes based upon the assumption that she can drive at 60 miles per hour. However, due to an accident, the speed along her route has been reduced to 20 miles per hour (MPH). Where Kova is driving, the police are known for giving speeding tickets for being as little as 3 MPH over the speed limit, so Kova had originally planned on driving at about 50 MPH. How much longer, in minutes, will it take Kova to reach her destination than what she originally planned?

17

$$v^3 - 7v^2 + 8v - 56 = 0$$

For what real value of v is the equation above true?

18

If $f(x) = 7\sqrt{7x}$, what is $f^{-1}(49)$?

19

$$y = (2x - 1)(x + 6)$$

$$y = 3(x + 12)$$

Given the system of equations above, find the y coordinate of one of the points where the graphs of the equations intersect.

20

At one concession stand, each corn dog has 50% more Calories than each hot dog. If 7 hot dogs and 2 corn dogs at this stand have a total of 3200 Calories, how many Calories are in one of its corn dogs?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

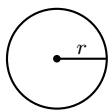
DIRECTIONS

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NOTES

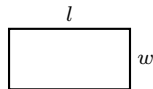
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3. Figures provided in this test are drawn to scale unless otherwise indicated.
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5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

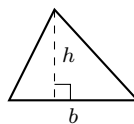


$$A = \pi r^2$$

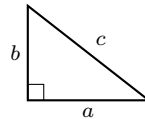
$$C = 2\pi r$$



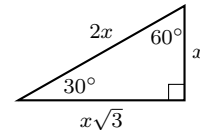
$$A = lw$$



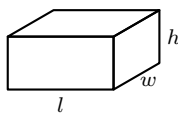
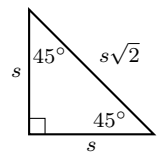
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



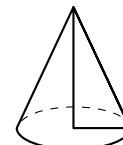
$$V = lwh$$



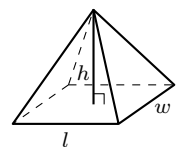
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



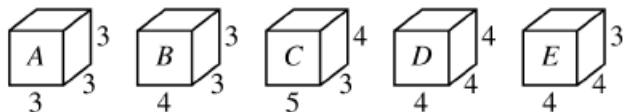
1

What is $\log_b(b^7)$?

- A) 1
- B) b
- C) 7
- D) $7b$

2

Which of the rectangular solids shown above has a volume closest to the volume of a right circular cylinder with radius 2 and height 4?



Note: Figures not drawn to scale.

- A) A
- B) B
- C) D
- D) E

3

The sum of four numbers is 2079. One of the numbers, n , is 10% more than the sum of the other three numbers. What is the value of n ?

- A) 330
- B) 990
- C) 1089
- D) 1143

4

Minh is on a strict diet, where he can only eat meat that's goat or lamb. He is also on a strict budget where he can only spend at most \$55 on meat a week. Last week Minh bought three pounds of goat and two pounds of lamb, which left him a surplus of \$4 in his meat budget. This week the prices for the meats hadn't changed, and he bought one pound of goat and four pounds of lamb with \$8 left in his weekly meat budget. If the prices stayed the same next week, what is the maximum amount of goat Minh could buy if he bought no lamb?

- A) $4\frac{3}{13}$ pounds
- B) 5 pounds
- C) $5\frac{1}{2}$ pounds
- D) $6\frac{1}{9}$ pounds



5

The monthly rent for a particular office space that comes with desks, internet access, telephones and computers is \$4000. The cost of the internet, electricity, local telephone, garbage, and water are included, but there is an additional fee of \$200 per time the building management are called upon to do IT work on their computers or any other maintenance on the facilities. For one year, Choke-On-A-Splinter Termite Control spent \$50,000 on renting out this particular office space. How many times did Choke-On-A-Splinter Termite Control ask building management for assistance with the facilities that year?

- A) 5
- B) 10
- C) 230
- D) 250

6

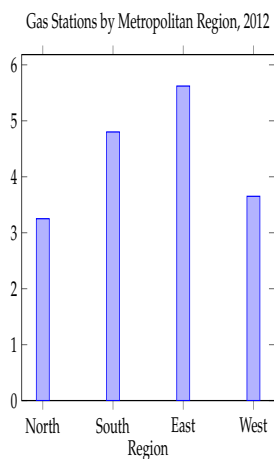
$$h = 2.3t^2 - 30t - 0.2$$

The equation above expresses the approximate depth d , in meters, of a bubble shot from an air cannon underneath the surface of a fluid tank with a downward velocity of 30 meters per second. After approximately how many seconds will the bubble escape the fluid?

- A) 12.0
- B) 12.5
- C) 13.0
- D) 13.5



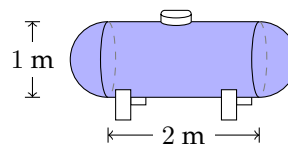
7



The number of gas stations in 4 different regions of a metropolitan area in 2012 is shown in the graph above. If the total number of gas stations is 1,732, what is an appropriate label for the vertical axis of the graph?

- A) Number of gas stations
- B) Number of gas stations (in tens)
- C) Number of gas stations (in hundreds)
- D) Number of gas stations (in thousands)

8



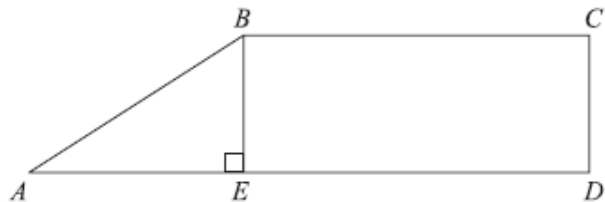
The storage component of a propane tank is built from two hemispheres and a right circular cylinder with internal measurements represented by the figure above. Of the following, which is the closest to the volume of the storage in the propane tank, in cubic meters?

- A) 2.09
- B) 3.67
- C) 10.47
- D) 16.76



9

In the figure below $AD = 16$, $ED = 11$, and AE is congruent to CD . What is the length of AB ?



- A) $5\sqrt{2}$
- B) 6
- C) $11\sqrt{2}$
- D) 25

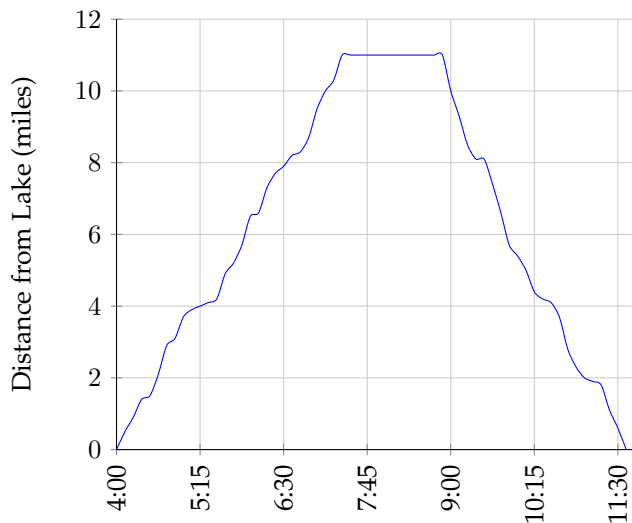
10

For what values of x is $|2x - 3| = -1$?

- A) $x = 1$
- B) $x = 2$
- C) $x = 1, 2$
- D) No value of x

11

Joanna's Hike



The graph above shows Joanna's distance from her base camp at a lake. On this day, she left her base camp to do a photo shoot early in the morning. Based upon the graph, about when did Joanna start setting up her photo shoot?

- A) 4:00
- B) 7:30
- C) 8:45
- D) 11:30



12

Simplify:

$$(x + 2)(x^2 - 2x + 4) - (x - 1)(x^2 - 2x + 1)$$

- A) $-3x^2 + 3x - 9$
- B) $3x^2 - 3x - 7$
- C) $x^2 - 2x + 7$
- D) 9

13

The solution set of $y - x < 0$ lie in which quadrant(s)?

- A) *I* and *II* only
- B) *I*, *II*, *III* only
- C) *I*, *II*, *IV* only
- D) none of these

14

Simplify:

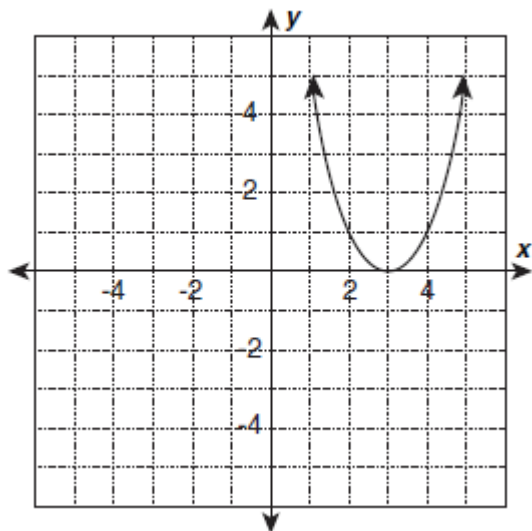
$$\frac{x + 7}{x + 1} - \frac{x - 1}{x + 7}$$

- A) $\frac{48}{x^2 + 7}$
- B) -1
- C) $\frac{2(7x + 24)}{(x + 1)(x + 7)}$
- D) $\frac{2(7x + 25)}{(x + 1)(x + 7)}$



15

The graph of a function $f(x)$ is shown in the diagram below.

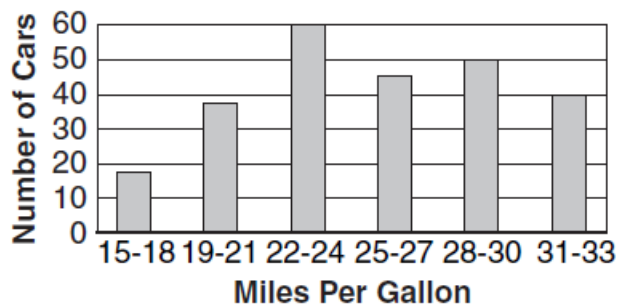


Which of the following formulas CANNOT describe the graph of $f(x)$

- A) $x^2 - 6x + 9$
- B) $\frac{x^3 - 6x^2 + 9x}{x}$
- C) $(x - 3)^2$
- D) $(x + 3)^2$

16

A car rental company monitored its fleet's gas mileage rates. The number of miles per gallon achieved by the company's 250 cars is depicted below.



What percent of the cars achieved a gas mileage rating between 22 and 24 miles per gallon?

- A) 24
- B) 30
- C) 42
- D) 60



17

Katie went to a store to buy two kinds of tomatoes. She bought 5 purple tomatoes for \$1.20 and 6 green tomatoes for \$1.80. At home, Katie put 3 purple tomatoes and 4 green tomatoes on a chair. What was the total cost in dollars of the tomatoes on the chair?

- A) \$1.86
- B) \$1.92
- C) \$2.00
- D) \$3.00

18

Which of the following lists all the asymptotes of the graph

$$y = \frac{2x^3 - 2x}{x^3 + x}?$$

- A) $y = 2$ and $x = 1$
- B) $y = 2$ and $x = 0$
- C) $y = 2$
- D) $x = 0$

19

$$C = 425 + 9(A - 15)$$

A restaurant charges for reserving a buffet according to the formula above, where C is the cost of the reservation in dollars and A is the number of people attending, and A must be at least 15 people. Which of the following is the best explanation of the meaning of 425 in the equation?

- A) The staff for a buffet event need to be paid \$425.
- B) \$425 is insurance so if the party cancels last minute, the restaurant isn't losing money.
- C) \$425 is the cost of renting the room the buffet is in.
- D) The restaurant usually charges \$425 for 15 people to eat at the buffet.



20

$$\begin{aligned} Ax + By &= C \\ y &= \frac{7}{3}x + 1 \end{aligned}$$

The system of equations above has no solution. Which of the following could be $Ax + By = C$?

- A) $7x - 3y = 3$
- B) $10x - 3y = 6$
- C) $14x - 6y = 3$
- D) $14x + 6y = 6$

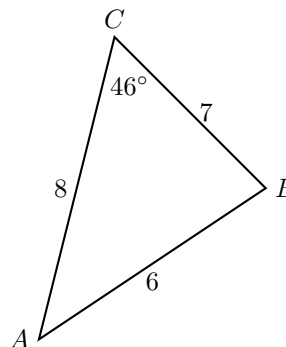
21

Which of the following has only complex roots?

- A) $x^2 - 1$
- B) $x^2 + 16$
- C) $x^2 - 2x + 1$
- D) $x^2 - 3x + 2$

22

Consider the figure below:



Which of the following equations can be used to solve for angle A ? (Note: For any triangle with sides a, b, c that are opposite angles A, B, C respectively, $\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$.)

- A) $\frac{\sin(46^\circ)}{8} = \frac{\sin(A)}{6}$
- B) $\frac{\sin(46^\circ)}{7} = \frac{\sin(A)}{8}$
- C) $\frac{\sin(46^\circ)}{7} = \frac{\sin(A)}{6}$
- D) $\frac{\sin(46^\circ)}{6} = \frac{\sin(A)}{7}$



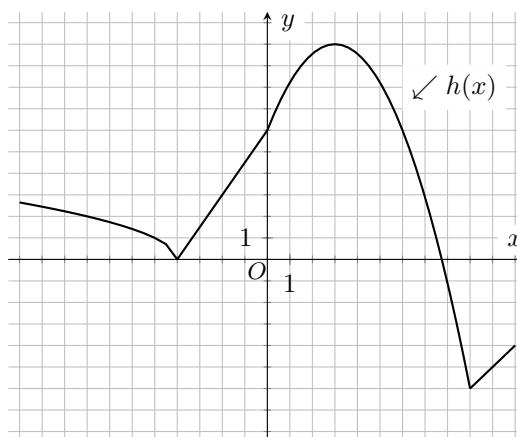
23

$$A_k = A_0 e^{-rk}$$

A simple model for land prices in a city is to presume the city has one center and land values are determined predominantly by distance from that center. In the equation above, A_k represents the land value per acre, A_0 represents the land value of an acre at the city's center, r is a constant "rate" of value lost, and k is the distance from the city center in kilometers. An urban economist wishing to analyze the effectiveness of this model looks at two properties in a given city. He finds a property at what he is modeling as the city center that has a land value of \$400,000 per acre. 5 kilometers away from that center he finds a property with a land value of \$200,000 per acre. If these properties are used to generate the model, which of the following is the closest to the value of r ?

- A) 0.04
- B) 0.14
- C) 0.60
- D) 0.69

24



The complete graph of the function h is shown in the xy -plane above. For what value of x is the value of $h(x)$ at its maximum?

- A) 3
- B) 9
- C) 10
- D) 11



25

The speed of a sound wave in dry air may be modeled by

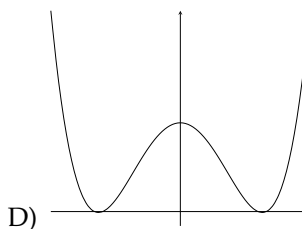
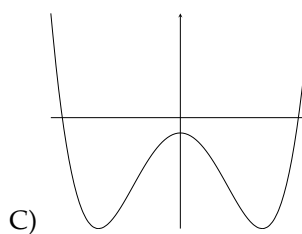
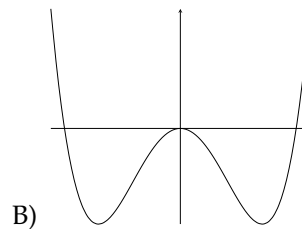
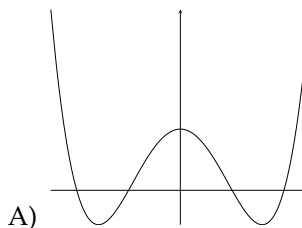
$$v_s = 20.05\sqrt{t}$$

where v_s is the speed of the sound in meters per second and t is the temperature of the air in degrees Kelvin. At which of the following air temperatures will the speed of a sound wave be closest to $350\frac{\text{m}}{\text{s}}$?

- A) 304K
- B) 305K
- C) 306K
- D) 307K

26

If the function p has four distinct zeros, which of the following could represent the complete graph of p in the xy -plane?





27

Line w in the xy -plane contains points from each of Quadrants I, III, and IV, but no points from Quadrant II. Which of the following must be true?

- A) The slope of line w is positive.
- B) The slope of line w is negative.
- C) The slope of line w is zero.
- D) The slope of line w is undefined.

28

Which of the following is a binomial?

- A) $\sqrt{x}$
- B) x
- C) x^2
- D) $x^2 - x^3$

29

If $n = \sqrt{p}$ for any positive integer p , and if $p = v^2 - 4v + 4$, what is n in terms of v ?

- A) $\sqrt{v}$
- B) $v - 2\sqrt{v} + 2$
- C) $2v\sqrt{v+1}$
- D) $v - 2$

30

If $\frac{14}{9}r = \frac{3}{7}$, what is the value of r ?

- A) $\frac{98}{27}$
- B) $\frac{3}{2}$
- C) $\frac{2}{3}$
- D) $\frac{27}{98}$

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

	7	/	1	2
.	.	.	.	.
	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

Answer: 2.5

← Decimal point

	2	.	5
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	7
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

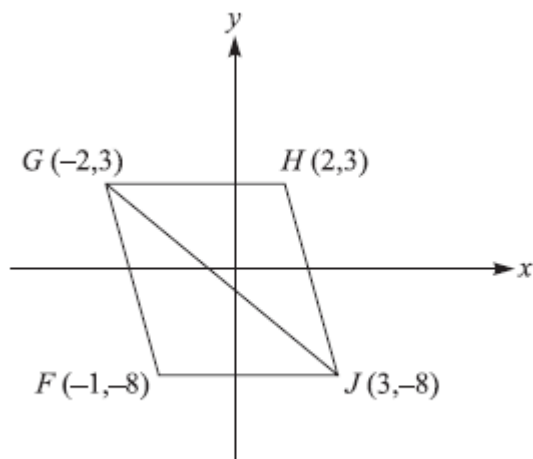
	2	0	1
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

Given the vertices of parallelogram $FGHJ$ in the standard (x, y) coordinate plane below, what is the area of triangle GHJ , in square units?



32

$$v = \frac{1.273q}{d^2}$$

The fluid velocity v through a pipe can be found using the formula above, where d is the diameter of the pipe and q is the volume flow. A civil engineer uses the formula to figure out the fluid velocity if the diameter of the pipe were d and if the diameter of the pipe were $0.4d$ given a constant volume flow. What is the ratio of the fluid velocity in the wider pipe to the fluid velocity in the narrower pipe?

33

In one quarter, Duc and Terri spend a combined 80 hours in the computer lab. If Duc spent 22 fewer hours than Terri did, how many hours did Terri spend in the lab?



34

A new restaurant is getting its first online customer satisfaction ratings on a website, which lets users grade restaurants between 0.0 and 5.0 stars, inclusive. In the first 8 ratings, the restaurant received an average grade of 3.6. What is the least value the restaurant can receive for its 9th rating and still be able to have an average of at least 4.1 for the first 15 ratings?

35

In the social sciences and in biology, game theory is used to predict the behavior of actors that have to interact with one another. Ordered pairs are used to communicate how much a particular actor prefers or dislikes given potential outcomes. The game theory table below was one speculation about the incentive structure just prior to China's offensive during the Korean War. The United States' perception of rewards are given as x coordinates while China's perception of rewards are given as y coordinates.

		China		
		Attack	Amass Defences	Stay Neutral
United States	Total Victory By Any Means	(2, 1)	(8, -2)	(6, -10)
	Total Victory Without Nukes	(-3, 5)	(10, -2)	(10, -6)
	Cease offensive	(4, -6)	(-6, 8)	(0, 9)

If China expects that the United States will not cease its offensive but can't determine the United States' willingness to use nuclear weapons, what would China's expected payoff be for attacking?

36

What is the smallest integer N so that N is in the range of $f(x) = \frac{\sqrt{x+4}}{3}$?



37

$$R = -1.5D + 40$$

A squirrel buried many stashes of small nuts for consumption during the winter. Each day after winter started, she consumed a fixed amount from one of her stashes. The equation above models the amount R , in pounds of nuts, that the squirrel has remaining in the first stash she eats from after the first D days of winter. According to the model, how many pounds of nuts does the squirrel eat per day?

38

If x hours and 45 minutes is equal to 405 minutes, what is the value of x ?

Answer Key

Calculator Off

1	D
2	C
3	D
4	C
5	A
6	A
7	C
8	B
9	A
10	D
11	C
12	D
13	A
14	B
15	C
16	36
17	7
18	7
19	15 or 45
20	480

Calculator On

1	C
2	D
3	C
4	C
5	B
6	D
7	C
8	A
9	A
10	D
11	B
12	B
13	D
14	D
15	D
16	A
17	B
18	C
19	C
20	C
21	B
22	D
23	B
24	A
25	B
26	A
27	A
28	D
29	D
30	D
31	22
32	0.16 or
	4/25
33	51
34	2.7
35	3
36	2
37	1.5
38	6



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

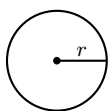
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

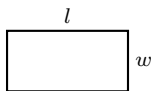
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

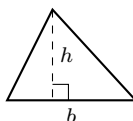


$$A = \pi r^2$$

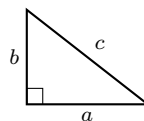
$$C = 2\pi r$$



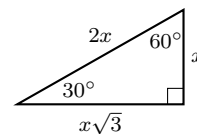
$$A = lw$$



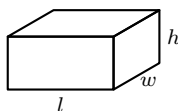
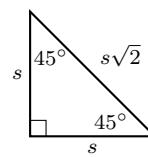
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



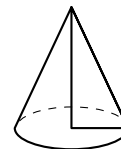
$$V = lwh$$



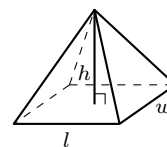
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

If $(ux + 2)(vx - 5) = 15x^2 + tx - 10$ for all values of x , and $u + v = 8$, what are the two possible values for t ?

- A) $-\frac{2}{5}$ and $\frac{5}{3}$
- B) -5 and -19
- C) 3 and 5
- D) 8 and 15

2

Solve for y :

$$\sqrt[3]{y} + \sqrt[3]{-27y} = -8$$

- A) $y = \frac{64+96\sqrt{3}}{23}$
- B) $y = 64$
- C) $y = -8$
- D) $y = \frac{-32}{117}$

3

A line in the xy -plane passes through $(4, 5)$ and has a slope of $\frac{-1}{3}$. Which of the following points is on the line?

- A) $(1, 2)$
- B) $(3, 8)$
- C) $(7, 4)$
- D) $(7, 6)$

4

$$\begin{aligned}a - 3b &= 4 \\ 3a + 4b &= 51\end{aligned}$$

Which of the following ordered pairs (a, b) satisfies the system of equations above?

- A) $\left(\frac{-89}{7}, \frac{-39}{7}\right)$
- B) $\left(\frac{-39}{7}, \frac{-89}{7}\right)$
- C) $(13, 3)$
- D) $(3, 13)$



5

In the xy -plane, the parabola with the equation $y = (x - 7)^2 - 11$ intersects the line with equation $y = 70$ at two points, P_1 and P_2 . What is the length of $\overline{P_1P_2}$?

- A) 18
- B) 14
- C) 8
- D) 7

6

At how many points do the graphs described by $2x - y = 7$ and $3x - y = 1$ intersect?

- A) 0
- B) 1
- C) 2
- D) 3

7

Which of the quadratics below intersect only the negative x -axis?

- A) $x^2 + 5x + 6$
- B) $x^2 - 1$
- C) $x^2 - 6x + 9$
- D) None of the above

8

Expand $(x - 1)(x^2 + x + 1)$

- A) x^3
- B) $x^3 - 1$
- C) $x^3 + 1$
- D) $x^3 + x$



9

A remodeling company estimates the price of a job, in dollars, using the equation $P = F + 45wt$, where P is the estimated price of the job, F is the estimated fixed cost for materials, w is the number of employees devoted to the job, and t is the number of hours spent completing the job. Which of the following is the best interpretation of the number 45 in the equation?

- A) Each construction worker works 45 hours a week.
- B) The price of a job increases by \$45 every hour.
- C) There will be 45 construction workers on the job.
- D) The company charges \$45 per hour for each worker on the job.

10

If $x^{\frac{3}{5}} = 7$, then what is x^9 ?

- A) $7^{\frac{27}{5}}$
- B) $7^{\frac{42}{5}}$
- C) 7^{15}
- D) 35^{15}

11

$$h(x) = kx^6 - 5$$

For the function h defined above, a is a constant and $h(5) = 620$. What is the value of $h(-5)$?

- A) 620
- B) 5
- C) -2
- D) -630

12

The line $y = wx + 4$, where w is a constant, is graphed in the xy -plane. If the line contains the point (g, h) , where $g \neq 0$ and $h \neq 0$, what is the slope of the line in terms of g and h ?

- A) $\frac{4-h}{g}$
- B) $\frac{g}{4-h}$
- C) $\frac{g}{h-4}$
- D) $\frac{h-4}{g}$



13

Given $x \neq -7$, simplify:

$$x - 5 \div \frac{x^2 - 2x - 15}{x + 7}$$

A) 2

B) $\frac{x+7}{x+3}$ C) $x^2 - 14x + 93 - \frac{576}{x+7}$ D) $\frac{1}{x^2+10x+21}$

14

$$d = \frac{180(n-2)}{n}$$

The measure d , in degrees, of an interior angle of a regular polygon is related to the number of sides, n , of the polygon by the formula above. If the measure of an interior angle is greater than 150° , what is the minimum number of sides it can have?

A) 10

B) 11

C) 12

D) 13

15

If $\frac{c+d}{c} = \frac{6}{7}$, which of the following must also be true?

A) $\frac{2c+d}{c} = 1$ B) $\frac{d-c}{c} = \frac{-1}{7}$ C) $\frac{d}{c} = \frac{-1}{7}$ D) $\frac{d}{c} = \frac{13}{7}$

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

5. **Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If  is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

6. **Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write answer in boxes. →

Grid in result. →

	7	/	1	2
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	☐	☐	☐	☐
	0	0	0	0
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8	8	8	8	8
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← Fraction line

Answer: 2.5

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← Decimal point

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
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Answer: 201 – either position is correct

	2	0	1
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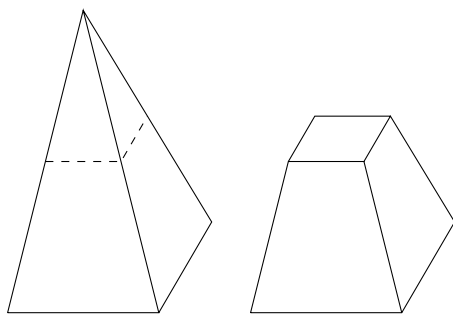
NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16

Evaluate $\log(100000)$.

17



A frustum is formed by cutting a smaller right square pyramid from a larger right square pyramid as shown above (although the diagram is not to scale). The height of the frustum is the half of the height of the square pyramid that was removed. If the frustum rests with the larger base down, as shown, its weight generates a pressure on the ground of 72 pounds per square inch. What pressure, in pounds per square inch, would the frustum's weight create if the frustum were turned upside down?

18

$$g = \frac{GM}{r^2}$$

The equation above represents the expected acceleration in a direction towards a mass due to its gravity, where G is a constant, M is the mass of the gravitational source, and r is the distance from the gravitational source. Suppose an astronaut calculated that his ship would experience a gravitational acceleration of $3.6 \frac{m}{s^2}$ due to proximity to a planet at a certain location. However, his estimate of the mass of the planet is two thirds of the actual mass of the planet. What would be the actual gravitational acceleration, in meters per second squared, that his ship would experience at that location?

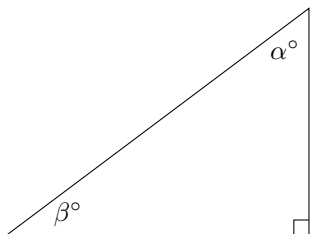
19

$$\begin{aligned} x^2 + y^2 &= 100 \\ y &= -x + 14 \end{aligned}$$

According to the system of equations above, what is one possible value for y ?



20



In the triangle above, the tangent of β° is 0.75. What is the sine of α° ?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

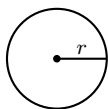
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

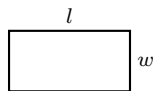
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

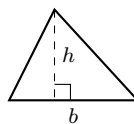


$$A = \pi r^2$$

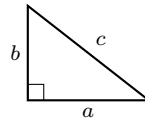
$$C = 2\pi r$$



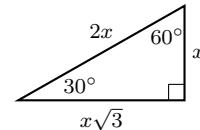
$$A = lw$$



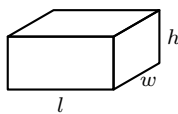
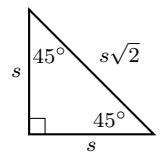
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



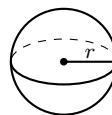
Special Right Triangles



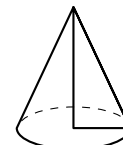
$$V = lwh$$



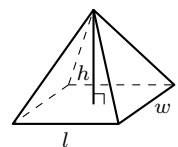
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.

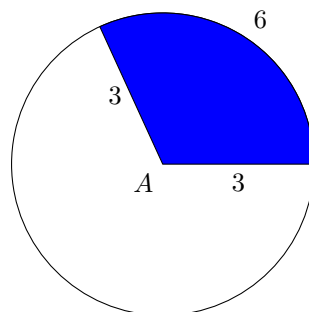


1

Which of the following statements would NOT be a reasonable inference based upon the survey data?

- A) A randomly selected guardian is likely to find a restricted government role valuable in social affairs.
- B) A randomly selected guardian likely places the same value on economic autonomy as they place on social autonomy.
- C) A randomly selected guardian likely opposes complete market deregulation.
- D) A randomly selected guardian likely opposes strict government-controlled uniform codes.

2



The sector of circle A above is bounded by two radii of length 3 and an arc of length 6. What is the sector's area?

- A) $\frac{\pi}{4}$
- B) 4
- C) 9
- D) 9π

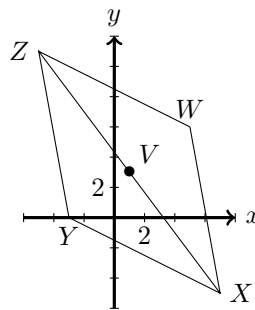


3

Which of the following functions is positive for all real x ?

- I. $3x^5 - x$
 - II. $x^2 + 3x$
 - III. $(2/3)^{-x-5} + 1$
- A) I only
 B) III only
 C) II and III
 D) I, II, and III

4



In the xy -plane above, $WXYZ$ is a kite and point V is the intersection of the diagonals of the kite. The coordinates of points V and W are $(1, 3)$ and $(4, 7)$, respectively. Which of the following is an equation of the line that passes through points X and Z ?

- A) $y = \frac{3}{4}x + \frac{9}{4}$
 B) $y = \frac{3}{4}x - 4$
 C) $y = \frac{-4}{3}x + \frac{13}{3}$
 D) $y = \frac{-4}{3}x - 4$



5

In order to determine if a particular medicine M is successful in reducing high blood pressure (hypertension), a research study was conducted. From 50,000 volunteers with hypertension willing to participate in the study, 500 were selected at random. They were further blocked by gender. They were then assigned the placebo group or the treated group at random. The resulting data showed that 72% female participants taking M had significantly reduced blood pressure compared to the female placebo group. 51% of the male participants taking M had significantly reduced blood pressure compared to the male placebo group. Based on the design and results of the study, which of the following is an appropriate conclusion?

- A) M must work well with estrogen.
- B) Treatment M reduces blood pressure in many but not all people with hypertension.
- C) Testosterone must inhibit some of the effectiveness of M.
- D) Treatment M is the best medication for reducing blood pressure in hypertensive women.

6

$$\begin{array}{r} A B \\ + B A \\ \hline 1 A 7 \end{array}$$
 A and B represent different digits in the correctly worked out addition problem above. What is A+B?

- A) 6
- B) 12
- C) 14
- D) 17



7

An economist is running a simulation with fictitious currencies to gain insights into international currency trading. After 19 rounds, the relationship between the value of two particular currencies, the krupit and the bilibruble, begins to stabilize. On round 20, a krupit was worth 2 bilibrubles. Each round after that, it took 0.03 more bilirubles than the previous round to trade for a krupit. If t represents the number of turns since round 20, which of the following inequalities describes the set of turns after turn 20 where it takes more than 4 bilirubles to trade for 1 krupit?

- A) $2 + 0.03t > 4$
- B) $4 > 2 + 0.03$
- C) $2 > 0.03t$
- D) $4 > 2 + 0.03t$

8

Mercury makes one complete orbit around the Sun in 88 days. Mercury travels about 360,103,000 kilometers in a complete orbit. Of the following, which is closest to the average speed of Mercury, in kilometers per hour, as it orbits the Sun?

- A) 170,500
- B) 986,600
- C) 4,092,100
- D) 15,004,300

9

Which of the following (if any) is equivalent to $2x^3y^6 \times x^{-2}y^{-4} \times zxy$?

- A) $6zxy^3x$
- B) $5x^{-6}y^{-24}z$
- C) $6xy^2$
- D) None of the above



10

Trevon surveyed a random sample of the junior class to determine whether the Spring Soirée should be held before or after AP exams. Of the 80 students surveyed, 81.3% preferred after the AP exams. Based on this information, about how many students in the entire 408 person class would be expected to prefer having the Spring Soirée after the AP exams?

- A) 33
- B) 233
- C) 323
- D) 332

11

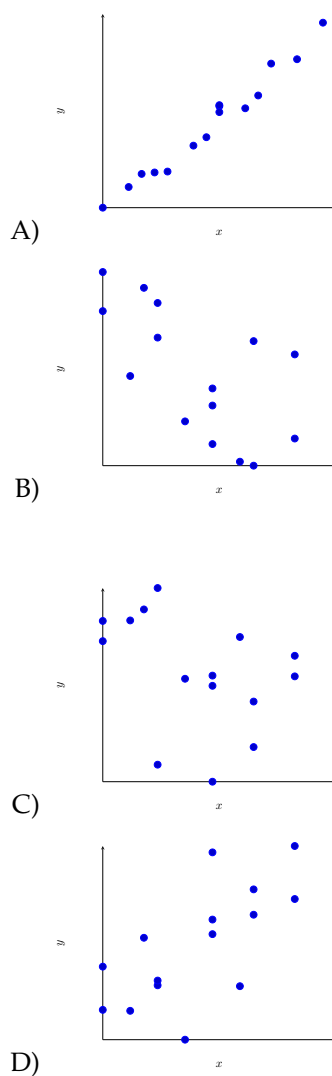
$$\begin{array}{l} 2t^2 - 7t - 4 \\ 4t^2 + 2t - 5 \end{array}$$

Which of the following is the sum of the two polynomials shown above?

- A) $6t^2 - 5t - 9$
- B) $6t^2 + 5t - 9$
- C) $6t^4 - 5t^2 - 9$
- D) $6t^4 + 5t^2 - 9$

12

Which of the following graphs best shows a weak positive association between x and y ?





13

If $a - b = 2$ and $a^2 + b^2 = 34$, what is ab ?

- A) -35
- B) 3
- C) 15
- D) $\sqrt{285} \approx 16.88$

14

If $9 + 5w$ is 17 less than 56, what is the value of $7w$?

- A) 6
- B) 19.5
- C) 32
- D) 42

15

The ratio of all-star baseball cards to regular baseball cards in Tony's collection is five to seven. After Tony bought four more all-star baseball cards, his all-star cards make up half of his new collection. How many cards did Tony have before his new purchase?

- A) 12
- B) 10
- C) 18
- D) 24



16

Number of Residents in Canada in 2011, in Thousands

Region	Age, in years				Total
	0-19	20-39	40-59	60+	
Atlantic Canada	494	551	731	552	2328
Central Canada	4,795	5,322	6,223	4,414	20,755
Northern Canada	34	32	30	11	107
Western Canada	2,463	2,789	3,016	2,020	10,287
Total	7,785	8,694	10,000	6,998	33,477

The table above shows the number of residents in 2011, in thousands, in four geographic regions and 4 age groups. Based on the table, if a resident of Canada was 0-19 years old in 2011, which of the following is closest to the probability that the resident was from Western Canada?

- A) 0.307
- B) 0.316
- C) 0.684
- D) 0.693

17

An elevator has a maximum capacity of 15 persons or 4000 pounds. A football team's linemen and backs are completely worn out from their second workout session of the day and want to get up to their room without taking the stairs. The linemen weigh about 300 pounds each and the backs weigh about 180 pounds each. Let l represent the number of linemen and b represent the number of backs. Which of the following sets of inequalities represents the ways the players could get into the elevator without exceeding the limits?

- A) $\begin{cases} 300l + 180b \leq 4000 \\ l + b \leq 15 \end{cases}$
- B) $\begin{cases} \frac{l}{300} + \frac{b}{180} \leq 15 \\ l + b \leq 4000 \end{cases}$
- C) $\begin{cases} 300l + 180b \leq 15 \\ l + b \leq 4000 \end{cases}$
- D) $\begin{cases} 300l + 180b \leq 4000 \\ l + b \leq 4000 \end{cases}$



18

Gopal is a nutritionist studying the production of antioxidants by two types of tomatoes. He determines that a typical type Q tomato in his sample contains 40% more micrograms(μg) of antioxidants than a typical type R tomato in his sample. Based upon Gopal's observation, if the type R tomatoes sampled had an average of $2900\mu\text{g}$ antioxidants per tomato, what was the average mass of the antioxidants in a sampled type Q tomato?

- A) $1740\mu\text{g}$
- B) $2071\mu\text{g}$
- C) $2940\mu\text{g}$
- D) $4060\mu\text{g}$

19

Super thin basalt fiber wool can be manufactured to have a density of $2.5 \frac{\text{kg}}{\text{m}^2}$. A metric ton is 1000 kg. How many metric tons of super thin basalt fiber wool would be needed to cover one square kilometer?

- A) 0.0025
- B) 2.5
- C) 2,500
- D) 2,500,000

20

If $|5x + 8| \leq -12$, what is the maximum possible value for x ?

- A) -4
- B) 4
- C) ∞
- D) There is no possible value for x .

21

x	-4	-3	-2	-1
$h(x)$	-11	-7	-3	1

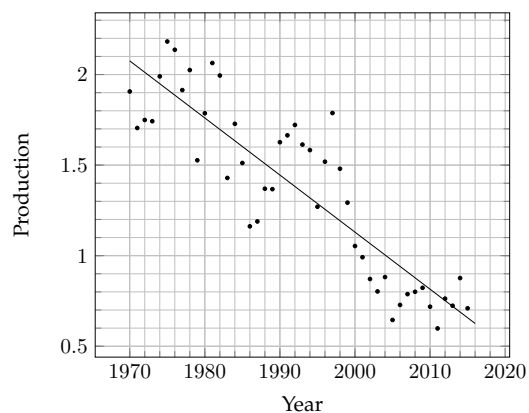
The table above shows some values of the linear function h . Which of the following defines h ?

- A) $h(x) = 4x + 5$
- B) $h(x) = x - 7$
- C) $h(x) = -x - 15$
- D) $h(x) = -4x - 27$



22

U.S. Tobacco production in billions of pounds



According to the line of best fit in the scatterplot above, which of the following best approximates the year in which the estimated tobacco production of the United States was 1 billion pounds?

- A) 2001
- B) 2002
- C) 2003
- D) 2004

23

$$B = \frac{m - 60}{3.24}$$

A military is randomly weighing trainees to ensure they are meeting fitness goals. The above equation represents the expected BMI of a 180-cm tall recruit carrying 60 kg of gear, where m is the mass of the trainee while carrying his gear in kg. What is m to the nearest kg for a 180-cm tall recruit when B is 21.605?

- A) 12
- B) 40
- C) 67
- D) 130



24

Gender	Age		Total
	0 to 24	25 to 40	
Male	28	17	45
Female	15	3	18
Total	43	20	63

The table above shows the distribution of age and gender for an amateur athletic event. Due to the number of contestants, somebody in the first round must receive a bye. If the recipient of the bye is selected at random, what is the probability that the recipient will either be a female under the age of 25 or a male over the age of 24?

- A) $\frac{15}{63}$
- B) $\frac{17}{63}$
- C) $\frac{31}{63}$
- D) $\frac{32}{63}$

25

If $y = \frac{k}{x}$ and $y = 15$ when $x = 4$, what is the value of x when $y = 10$?

- A) -7
- B) -1
- C) 6
- D) $2\frac{2}{3}$

26

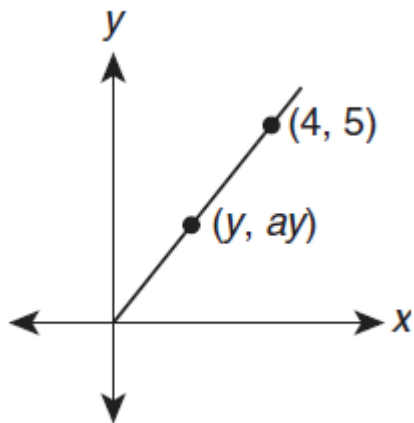
What is $\log_3(12)$?

- A) $\log_3(2)$
- B) $1 + \log_3(2)$
- C) $1 + 2\log_3(2)$
- D) $2 + 2\log_3(2)$



27

In the diagram, the line represents a direct variation. Which of the following is the value of the constant a ?



- A) 1
- B) $4/5$
- C) $5/4$
- D) 5

28

Which of the following statements best describes the difference between the restaurant's prices last year versus this year?

- A) The ratio of their prices changed as the meat dishes became cheaper and the vegetable sides became more expensive.
- B) The ratio of their prices changed as the meat dishes became more expensive and the vegetable sides became cheaper.
- C) Both vegetable sides and meat dishes became more affordable, although the ratio of their prices remained unchanged.
- D) Both vegetable sides and meat dishes become more expensive, although the ratio of their prices remained unchanged.



29

Let n and k be numbers such that $-k > n > k$. Which of the following must be true?

- I. $|n| > k$
 - II. $k < 0$
 - III. $k < n$
- A) I only.
B) I and III only.
C) II and III only.
D) I, II, and III.

30

Rocio walks 40 yards in 27.1 seconds. If she walks at this same rate, which of the following is the closest to the distance she will walk in 10 minutes?

- A) 15 yards
B) 150 yards
C) 400 yards
D) 900 yards

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

- Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If  is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

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	☐	☐	☐	☐
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9	9	9	9	

Answer: 2.5

← Decimal point

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Answer: 201 – either position is correct

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NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

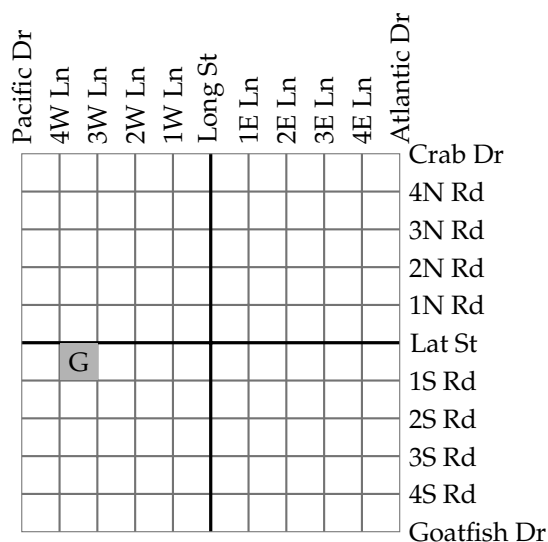
The terms of a sequence are defined by $a_n = a_{n-2} + 5a_{n-1}$ for $n > 2$. What is the value of a_4 if $a_1 = 2$ and $a_2 = 4$?

32

$$f(x) = \frac{8x}{(2x-1)^2 - 2(2x-1) - 24}$$

What is one of the x values excluded from the domain of f ?

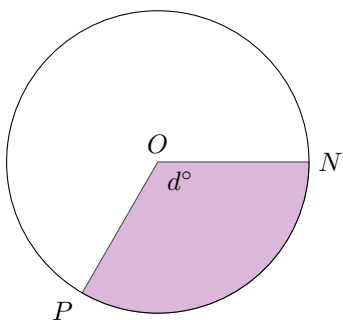
33



In the town of Gogruffy, the city planners designed downtown to have square blocks of equal size, as shown in the map above. This way they could use the term “block” as a uniform measure of distance. Sumaya wants to establish a deli on an intersection within one block of Long St for customer traffic and within 4 blocks of the grocery store (G) as the crow flies just in case she has to go make a quick supply run. What is the distance, to the nearest whole block, between the most northern and the most southern locations she could use for her deli?



34



Note: Figure not drawn to scale.

In the figure above, the circle has center O and has radius 7. If the sector bounded by the arc $\widehat{NP}$ and radii $\overline{OP}$ and $\overline{NO}$ has an area between 44 and 45, what is one possible integer value for d ?

35

To both protect pangolins from poaching and to study their behavior, a scientist proposes putting microchips inside 250 pangolins. For demonstration purposes, the scientist placed chips inside 4 male pangolins and 1 female pangolin already. Currently the scientist has easy access to 108 pangolins that don't have microchips, of which 44 are male. Based upon poacher preferences, the scientist proposes having males compose at least 70% of the group of pangolins implanted with microchips. At least how many additional male pangolins must be found to be able to achieve the desired gender ratio for the microchip program?

36

A climate scientist estimates that sea level rise will average 1.1 cm per year over the next century. According to the climate scientist's estimate, how long will it take, in years, for the sea level to rise 99 cm?



37

If $f(2x) = 6x + 7$, what is $f^{-1}(10)$?

38

In a circle with center C , central angle BCD has a measure of $\frac{7\pi}{9}$ radians. The length of $\widehat{BD}$ is what fraction of the circumference of the circle?

Answer Key

Calculator Off	
1	B
2	B
3	C
4	C
5	A
6	B
7	A
8	B
9	D
10	C
11	A
12	D
13	B
14	C
15	C
16	5
17	162
18	5.4
19	Either 6 or 8.
20	$\frac{4}{5}$ or 0.8

.....

Calculator On	
1	B
2	C
3	B
4	C
5	B
6	D
7	A
8	A
9	A
10	D
11	A
12	D
13	C
14	D
15	D
16	B
17	A
18	D
19	C
20	A
21	A
22	D
23	D
24	D
25	C
26	C
27	C
28	C
29	D
30	D
31	114
32	Any from the set $x \in \{-1.5, -\frac{3}{2}, 3.5, \frac{7}{2}\}$
33	8
34	103 or 104 or 105
35	127
36	90 years.
37	1
38	$\frac{7}{18}$ or .388 or .389



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

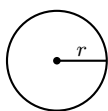
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

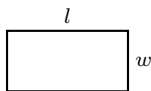
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

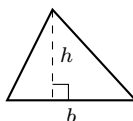


$$A = \pi r^2$$

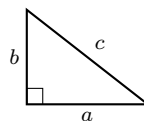
$$C = 2\pi r$$



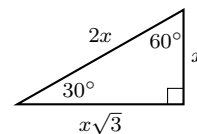
$$A = lw$$



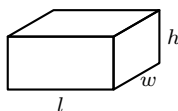
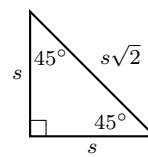
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



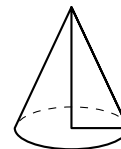
$$V = lwh$$



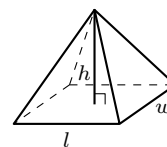
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

$$M + Y \leq 20$$

$$40 \leq M + C \leq 100$$

$$M \geq C - 40$$

A graphic designer is given curious instructions for picking a main color for a logo. The designer is told to use an MCY method for defining colors, where M is a value for concentration of magenta, C is a value for cyan, and Y is for yellow. Moreover, the designer must use a main color that conforms to the equations above. What is the maximum value for cyan the designer can use for the main logo color?

- A) 80
- B) 60
- C) 40
- D) 20

2

$$g(x) = \frac{2}{11}x + b$$

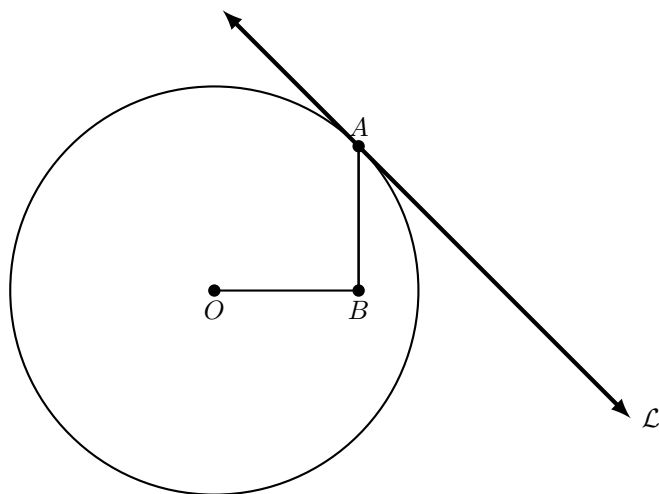
In the function above, b is a constant. If $g(44) = 20$, what is the value of $g(-33)$?

- A) 12
- B) 6
- C) -6
- D) -7



3

For a circle centered at O , line $\mathcal{L}$ is tangent to this circle at A and has a slope of $-\frac{1}{2}$. If the length of $\overline{OB}$ is 2 and $\overline{OB} \perp \overline{AB}$, then what is the radius of the circle?



- A) 2
- B) $3\sqrt{2}$
- C) $2\sqrt{5}$
- D) 6

4

Which of the following (x, y) pairs is a solution to the equation $xy + y = 1$?

- A) $(1, -1)$
- B) $(-1, 1)$
- C) $(2, 2)$
- D) None of the above



5

On an internship, Tammy was assigned a schedule where the number of hours she spends a week on training would decrease by a constant amount each week that passed of the internship. Tammy's internship schedule requires her to spend 30 hours in training on week 3 and spend 10 hours in training on week 28. Which of the following best describes how the time Tammy must spend in training changes between week 3 and week 28 on her internship?

- A) Tammy's training requirement decreases by 5 hours every 4 weeks.
- B) Tammy's training requirement decreases by 4 hours every 5 weeks.
- C) Tammy's training requirement decreases by 3 hours every 2 weeks.
- D) Tammy's training requirement decreases by 1 hour every 8 days

6

Which of the following is the same as $\sqrt{x^2}$

- A) $|x|$
- B) x^2
- C) x^4
- D) None of the above

7

$$7y - 5x = 39$$

$$3x - 5x = 11$$

If (x, y) is a solution to the system of equations above, what is the value of $x - y$?

- A) -5
- B) 3.32
- C) 5
- D) 5.8



8

Which of the following equations represents the *commutative law of addition*?

- A) $x + 0 = x$
- B) $x + y = y + x$
- C) $x + (y + z) = (x + y) + z$
- D) $x \cdot y = y \cdot x$

9

Which of the following equations represents a line that is parallel to the line with the equation $y - 6 = \frac{9}{5}(x + 13)$?

- A) $18x - 10y = 5$
- B) $10x - 18y = 5$
- C) $18x + y = 2$
- D) $y - 4x = 7$

10

If $3x + y = 6$, what is the value of $2^y 8^x$?

- A) 2^4
- B) 4^3
- C) 8^0
- D) The value cannot be determined from the information given.

11

$$y + 1 = 4(x - 4)$$

$$3x + 2y = 32$$

What is the value of y for the solution to the system of equations above?

- A) $3\frac{9}{11}$
- B) $5\frac{7}{11}$
- C) 6
- D) 7



12

Which of the following is not the same as $\sqrt{432}$?

- A) $6\sqrt{12}$
- B) $4\sqrt{17}$
- C) $4\sqrt{27}$
- D) $2\sqrt{108}$

13

If $\frac{4}{a} = \frac{8}{3a-12}$, what is the value of $3a - 12$?

- A) -3
- B) 0
- C) 3
- D) 12

14

Which of the following functions lie in only the first two quadrants?

- A) $f(x) = 3x - 1$
- B) $f(x) = 2x$
- C) $f(x) = -2x$
- D) $f(x) = x^2 + 1$

15

$$\sqrt{2x^2 + 31} - y = 6$$

If $x < 0$ and $y = 3$ in the equation above, what is the value of x ?

- A) -5
- B) $\frac{-5\sqrt{2}}{2}$
- C) $\frac{-\sqrt{38}}{2}$
- D) 25

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

7	/	1	2
•	•	•	•
0	0	0	0
1	1	•	1
2	2	2	•
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
•	7	7	7
8	8	8	8
9	9	9	9

Answer: 2.5

← Decimal point

	2	.	5
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	3
4	4	4	4
5	5	5	•
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	•
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	7
8	8	8	8

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	•
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
•	•	•	•
0	•	0	0
1	1	1	•
2	•	2	2
3	3	3	3

	2	0	1
•	•	•	•
0	•	0	0
1	1	•	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16

If $f(x) = \frac{1}{x}$ and $g(x) = 2x - x^2$, then what is $f(g(\frac{1}{2}))$?

17

Evaluate $(-1)^{99} + (-1)^{100} + (-1)^{101} + (-1)^{102}$

18

From the standpoint of an amateur astronomer on earth, the width of Jupiter spans approximately $\frac{\pi}{12960}$ radians of angular measure. An arcminute is $\frac{1}{60}$ of a degree. Let x be the approximate width of Jupiter measured in arcminutes. Find x .

19

A radio show always awards \$20,000 in prizes to its contestants. The prizes are awarded in amounts of \$1,000 or \$3,000. If at least three \$1,000 prizes and at least three \$3,000 prizes are given, what is one possible number of the total number of prizes awarded?

20

In isosceles $\triangle ABC$, $AB = BC$ and $\tan m\angle C = \sqrt{15}$. What is $\frac{AB}{AC}$?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

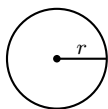
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

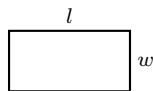
1. The use of a calculator **is permitted**.
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4. All figures lie in a plane unless otherwise indicated.
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REFERENCE

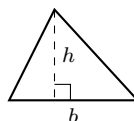


$$A = \pi r^2$$

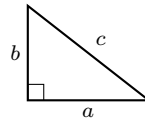
$$C = 2\pi r$$



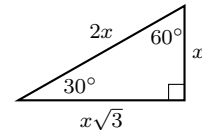
$$A = lw$$



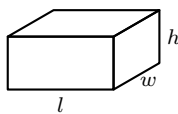
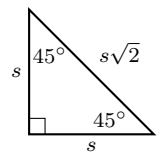
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



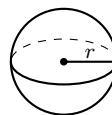
Special Right Triangles



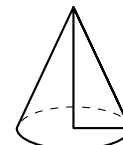
$$V = lwh$$



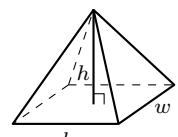
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

$$f(x) = \frac{2}{5}(x - 3) + 1$$

Which of the following tables of values could be a table of values for the function above?

A)

x	3	8	13	18
$f(x)$	1	3	5	7

B)

x	0	5	10	15
$f(x)$	-2	0	2	4

C)

x	-2	0	2	4
$f(x)$	0	5	10	15

D)

x	0	5	10	15
$f(x)$	-5	-3	-1	1

2

In the xy -plane, the line determined by the points $(\alpha, 4)$ and $(9, \alpha)$ passes through the origin. Which of the following could be the value of α ?

A) -6

B) 0

C) $\frac{8}{3}$

D) 6.5

3

For any integer n , which of the following is always 1 more than double an even number?

A) $n + 1$

B) $2n + 1$

C) $4n + 1$

D) $4n + 3$

4

If a 20 pound circular cheese round is sliced through its center into 18° wedges, and then each of those wedges is cut into halves to be individually wrapped for retail, what is the weight, in ounces, of a half wedge? (1 pound = 16 ounces)

A) 4

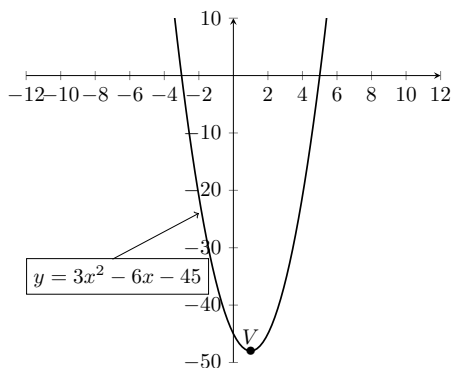
B) 8

C) 9

D) 16



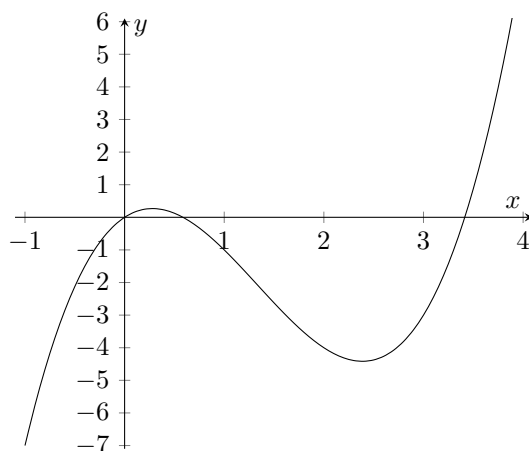
5



Which of the following is an equivalent form of the equation of the graph shown in the xy -plane above, from which the coordinates of vertex V can be identified as constants in the equation?

- A) $3(x + 3)(x - 5)$
- B) $3(x - 3)(x + 5)$
- C) $3x(x - 2) - 45$
- D) $3(x - 1)^2 - 45$

6



The function $g(x) = x^3 - 4x^2 + 2x$ is graphed in the xy -plane above. If n is a constant such that $g(x) = n$ has only 1 solution, which of the following could be the value of n ?

- A) -6
- B) -4
- C) -2
- D) 0



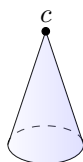
7

What is the non-zero polynomial of least degree that contains $1 - i$ as a root?

- A) $x^2 - 1$
- B) $x^2 - 2x + 2$
- C) None of these.
- D) The answer cannot be determined from the given information.

8

In a right circular cone, suppose that r is the radius of the base. If the diameter of the base and height of the cone have the same length, what is the distance (in terms of r) from C to any point on the circumference of the base?



- A) $r\sqrt{3}$
- B) $r\sqrt{5}$
- C) $2r$
- D) $4r$

9

Consider the following system of equations:

$$\begin{aligned} 2x + 3y &= -4 \\ 6x + 9y &= -12 \end{aligned}$$

How many solutions does this system have?

- A) 0
- B) 1
- C) 4
- D) Infinitely many

10

$$\kappa(x) = -\frac{1}{3}(x+3)(x+9)$$

Which of the following is an equivalent form of the function f above in which the maximum value of κ appears as a constant or coefficient?

- A) $\kappa(x) = -\frac{1}{3}x^2 - 9$
- B) $\kappa(x) = -\frac{1}{3}x^2 - 4x - 9$
- C) $\kappa(x) = -\frac{1}{3}(x+6)^2 + 3$
- D) $\kappa(x) = -\frac{1}{3}(x-6)^2 - 45$



11

A printer can ink 100 pages of newsprint in one minute. At this rate, how many pages of newsprint will be inked in 15 seconds?

- A) Six
- B) Fifteen
- C) Twenty
- D) Twenty-Five

12

What is $\log(\ln(e^{1,000.0}))$?

- A) 3
- B) 2
- C) 3.01
- D) 10

13

Last week Peter worked 8 fewer hours than Dmitri. If they worked a combined total of 76 hours, how many hours did Dmitri work last week?

- A) 34
- B) 38
- C) 42
- D) 68

14

For all x , let the function f be defined as

$$f(x) = -(3x - 1)^3$$

Which of the following is mathematically equivalent to the function above?

- A) $f(x) = -27x^3 - 1$
- B) $f(x) = -27x^3 + 1$
- C) $f(x) = -27x^3 + 27x^2 - 9x + 1$
- D) $f(x) = -27x^3 - 27x^2 + 9x - 1$



15

A quality control manager at a chip factory inspects 3 bags of potatoes for every 100 bags of potatoes brought in before deciding to keep or reject a shipment of potatoes. At this rate, how many bags of potatoes will be inspected if the factory orders 30,000 bags of potatoes?

- A) 300
- B) 900
- C) 2400
- D) 6400

16

In the xy -plane, the graph of function j has x -intercepts at -4 , 2 , and 4 . Which of the following could define j ?

- A) $j(x) = (x - 4)(x - 2)^2$
- B) $j(x) = (x + 2)(x + 4)^2$
- C) $j(x) = (x - 4)(x + 2)(x + 4)$
- D) $j(x) = (x - 4)(x - 2)(x + 4)$

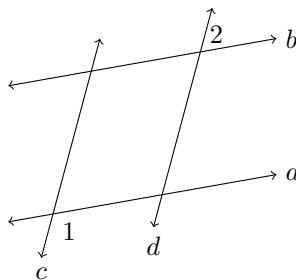
17

$$x^2 + y^2 + 6x - 14y = -22$$

The equation of a circle in the xy -plane is shown above. What is the radius of the circle?

- A) 6
- B) 7
- C) 8
- D) 36

18



In the figure above, lines a and b are parallel to each other, as are lines c and d . If $m\angle 2 = 65^\circ$, what is the measure of $\angle 1$?

- A) 25°
- B) 50°
- C) 65°
- D) 115°



19

Consider the following list of numbers where p represents a real number.

$$0.25p, p/3, -4p, 2p$$

If $p < 0$, which of the numbers above is the least?

- A) $.25p$
- B) $p/3$
- C) $-4p$
- D) $2p$

20

Glacial Force Junior High has to make cuts to its winter athletic budget. To determine which of the sports could be cut with the least backlash, the director arranges to have 50 randomly selected parents ask their students their opinion on several school sports. The data collected is summarized in the two tables below.

28 Boys

	Love	Like	Don't Care	Dislike	Hate
Curling	4	0	9	10	5
Hockey	11	7	4	4	2
Ice Skate Events	1	4	12	6	5
Snow Ski Events	6	3	4	5	10
Snowboard Events	10	5	4	3	6

22 Girls

	Love	Like	Don't Care	Dislike	Hate
Curling	3	2	9	4	4
Hockey	9	1	6	0	6
Ice Skate Events	8	3	2	4	5
Snow Ski Events	3	3	10	2	4
Snowboard Events	4	2	10	3	3

Given Glacial Force Junior High has a population of 271 students, of which 157 are male, which of the following statements would be a reasonable inference based upon the survey data?

- A) People who like snowboarding are a clic on campus just like people who like skiing, and the two clics don't like each other.
- B) Boys and girls tend to think ice skate events are for girls, making for a stereotype that's difficult to dispel.
- C) A large portion of campus doesn't care for or dislikes all of the campus winter sports.
- D) The sample size is too small to make a reliable inference about student preferences.



21

If $x = -3$, which of the following is least in value?

- A) $(x - 1)(x - 2)$.
- B) $(x - 4)(x + 5)$.
- C) $(x - 3)(x + 3)$.
- D) $(x + 1)(x + 5)$.

22

$$I = \frac{S}{4\pi d^2}$$

The intensity of a sound wave at a given location is approximated by the formula above, where I is the intensity, S is the loudness of the sound at its source, and d is the distance from the source of the sound. Bryan is trying out new sound measuring devices he calls Mjollnir and Excalibur. Mjollnir measures the intensity of a sound as 100 times the intensity of the same sound measured by Excalibur. If both machines are working properly, the distance of Mjollnir from the source of the sound should be what fraction of the distance that Excalibur is from the source of the sound?

- A) $\frac{1}{10}$
- B) $\frac{1}{25}$
- C) $\frac{1}{40}$
- D) $\frac{1}{100}$



23

What is $(1, \sqrt{3})$ in polar coordinates?

- A) $(2, 30^\circ)$
- B) $(2, 60^\circ)$
- C) $(3, 30^\circ)$
- D) $(3, 60^\circ)$

24

$$5000e^{3r}$$

The expression above gives the amount of money, in dollars, generated in three years by a \$5,000 deposit in a bank account that pay an annual interest rate of $r\%$ compounded continuously. Which of the following expressions shows how much additional money is generated at an interest rate of 3.7% than at an interest rate of 3.2% ?

- A) $\frac{5000e^{11.1}}{5000e^{9.6}}$
- B) $5000e^{11.1-9.6}$
- C) $5000e^{\frac{11.1}{9.6}}$
- D) $5000e^{11.1} - 5000e^{9.6}$

25

If $\frac{-5}{9}\Delta = \frac{2}{3}$, what is the value of Δ ?

- A) $\frac{-6}{5}$
- B) $\frac{-10}{27}$
- C) $\frac{1}{9}$
- D) $\frac{11}{9}$

26

If $x + 3\sqrt{x} - 4 = 0$, which of the following could be the value of x ?

- A) -16
- B) 1
- C) 1.6
- D) 16



27

In planning maintenance for a county's electrical grid, a utility company estimates that, starting from the present, the population of the county will increase by 3% every 2 years. If the present population of the county is 985,000, which of the following expressions represents the utility company's estimate of the population of the city y years from now?

- A) $985,000(1.03)^{\frac{y}{2}}$
- B) $985,000(1.03)^{2y}$
- C) $985,000(0.97)^{\frac{y}{2}}$
- D) $985,000(0.97)^{2y}$

28

Define $f(a, b) = a^b$. Which of the following must be true for every x, y ?

- A) $f(1, y) = f(0, y)$
- B) $f(\sqrt{x}, y) = f(x, .5y)$
- C) $f(.5x, y) = f(, 2y)$
- D) $f(0, y) = f(x, 0)$

29

Suppose that N is the cost of k CD's at jTones web-store. Which of the following expressions best describes the price of x CD's?

- A) N/x
- B) N/k
- C) kN
- D) None of the above

30

Of the following four situations, which option would yield an exponential decay in the population of cacti in a given area?

- A) Gatherers collect most of the cactus fruit the cacti produce every year during harvest season.
- B) Each successive year, travelers cut down 100 cacti for food and drink.
- C) A burgeoning population of desert rodents cause the death of 1% of the adult cacti each year.
- D) Each successive year, the cactus population nets a loss of 2% of its members due to a currently incurable disease.

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

7	/	1	2
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 2.5

← Decimal point

2	.	5
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

2	/	3
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

2	0	1
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

2	0	1
•	•	•
0	0	0
1	1	1
2	2	2
3	3	3
4	4	4
5	5	5
6	6	6
7	7	7
8	8	8
9	9	9

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.

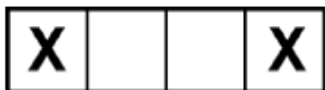


31

$$(8p^2 - 4p + 9) - 3(-2p^2 + p + 5)$$

If the expression above is rewritten in the form $ap^2 + bp + c$, where a , b , and c are constants, what is the value of a ?

32



The figure above represents four offices that will be assigned randomly to four employees, one employee per office. If Pat and Sam are two of the four employees, what is the probability that each will be assigned an office indicated with an X?

33

1 cubic meter	=	1,000 liters
1,000 milliliters	=	1 liter

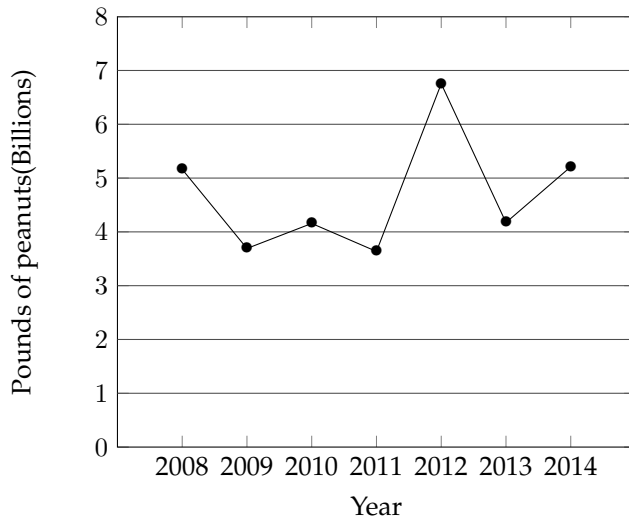
A chemical factory makes deionized water in 10 cubic meter batches, which then are packaged in bottles containing 250 milliliters each. Based on the information given in the box above, how many 250 milliliter bottles are there in one 10 cubic meter batch?

- A) 400,000
- B) 40,000
- C) 4,000
- D) 0.0004



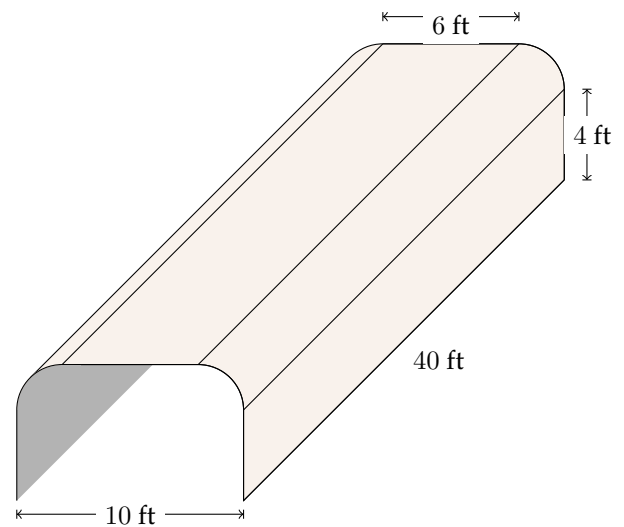
34

U.S. Peanut Production by Year



According to the line graph above, what integer is the best approximation for the rate of change in peanut production per year from 2008 to 2014?

35



Jeremy has a frame in the shape above he wishes to paint on both the inside and the outside. The frame has 90 degree partial cylinder sides connecting three rectangles. The paint he wishes to use has a coverage of about 38 square yards per gallon. If Jeremy can only buy his desired paint in whole gallons, how many gallons should he buy?



36

To the nearest Syrian pound, what does the analyst believe the value of the U.S. dollar will be at the end of four months?

37

The normal diastolic blood pressure D , in millimeters of mercury, for an adult y years old, where y is between 20 and 65, can be modeled by the equation $y = \frac{x+373}{5}$. According to the model, for every increase of 10 years in age, by how many millimeters of mercury will the normal diastolic pressure for a person between the ages of 20 and 65 increase?

38

$$C_m(x) = 10x + 90$$

$$C_n(x) = 15x$$

The functions above represent the cost of visiting a skating rink x times over the course of a year, where $C_m(x)$ represents the cost to members and $C_n(x)$ represents the cost to guests who aren't members. If Wyatt doesn't want to become a member unless it would save him money, what is the minimum number of times Wyatt would have to plan on going to the skating rink this year to convince him to become a member?

Answer Key

Calculator Off	
1	B
2	B
3	C
4	D
5	C
6	B
7	A
8	B
9	A
10	B
11	D
12	B
13	C
14	D
15	A
16	$\frac{4}{3}$
17	0
18	$\frac{5}{6}$ or .833
19	Any integer from 10 to 14 inclusive.
20	2

Calculator On	
1	A
2	A
3	C
4	B
5	D
6	A
7	C
8	B
9	D
10	C
11	D
12	A
13	C
14	C
15	B
16	D
17	A
18	D
19	D
20	C
21	B
22	A
23	B
24	D
25	A
26	B
27	A
28	B
29	D
30	D
31	14
32	1/6
33	B
34	0
35	5
36	219
37	2
38	19



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

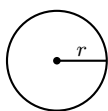
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

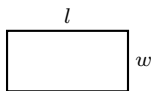
1. The use of a calculator **is not permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

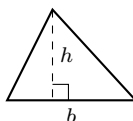


$$A = \pi r^2$$

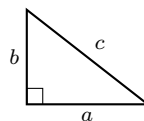
$$C = 2\pi r$$



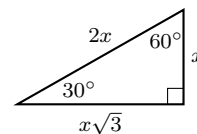
$$A = lw$$



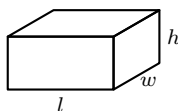
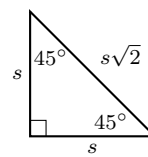
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



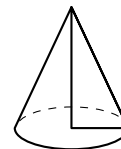
$$V = lwh$$



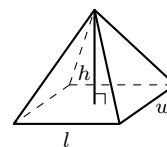
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

If $\Phi(x) = -5x + 11$, what is $\Phi(-3y)$ equal to?

- A) $-15y + 11$
- B) $15y + 11$
- C) $-15y - 11$
- D) $15y - 33$

2

$$\begin{aligned} 3x - 4y &= 1 \\ 2x + y &= -14 \end{aligned}$$

What is the solution (x, y) to the system of equations above?

- A) $(\frac{5}{3}, -\frac{52}{3})$
- B) $(-5, 4)$
- C) $(-5, -4)$
- D) $(\frac{57}{11}, \frac{40}{11})$

3

$$-4(4t + 5)(-t + 4)$$

Which of the following is equivalent to the expression above?

- A) $16t^2 + 44t + 80$
- B) $16t^2 - 80$
- C) $16t^2 - 44t - 80$
- D) $8t^2 - 37t + 20$

4

If

$$\frac{x-5}{2} + \frac{x-5}{3} = -x-1$$

what is the value of x ?

- A) $x = 9$
- B) $x = \frac{19}{18}$
- C) $x = \frac{3}{2}$
- D) $x = 4$



5

The sign below shows the advertised prices for car washes and gasoline at a particular gas station with a touchless car wash service.

Per gallon	CAR WASH \$14.99
87 Regular \$2.06	
89 Premium \$2.29	
91 Supreme \$2.39	Save \$0.12 per gallon with any car wash purchase!

If x represents the number of gallons of gasoline purchased, which of the following expressions represents the cost, in dollars, of getting grade 89 gasoline and a car wash?

- A) $2.06x + 14.99$
- B) $2.17x + 14.99$
- C) $2.29x + 14.99$
- D) $14.87x + 2.29$

6

If

$$\frac{1}{x+1} - \frac{1}{x+2} = \frac{1}{2x+8}$$

which of the following statements must be true?

- A) $x < 0$
- B) $x > 0$
- C) There are two possible values for x .
- D) There are no possible values for x .

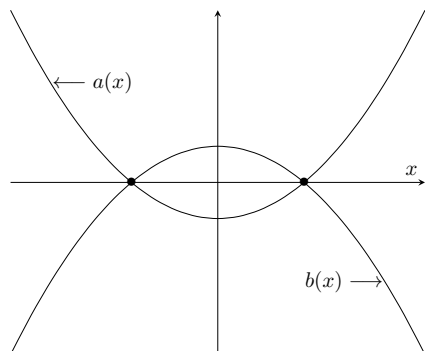
7

Which of the following sets contains a number that is smaller than any other number?

- A) The Rationals
- B) The Integers
- C) The Positive Integers
- D) The Negative Integers



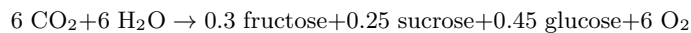
8



The functions p and q are defined by the equations $a(x) = 8x^2 - 2$ and $b(x) = -8x^2 + 2$ and graphed in the xy -plane above. The graphs of a and b intersect at two points as shown above. What is the distance between the two points of intersection?

- A) 2
- B) 1
- C) $\frac{1}{2}$
- D) $\frac{1}{4}$

9



In the chemical reaction above, the requirements to produce 1 fructose are the same as the requirements to produce 1 glucose or 1 sucrose. In this reaction, how much of the H_2O input is reserved for glucose production?

- A) $0.45 \text{ H}_2\text{O}$
- B) $0.55 \text{ H}_2\text{O}$
- C) $2.7 \text{ H}_2\text{O}$
- D) $3.3 \text{ H}_2\text{O}$

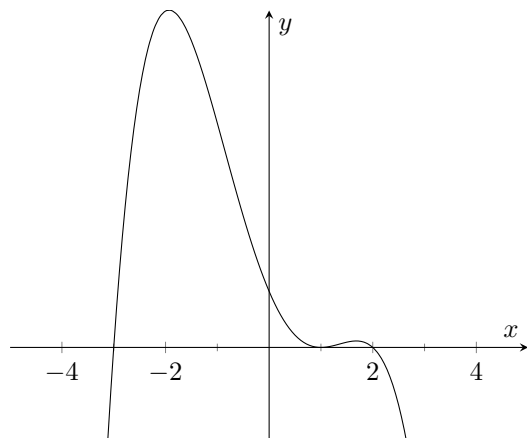
10

The expression $\frac{16p-7}{4p+3}$ is equivalent to which of the following?

- A) $4 - \frac{7}{3}$
- B) $\frac{4+7}{-3}$
- C) $4 - \frac{19}{4p+3}$
- D) $4 + \frac{5}{4p+3}$



11



If the graph above represents a polynomial function $p(x)$ of degree 4, which of the following equations could be the equation of $p(x)$?

- A) $p(x) = -2(x + 3)(x - 2)(x - 1)^2$
- B) $p(x) = -\frac{1}{2}(x + 3)^2(x - 1)(x - 2)$
- C) $p(x) = (x + 3)(x - 2)(x - 1)^2$
- D) $p(x) = \frac{1}{2}(x + 3)^2(x - 1)(x - 2)$

12

Which of the following is equal to $n^{\frac{3}{4}}$, for all values of n ?

- A) $\sqrt[4]{n^3}$
- B) $\sqrt[3]{n^4}$
- C) $\sqrt[3]{n^{\frac{1}{4}}}$
- D) $\sqrt[3]{n^4}$

13

Find all x such that $\sqrt{(x - 4)^2} = 5$.

- A) 5 and 9
- B) -1
- C) -1 and 9
- D) -1, 5 and 9



14

Which of the following functions goes through the origin?

- A) $f(x) = 3x - 1$
- B) $f(x) = |x + 1| - 1$
- C) $f(x) = \frac{1}{x}$
- D) $f(x) = x^2 + 2x + 1$

15

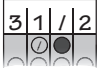
Estelle is using the following equation as a simple model to demonstrate how an object in orbit around a star may have different distances from the star at various points in its orbit. For her model, the star is located at $(0, 1)$. In the model, what is the distance the orbiting object is from the star when the y coordinate of its position is 1?

$$\frac{x^2}{64} + \frac{y^2}{81} = 1$$

- A) $\frac{2528}{81}$
- B) $\frac{27\sqrt{7}}{8}$
- C) $\frac{32\sqrt{5}}{9}$
- D) $\frac{2521}{64}$

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

7	/	1	2
☐	☒	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 2.5

← Decimal point

	2	.	5
☐	☐	☒	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
☐	☒	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	6
☒	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
☒	☐	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

	2	0	1
☐	☒	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

2	0	1	
☐	☒	☐	☐
☐	☐	☐	☐
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

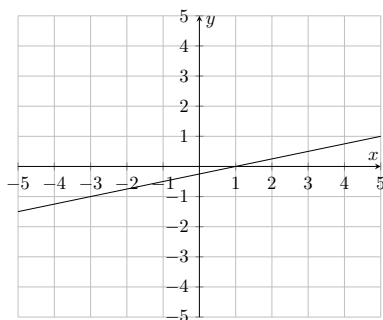
NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



16

On a recent Mars mission, sensors indicated that the temperature of Mars' atmosphere 5.00 miles above the surface was approximately -42.4°F . The temperature 10.00 miles above the Martian surface was approximately -74.7°F . At altitudes above 4.35 miles before reaching open space, for every .5 miles, the temperature of the Martian atmosphere decreases by $f^{\circ}\text{F}$, where f is a constant. What is the value of f ?

17



What is the slope of the line in the graph above?

18

If $t = 3\sqrt{7}$ and $4t = \sqrt{7}q$, what is the value of q ?

19

How many lines go through the points $(a, 0)$, $(b, 0)$ and $(0, c)$ if a, b, c are distinct fixed numbers greater than 0?

20

If $t < 0$ and $t^2 - 2t = 35$, what is $|t|$?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

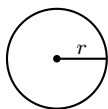
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

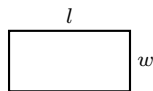
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

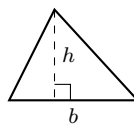


$$A = \pi r^2$$

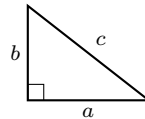
$$C = 2\pi r$$



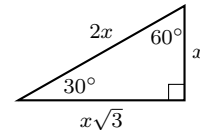
$$A = lw$$



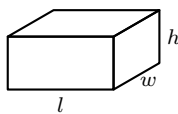
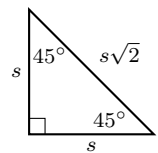
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



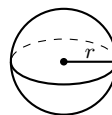
Special Right Triangles



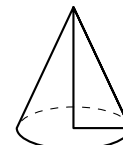
$$V = lwh$$



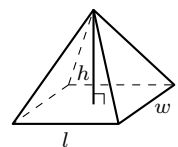
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

A rosticceria sells a dark meat plate with sides for \$8.50 and a white meat plate with sides for 9.25. One day the rosticceria sold 199 plates for \$1760.50. How many white meat plates were sold that day?

- A) 69
- B) 92
- C) 107
- D) 119

2

Two perpendicular lines are defined by the solution set to equation

$$3x - ay = -8$$

and to equation

$$8x + \frac{3}{2}ay = 4.$$

Which of the following could be a possible value of the coefficient a ?

- A) 1
- B) 4
- C) 8
- D) 12

3

Suppose m and n are positive integers and $m + n = mn$. Which of the following could be true?

- I. $n - 1$ is odd.
- II. m and n are odd.
- III. m and n are even.

- A) I and II.
- B) I and III.
- C) II only.
- D) III only.

4

Which of the following is an equation of a circle in the xy -plane with center $(-3, 5)$ and a radius with endpoint $(9, 0)$?

- A) $(x - 3)^2 + (y - 5)^2 = 13$
- B) $(x + 3)^2 + (y - 5)^2 = 13$
- C) $(x - 3)^2 + (y - 5)^2 = 169$
- D) $(x + 3)^2 + (y - 5)^2 = 169$



5

$$\begin{aligned} y &\leq tx \\ y &\geq ux \end{aligned}$$

In the xy -plane, if $(-4, 0)$ is a solution to the system of inequalities above, which of the following relationships between t and u must be false?

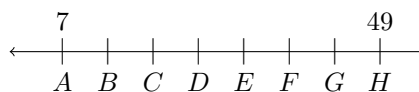
- A) $t = u$
- B) $t > u$
- C) $t < u$
- D) $ut \leq 0$

6

Find all x such that $x^2 + x - 12 < 0$.

- A) $-1 < x < 1$
- B) $x < -4$ or $x > 3$
- C) $-4 < x < 3$
- D) $x < -2$ or $x > 3$

7



On the number line above, all the intervals between points are the same. Point A is at 7 and point H is at 49. What is the coordinate of C ?

- A) 6
- B) 12
- C) 13
- D) 19

8

If $x - 9 = \sqrt{x + 3}$, what are the possible values for x ?

- A) $x \in \emptyset$
- B) $x \in \{6\}$
- C) $x \in \{6, 13\}$
- D) $x \in \{13, 4\}$



9

The tables below give the distribution of low temperatures in degrees Celsius ($^{\circ}\text{C}$) for a South Dakota location and a Montana location over February of 2010.

South Dakota Location

Temperature ($^{\circ}\text{C}$)	Frequency
-16.1 to -8.1	5
-8.0 to 0.0	5
0.1 to 8.0	5
8.1 to 16.0	9
16.1 to 24.0	1
24.1 to 27.0	3

Montana Location

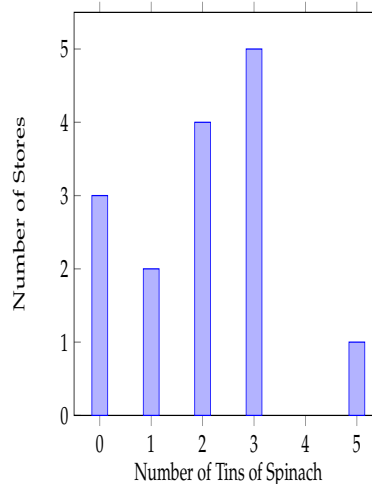
Temperature ($^{\circ}\text{C}$)	Frequency
1.0 to 5.9	1
6.0 to 10.9	4
11.0 to 15.9	3
16.0 to 20.9	7
21.0 to 25.9	9
26.0 to 30.9	4

Which of the following is true about the data shown for the month of February?

- A) The standard deviation of low temperatures at the South Dakota location is larger.
- B) The standard deviation of low temperatures at the Montana location is larger.
- C) The standard deviation of low temperatures at both locations is about the same.
- D) The data provided is too vague to get any basis for comparing the standard deviation of low temperatures between the two locations.

10

Number of Tins of Spinach in Each of 15 Dungeon Stores



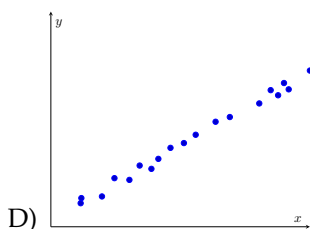
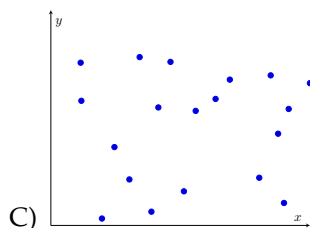
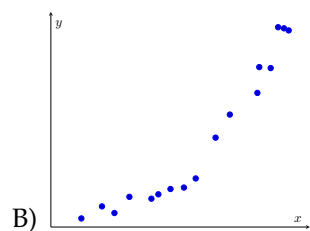
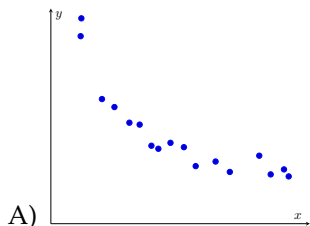
Based on the histogram above, which of the following are equivalent with regards to tins of spinach in a store?

- A) Arithmetic mean and median
- B) Arithmetic mean and mode
- C) Median and mode
- D) Arithmetic mean, median and mode



11

Which of the following scatterplots shows a relationship that is appropriately modeled with the equation $y = ax^b$, where a is positive and b is greater than 1?



12

$$y = 2x^2 - x - 6$$

$$y = -x^2 + 2x + 12$$

What is/are the solution(s) to the system above?

- A) $(x, y) \in \{(-2, 3)\}$
- B) $(x, y) \in \{(-2, 4)\}$
- C) $(x, y) \in \{(3, 9)\}$
- D) $(x, y) \in \{(-2, 3), (3, 9)\}$

13

What fraction lies exactly halfway between $\frac{1}{7}$ and $\frac{1}{5}$?

- A) $\frac{1}{6}$
- B) $\frac{6}{35}$
- C) $\frac{12}{35}$
- D) $\frac{1}{4}$



14

Let x represent points scored and y represent points allowed. Given this information, which of the following statements is clearly TRUE?

- A) There is a negative correlation between points scored and points allowed.
- B) The Supersonics lost all the games they allowed more than 100 points.
- C) $y = \frac{2}{3}x + 35$ is close to a line of best fit for the data.
- D) The Supersonics never lost by more than 20 points.

15

The graph of the linear function g has intercepts at $(x_0, 0)$ and $(0, y_0)$ in the xy -plane. If $x_0 y_0 = x_0^2$ and $y_0 \neq 0$, which of the following is true about the slope of the graph of g ?

- A) It is negative.
- B) It is zero.
- C) It is positive.
- D) It could be positive or negative.

16

The difference between ϕ and ζ is 7. The difference between ϕ squared and ζ squared is 21. What is $\phi^2 + 2\phi\zeta + \zeta^2$?

- A) 1
- B) 9
- C) 35
- D) 49

17

Evaluate $x^2 - 4xy^2 + 4y^4$ when $x = 17$ and $y = 2$.

- A) $12\sqrt{3}$
- B) 17
- C) 81
- D) 9



18

$$\begin{aligned}Ax + By &= C \\ y &= \frac{2}{3}x + 5\end{aligned}$$

The system of equations above has only one solution. Which of the following could not be $Ax + By = C$?

- A) $2x - 3y = 10$
- B) $2x + 3y = 7$
- C) $3x - 2y = 11$
- D) $x - 5y = 2$

19

At Cesar Chavez High School, approximately 16% of the sophomores and 9% of the juniors tried out for the school soccer teams last year. If there were 437 sophomores and 398 juniors enrolled at Cesar Chavez High School last year, which of the following is closest to the total number of juniors and seniors at Cesar Chavez High School last year who tried out for soccer?

- A) 36
- B) 46
- C) 70
- D) 106



20

Which of the following is a reasonable inference to make judging from the data collected?

- A) Randomness assigned people with less stamina to Group 1.
- B) A large number of the students in group 2 tried to save energy so that they could run well in their second mile.
- C) Someone believing he or she has taken an energy drink likely has a positive effect on his or her running speed.
- D) The differences between the performance of the first group and second group are statistically insignificant.

21

$$y = x^2 - 4$$

$$y = 6x - 9$$

What are the solutions to the system above?

- A) $(1, -3)$
- B) $(1, 5)$
- C) $(3 + \sqrt{22}, 3 - \sqrt{22})$
- D) $(1, -3)$ and $(5, 21)$

22

Suppose $\sin(90^\circ - x) = \frac{b}{a}$. Then what is $\cos(x)$?

- A) $\frac{b}{a}$
- B) $\frac{a}{a^2 + b^2}$
- C) $\frac{a}{\sqrt{a^2 + b^2}}$
- D) 1

23

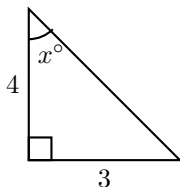
Which of the following inequalities yields all b such that $2x^2 + b + 18$ has no real roots?

- A) $b < 6$
- B) $b > 12$
- C) $b < 12$
- D) $-12 < b < 12$



24

In the triangle below, which of the following is the greatest?



- A) $\sin(x)$
- B) 0
- C) $\tan(x)$
- D) 1

25

$$v(x) = 2x^2 + x - 6$$

$$w(x) = 6x^3 + 3x^2 - 18x$$

The polynomials $v(x)$ and $w(x)$ are defined above. Which of the following polynomials is divisible by $3x - 2$?

- A) $a(x) = w(x) - 2v(x)$
- B) $b(x) = w(x) + 2v(x)$
- C) $c(x) = 3w(x) - v(x)$
- D) $d(x) = 3w(x) - 2v(x)$

26

Translate the following verbal mathematical expression into their equivalent symbolic counterparts. The product of three numbers is five times their sum.

- A) $xy = 2(x - y)$
- B) $xyz = 5(x + y + z)$
- C) None of these.
- D) The answer cannot be determined from the information given.



27

In a circle, diameter $\overline{AB}$ bisects chord $\overline{CD}$ at M . If the length of $\overline{AB}$ is 10 and the length of $\overline{CD}$ is 8, then what is the length of $\overline{MB}$?

- A) 5
- B) 4
- C) 2
- D) Cannot be determined

28

For positive real numbers a, b, c and d , we know that a percent of b percent is the same as c percent of d . What is d in terms of a, b and c ?

- A) $\frac{ab}{100c}$
- B) $\frac{ab}{1000c}$
- C) $\frac{ab}{10000c}$
- D) $\frac{ac}{10000b}$

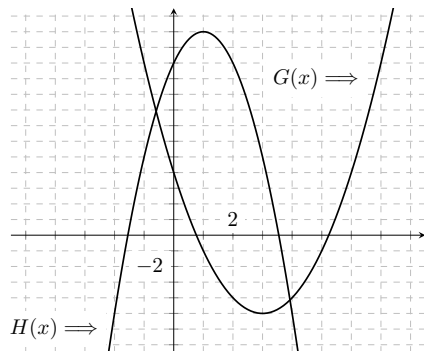
29

If $\log_2(13) = x$ and $\log_2(26) = y$, what is $x - y$?

- A) -1
- B) 2
- C) 2.5
- D) 5.9



30



Graphs of the functions $G(x)$ and $H(x)$ are shown in the xy -plane above. For which of the following values of x does $G(x) + H(x) = 0$?

- A) 4
- B) 3
- C) 2
- D) 1

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.

5. **Mixed numbers** such as $3\frac{1}{2}$ must be gridded

as 3.5 or $7/2$. (If $\begin{array}{|c|c|c|c|} \hline 3 & 1 & / & 2 \\ \hline \bullet & \bullet & \bullet & \bullet \\ \hline \end{array}$ is entered into the

grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)

6. **Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Answer: $\frac{7}{12}$

Write answer in boxes. →

7	/	1	2
•	•	•	•
0	0	0	0
1	1	•	1
2	2	2	•
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
•	7	7	7
8	8	8	8
9	9	9	9

← Fraction line

Grid in result.

Answer: 2.5

	2	.	5
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	3
4	4	4	4
5	5	5	•
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

← Decimal point

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	•
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	7
8	8	8	8
9	9	9	9

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	•
8	8	8	8
9	9	9	9

Answer: 201 – either position is correct

	2	0	1
•	•	•	•
0	•	0	0
1	1	1	•
2	•	2	2
3	3	3	3

2	0	1	
•	•	•	•
•	0	0	0
1	1	•	1
•	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

$$\begin{aligned}y &\geq 5x - 465 \\y &\leq -3x + 1879\end{aligned}$$

In the xy -plane, if a point with coordinates (α, β) lies in the solution set of the system of inequalities above, what is the maximum possible value of α ?

32

Bonnie can make at least 20 marzipan flowers an hour, and at most 36 marzipan flowers an hour. Based upon this information, what is a possible amount of time, in hours, that it could take Bonnie to make 180 marzipan flowers?

33

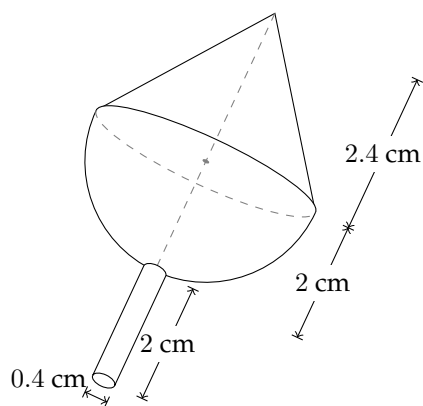
A sporting event is nearly sold out, and the only remaining are the corner seats that cost \$55 near the away team's reserved seats and the adjacent endzone seats that cost \$70. If George spends at least \$225 but no more than \$400 on n endzone tickets and 2 corner seats, what is one possible value of n ?

34

In a geometric series consisting of only positive terms, the second term is 6 and the fourth term is 54. What is the first term?



35



Note: Diagram is not to scale.

An aspiring fast food entrepreneur is designing a top for his first kid's meal toy. The top is comprised of a cylindrical handle which connects to a hemisphere that shares its base with a right cone. He plans to make the top using the dimensions as marked in the diagram. His materials cost will be \$0.14 per cubic centimeter. He wanted to make the tops solid until he discovered how much that would cost per toy, so instead he's hollowed out 80% of the material from the design. How much, in dollars, will it cost him to make each toy?

36

Dan wants to examine why it seems he can't slow down when ice skating. He finds that Sally, a figure skater that weighs $540N$, only generates about $16N$ of friction when she's on the ice in her skates. The friction coefficient for steel on ice is approximately what percent of the friction coefficient for leather on steel? (Note: Ignore the percent symbol when entering your answer. For example, if the answer is 30.9%, enter 30.9)

37

If $\frac{2x^2+7x-1}{x-2} = Ax + B + \frac{C}{x-2}$, find $A + B + C$.



38

Practice #1	
Points Scored	Number of Possessions
2	9
3	11
4	3
5	1
6	0
7	1

Practice #2	
Points Scored	Number of Possessions
2	6
3	14
4	5
5	2
6	2
7	1

Coach B., in an effort to improve his team's ball movement and ball security, has his team play a game he calls "Dimeball." The charts above represent the points scored on made baskets in "Dimeball" over two practices. What was the average number of points made on a scoring possession by Coach B.'s players over the course of the two "Dimeball" games?

Answer Key

Calculator Off	
1	B
2	C
3	C
4	D
5	B
6	C
7	C
8	C
9	C
10	C
11	A
12	C
13	C
14	B
15	C
16	3.23
17	$\frac{1}{4}$ or .25
18	144
19	0
20	7

Calculator On	
1	B
2	B
3	B
4	D
5	B
6	C
7	D
8	A
9	A
10	A
11	B
12	D
13	B
14	C
15	A
16	B
17	C
18	A
19	D
20	D
21	B
22	A
23	D
24	D
25	A
26	B
27	C
28	A
29	A
30	B
31	293
32	Some number between 5 and 9 hours inclusive.
33	2 or 3 or 4
34	2
35	.758
36	7.41
37	34
38	3.24



Math Test – No Calculator

25 MINUTES, 20 QUESTIONS

Turn to Section 3 of your answer sheet to answer the questions in this section.

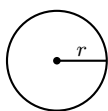
DIRECTIONS

For questions 1-15, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 16-20, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 16 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

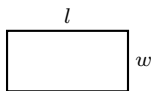
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5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

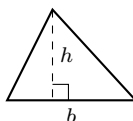


$$A = \pi r^2$$

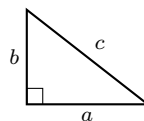
$$C = 2\pi r$$



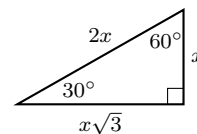
$$A = lw$$



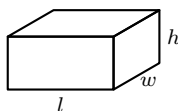
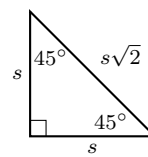
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



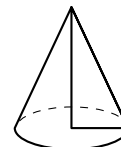
$$V = lwh$$



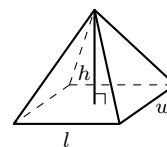
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

$$y = \frac{1}{2}(x+8)(x+2) + 6$$

Which of the following mathematically identical equations to the equation above states the coordinates for the vertex of the parabola explicitly as constants in the equation?

- A) $y = \frac{1}{2}(x+5+i\sqrt{3})(x+5-i\sqrt{3})$
- B) $y = \frac{1}{2}(x+5)^2 + \frac{3}{2}$
- C) $y = \frac{1}{2}[(x+5)^2 + 3]$
- D) $y = \frac{1}{2}x^2 + 5x + 14$

2

$$x^2 - 2\Theta x = \frac{-1}{9}\Omega$$

In the quadratic equation above, β and κ are constants. What are the solutions for x ?

- A) $x \in \{\Theta \pm \frac{\sqrt{4\Theta^2 - \Omega}}{2}\}$
- B) $x \in \{\Theta \pm \frac{\sqrt{9\Theta^2 - \Omega}}{3}\}$
- C) $x \in \{\Theta \pm \frac{\sqrt{9 - 5\Omega}}{3}\}$
- D) $x \in \emptyset$

3

Suppose a and b are both non-zero rational numbers. Which must always be rational?

- A) ab
- B) $\frac{a}{b}$
- C) $\frac{b}{a}$
- D) All of the above

4

The equation $\frac{6x^2 - 7x - 3}{nx + 7} = 2x - 7 + \frac{46}{nx + 7}$ is true for all values of $x \neq \frac{-7}{n}$, where n is a constant. What is the value of n ?

- A) 3
- B) 2
- C) $\frac{1}{3}$
- D) -3



5

$$\begin{aligned}x^2 + y^2 &= 121 \\ y &= k|x|\end{aligned}$$

If k is a real constant, which of the following statements must be false about the solutions to the system?

- A) The x values of the solutions are opposites of one another.
- B) The y values of the solutions are equal.
- C) There are two solutions to the system.
- D) $x < 11$ for all solutions to the system.

6

$$\frac{\zeta + \frac{1}{5}}{\zeta - \frac{1}{5}} = 2$$

If the equation above is true, what is the value of ζ ?

- A) $1\frac{-1}{5}$
- B) $\frac{1}{5}$
- C) $\frac{3}{5}$
- D) $\frac{4}{5}$

7

If $\sin \theta = 0.6$, what is the value of $\frac{2 \cos \theta}{\sin \theta}$?

- A) 6
- B) $\frac{25\sqrt{34}}{51}$
- C) $\frac{4}{3}$
- D) $\frac{8}{3}$

8

$$f_h = \left(\frac{c}{c - v} \right) f_e$$

The formula above gives the frequency heard (f_h) at a stationary point from a moving object given the velocity of sound in the pertinent medium (c), the speed the object is moving towards the listener relative to the medium (v), and the actual frequency of the sound at the source (f_e). Which of the following is a correct formula for v ?

- A) $v = c \left(1 - \frac{f_e}{f_h} \right)$
- B) $v = c(f_h - f_e)$
- C) $v = f_e - f_h$
- D) $v = 1 - \frac{f_e}{f_h}$



9

A square's area in square feet has the same numeric value as its perimeter in inches. What is the length of a side of the square, in feet?

- A) 48
- B) 12
- C) 3
- D) $\frac{1}{36}$

10

Which of the following quadratics cannot be written as a product of binomials with integer coefficients?

- A) $x^2 + 4$
- B) $x^2 - 3x + 2$
- C) $x^2 + 3x + 2$
- D) $9x^2 - 25$

11

$$(j + k)^2 - (k - j)^2$$

- A) 0
- B) $2j^2$
- C) $4jk$
- D) $2j^2 + 2k^2$

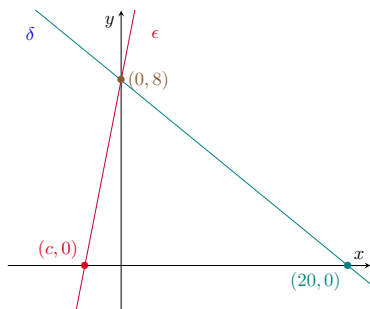
12

Twice as many countries adopted the Euro from 1999 to 2001 as had adopted the currency from 2002 to 2014, with no countries deciding to leave the currency. If n countries adopted the currency from 1999 to 2001, which of the following expressions represents the total number of countries that were using the Euro in 2014?

- A) $\frac{1}{2}n$
- B) $\frac{3}{2}n$
- C) $2n$
- D) $3n$



13



In the xy -plane above, line δ is perpendicular to line ϵ when the x and y axes are using the same scale. In the diagram the x and y axes are scaled differently. What is the value of c ?

- A) $-\frac{2}{5}$
- B) $-\frac{16}{5}$
- C) $-\frac{5}{2}$
- D) $\frac{5}{2}$

14

Bob comes up with his own strength indicator for evaluating offensive guards based upon their NFL Scouting Combine results in the bench press and the broad jump. He lets p represent $\frac{1}{20}$ of their bench press score and j represent $\frac{1}{90}$ of their broad jump score. He determines the following inequalities all hold true for all the offensive guard draft picks he had combine data for:

$$\begin{aligned} j &\geq p + 0.9 \\ p &\geq 1.05 \\ j &\geq 0.9 \\ p + j &\leq 3.1 \end{aligned}$$

Based upon this information, what is the largest broad jump score an offensive guard could have earned in Bob's data set?

- A) 279
- B) 184.5
- C) 2.05
- D) 1.95



15

On Friday evening, Lulu posted an average of u times on social media each hour for 4 hours, and D.J. posted an average of v times on social media each hour for 7 hours. Which of the following represents the average number of social media posts put online Friday evening between the two people?

- A) $\frac{7u+4v}{2}$
- B) $\frac{4u+7v}{2}$
- C) $\frac{11}{2}uv$
- D) $11uv$

**DIRECTIONS**

For questions 16–20, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

7	/	1	2
•	•	•	•
0	0	0	0
1	1	•	1
2	2	2	•
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
•	7	7	7
8	8	8	8
9	9	9	9

Answer: 2.5

← Decimal point

	2	.	5
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	3
4	4	4	4
5	5	5	•
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
•	•	•	•
0	0	0	0
1	1	1	1
2	•	2	2
3	3	3	•
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	7
8	8	8	8

.	6	6	7
•	•	•	•
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	•	•	•
7	7	7	•
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
•	•	•	•
0	•	0	0
1	1	1	•
2	•	2	2
3	3	3	3

	2	0	1
•	•	•	•
0	•	0	0
1	1	•	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.

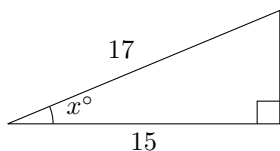


16

$$(4x - 15)(x - 1) = (x - 3)(x - 2) - 2$$

Find a value of x that is a solution of the equation above.

17



In the diagram above, what is $\tan(90 - x)^\circ$?

18

$$8y(2y - 1) - 3(4y - 1) = ay^2 + by + c$$

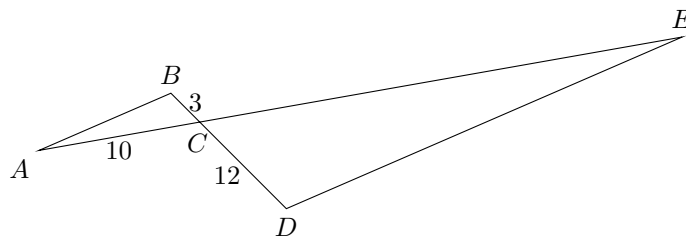
In the equation above, a , b and c are constants. If the equation above is true for all values of y , what is the value of $|b|$?

19

$$\omega^5(\omega^2 - 82) = -81\omega^2$$

If $\omega > 0$, what is one possible solution to the equation above?

20



In the figure above, $\overline{AB} \parallel \overline{DE}$ and C is on both $\overline{AE}$ and $\overline{BD}$. What is CE ?



Math Test – Calculator

55 MINUTES, 38 QUESTIONS

Turn to Section 4 of your answer sheet to answer the questions in this section.

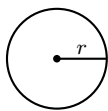
DIRECTIONS

For questions 1-30, solve each problem, choose the best answer from the choices provided, and fill in the corresponding circle on your answer sheet. For questions 31-38, solve the problem and enter your answer in the grid on the answer sheet. Please refer to the directions before question 31 on how to enter your answers in the grid. You may use any available space in your test booklet for scratch work.

NOTES

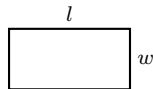
1. The use of a calculator **is permitted**.
2. All variables and expressions used represent real numbers unless otherwise indicated.
3. Figures provided in this test are drawn to scale unless otherwise indicated.
4. All figures lie in a plane unless otherwise indicated.
5. Unless otherwise indicated, the domain of a given function f is the set of all real numbers x for which $f(x)$ is a real number.

REFERENCE

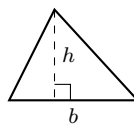


$$A = \pi r^2$$

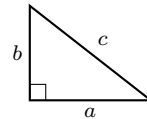
$$C = 2\pi r$$



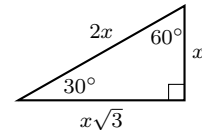
$$A = lw$$



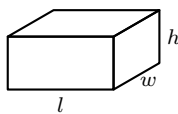
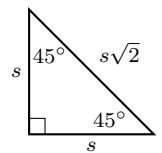
$$A = \frac{1}{2}bh$$



$$c^2 = a^2 + b^2$$



Special Right Triangles



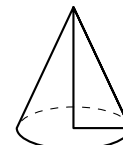
$$V = lwh$$



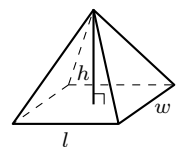
$$V = \pi r^2 h$$



$$V = \frac{4}{3}\pi r^3$$



$$V = \frac{1}{3}\pi r^2 h$$



$$V = \frac{1}{3}lwh$$

The number of degrees of arc in a circle is 360.

The number of radians of arc in a circle is 2π .

The sum of measures in degrees of the angles of a triangle is 180.



1

2, 3, 6, n , 42, ...

The first term of the sequence above is 2, and each term after the first is 3 less than 3 times the previous term. What is the value of n ?

- A) 9
- B) 12
- C) 15
- D) 21

2

Suppose the quadrilateral $ABCD$ has sides that are integers and $AD = 5$ and $BC = 3$. If the diagonal $\overline{AC}$ has a length of 4, what is the greatest possible perimeter that $ABCD$ can have?

- A) 19
- B) 21
- C) 22
- D) 30

3

If $\frac{x}{4} = 4 - \sqrt{x-4}$, what are the possible values for x ?

- A) $x \in \{8\}$
- B) $x \in \{40\}$
- C) $x \in \{8, 40\}$
- D) $x \in \{24 \pm 8\sqrt{6}\}$

4

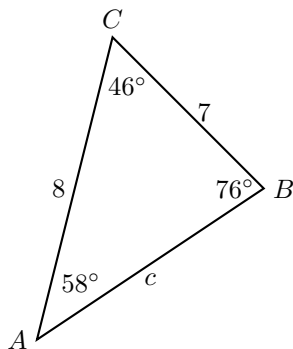
When is $\frac{\sqrt{4x-4}}{x-1}$ undefined?

- A) $x < 4$
- B) $x \leq 1$
- C) $x \geq 4$
- D) $x = 0$



5

Consider the figure below:



Which equation holds true?

(Note: For any triangle with sides a, b, c that are opposite angles A, B, C respectively, $\frac{\sin(A)}{a} = \frac{\sin(B)}{b} = \frac{\sin(C)}{c}$.)

- A) $\frac{\sin(58^\circ)}{7} = \frac{\sin(46^\circ)}{c}$
- B) $\frac{\sin(46^\circ)}{7} = \frac{\sin(58^\circ)}{c}$
- C) $\frac{\sin(76^\circ)}{8} = \frac{\sin(46^\circ)}{c}$
- D) $\frac{\sin(76^\circ)}{6} = \frac{\sin(58^\circ)}{c}$

6

Two weeks ago, buying a shirt from Store X cost the same as a shirt from Store Y. In Store X, the price of a shirt increased by 30% and then decreased by 40% the following week. In Store Y, the price of a shirt decreased by 40% and then increased by 30% the following week. Which of the following must be true (at the present time)?

- A) A shirt from Store X is more expensive
- B) A shirt from Store Y is more expensive
- C) A shirt from Store X costs the same as a shirt from Store Y
- D) If the original cost of a shirt is less than \$1, then a shirt from Store X is more expensive.



7

Weights of American Male Voters

138	146	153	160	168	169
174	174	184	184	191	204
209	214	215	231	243	374

The table above lists the weights of a random sample of 18 American male voters. The 374-pound gentleman is an outlier. Of the arithmetic mean, median, and range of the values listed, which would change the most if the 374-pound gentleman were removed from the data?

- A) Arithmetic mean
- B) Median
- C) Range
- D) Removing an outlier doesn't change measures of central tendency.

8

Which of the following numbers is NOT a solution of the inequality $t^2 - 5 \geq 4t$?

- A) -1
- B) 2
- C) 5
- D) 8

9

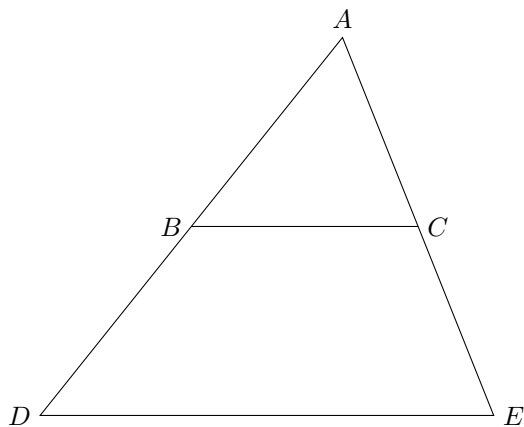
Which of the following lists all of the positive factors of 81?

- A) 1, 81
- B) 1, 9, 81
- C) 1, 3, 27
- D) 1, 3, 9, 27, 81



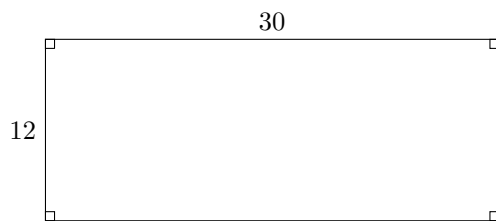
10

Let $\overline{BC}$ be parallel to $\overline{DE}$. Moreover, $BC = 5$ and $DE = 10$. Then what is the ratio of AB to BD ?



- A) 1
- B) 2
- C) 3
- D) 4

11



The above piece of metal is going to be rolled into the lateral surface of a can. Which of the following could be the radius of the can?

- A) $\frac{2\sqrt{3\pi}}{\pi}$
- B) $\frac{12}{\pi}$
- C) $\frac{15}{\pi}$
- D) $\frac{18}{\pi}$

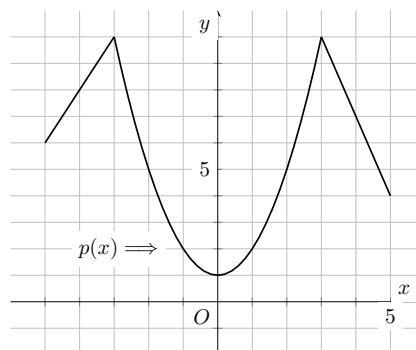


12

An engineering firm estimates that a project will cost d dollars to complete, where $d > 2,000,000$. The goal is for the actual cost to be within 10% of the estimated cost. If the firm meets the goal and takes a to complete the project, which of the following inequalities represents the relationship between the estimated cost and the actual cost?

- A) $d < \frac{9a}{10}$
- B) $a < \frac{9d}{10}$
- C) $-\frac{a}{10} < d - a < \frac{a}{10}$
- D) $-\frac{d}{10} < d - a < \frac{d}{10}$

13



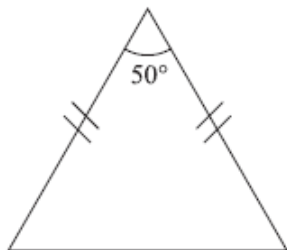
The complete graph of the function p is shown in the xy -plane above. Which of the following are equal to 5?

- I. $p(-2)$
 - II. $p(2)$
 - III. $p(4)$
- A) I only.
 - B) II only.
 - C) III only.
 - D) I and II only.



14

The isosceles triangle below has one angle measure as shown. What is the measure of each of the other angles?



- A) 30°
- B) 45°
- C) 50°
- D) 65°

15

For all $x \leq -2$, let the function g be defined as $g(x) = \sqrt{-6 - 3x}$. Which of the following correctly represents the inverse function $g^{-1}(x)$?

- A) $g^{-1}(x) = \frac{x-6}{3}, x \geq 0$
- B) $g^{-1}(x) = -\sqrt{-6 - 3x}, x \leq -2$
- C) $g^{-1}(x) = \frac{x^2+6}{-3}, x \geq 0$
- D) $g^{-1}(x) = \frac{x^2-6}{3}, x \geq 0$



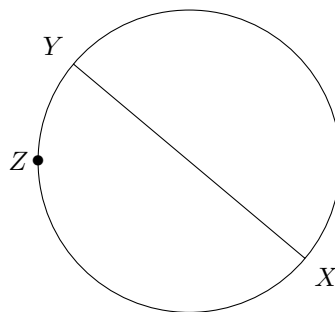
16

$$h_t = -\frac{1}{2}gt^2 + v_0t + h_0$$

The equation above gives the height, h_t , in meters, of a ball t seconds after it is thrown with an initial vertical speed (positive going up and negative going down) of v_0 meters per second from a starting height of h_0 meters from the surface of a planet or asteroid. g represents the acceleration due to gravity on the planet or asteroid. Which of the following gives g in terms of v_0 , h_0 , and h_t ?

- A) $g = \frac{-2h_t - h_0 - v_0t}{t^2}$
- B) $g = \frac{-2h_t - 2h_0 - 2v_0t}{t^2}$
- C) $g = \frac{-2h_t + 2h_0 + 2v_0t}{t^2}$
- D) $g = \sqrt{\frac{2h_0 - 2h_t}{t}} + 2v_0$

17



In the diagram above, $\overline{XY}$ passes through the center of the circle. If the length of $\widehat{XZY}$ is 24π , what is the length of the radius of the circle?

- A) 6
- B) 12
- C) 24
- D) 48

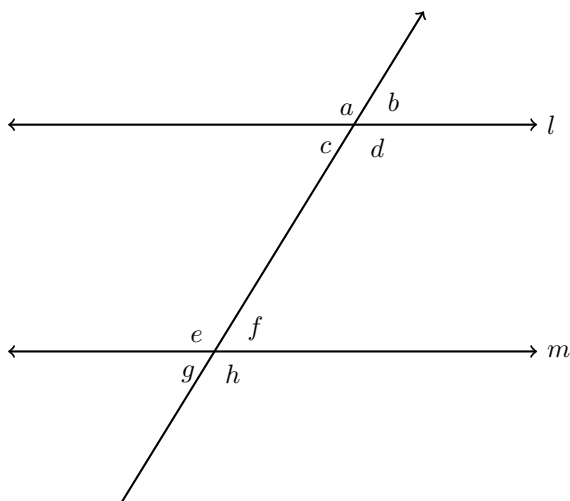
18

What does the solution set of $x + \pi y > 100$ look like?

- A) The empty set
- B) A half plane
- C) The inside of a circle
- D) The outside of a circle



19



Which of the following lists the interior angles where $l \parallel m$?

- A) a, b, e, f
- B) e, f, g, h
- C) b, d, f, h
- D) c, d, e, f

20

A function κ satisfies $\kappa(1) = 4$ and $\kappa(8) = 3$. A function λ satisfies $\lambda(8) = 1$ and $\lambda(11) = 8$. What is the value of $\kappa(\lambda(\lambda(11)))$?

- A) 1
- B) 3
- C) 4
- D) 8

21

The speed of a sound wave in dry air may be modeled by

$$v_s = 20.05\sqrt{t}$$

where v_s is the speed of the sound in meters per second and t is the temperature of the air in degrees Kelvin. Which of the following best expresses the air temperature in terms of the speed of the sound wave?

- A) $t = \frac{v_s^2}{402.0}$
- B) $t = \frac{v_s^2}{20.05}$
- C) $t = 402.0v_s^2$
- D) $t = \frac{\sqrt{v_s}}{20.05}$



22

If $\log_x(81) = 4$, then x is closest to which of the following values?

- A) 2
- B) 4
- C) 1.19
- D) 3.001

23

What is the degree of the minimum non-zero polynomial with real coefficients that contains $1 - i$ and $3 + \sqrt{2}\pi i$ as roots?

- A) 2
- B) 3
- C) 4
- D) 5

24

$$4a + u = 7a + 11$$

$$4b + v = 7b + 11$$

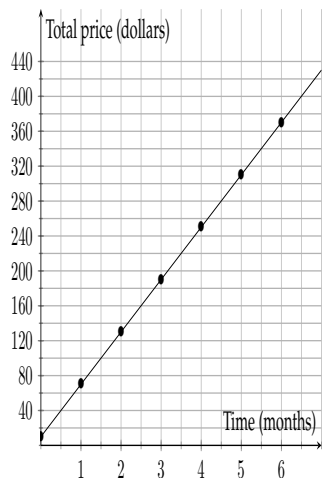
In the equations above, u and v are constants. If u is $\frac{5}{6}$ plus v , which of the following is true?

- A) $a = b + \frac{5}{6}$
- B) $a = b + \frac{5}{18}$
- C) $a = b - \frac{5}{6}$
- D) $a = b - \frac{5}{2}$



25

Total Cost of Belonging to an Organic Cooperative by the Month



The graph above displays the total price p paid by a customer to belong to an organic cooperative for m months. Which of the following represents the relationship between p and m ?

- A) $p = \frac{3}{2}m + 1$
- B) $p = 10m + 60$
- C) $p = 60m + 10$
- D) $p = 70m$

26

If n is a prime number, then $6n$ could not be

- A) An even number.
- B) A multiple of 3.
- C) A perfect square.
- D) A whole number.

27

The set of all real number x satisfying $\sqrt{x^2} = |x|$ is

- A) All real numbers
- B) All negative reals
- C) All non-positive reals
- D) All non-negative reals



28

Translate the following verbal mathematical expression into their equivalent symbolic counterparts.

The product of two numbers is twice the difference between them.

- A) $x + y = (1/2)(x - y)$
- B) $xy = 2(x - y)$
- C) $xyz = 5(x + y + z)$
- D) None of these.

29

x	1	3	5	7	9
$g(x)$	5	9	13	17	21

The table above shows some values of the linear function f . Which of the following defines f ?

- A) $f(x) = -x + 6$
- B) $f(x) = (x - 2)(x - 6)$
- C) $f(x) = 2x + 3$
- D) $f(x) = 2(x - 3)^2 + 9$

30

In an election, 3.6 million votes were cast. Everyone voted for either Xavier, Javier, or Zavir. Xavier received 72,000 fewer votes than Javier and Zavir combined. What percent of the 3.6 million votes were cast for Xavier?

- A) 48%
- B) 49%
- C) 49.5%
- D) 50.5%

**DIRECTIONS**

For questions 31-38, solve the problem and enter your answer in the grid, as described below, on the answer sheet.

- Although not required, it is suggested that you write your answer in the boxes at the top of the columns to help you fill in the circles accurately. You will receive credit only if the circles are filled in correctly.
- Mark no more than one circle in any column.
- No question has a negative answer.
- Some problems may have more than one correct answer. In such cases, grid only one answer.
- Mixed numbers** such as $3\frac{1}{2}$ must be gridded as 3.5 or 7/2. (If  is entered into the grid, it will be interpreted as $\frac{31}{2}$, not $3\frac{1}{2}$.)
- Decimal answers:** If you obtain a decimal answer with more digits than the grid can accommodate, it may be either rounded or truncated, but it must fill the entire grid.

Write answer in boxes. →

Answer: $\frac{7}{12}$

← Fraction line

Grid in result.

	7	/	1	2
.	.	.	.	.
	0	0	0	0
1	1	1	1	1
2	2	2	2	2
3	3	3	3	3
4	4	4	4	4
5	5	5	5	5
6	6	6	6	6
7	7	7	7	7
8	8	8	8	8
9	9	9	9	9

Answer: 2.5

← Decimal point

	2	.	5
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Acceptable ways to grid $\frac{2}{3}$ are:

	2	/	3
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	6
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

.	6	6	7
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8

Answer: 201 – either position is correct

	2	0	1
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

	2	0	1
.	.	.	.
	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3

NOTE: You may start your answers in any column, space permitting. Columns you don't need to use should be left blank.



31

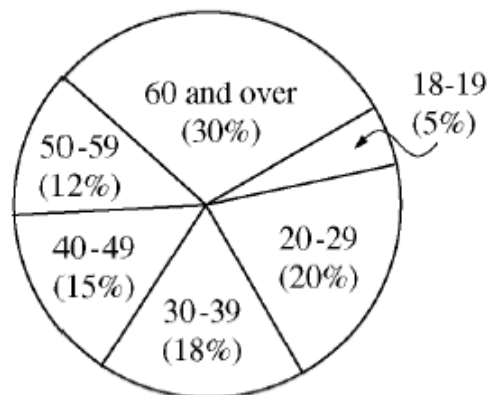
What value should the analyst use for r ?

32

What is the sum of the first 15 terms of an arithmetic sequence with 8 as the first term, and 13 as the second?

33

AGE DISTRIBUTION OF REGISTERED VOTERS IN WASHINGTON COUNTY



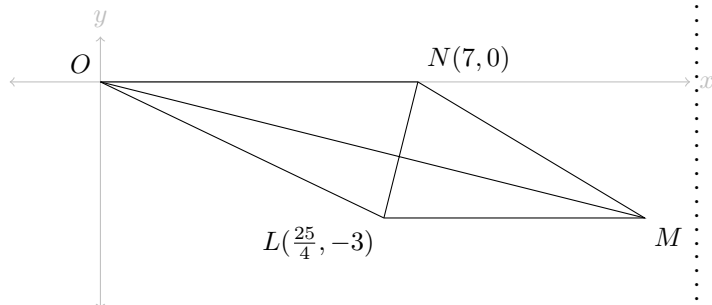
According to the graph above, if there are 6,000 registered voters aged 60 and over in Washington County, how many registered voters are under the age of 30?

34

If $x = 5$, what is the value of $\log_{\pi}(\pi^x)$?



35



In the diagram above, the diagonals of trapezoid $LMNO$ are perpendicular. What is the x coordinate of M ?

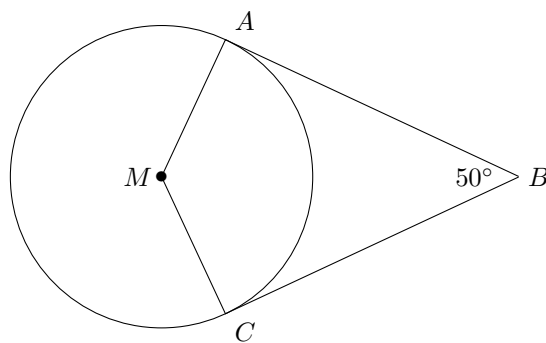
36

A partially filled toxic waste drum contains 15 gallons of chemical waste. A disposal tap is opened, and chemical waste flows into the drum at the rate of 12 gallons per minute. How many gallons of waste will be in the drum after 6 minutes?

37

In the xy -plane, the point $(-1, 14)$ lies on the graph of the function $h(x) = -2x^3 + jx + 10$. What is the value of j ?

38



In the figure above, M is the center of the circle and $\overline{AB}$ is tangent to circle M , as is $\overline{BC}$. If the circumference of the circle is 108, what is the measure of $\widehat{AC}$?

Answer Key

Calculator Off	
1	B
2	B
3	D
4	A
5	D
6	C
7	D
8	A
9	A
10	A
11	C
12	C
13	B
14	B
15	B
16	1 or $\frac{11}{3}$ or 3.67 or 3.66
17	$\frac{15}{8}$
18	20
19	1 or 9
20	40

.....

Calculator On	
1	C
2	C
3	B
4	B
5	A
6	C
7	C
8	B
9	D
10	A
11	C
12	D
13	D
14	D
15	D
16	C
17	C
18	B
19	D
20	C
21	A
22	D
23	C
24	B
25	C
26	C
27	A
28	B
29	D
30	B
31	1.05
32	585
33	5,000
34	5
35	12
36	87
37	2
38	39